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**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Inv. No. 337-TA-1478

RESPONDENTS' PRE-HEARING BRIEF

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TABLE OF ABBREVIATIONS

Abbreviation	Description
'359 Patent	U.S. Patent No. 10,642,359
'675 Patent	U.S. Patent No. 10,231,675
'677 Patent	U.S. Patent No. 10,945,677
'789 Patent	U.S. Patent No. 10,978,789
Oura	Respondents Ouraring Inc. and Oura Health Oy, collectively
Samsung	Complainants Samsung Electronics America, Inc. and Samsung Electronics Co., Ltd., collectively
Asserted Patents	The '359 Patent, '675 Patent, '677 Patent, and '789 Patent, collectively
POSITA	A person of ordinary skill in the art at the time of the claimed invention
PCB	Printed Circuit Board
ALJ	Administrative Law Judge
CAD	Computer-Aided Design
ID	Initial Determination
IPR	Inter Partes Review
LED	Light Emitting Diode
LEO	Limited Exclusion Order
MCU	Microcontroller Unit
PGR	Post Grant Review
PPG	Photoplethysmography
Tr.	Deposition Transcripts
U.S. / US	United States

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Abbreviation	Description
USPTO / PTO	United States Patent and Trademark Office
CSUCE	Complainant's Statement of Undisputed Claim Elements

TABLE OF SAMSUNG'S ASSERTED¹ CLAIMS

'675 Patent (claims 13 and 15)	
[13.pre]	A method comprising:
[13.a]	determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device;
[13.b]	acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body
[13.c]	activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;
[13.d]	detecting a biometric information of the user using the activated at least one second sensor;
[13.e]	during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and
[13.f]	in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.
[15]	The method of claim 13, further comprising: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information.

'677 Patent (claims 1-3 and 9)	
[1.pre]	A wearable electronic device comprising:
[1.a]	a motion sensor;
[1.b]	one or more optical sensors includes a light emitting unit and a light receiving unit; and
[1.c]	at least one processor configured to:
[1.d]	obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,
[1.e]	emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition,
[1.f]	obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user,

¹ Claim 6 of the '789 Patent is not "asserted" as a basis of alleged infringement, but is the basis of Samsung's allegations of domestic industry for the '789 Patent. Claim 1 of the '789 Patent is not asserted in any sense, yet is the independent claim on which claim 6 depends.

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'677 Patent (claims 1-3 and 9)	
[1.g]	obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and
[1.h]	refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.
[2.a]	The wearable electronic device of claim 1, wherein the at least one processor is further configured to: determine whether the wearable electronic device is attached to the user using the one or more optical sensors; and
[2.b]	perform the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user.
[3]	The wearable electronic device of claim 2, wherein the at least one processor is further configured to: perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.
[9.pre]	A wearable electronic device comprising:
[9.a]	a first sensor;
[9.b]	a second sensor; and
[9.c]	at least one processor configured to:
[9.d]	obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,
[9.e]	activate the second sensor based at least in part of the movement,
[9.f]	obtain, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and
[9.g]	refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range.

'359 Patent (claims 1 and 4)	
[1.pre]	A wearable device comprising:
[1.a]	a motion sensor;
[1.b]	a biological sensor; and
[1.c]	a controller configured to:
[1.d]	obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor,
[1.e]	determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, and
[1.f]	perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition.
[4]	The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.

'789 Patent (claims 6, 7, and 9)	
[1.pre]	A portable communication device comprising:
[1.a]	a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;
[1.b]	a wireless communication circuitry; and
[1.c]	a first printed circuit board (PCB) including:
[1.d]	a first PCB portion;
[1.e]	a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion;
[1.f]	a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion;
[1.g]	a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and
[1.h]	a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.
[6.pre]	The portable communication device of claim 1, further comprising:
[6.a]	a second PCB on which the wireless communication circuitry is located,
[6.b]	wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.
[7.pre]	A portable communication device comprising:
[7.a]	a first electronic component;
[7.b]	a second electronic component; and
[7.c]	a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:
[7.d]	a first rigid PCB portion;
[7.e]	a first flexible PCB portion in contact with the first rigid PCB portion;
[7.f]	a second rigid PCB portion in contact with the first flexible PCB portion; and
[7.g]	a second flexible PCB portion in contact with the second rigid PCB portion,
[7.h]	wherein each of the first rigid PCB portion and the second rigid PCB portion includes: a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion;
[7.i]	a first conduction layer located on or above the first insulating layer;

'789 Patent (claims 6, 7, and 9)	
[7.j]	a second insulating layer located on or above the first conduction layer and having a first thickness;
[7.k]	a second conduction layer located on or above the second insulating layer; and
[7.l]	a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion, and
[7.m]	wherein each of the first flexible PCB portion and the second flexible PCB portion includes: a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;
[7.n]	a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of first rigid PCB portion and the first conduction layer of the second rigid PCB portion;
[7.o]	a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;
[7.p]	a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and
[7.q]	a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.
[9]	9. The portable communication device of claim 7, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

I. INTRODUCTION

Founded in 2013, Oura was instrumental in creating the smart ring market. Oura has released five generations of the Oura Ring since 2015, most recently launching its latest generation (the Ring 5) on May 28, 2026. Oura's innovation has been widely recognized across the industry, and it continues to innovate to provide cutting edge smart rings to consumers.

Samsung was a comparatively later entrant into the smart ring market. Prior to launching its first Galaxy Ring in July 2024, Samsung preemptively attacked Oura's intellectual property by first filing an unsuccessful declaratory judgment action against Oura in May, 2024, followed by twelve post-grant proceedings in the Patent Office. These post-grant proceedings were overwhelmingly unsuccessful, and Oura's intellectual property has withstood multiple challenges. As Samsung's barrage dissipated, Oura sued Samsung in the ITC to protect its intellectual property and domestic industry from Samsung's importation of infringing smart rings, a proceeding that is currently pending. *See* Inv. No. 337-TA-1468.

By all indications, the present Investigation was filed as a retaliatory suit in the ongoing dispute between the parties. But, as might be unsurprising from a later market entrant, Samsung asserts weak patents. The Oura Ring Gen. 1, released in the United States in 2015, anticipates or renders obvious three of the four asserted patents and has closely similar features as are now accused by Samsung of infringement. Additionally, three of the four asserted patents are directed to abstract concepts applicable to generic wearable devices and are therefore patent ineligible under 35 U.S.C. § 101. Published patents, patent applications, and publications furthermore act as prior art to anticipate or render obvious each of the asserted patents. And given differences between claims written to generic wearable devices and the accused Oura Gen. 4 ring, Samsung cannot carry its burden to show infringement; it did not even assert infringement against Oura's alternative Ring 5 and [REDACTED] rings.

Samsung's economic domestic industry case fares no better. Samsung frames its EDI investments in terms of the Galaxy Watches and Ring for the '359, '677, and '675 Patents, and smartphones for the '789 Patent (but not the Galaxy Watches or Ring). But Samsung claims an EDI for these categories based on investments made by SEA and under only subsections (A) (plant and equipment) and (B) (labor and capital). To quantify its EDI, Samsung utilizes a flawed allocation based on overstated investments by SEA. SEA is the U.S. sales arm for SEC. SEA's work only comes in for the [REDACTED]

[REDACTED]. Samsung admits that the investments it claims for EDI are [REDACTED]

[REDACTED]. And those efforts are very limited; for example, SEA [REDACTED] before their U.S. launch. Most of these [REDACTED] are related to [REDACTED] which means they do not apply to the DI Products that are [REDACTED], such as all DI Rings and about [REDACTED] of DI Watches. Thus, Samsung will not be able to prove a sufficient domestic industry.

In view of the myriad flaws in Samsung's case, following the evidentiary hearing, Oura expects to argue that no violation of Section 337 has occurred.

II. BACKGROUND

A. Procedural History

On December 15, 2025, Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. filed a complaint under Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337 ("Section 337"), against the Oura Respondents. (EDIS Doc. ID 866474). Samsung filed a supplement to its complaint on December 31, 2025 (EDIS Doc. ID 867722), and an amended complaint on January 5, 2026. (EDIS Doc. ID 867832).

On January 14, 2026, the Commission instituted this Investigation based on allegations of violations of Section 337 the importation into the United States, the sale for importation, or the sale within the United States after importation of certain products identified in paragraph (2) by reason of infringement of one or more of claims 1, 2, 4, 7, 8, 10, 14, 15, and 20 of the '359 patent; claims 1-7, 9-14, and 16-20 of the '677 patent; claims 13-20 of the '675 patent; and claims 7-10 of the '789 patent. (EDIS Doc. ID 868875).

On July 29, an Initial Determination was issued terminating all claims except claims 1 and 4 of the '359 patent; claims 1-3 and 9 of the '677 patent; claims 13 and 15 of the '675 patent, and claims 7 and 9 of the '789 patent. (EDIS Doc. ID 889677).

The evidentiary hearing is scheduled for September 16-18, 21, and 22, 2026, with a target date set for May 17, 2027. (EDIS Doc. ID 881268).

B. The Parties

1. The Samsung Complainants

Complainant Samsung Electronics Co., Ltd. ("SEC") is a South Korean corporation and is the parent company of a multinational conglomerate operating under the name "Samsung." *See, e.g.,* RX-0390.4.

Complainant Samsung Electronics America, Inc. ("SEA") is a wholly owned subsidiary of SEC, incorporated under the laws of New York, and headquartered in Englewood Cliffs, New Jersey, whose business relates to US sales and marketing of electronic devices in the US, including the accused Samsung Galaxy Ring. Complaint ¶¶9, 11 (EDIS Doc. ID 866474).

Together, SEC and SEA are referred to herein as "Complainants" or "Samsung."

2. The Oura Respondents

Respondent Ouraring Inc. is a Delaware corporation headquartered in San Francisco, California, with research, development, and testing facilities in San Diego and Los Angeles. *See,*

e.g., RX-0318, RX-0314, RX-0213C. Ouraring Inc. is a wholly owned subsidiary of Respondent Oura Health Oy, a Finnish corporation founded in 2013 in Oulu, Finland. RX-0318. In January 2026, as part of Oura's continued efforts to shift its center of gravity to the United States, Oura formed a new ultimate parent: Oura Inc., a Delaware corporation. *Id.*

Together, Ouraring Inc. and Oura Health Oy are referred to herein as "Oura."

C. Samsung's Asserted Patents

Samsung asserts four patents in this Investigation. While the patents will be discussed in greater detail in relation to the specific disputes between the parties, a brief introduction is as follows:

1. '675 Patent

U.S. Patent No. 10,231,675 is entitled "Method For Detecting Biometric Information And Electronic Device Using the Same," and was issued on March 19, 2019. JX-0003. The '675 Patent claims priority to Korean Patent Application KR 10-2015-0113972, filed August 12, 2015. *Id.* The '675 Patent relates generally to improving electronic devices know in the art that are "equipped with biometric sensors (e.g., health care sensors) [that] measure biometric information (e.g., health data) ... includ[ing] blood pressure, a heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen." JX-0003, 1:29-35.

A POSITA for the '675 Patent would have had at least a four-year degree in electrical engineering, mechanical engineering, biomedical engineering, computer engineering, or a related field of study, or equivalent experience, and at least two years of experience in academia or industry studying or developing wearable monitoring devices such as non-invasive biosensors. Additional work experience can make up for a lack of education, and additional education can make up for a lack of work experience. A POSITA could have also worked as part of a

multidisciplinary team and would have drawn upon not only his or her skills, but also taken advantage of specialized skills of others on the team. JDX-0001C ¶¶70-71.

2. '677 Patent

U.S. Patent No. 10,945,677 is entitled “Method For Detecting Biometric Information And Electronic Device Using the Same,” and was issued on March 16, 2021. JX-0002. The '677 Patent is a continuation of U.S. Patent Application No. 15/173,099, filed on June 3, 2016—which is now the '675 Patent—and claims priority to Korean Patent Application KR 10-2015-0113972, filed August 12, 2015. *Id.* The '677 Patent also relates generally to improving electronic devices known in the art that are “equipped with biometric sensors (e.g., health care sensors) [that] measure biometric information (e.g., health data) ... includ[ing] blood pressure, a heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen.” JX-0002, 1:32-37.

A POSITA for the '677 Patent would have had the same level of skill as for the '675 Patent. JDX-0001C ¶¶70-71.

3. '359 Patent

U.S. Patent No. 10,642,359 is entitled “Wearable Biosignal Interface and Method Thereof,” and was issued on March 5, 2020. JX-0001. The '359 Patent claims priority to Korean Patent Application KR 10-2013-0098476, filed August 20, 2013. *Id.* The '359 Patent relates to the field of “wearable-type mobile devices” for “operating an external device.” JX-0001, Abstract, 1:28-35. As disclosed by the '359 Patent, wearable mobile devices can recognize gestures “by an accelerometer,” which “may be used to operate the wearable mobile device.” JX-0001, 1:34-35. The '359 Patent purports to provide a solution in order to “recognize[] an intention of a user with a greater precision,” specifically using a “controller configured to determine a validity” of the movement signal. JX-0001, 1:43-55.

A POSITA for the '359 Patent would have had the same level of skill as for the '675 Patent. JDX-0001C ¶¶70-71.

4. '789 Patent

U.S. Patent No. 10,978,789 is entitled “Electronic Device Including High-Frequency Transmission Circuit,” and was issued on April 13, 2021. JX-0004. The '789 Patent claims priority to U.S. Patent Application No. 15/621,687, filed June 13, 2017. *Id.* The '789 Patent “relates generally to a high-frequency transmission circuit included in an electronic device” and “to a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.” JX-0004, 1:19-24. Specifically, the '789 Patent attempts to address the problem of replacing traditional coaxial cables with integrated PCB-based transmission lines, including flexible PCBs (FPCBs) and rigid PCBs. JX-0004, 1:31-65.

A POSITA for the '789 Patent would also have had the same level of skill as for the '675 Patent. JDX-0001C ¶¶70-71. The skill level of at least a bachelor's degree in electrical engineering or a comparable field of study and approximately two years of professional experience working with printed circuit boards; or a four-year degree in electrical engineering, computer engineering, electronics engineering, or a related field of study, or equivalent experience with at least two years of academic or industry experience in the design or analysis of printed circuit boards, RF transmission lines, antenna feed structures, and/or wireless communication subsystems for portable electronic devices would also be consistent with this definition. JDX-0001C ¶¶3114-17; JDX-0007C ¶¶57-60; JDX-0001C ¶¶77-80.

D. The Products at Issue

1. Oura's Products

The parties do not dispute that Oura's Gen. 4 ring is at issue in this Investigation. Oura's Gen. 4 ring is a wearable electronic device for detecting biometric information. Specifically, it is

a smart ring that uses three types of sensors (a temperature sensor, an optical/photoplethysmography (PPG) sensor, and an accelerometer) to gain information about a user and the user's activities in order to provide information that may be useful, such as sleep state and activity information.

The parties dispute whether Oura's Ring 5 (also referred to as "[REDACTED]" or Oura's "Gen. 5 ring") and [REDACTED] smart rings are eligible for adjudication in this Investigation. While the Ring 5 and [REDACTED] are unaccused by Samsung, a respondent may put forth unaccused products for adjudication, and Oura more than meets its burden to show that the [REDACTED] and [REDACTED] rings are eligible for adjudication. *See Certain Human Milk Oligosaccharides and Methods of Producing the Same*, Inv. No. 337-TA-1120, Comm'n Op. at 18 (June 8, 2020) (setting forth four factors to determine the eligibility for adjudication of non-accused products). Thus, the Ring 5 and [REDACTED] rings should also be adjudicated alongside Oura's Gen. 4 ring in this Investigation.

a) *Oligosaccharides* Factor 1: Ring 5 and [REDACTED] are Within the Scope of the Investigation

There can be no reasonable dispute that Ring 5 and [REDACTED] are within the scope of this Investigation. As set forth by the Notice of Institution, the scope of this investigation encompasses "wearable devices, specifically wearable biosignal interfaces and an operation method thereof as well as wearable electronic devices for detecting biometric information." Notice of Institution of Investigation (Jan. 14, 2026) (EDIS Doc. ID 868875).

Just like the Oura Gen. 4 ring, Ring 5 and [REDACTED] are wearable electronic devices for detecting biometric information, and thus within the scope of this Investigation:

Oura Gen. 4 ring	Ring 5	
Biometric Sensors: • [REDACTED] [REDACTED] [REDACTED] [REDACTED]	Biometric Sensors: • [REDACTED] [REDACTED] [REDACTED] [REDACTED]	Biometric Sensors: • [REDACTED] [REDACTED] [REDACTED] [REDACTED]
RX-0238C at OURA1468_00029607-10	RX-0239C at OURA1468_00023858-59	RX-0237C at OURA1468_00255137

Additionally, hearing testimony from Mr. Larson, Mr. Christophy, and/or Dr. Mortazavi will establish that Ring 5 and [REDACTED] are wearable electronic devices for detecting biometric information. JDX-0007C ¶¶130-31, 382-83, 552-53, 652.

b) *Oligosaccharides* Factor 2: Ring 5 and [REDACTED] Have Been Imported

Both Ring 5 and [REDACTED] have been imported. Oura made available to Samsung physical samples of Ring 5 and [REDACTED] in the United States on March 31, 2026, and these samples had previously been imported. RX-0229 (May 29, 2026 Production Email).

Furthermore, the parties have stipulated that Ring 5 (as “[REDACTED] and [REDACTED] have been imported. JX-0074C (importation stipulation) at 1, Exhibit A.

Additionally, Mr. Christophy may testify that Ring 5 and [REDACTED] have been imported, including between February 26 and March 20, 2026 for Ring 5 and by February 9, 2026 for [REDACTED] and that Oura does not yet produce ring products (including Ring 5 and [REDACTED] in the United States.

c) *Oligosaccharides* Factor 3: Ring 5 and [REDACTED] are Sufficiently Fixed in Design

Both Ring 5 and [REDACTED] are fixed in design. As Mr. Christophy is expected to testify, Ring 5 went on sale in the United States on May 28, 2026 with shipments to consumers starting less than a week later by June 4, 2026. Mr. Christophy is also expected to testify that Ring 5 had already entered mass production [REDACTED]. Even before that, the relevant features of Ring 5 were substantially fixed as of [REDACTED] which is expected to be supported by testimony by Mr. Christophy. RX-0236C.

[REDACTED] is also fixed in design for purposes of this inquiry. The relevant question is not whether the products-at-issue are final in every possible respect, but rather, whether the products “are sufficiently fixed for adjudication.” *Certain Photovoltaic Trunk Bus Cable Assemblies and Components Thereof*, Inv. No. 337-TA-1438, ID at 105-06 (Feb. 6, 2026) (EDIS Doc. ID 873421). As the experts of both parties agree, the ’789 Patent concerns a “rigid-flex PCB” design. JDX-0003C ¶¶40-41, 47; JDX-0007C ¶¶50-51. Thus, the inquiry focuses most particularly on the fixedness of the printed circuit board (“PCB”) design, which in the case of [REDACTED] is entirely fixed.

As Mr. Christophy and Mr. Larson are expected to testify, Oura started the development of [REDACTED] prior to [REDACTED] (RX-0240C), built prototypes [REDACTED] (*id.*), and by [REDACTED], Oura had reached “[REDACTED]” with expected launch of [REDACTED] in [REDACTED]. While not yet commercially released [REDACTED]

[REDACTED] has been fixed in design since at least [REDACTED] with respect to its relevant PCB, the component at issue for the '789 patent. RX-0241C at OURA1468_00255164-65, -180 (providing "[REDACTED]
[REDACTED]
[REDACTED]"); RX-0242C at OURA1468_00255210 (noting [REDACTED]
[REDACTED]). [REDACTED] has been described in [REDACTED]
[REDACTED], and other technical specifications. RX-0237C.

Indeed, further evidencing the fixed nature of the [REDACTED] rings, multiple physical [REDACTED] rings—both as new and torn down—were offered to Samsung for inspection on March 31, 2026. RX-0223 (Production Email); *Certain Audio Players and Controllers, Components Thereof, and Products Containing Same*, Inv. No. 337-TA-1191, Order No. 19 at 4 (Sept. 17, 2020) (EDIS Doc. ID 719871) (providing “physical devices” shows fixedness of design).

That certain unrelated aspects of [REDACTED] does not show that [REDACTED] is insufficiently fixed in design. *See Certain Electrical Connectors and Cages*, Inv. No. 337-TA-1241, ID at 35-36 (noting that products can be sufficiently fixed in design even without completing “testing necessary to be released to the market”); *Certain Video Security Equipment and Systems, Related Software, Components Thereof, and Products Containing Same*, Inv. No. 337-TA-1281, ID at 22 (Oct. 24, 2022) (EDIS Doc. ID 784112) (indicating that the fact “a change could be made in the future . . . is not sufficient for a finding that a redesign cannot be adjudicated”).

Consistent with the evidence above, Mr. Christophy and Mr. Larson may testify to the development status of the Ring 5 and [REDACTED] as of the time of the devices' importation and

production to Samsung, and provide further testimony showing that these devices were fixed in design as of this time.

Thus, given that Ring 5 is on sale with a completely final design, and [REDACTED] is final with respect to its relevant PCB components (*i.e.*, the “claimed features”), both Ring 5 and [REDACTED] “are sufficiently fixed for adjudication” with regard to the ’789 Patent. *See Certain Photovoltaic Trunk Bus Cable Assemblies*, Inv. No. 337-TA-1438, ID at 105-06.

d) *Oligosaccharides* Factor 4: Sufficient Discovery Regarding Ring 5 and [REDACTED] has been Provided

The fourth *Oligosaccharides* factor (regarding the sufficiency of discovery) does not present a high bar: “the key question is whether Respondents provided ‘sufficient (not extensive) fact and expert discovery to put [complainant] on notice of that [product] and its relevant features.’” *Certain Outdoor and Semi-Outdoor Electronic Displays*, Inv. No. 337-TA-1331, ID at 10.

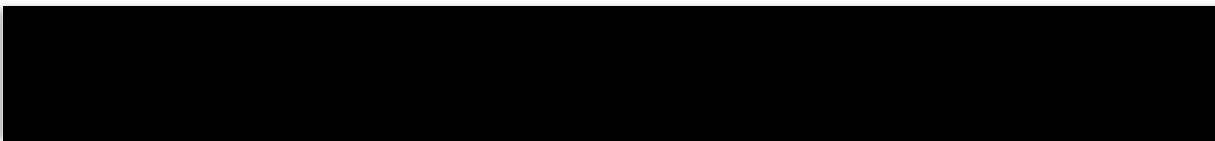
Oura began producing technical documents for Ring 5 on January 26, 2026, only 10 days after this Investigation was instituted. RX-0215. Oura produced the source code, including firmware and application code, for Ring 5 and [REDACTED] on February 11, 2026. RX-0219. Oura began producing technical documents for [REDACTED] a month later on February 25, 2026, after having identified Ring 5 (as “[REDACTED] and [REDACTED] as domestic industry products that would be relied upon by Oura in Investigation No. 337-TA-1468. RX-0218C at 10; RX-0220 (Feb. 25, 2026 1468 Investigation Production Letter). On March 20, 2026, Samsung deposed Oura’s VP of Hardware Engineering extensively on technical details of both Ring 5 and [REDACTED] including the relevant PCB structure. RX-0224C (Christophy Dep. Tr.) at 61:10-15 (confirming that Ring 5 is expected to be released in spring 2026); 64:25-65:3 (confirming that the release date of [REDACTED]); 62:24-64:19 (explaining the hardware and sensors of Ring 5); 65:10-72:10 (discussion of [REDACTED]

hardware and design); 131:4-146:15 (testifying regarding the CAD and hardware of Ring 5); 146:16-150:7 (manufacturing process of Ring 5); 150:8-15 (testimony on the PCB structure of Ring 5); 154:9-155:15 (discussing the [REDACTED] regarding Ring 5); 157:15-178:17 (testifying on physical design of [REDACTED] Ring, including hardware components and PCB structure).

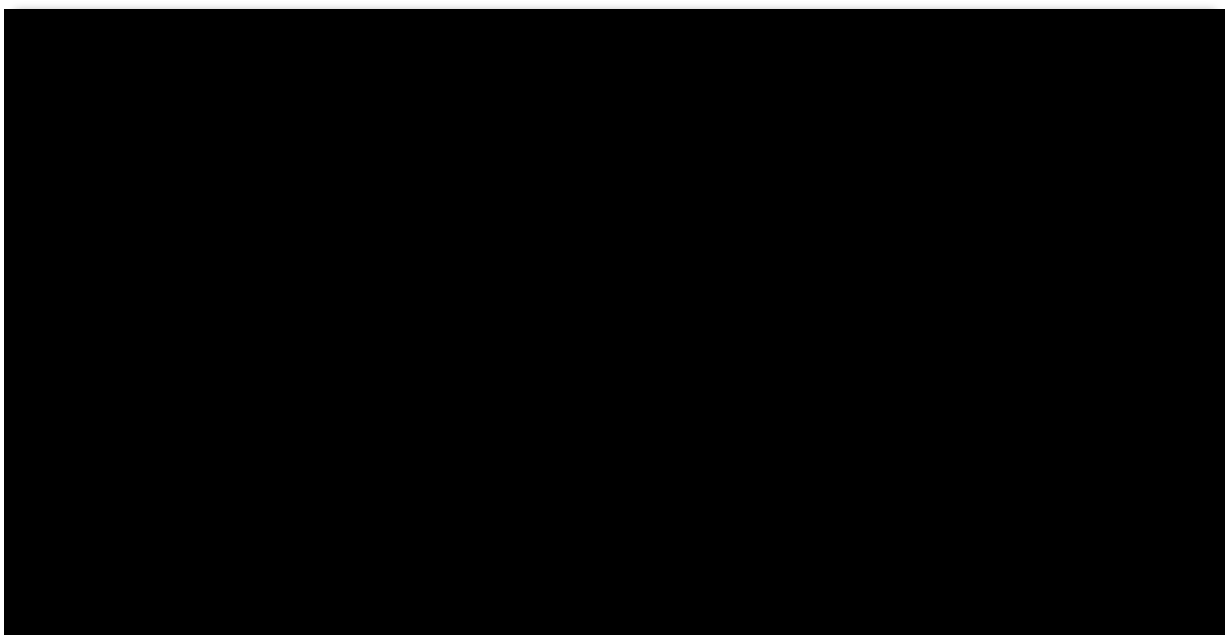
On April 29, 2026, Oura's interrogatory responses identified Ring 5 and [REDACTED] as products for which Oura would seek a determination of noninfringement in this Investigation. RX-0226C (April 29 Interrogatory Responses) at 12-13, 79-81. Oura then continued to supplement its interrogatory responses to identify further relevant documents relating to Ring 5 and [REDACTED] and noted Samsung's clear failure of proof with regard to any product besides the Oura Gen. 4 ring. RX-0230C (May 13, 2026 Supplemental Interrogatory Responses) at 13, 16, 18, 21, 23; RX-0231C (May 27, 2026 Supplemental Interrogatory Responses) at 82-84. On May 15 and 19, 2026, Samsung deposed two Oura corporate witnesses designated on technical details of Accused Products, including Ring 5 and [REDACTED] JDX-0025C (Christophy Dep. Tr.) (May 15, 2026); JDX-0026C (Larson Dep. Tr.) (May 19, 2026). On May 21, 2026, Samsung deposed the senior director of sourcing, Mr. Vincent, who testified extensively about the development process of Ring 5 and [REDACTED] along with the release dates for both rings. JDX-0028C. In fact, Oura has complied with every discovery request from Samsung regarding Ring 5 and [REDACTED]. Not once did Samsung ever suggest that Oura's production, whether related to technical documents, native CAD files, product images, teardown images, physical samples, components, build of materials, or deposition witnesses was lacking in any capacity.

The discovery provided by Oura (well over two hundred technical documents spanning several thousand pages as well as source code for Ring 5 and [REDACTED]) was more than sufficient

for Samsung to be on notice of the relevant features of these products. Oura produced on February 25, 2026 the [REDACTED] schematic:

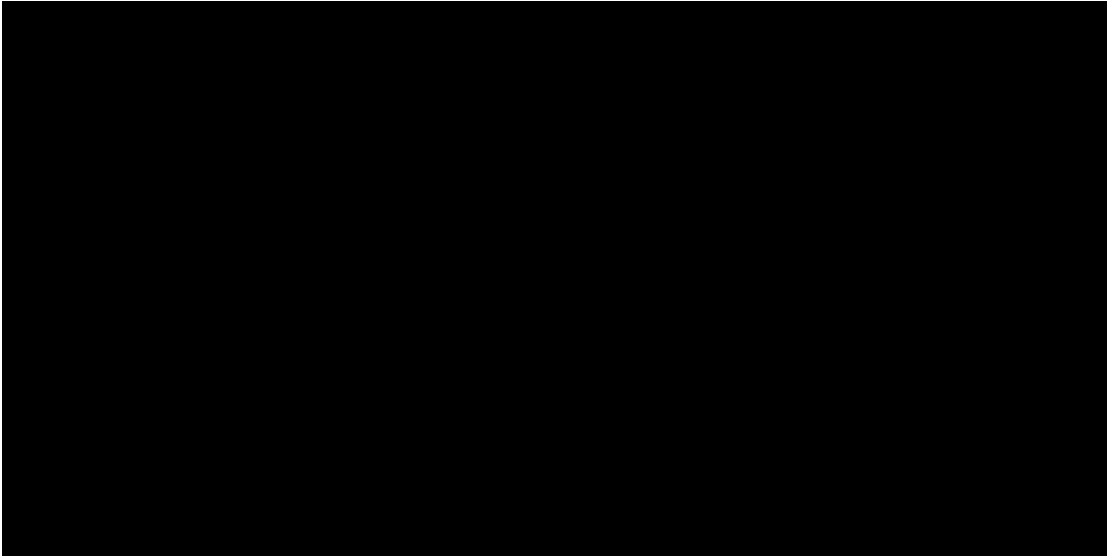


RX-0243C; *see also* RX-0244C (similar schematics noting [REDACTED]). Oura also produced the PCB stackup diagram for Ring 5 in the same production:

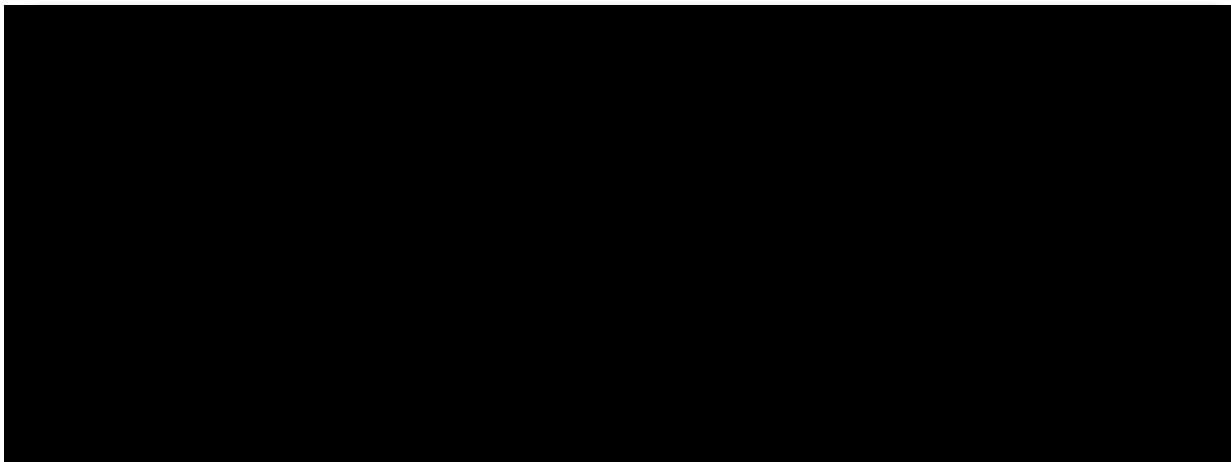


RX-0245C.

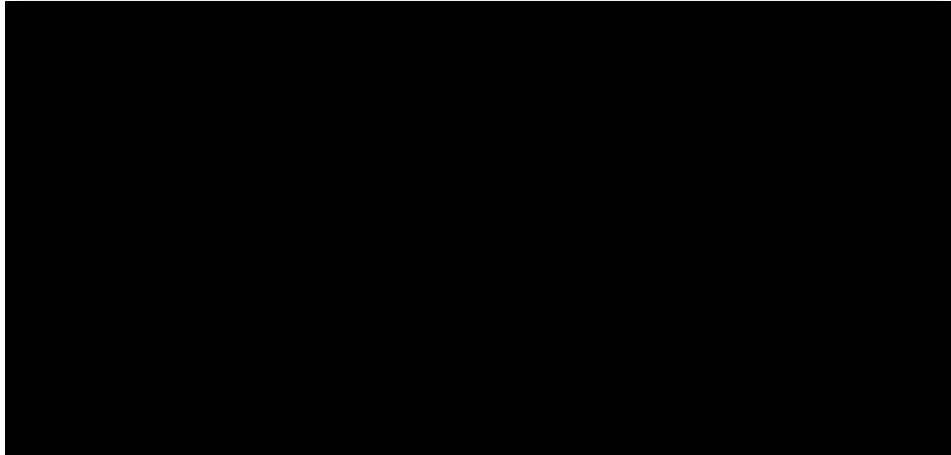
On March 17, 2026, Oura produced teardown photos of Ring 5 and [REDACTED]



RX-0163C (OURA1468_00254628) ([REDACTED] left); RX-0162 (OURA1468_00254646) (Ring 5, right). Within this production, Oura also produced the [REDACTED] PCB schematic and PCB stackup diagram:

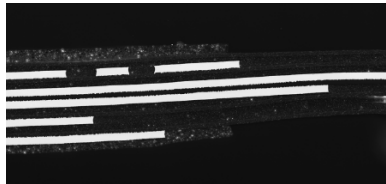


RX-0248C (OURA1468_00255339 – 341).

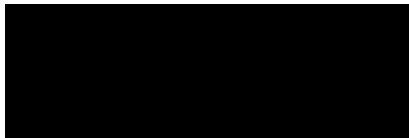


RX-0249C (OURA1468_00255363 – 364).

Furthermore, on March 31, 2026, Oura made available for inspection physical samples of Ring 5 (OURA_PHYS_0000006) and [REDACTED] (OURA_PHYS_0000007). RX-0223. And on May 29, 2026, Oura produced extensive Ring 5 and [REDACTED] testing photographs/documents, including cross-sectional views of their respective PCBs:



RX-0250 (Ring 5)



RX-0251C ([REDACTED])



RX-0252 (Ring 5)

[REDACTED]

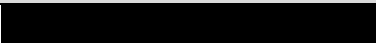
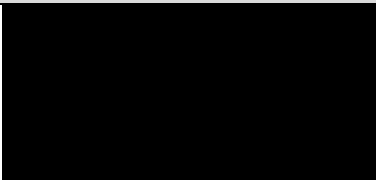
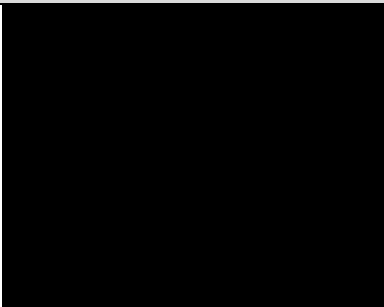
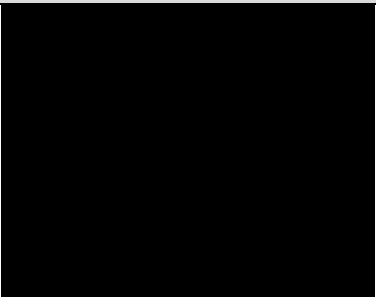
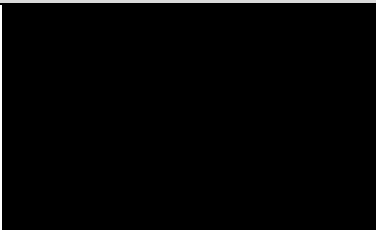
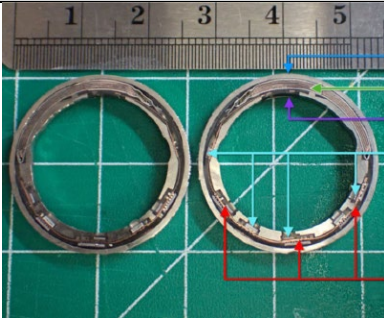
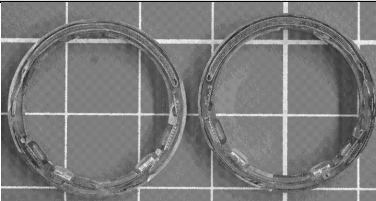
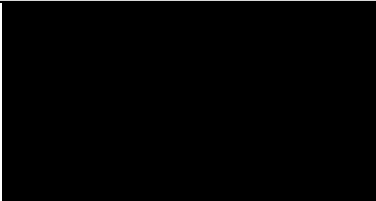
RX-0253C ([REDACTED] Dr. Mortazavi was then able to opine on noninfringement (JDX-0007C ¶¶652-80) and answer deposition questions (RDX-0032C at 532:15-569:20) regarding Ring 5 and [REDACTED]

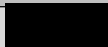
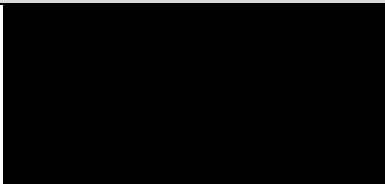
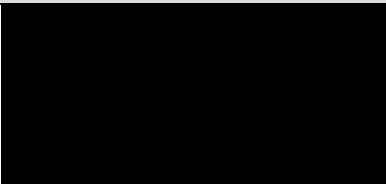
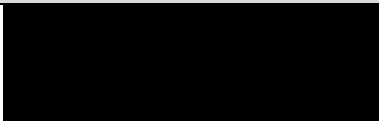
Even Samsung's expert on the '789 Patent, Dr. Warren, provided testimony on redirect examination from Samsung's counsel that demonstrated that he understood the structure of Ring 5 and [REDACTED] RDX-0041C (Warren Dep. Tr.) at 206:22-207:4; *see also id.* at 35:14-15, 41:5-10, 135:7-9, 200:10-215:2 (further testimony concerning Ring 5 and [REDACTED])²

The quantity and timing of discovery more than exceeds Commission precedent for providing sufficient discovery. *See Oligosaccharides*, Inv. No. 337-TA-1120, Comm'n Op. at 23, n. 25; *Certain Mobile Devices with Multifunction Emulators*, Inv. No. 337-TA-1170, ID at 9-10 (March 16, 2021) (EDIS Doc. ID 738549).

Indeed, the discovery produced was of the same type as relied upon by Samsung's expert with regard to alleged infringement of the '789 Patent by the Oura Gen. 4 ring. A brief comparison of the principal evidence relied upon by Dr. Warren when opining on alleged infringement of the Oura Gen. 4 ring to the evidence produced by Oura for Ring 5 and [REDACTED] follows:

² This surprise testimony on redirect examination should be struck—*see* Oura's Motion to Strike Deposition Testimony (EDIS Doc. ID 887862) (Motion No. 1478-008)—but nonetheless demonstrates that sufficient discovery had been provided.

PCB Schematics		
Oura Gen. 4 ring	Ring 5	
 RX-0254C (relied upon at JDX-0003C (Warren Inf. Rep.) ¶¶71, 104, 111, 122, 124, 128, 132)	 RX-0243C	 RX-0249C
PCB Stackup Diagram		
Oura Gen. 4 ring	Ring 5	
 RX-0255C (relied upon at JDX-0003C ¶¶74, 107, 112, 113, 115, 137-38, and generally <i>passim</i>)	 RX-0245C	 RX-0248C
Cross-Section Teardowns		
Oura Gen. 4 ring	Ring 5	
 RX-0256 (relied upon at JDX-0003C ¶105)	 RX-0162	 RX-0163C

Scanning Electron Microscope Images		
Oura Gen. 4 ring	Ring 5	
 RX-0257C (relied upon at JDX-0003C ¶117, 119, 129, and generally <i>passim</i>)	 RX-0250	 RX-0251C

Indeed, Oura proactively provided extensive scanning electron microscope images of the PCBs of Ring 5 and  (e.g., RX-164C, RX-0253C, RX-0259C, RX-0165C, RX-0251C, RX-0166C, RX-0169, RX-0252, RX-0167, RX-0250, RX-0168), whereas it was Samsung's expert who obtained teardown photos for the Oura Gen. 4 ring.

Likewise, for the '359, '675, and '677 patents, Samsung's principle evidence of alleged infringement was the source code of the Oura Gen. 4 ring—Oura produced the source code, including firmware and application code, for Ring 5 and  on February 11, 2026. RX-0219.

Oura did not withhold discovery, and should Samsung have believed that there were deficiencies, it could have raised any such concerns. *See Oligosaccharides*, Inv. No. 337-TA-1120, Comm'n Op. at 23.

2. Samsung's Asserted Domestic Industry Products

For the '359 Patent, '677 Patent, and the '675 Patent, the asserted Domestic Industry Products are Samsung's Galaxy Watch4, Watch5, Watch6, Watch7, Watch8, Watch Ultra, and Galaxy Ring. These are wearable devices that are either worn on a user's wrist (in the case of the Galaxy Watch products) or finger (in the case of the Galaxy Ring).

For the '789 Patent, the asserted Domestic Industry Products are the Samsung Galaxy S22, S22+, S22 Ultra, S23, S23+, S23 Ultra, S24, S24+, S24 Ultra, S25, S25+, and S25 Edge. These are not wearable devices, but rather smartphones.

E. Claim Construction

There are no agreed constructions. Order No. 10 at 1.

The constructions of the disputed claim terms in claims 13 and 15 of the '675 Patent are as follows:

Claim Term	Adopted Construction
“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body” (claim 13)	Not indefinite. The second time period will not begin if there is no wear determination, and “shortly before” the first time period means that the second time period can start before the first time period ends. What qualifies as “shortly before” or “immediately after” will be addressed with the parties’ infringement, domestic industry, and prior-art invalidity arguments.
“third period of time” (claim 15)	Not indefinite; no further construction.

The parties previously disputed a construction concerning claim 14 of the '359 Patent (*see id.* at 13-21), but claim 14 has been terminated from the Investigation. (EDIS Doc. ID 889677).

III. JURISDICTION

Oura does not contest the Commission’s statutory authority under Section 337 to investigate alleged unfair acts in importation. Oura has stipulated to the importation into the United States of its Gen. 4 ring as well as the Ring 5 and [REDACTED] rings. JX-0074C.

IV. '675 PATENT

A. Oura's Products Do Not Infringe the Asserted Claims

As set forth in Oura's opposition to Samsung's motion for summary determination (EDIS Doc. ID 889195), Oura contends that Samsung will not be able to carry its burden to show infringement of the '675 Patent by any product at issue (Oura's Gen. 4, Ring 5, and [REDACTED] rings).

1. Oura's Gen. 4 Ring Does Not Infringe the Asserted Claims

First, Samsung will not be able to show that the Oura Gen. 4 ring "inactivat[es] the at least one second sensor" in response "identifying that the motion of the electronic device included in the second information is more than a threshold motion," as required by claim element [13.f]. As Dr. Welch must admit upon cross-examination, the state transition alleged by Samsung to meet this limitation (from the [REDACTED]) at most [REDACTED] [REDACTED] of the Oura Gen. 4 ring—thereby potentially [REDACTED] [REDACTED]—but the [REDACTED]. Thus, as a result of this state transition, the optical sensor does not "inactivat[e] the at least one second sensor."

Second, Samsung's expert, Dr. Welch, cannot provide credible opinions sufficient to sustain Samsung's burden of proof because he has contradictory opinions with regard to invalidity in view of Oura's Gen. 1 ring, which has substantially similar state transitions as are accused. Specifically, Dr. Welch opines that the Oura's Gen. 1 ring's transition from the [REDACTED] state (or potentially [REDACTED] state) does not meet the requirements of claim limitation [13.c], whereas the Gen. 4 ring's transition from the [REDACTED] state (or potentially [REDACTED] state) does—a contradictory opinion, given that, as Dr. Mortazavi and Mr. Larson will testify, the [REDACTED]. JDX-0007C ¶¶154, 191. Dr. Welch's invalidity opinions undermine his infringement opinions, and he will not be able to help Samsung carry its burden of proof.

2. Ring 5 and [REDACTED] Do Not Infringe the Asserted Claims

Samsung has put forth no contention or evidence of infringement by these rings and accordingly cannot carry its burden of proof. *See Certain Crafting Machines and Components Thereof*, Inv. No. 337-TA-1426, ID at 40 (Jan. 21, 2026) (Cheney, Chief ALJ) (EDIS Doc. ID 869643) (a complainant cannot carry its burden of proof on redesigned products without evidence of infringement). Dr. Mortazavi and Mr. Larson will testify that although Ring 5 and [REDACTED] have broadly similar functionality to Oura's Gen. 4 ring, the software and hardware functionality is not the same; given this, and given the lack of any theory of representativeness, Samsung will not be able to carry its burden of proof for these reasons as well.

B. Claims 13 and 15 of the '675 Patent Are Invalid in View of the Prior Art

There are five asserted grounds of prior art-based invalidity that render invalid claims 13 and 15 of the '675 Patent:

Invalidity Ground	Basis	Applicable Claims
Ground 1	Anticipation in View of Oura Ring. Gen 1	13, 15
Ground 2	Anticipation in View of Hong	13, 15
Ground 3	Obviousness in View of Hong and Utter II	13, 15
Ground 4	Anticipation in view of Utter II	13
Ground 5	Obviousness in View of Utter II and Hong	13, 15

These five grounds of invalidity are explained in detail herein.

1. Overview of Prior Art References

The Oura Ring Gen. 1 was Oura's first generation of smart ring, and was a wearable device worn on a user's finger. JDX-0001C ¶¶1350-1353. This smart ring used physiological data signals to calculate deep sleep, light sleep, REM sleep, and awake states, including by using an optical sensor, an accelerometer, and a temperature sensor. RX-0023. As will be established in Section

IV.B.2, *infra*, the Oura Ring Gen. 1 was in public use and/or available to the public at least prior to August 12, 2015, and accordingly qualifies as prior art to the '675 Patent under at least 35 U.S.C. § 102(a).

U.S. Patent No. 8,948,832 to Hong ("Hong") (RX-0072) was filed on May 30, 2014, and issued February 3, 2015, and therefore qualifies as prior art to the '675 patent under at least 35 U.S.C. § 102(a)(1). Hong discloses a "wearable fitness monitoring device" (and specifically a wrist-worn biometric monitoring device) including "a motion sensor and a photoplethysmographic (PPG) sensor," wherein "[t]he PPG sensor includes (i) a periodic light source, (ii) a photo detector, and (iii) circuitry determining a user's heart rate from an output of the photo detector." JDX-0001C ¶¶2368-70 (citing RX-0072).

U.S. Patent Application Publication No. 2015/0186609 to Utter ("Utter II") (RX-0071) was filed on February 28, 2014, and published July 2, 2015, and therefore qualifies as prior art to the '675 patent under at least 35 U.S.C. § 102(a)(1). Utter II discloses a "wireless data capable strapband" that includes sensors "for generating motion signals in response to body motion (*e.g.*, accelerometry), for generating force signals in response to force exerted by a body portion, for generating biometric signals indicative of biometric activity in the body (*e.g.*, heart rate, respiration, arousal), and for generating location signals based on user location." JDX-0001C ¶¶2368-70 (citing RX-0071).

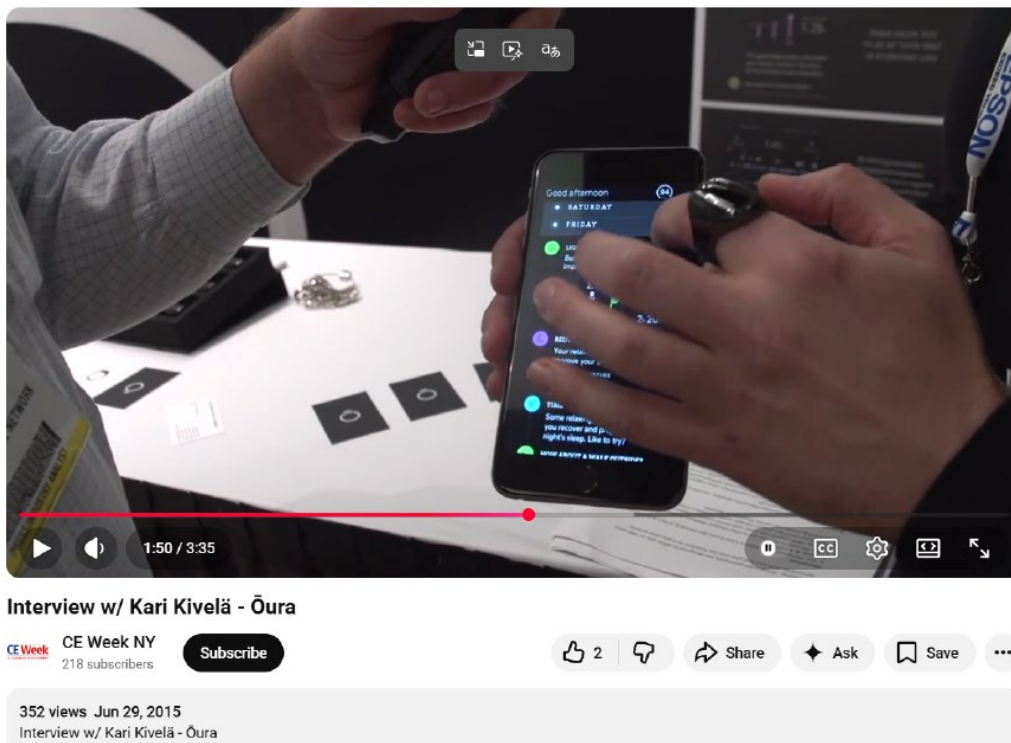
2. *Availability of the Oura Ring Gen. 1 as Prior Art and the Evidentiary Status of Exhibits Relating to the Oura Ring Gen. 1*

The Oura Ring Gen. 1 was publicly launched by Oura in San Francisco, CA on March 3, 2015. RX-0020; RX-0021 (images and showing Mr. Petteri Lahtela, one of Oura's co-founders, speaking on stage at the March 3, 2015 launch event in San Francisco, CA, wherein RX-0020 is dated March 4, 2014).

As Dr. Mortazavi will testify, and Mr. Christophy and Mr. Larson will corroborate, the Oura Ring Gen 1 was fixed in its design and functionality at least as early as June 2015. According to a Wayback Machine archived capture from August 26, 2015, “[u]ser testing has been done in our team from the beginning of 2015 onward [and] [d]esign of the ring and ring box were frozen in June 2015 after extensive electrical, reliability and environmental testing.” RX-0023 at OURA1478_00001534. Substantial evidence from technical documentation confirms that the design and functionality of the Oura Ring Gen 1 was fixed at least as early as June 2015. *See, e.g.*, RX-0026C (Oura Ring Gen 1 [REDACTED]); RX-0031C (Oura Ring Gen 1 [REDACTED]); RX-0028C (Oura Ring Gen 1 [REDACTED]); RX-0029C (Oura Ring Gen 1 [REDACTED]); RX-0030C (Oura Ring Gen 1 [REDACTED]); RX-0033C (Oura Ring Gen 1 f [REDACTED]). The Oura Ring Gen. 1 also went through substantial product safety testing, including testing required by the FCC, in late May to June, 20215. *See* RX-0036C (emissions test report, June 4, 2015); RX-0037C (emissions test report, June 1, 2015); RX-0038C (emissions test report, May 29, 2015); RX-0039C (emissions test report, June 3, 2015); RX-0040C (emissions test report, May 28, 2015); RX-0041C (voltage dips, short interruptions and voltage variations immunity test, June 9, 2015); RX-0042C (electrical fast transient/burst immunity test, June 8, 2015); RX-0043C (surge immunity test, June 8, 2015); RX-0044C (conducted immunity test report, June 9, 2015); RX-0045C (FCC emissions test report, June 11, 2015); RX-0046C (FCC radiated emissions test report, June 12, 2015); RX-0047C (FCC radiated emissions test report, June 11, 2015); RX-0048C (Environmental and mechanical testing, June 27, 2015) RX-0049C (ESD testing, June 10, 2015). *See also*, JDX-0001C ¶¶1302-21 (opining on evidence).

Furthermore, as noted by the Wayback Machine archived capture from August 26, 2015, “[t]housands of people have tried and tested OURA Ring at Wearable Conference, Quantified Self Expo, Decoded Fashion, Sleep Concerence [sic] and CE Week.” RX-0023 at OURA1478_00001532. Each of these events qualifies as a instance of public use and/or availability to the public.

CE Week 2015 occurred on June 23-25, 2015 in New York, NY, and the Oura Ring Gen 1 was demonstrated at this event:



RX-0058³ (video interview of Mr. Kari Kivelä, then of Oura, at CE Week 2015, showing the fully functional Oura Ring Gen. 1, video dated June 29, 2015); RX-0060C, RX-0061, RX-0062

³ RX-0058 and RX-0063 (discussed later in this paragraph) are not being offered to establish the truth of any statement by any declarant within this video, and thus the use of RX-0058 and RX-0063 will not trigger any rule against hearsay. Instead, they serve as a visual record of the ring’s availability. However, in the alternative, Oura gives notice that it may rely on the residual exception to hearsay on the basis that RX-0058 and RX-0063 have sufficient guarantees of trustworthiness (including, at least, that it is a published video and its contents are consistent with

([REDACTED]); RX-0022C (Oura Story 2012-2018 demonstrating display at CE Week 2015); RX-0063 (ABC News covering Oura's display of the Oura Ring Gen 1 at CE Week 2015 in video dated July 9, 2015, and posted to ABCnews.com).



RX-0063. Dr. Mortazavi also spoke with Ms. Uusitalo, depicted in RX-0022C and RX-0063, to confirm that the Oura Ring Gen. 1 was made available to the public at CE Week 2015. JDX-0001C ¶1329.

The Oura Ring Gen. 1 operates and controls its sensors using firmware source code. RX-0033C (Oura Ring Gen 1 [REDACTED]). Oura produced the Oura Ring Gen. 1 firmware source code at RX-0034C. Dr. Mortazavi will testify that the produced Oura Ring Gen. 1 firmware source code is consistent with RX-0033C. JDX-0001C ¶1318. Moreover, Mr. Larson is expected to testify that there were no functional changes to Oura Ring Gen 1 firmware from what was Oura Ring Gen. 1 made available to the public at CE Week

Oura company records which Mr. Larson or Mr. Christophy will testify to, along with the evidence cited within this paragraph), and it is more probative to establish the public availability of the Oura Gen. 1 ring at this convention than any other evidence that Oura can obtain through reasonable efforts.

2015 and later sold to the public, and thus the firmware source code at RX-0034C is the same as, or else has no functional differences relevant to any functionality relevant to any invalidity theory, with the firmware source code controlling the Oura Ring Gen. 1 at CE Week 2015.

This evidence renders the Oura Ring Gen. 1 in public use and/or available to the public prior to August 12, 2015, and accordingly qualifies as prior art to the '675, '677, and '789 Patents under at least 35 U.S.C. § 102(a).

3. Ground 1: Anticipation in View of Oura Ring Gen 1

Claims 13 and 15 of the '675 Patent are invalid as anticipated by the Oura Ring Gen. 1, which meets each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi. This is especially clear should the opinions of Dr. Welch regarding alleged infringement by the Oura Gen. 4 ring be afforded weight, as claims should be construed in the same for purposes of both anticipation and infringement.

i. *[Uncontested]⁴ Claim element [13.pre]*

Claim element [13.pre], to the extent it is limiting, requires performance of “a method.” The Oura Ring Gen. 1 performs a method, in particular the performance of a state-machine-driven method that sequences sensor operations across states [REDACTED]

[REDACTED].
JDX-0001C ¶¶2384-86.

ii. *[Uncontested] Claim element [13.a]*

Claim element [13.a] is performed by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 is a smart ring that is an electronic device, and uses an accelerometer (a first sensor) to determine whether it is being worn on a user’s finger. JDX-0001C ¶¶2387-91

⁴ Statements of “Uncontested” with regard to the Oura Ring Gen. 1 are made while acknowledging that Samsung challenges the prior art status of the Oura Ring Gen. 1. *See* CSUCE at 29.

(citing RX-0069; RX-0033C at 6-10; RX-0026C; RX-0028C; RX-0034C (Oura Ring Gen. 1 source code)). Specifically, the Oura Ring Gen. 1 includes hardware such as “3D accelerometers”:

Hardware

The Gen1 ring was available in one model — in Stealth Black, Arctic White, or Mirror Black colors — and could be used to monitor sleep and daily activity.



Gen1 hardware and sensors included:

- Infrared (IR) LEDs, for nighttime biosignals and sleep stages detection
- 3D accelerometers, for daily activity and nighttime movement detection
- Temperature sensor, for nighttime skin temperature differences
- Made of Zirconia (ZrO₂), with a medical grade, 100% non-allergenic, nickel free, and non-metallic inside molding
- USB connected charger with white LED

RX-0069.0004; RX-0033C.0006-10 (showing that t

).

The source code shows that the Oura Ring Gen. 1

. JDX-0001C ¶2390 (citing RX-0034C). In state

. *Id.* In state

. *Id.* When

[REDACTED]. *Id.* This transition [REDACTED] constitutes determining whether the electronic device is worn on a part of the user's body using at least one first sensor. *Id.* This transition [REDACTED]
[REDACTED]. RX-0033C.0005.

iii. Claim element [13.b]

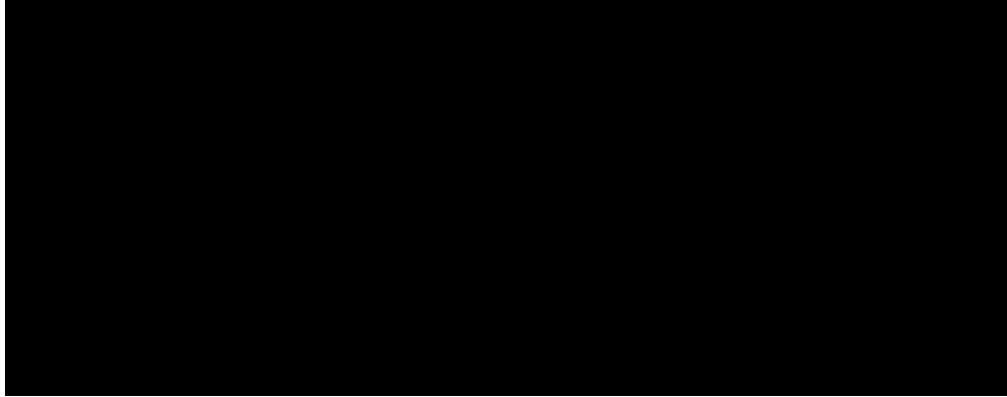
Claim element [13.b] is performed by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 acquires first information on a motion of the electronic device using at least a part of the at least one first sensor (*i.e.*, [REDACTED]) for a second period of time (a period of time [REDACTED]) immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body (the period of time that [REDACTED]). JDX-0001C ¶¶2392-400 (citing RX-0033C; RX-0034C).

Specifically, when operating in state [REDACTED] (the first period of time), the device

[REDACTED]
[REDACTED]. From state [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]. JDX-

0001C ¶¶2398 (citing RX-0034C). This relationship between the 1st period of time (in which the device [REDACTED]) and the 2nd period of time (in which the device [REDACTED]), and that the 2nd period of time occurs immediately after the 1st period of time, is illustrated by Dr. Mortazavi:



JDX-0001C ¶2396

iv. Claim element [13.c]

Claim element [13.c] is performed by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 includes a heart rate sensor that includes “LEDs” and a “phototransistor,” which together function as an optical sensor (“second sensor”). JDX-0001C ¶¶2401-2405 (citing RX-0033C.0006-10; RX-0069; RX-0023.0003 (“the ÕURA ring analyzes the quality of your rest and recovery by measuring your heart rate (optically)”; RX-0026C; RX-0028C).

The heart rate sensor (“second sensor”) of the Oura Ring Gen. 1 is activated [REDACTED]. Specifically, [REDACTED] state [REDACTED]. [REDACTED] [REDACTED]. JDX-0001C ¶2406. When [REDACTED] [REDACTED] [REDACTED] [REDACTED]. JDX-0001C ¶2408.

Thus, the optical sensor (second sensor) is [REDACTED]

[REDACTED]

Dr. Welch’s opinion that the Oura’s Gen. 1 ring’s transition from [REDACTED] state (or potentially [REDACTED] state) does not meet the requirements of claim limitation [13.c] is undermined by his infringement opinion that the Gen. 4 ring’s transition from the [REDACTED] state (or potentially [REDACTED] state) does meet this claim requirement—as Dr. Mortazavi and Mr. Larson will testify, the code is [REDACTED] JDX-0007C ¶¶154, 191.

v. *[Uncontested] Claim element [13.d]*

Claim element [13.d] is performed by the Oura Ring Gen. 1.

As discussed for claim element [13.c], Oura Ring Gen. 1 includes a heart rate sensor that includes “LEDs” and a “phototransistor,” which together function as an optical sensor (“second sensor”). JDX-0001C ¶¶2401-2405, 2410-2411.

This second (optical) sensor [REDACTED]
[REDACTED]. JDX-0001C ¶¶2412-13 (citing RX-0033C; RX-0034C). Specifically, [REDACTED]
[REDACTED]
[REDACTED]. *Id.*

vi. *[Uncontested] Claim element [13.e]*

Claim element [13.e] is performed by the Oura Ring Gen. 1.

While the heart rate sensor (“second sensor”) of the Oura Ring Gen. is detecting the biometric information (e.g., obtaining heart rate information) in [REDACTED]
[REDACTED]
[REDACTED]. JDX-0001C ¶¶2415-18 (citing RX-0033C; RX-0034C).

[REDACTED]

Specifically, [REDACTED]

[REDACTED]

[REDACTED]. *Id.*

vii. [Uncontested] Claim element [13.f]

Claim element [13.f] is performed by the Oura Ring Gen. 1.

In [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. JDX-0001C ¶¶2421. The function

[REDACTED]

[REDACTED]

[REDACTED] JDX-

0001C ¶¶2422.

From [REDACTED]

[REDACTED]

[REDACTED]. JDX-0001C ¶¶2420-22 (citing

OURA1478_00001496, 1500-1501; RX-0034C). Both [REDACTED]

[REDACTED]

[REDACTED]. *Id.*

Thus, in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion (*i.e.*, [REDACTED] [REDACTED]), the optical sensor is inactivated (the device [REDACTED] [REDACTED]).

viii. Claim element [15]

The added requirement of Claim 15 is performed by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 source code demonstrates that [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. [REDACTED]

[REDACTED]. JDX-0001C ¶¶2432-34 (citing RX-0033C; RX-0034C).

At least two measurements [REDACTED]

[REDACTED]. After [REDACTED]

[REDACTED]. The [REDACTED]

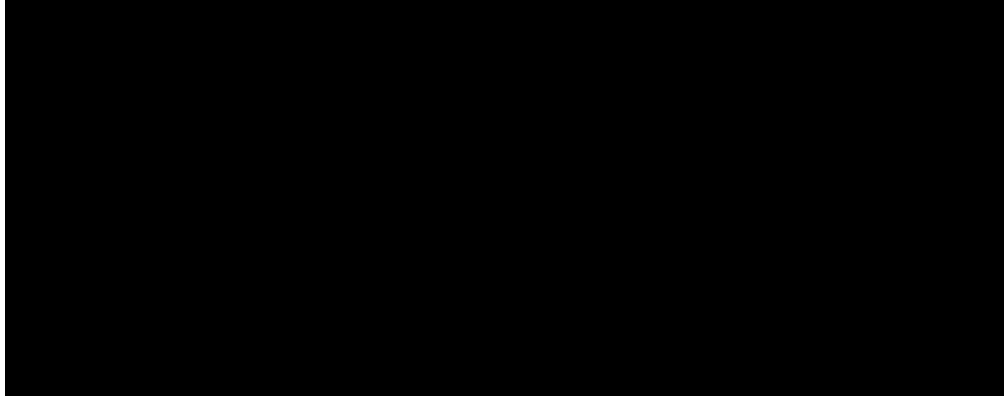
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. *Id.*



The

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. *Id.*

4. Ground 2: Anticipation in View of Hong

Claims 13 and 15 of the '675 Patent are invalid as anticipated by the teachings of Hong, which meet each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi.

i. *[Uncontested] Claim element [13.pre]*

Claim element [13.pre], to the extent it is limiting, requires performance of “a method.” Hong teaches a method, and specifically “wearable devices, typically of a small size, that are

designed to be worn relatively continuously by a person,” which “gather data regarding activities performed by the wearer or the wearer’s physiological state.” JDX-0001C ¶¶2750-2754 (citing RX-0072 at 11:3-13, 20:27-29, Fig. 2B).

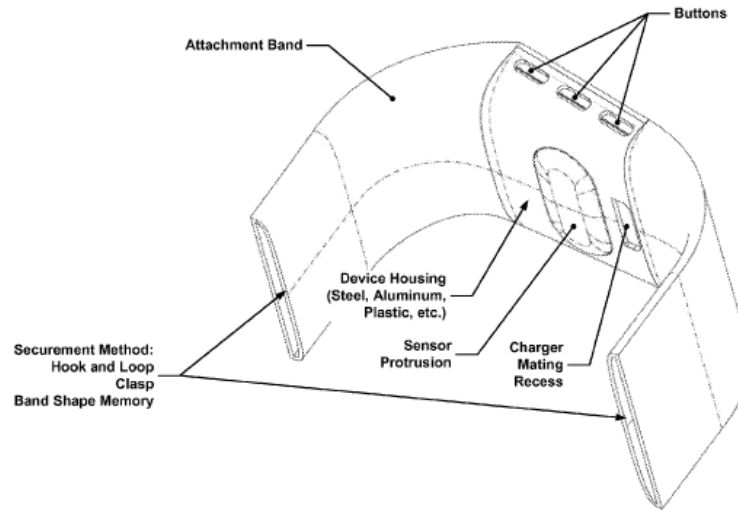


Figure 2B

Hong at Fig. 2B

ii. *[Uncontested] Claim element [13.a]*

Claim element [13.a] is taught by Hong.

Hong teaches the use of an “accelerometer” accelerometer that corresponds to the claimed “first sensor.” JDX-0001C ¶¶2756-2757 (citing RX-0072 at 3:12-16, 16:51-67, Figs. 12A-B). Furthermore, Hong discloses that upon detecting motion, the device “begins operating the heart rate monitor in a ‘worn detection mode’ that is configured to detect the device has transitioned from an unworn to a worn state.” JDX-0001C ¶¶2757-61 (citing RX-0072 at 72:44-60, 72:33-73:21, Figs. 12A-B, 18A). Thus, the motion sensor serves as the gatekeeper for the worn determination—without sufficient motion detected by the accelerometer, the device never advances to confirm a worn state. *Id.* This is demonstrated with blocks 1802, 1804, and 1806 of the flow diagram at Figure 18A of Hong:

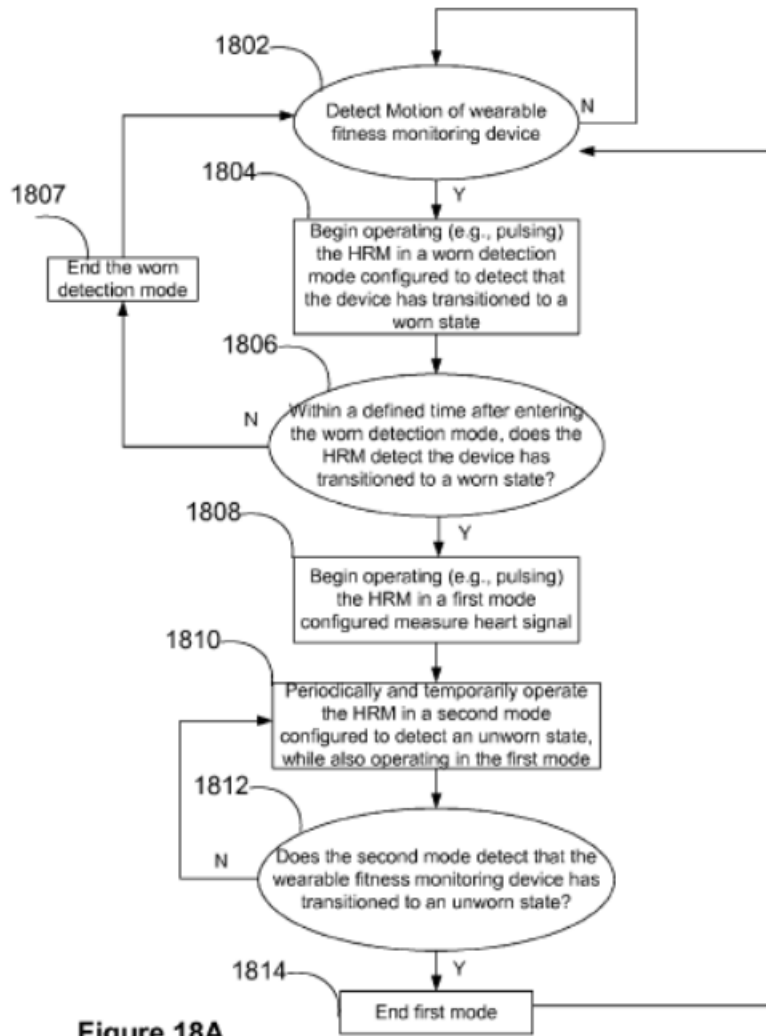


Figure 18A

Hong at Fig. 18A

iii. Claim element [13.b]

Claim element [13.b] is taught by Hong.

Hong discloses the “first period of time” as the wear-determination loop depicted in FIG. 18A, at blocks 1802-1807, during which the device repeatedly detects motion and enters worn-detection mode until it confirms an on-wrist state. JDX-0001C ¶¶2765 (citing RX-0072 at 72:33-73:21, Fig. 18A); JDX-0001C ¶2765. This motion detection loop is an extended temporal window

during which the device applies its motion-based algorithm to determine whether the device is being worn.

Hong discloses the “second period of time” as the post-confirmation window, beginning once worn state is confirmed at block 1806, during which the device continues to monitor motion via the accelerometer to determine whether the user has become sedentary or asleep. JDX-0001C ¶2766 (citing RX-0072 at 16:34-40, Fig. 18A). That sedentary/asleep determination is itself made from the first sensor’s motion data, as Hong discloses that “the biometric monitoring device determines the user is sedentary or asleep” based on whether “the device wearer’s motion is below a threshold.” RX-0072 at 16:34-40. During this “second period of time,” the accelerometer continues to generate motion data that the device uses to classify the user’s activity state, such as to determine if the user has fallen asleep. JDX-0001C ¶2767 (citing RX-0072 at 18:53-56, 19:2-5). The motion data acquired by the accelerometer during this sleep-determination window corresponds with the claimed “first information on a motion” that is acquired during the “second period of time.” *Id.*

Because Hong’s device transitions directly from the motion detection loop (the first period of time) into this motion-monitoring window (the second period of time) with no intervening gap, the second period begins immediately after the first period ends. JDX-0001C ¶¶2765-70.

iv. Claim element [13.c]

Claim element [13.c] is taught by Hong.

Hong teaches the use of a PPG sensor (heart rate monitor) that corresponds to the claimed “second sensor,” and is activated based on motion data gathered by the accelerometer (“first sensor”). JDX-0001C ¶¶2772-73 (citing RX-0072 at Abstract, 18:19-25, Fig. 18A).

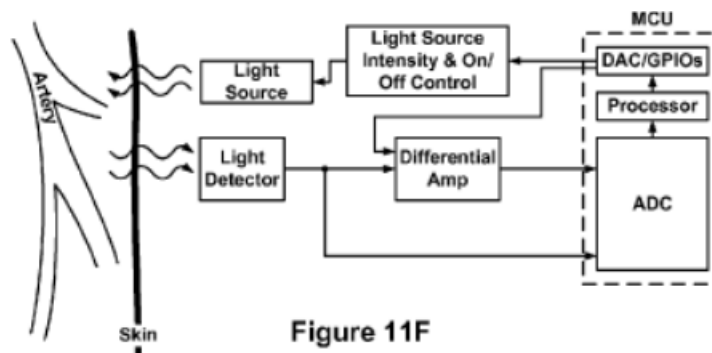
When the device determines the user is sedentary or asleep based on motion data (the first information), it activates the higher-rate sampling mode that activates the PPG sensor: “the

biometric monitoring device may operate the sensors in a higher-rate sampling mode (relative to non-sedentary periods or periods of user activity that exceed a predetermined threshold) to calculate heart rate variability.” RX-0072 at 18:19-25. The heart rate monitor and therefore the second sensor transitions from an off/low-power state to an active state only when motion data from the first sensor (accelerometer) indicates appropriate conditions, namely that the device is being worn and then is in a sleep state. JDX-0001C ¶¶2773-74 (citing RX-0072 at 70:7-11, 70:15-20).

v. *[Uncontested] Claim element [13.d]*

Claim element [13.d] is taught by Hong.

Hong teaches that the PPG sensor (the second sensor) detects the user’s heart rate, which is biometric information of the user. JDX-0001C ¶¶2776-2783 (citing RX-0072 at Abstract, 14:65-15:3, 21:49-55, 73:4-7, Figs. 11F, 18A). Specifically, the PPG sensor “detects pulses of light that interact with the user’s tissue, capturing signal variation caused by heartbeat related blood flow in the capillary vessels of the user,” which is then “processed or analyzed (for example, by resident processing circuitry) to obtain data that is representative of, for example, a user’s heart rate.” RX-0072 at 14:65-15:3, 21:49-55, 73:4-7.



Hong at Fig. 11F

vi. [Uncontested] Claim element [13.e]

Claim element [13.e] is taught by Hong.

Hong teaches that while the PPG sensor detects biometric information, the accelerometer (the first sensor) acquires motion information that is second information on a motion of the electronic device. JDX-0001C ¶¶2785-2788 (citing RX-0072 at 16:51-56, 76:25-28, Fig. 18A). Specifically, Hong teaches that “the raw heart rate signal measured by a PPG sensor may be improved by using one or more algorithms to remove motion artifacts [wherein] [m]ovement of the user (for determining motion artifacts) may be measured using sensors including, but not limited to, accelerometers.” RX-0072 at 16:51-56. This disclosure confirms the accelerometer operates concurrently with the PPG sensor during biometric information detection.

vii. Claim element [13.f]

Claim element [13.f] is taught by Hong.

Hong teaches that when the motion of the electronic device determined by the second information is more than a threshold motion, the PPG sensor is inactivated. JDX-0001C ¶¶2787-93 (citing RX-0072 at 15:33-45, 16:33-37, 19:10-13, Fig. 18A). Specifically, Hong first discloses that the device performs optical heart rate sampling when a user is asleep. RX-0072 at 16:33-37. During this time, the device monitors for user awakening using motion data (the second information): “[m]otion by the user and/or an elevated heart rate and/or a change in heart rate variability may be used by the biometric monitoring device to determine that the user has awoken.” RX-0072 at 19:10-13. When the device detects motion that “is not suitable or optimum to ... obtain desired heart rate measurement ... the sensor(s) employed to ... obtain a desired heart rate measurement may be placed in, or remain in, a low-power state.” RX-0072 at 15:33-45. In other words, Hong teaches that excessive motion that would otherwise degrade the reliability of the biometric measurement causes the second sensor to be reduced to, or held in, a low power state ceasing the

sensor's elevated biometric measurement operation as a direct result of the identified motion. Thus, when the user's motion rises above the threshold, the device returns the heart rate sensor to the low-power state which is inactivation.

viii. Claim element [15]

The added requirement of Claim 15 is taught by Hong.

Hong discloses a third period of time corresponding to the motion-monitoring window during the higher-rate sampling mode: once the device determines the user is sedentary or asleep (the second period of time), it continues to monitor motion via the accelerometer while the PPG sensor operates in the higher-rate mode. RX-0072 at 16:34-40; JDX-0001C ¶2803. This third period is shorter than the first period, since it may last only minutes, whereas the wear-detection loop may span multiple measurement intervals if the device remains unworn. RX-0072 at 72:44-62; JDX-0001C ¶2804. It also occurs immediately after the second period passes, because the higher-rate sampling mode begins the moment the device determines the user is sedentary or asleep. JDX-0001C ¶2804. The additional motion data acquired during this window in which the user is initially asleep supports the device's continued operation of the PPG sensor in the higher-rate mode. RX-0072 at 18:20-25; JDX-0001C ¶2805.

5. Ground 3: Obviousness in View of Hong and Utter II

Claims 13 and 15 of the '675 Patent are invalid as obvious by the teachings of Hong in view of Utter II, which together meet or render obvious each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi.

a) Graham Factors 1-3

Hong discloses a wrist-worn device that uses an accelerometer to detect whether the device is worn, activates heart rate and pulse sensors based on motion information, reduces power to those

sensors when undesired motion is detected, and acquires motion information while the sensors are active. JDX-0001C ¶¶2885-86 (citing RX-0072 at 15:33-45, 19:10-13, 72:33-73:21, Fig. 18A).

Hong also teaches that when the motion sensor indicates activity unsuitable for heart rate measurement, the device may be placed in a low power state, and that motion information is acquired concurrently with biometric measurement. RX-0072 at 15:33-45, 16:33-40, 19:10-13, 77:40-78:20, Figs. 18A, 19A-19D; JDX-0001C ¶¶2889, 2896.

Thus, as demonstrated by Ground 2, Hong teaches all elements of claims 13 and 15. But to the extent Hong does not expressly disclose a threshold comparison for whether motion exceeds a level and then to inactivate the second sensor (claim element [13.f]), this is rendered obvious by the teachings of Utter II.

Utter II discloses a wrist-worn device that performs heart-rate sensing, uses motion signals to alter sensors, jointly processes motion with heart rate data, determines not to use heart rate signals during excessive motion, and changes activation conditions based on motion, sleep, and stress data. RX-0071 ¶¶0083, 0107-0115, 0118-0119; Fig. 14; JDX-0001C ¶2888. Because both references address managing biometric sensors amid motion measurements, a POSITA would naturally look to both.

Utter II discloses a threshold-based deactivation. At stage 1403, “[a]n indication of motion (e.g., above a threshold value in G’s or G’s/unit of time G’s/sec) may mean the user 800 is not at rest. If a YES determination is made, then flow 1400 may transition to another stage, such as cycling back to the stage 1401.” RX-0071 ¶0108; JDX-0001C ¶2897. When motion exceeds the threshold (YES branch), the flow cycles back to stage 1401, bypassing stages 1405-1409 where biometric sensors are parsed thereby inactivating them. JDX-0001C ¶2897. Utter II confirms this:

“high accelerometry at the stage 1454 may forego parsing biometric sensors.” RX-0071 ¶¶0119. Utter II defines “parsed” to include “activated.” RX-0071 ¶¶0107; JDX-0001C ¶¶2897.

Thus, incorporating Utter II’s threshold-based deactivation into Hong to implement Hong’s teachings of placing the device in a low power state when the motion sensor indicates activity unsuitable for heart rate measurement (JDX-0001C ¶¶2889, 2896) would satisfy all limitations of claims 13 and 15, including claim element [13.f], without otherwise affecting the device taught by Hong.

A POSITA would have found it obvious to incorporate Utter II’s threshold-based deactivation into Hong. Threshold-based motion screening is well-known for managing biometric sensors in wrist-worn devices. JDX-0001C ¶¶2891. Both references address managing biometric sensors amid motion artifacts and recognize that motion affects measurement quality. RX-0072 at 15:33-45; RX-0071 ¶¶0108]. Incorporating Utter II’s deactivation into Hong would achieve deactivation based on motion screening during biometric measurement. JDX-0001C ¶¶2891. Furthermore, both references disclose wrist-worn devices with accelerometers and heart rate sensors that manage sensor operation based on detected motion. JDX-0001C ¶¶2887-88; RX-0072 at 15:33-45, 19:10-13, 72:33-73:21, Fig. 18A; RX-0071 ¶¶0083, 0107-0110, 0118-0119; Fig. 14. These similarities render the combination obvious to try. JDX-0001C ¶¶2891. A POSITA would have had a reasonable expectation of success. JDX-0001C ¶¶2892.

b) *Graham* Factor 4

Samsung has chosen to present its alleged secondary considerations of nonobviousness in a single argument common to the ’359, ’677, and ’675 Patents. Samsung fails to separate out the alleged inventions of the ’359, ’677, and ’675 Patents and thereby fails its burden to demonstrate a nexus to the claimed inventions, as was its burden to do. As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the

alleged inventions of the '675 Patent. JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

6. Ground 4: Anticipation in view of Utter II

Claim 13 of the '675 Patent is invalid as anticipated by the teachings of Hong, which meet each limitation of this claims as explained herein and as will be shown by the testimony of Dr. Mortazavi.

i. *[Uncontested] Claim element [13.pre]*

Claim element [13.pre], to the extent it is limiting, requires performance of “a method.” Utter II teaches a method, and specifically discloses a wearable device that performs method steps involving motion detection, physiological state determination, and sensor management, including a flow for passively determining a user’s true resting heart rate (TRHR). JDX-0001C ¶¶2942-45 (citing RX-0071 ¶¶[0035], [0038], [0107], Fig. 14).

ii. *Claim element [13.a]*

Claim element [13.a] is taught by Utter II.

Utter II discloses the use of motion sensors 160 (at least one first sensor) that are used to detect whether the device is worn, and are generically referred to by Utter II as sensors 110. JDX-0001C ¶¶2946-49 (citing RX-0071 ¶¶[0035], [0037], [0039], Fig. 14). Specifically, Utter II teaches that device 100 includes “motion sensors 160 (e.g., single or multi-axis accelerometer, gyroscope, piezoelectric device)” which generate signals indicative of the device’s coupling with the user’s body. RX-0071 ¶¶[0035], [0037]. Utter II describes determining whether the device is worn by using force and motion signals generated through coupling with the user’s body.

iii. *Claim element [13.b]*

Claim element [13.b] is taught by Utter II.

Utter II discloses a first period of time corresponding to wear determination, since device 100 continuously determines whether the user is coupled with the device through the Activity and Sampling cycle, which “continuously cycle[s] over Time such that data from sensors and activity may be gathered, analyzed, and acted on.” JDX-0001C ¶¶2952-53 (citing RX-0071 ¶¶[0036], [0088], Fig. 14).

Utter II also discloses a second period of time corresponding to the motion evaluation at stage 1403 of flow 1400, at which “signals from one or more sensors and/or sensor types for sensing motion may be analyzed to determine whether or not the user 800 is in motion.” JDX-0001C ¶2954 (citing RX-0071 ¶[0108]). This motion evaluation occurs immediately after wear determination, since once device 100 confirms coupling with the user’s body, flow 1400 parses relevant sensors at stage 1401 and immediately evaluates motion at stage 1403. JDX-0001C ¶¶2954-55 (citing RX-0071 ¶[0048], Fig. 14).

Accordingly, Utter II teaches acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor (data from the motion sensors 160) for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body.

iv. Claim element [13.c]

Claim element [13.c] is taught by Utter II.

Utter II’s teaches the use of biometric sensors 140 that correspond to the claimed “second sensor,” and are activated based on motion data gathered by the first sensor. JDX-0001C ¶2957-58 (citing RX-0071 ¶[0037]). Specifically, at stage 1403 of flow 1400 in the diagram of Figure 14, if a NO determination is made regarding motion (i.e., the user is not in motion), the flow proceeds to analyze sleep and stress states using biometric sensors such as at stage 1409: “If the NO branch is taken, then flow 1400 may transition to a stage 1405 where a determination may be made as to

whether or not one signals from sensors in 340 indicate that the user 800 is asleep.” JDX-0001C ¶2958 (citing RX-0071 ¶[0109], Fig. 14). Thus, the biometric sensors are activated only after the motion evaluation at stage 1403 returns a NO result, meaning activation of the second sensor is based on the first information (motion data). JDX-0001C ¶2958. Additionally, Utter II teaches that “high accelerometry at the stage 1454 may forego parsing biometric sensors” (RX-0071 ¶[0118]), and that “parsed” includes “activated” (RX-0071 ¶[0107]), and because the biometric sensors are parsed, and therefore activated, only once the stage 1403 motion evaluation returns a NO result, Utter II discloses activation of the second sensor based on the first information on motion. JDX-0001C ¶2958.

v. *[Uncontested] Claim element [13.d]*

Claim element [13.d] is taught by Utter II.

Utter II discloses that stage 1409 is only entered after user motion is not sensed a stage 1403 of Figure 14, but that once the conditions for entry into stage 1409 are met, sensor signals relevant to a passive determination of TRHR are analyzed using dedicated biometric sensors, including heart rate, heart rate variability, and TRHR sensors, to detect the user’s biometric information. JDX-0001C ¶¶2960-64 (citing RX-0071 ¶¶[0073], [0110]-[0111], Fig. 14). Thus, Utter II describes detecting biometric information of the user using the activated biometric sensors (second sensor).

vi. *[Uncontested] Claim element [13.e]*

Claim element [13.e] is taught by Utter II.

Utter II discloses that motion sensors (the first sensor) acquire motion data concurrently with biometric sensor operation. JDX-0001C ¶¶2966-67 (citing RX-0071 ¶¶[0069], [0076-77], [0110], [0123]). For example, at stage 1409, the heart rate analysis occurs “while the conditions precedent to entry into the stage 1409 are still present, that is the NO branches of group 1450 are

still valid (e.g., user 800 is at rest, is not asleep, and is not stressed),” demonstrating that motion sensors continue monitoring during biometric sensing. RX-0071 ¶0110. Indeed, Utter II teaches that motion signals are continuously needed during biometric sensing. JDX-0001C ¶2966 (citing RX-0071 ¶¶[0069]).

vii. Claim element [13.f]

Claim element [13.f] is taught by Utter II.

Utter II teaches at stage 1403 that “[a]n indication of motion (e.g., above a threshold value in G’s or G’s/unit of time G’s/sec) may mean the user 800 is not at rest. If a YES determination is made, then flow 1400 may transition to another stage, such as cycling back to the stage 1401.” JDX-0001C ¶2969 (citing RX-0071 ¶[0108], Fig. 14). Thus, when motion exceeds the threshold at stage 1403 (the YES branch), the flow cycles back to stage 1401, bypassing stages 1405-1409 where biometric sensors are parsed (activated) and analyzed. RX-0071 ¶[0107], Fig. 14. Thus, when the biometric sensors are not parsed due to threshold motion at stage 1403, they are, by Utter II’s own definition, not activated, i.e., inactivated. JDX-0001C ¶2969. Utter II further confirms this in group 1500b, teaching that “high accelerometry at the stage 1454 may forego parsing biometric sensors.” JDX-0001C ¶2969 (citing RX-0071 ¶[0119]).

Accordingly, Utter II teaches that if the detected motion information exceeds a threshold, the biometric sensors are inactivated.

7. Ground 5: Obviousness in View of Utter II and Hong

Claims 13 and 15 of the ’675 Patent are invalid as obvious by the teachings of Utter II in view of Hong, which together meet or render obvious each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi.

a) *Graham* Factors 1-3

Utter II discloses a wrist-worn activity monitoring device with sensors generate signals indicative of force via coupling or contact with a body portion, where the device acquires motion sensor data to distinguish between activity and sleep, and parses motion sensors to determine motion, sleep, and stress states before activating heart rate sensing. RX-0071 ¶¶[0035]-[0039], [0056]-[0058], [0068]-[0071], [0083], [0107]-[0115], [0118]-[0119], Fig. 14; JDX-0001C ¶¶3020-21.

Thus, as demonstrated by Ground 2, Utter II teaches all elements of claim 13. But to the extent it does not expressly disclose determining that the device is worn using an accelerometer as the first sensor (claim elements [13.a] and [13.b]), this is rendered obvious by Hong. The teachings of Hong also render obvious the added limitation of claim 15 when implemented with the teachings of Utter II.

Hong discloses a wrist-worn device that uses an accelerometer to detect whether the device is worn, activates biometric sensors based on motion information, and adjusts sensor activation based on detected user state. RX-0072 at 15:33-45, 16:33-40, 19:10-13, 77:40-78:20, Figs. 18A, 19A-19D; JDX-0001C ¶¶3020-24. Because both references address determining wear status and controlling biometric sensor activation through motion evaluation, a POSITA would naturally look to both. JDX-0001C ¶3020.

In this similar context, Hong teaches the use of the motion sensor for detecting whether the device is being worn. Specifically, Hong's FIG. 18A illustrates the accelerometer detecting a worn state: at block 1802, the device detects motion; at block 1804, it enters worn detection mode; and at block 1806, it determines whether a worn state is confirmed. RX-0072 at 72:33-73:21. Hong also describes a second period of time after worn state is confirmed, during which the device

monitors motion via the accelerometer and determines that the user has become sedentary or asleep, activating a higher-rate sampling mode. RX-0072 at 16:34-40; JDX-0001C ¶¶3020-24.

It would be obvious to incorporate Hong's wear detection and multi-period evaluation into Utter II's framework. Wear detection should precede biometric sensor activation to ensure reliable operation and avoid wasting power on false activations when unworn. JDX-0001C ¶3023. Because both Utter II and Hong address this challenge, a POSITA would be motivated to implement Hong's wear detection and multi-period evaluation within Utter II's framework. JDX-0001C ¶3023. The combination would yield predictable results. JDX-0001C ¶3024. Applying a known wear detection technique to a known biometric monitoring system achieves the predictable result of a device that determines wear with an accelerometer, acquires motion for a second period around wear determination, and takes another motion measurement for a third period. JDX-0001C ¶3024. A POSITA would have had a reasonable expectation of success. JDX-0001C ¶3025.

The incorporation of Hong's wear detection and multi-period evaluation into Utter II's framework would meet the limitations of claim elements [13.a] and [13.b], as it would yield a device that: (1) determines wear via accelerometer as the first sensor (claim 13.a) (RX-0072 at 3:12-16, 16:51-67, Figs. 12A-B; JDX-0001C ¶¶3026-30), and (2) acquires motion for a second period around wear determination (claim 13.b) (RX-0072 at 72:33-73:21, Fig. 18A; JDX-0001C ¶¶3031-38).

Furthermore, Hong describes a motion-monitoring window during the higher-rate sampling mode, constituting a third period shorter than the first. RX-0072 at 18:20-25, 72:44-62; JDX-0001C ¶¶3039-45. Incorporating Hong's wear detection and multi-period evaluation into Utter II's framework would yield a device that acquires additional motion information for a shorter third period immediately after the second period and activates biometric sensing based on both

information, and therefore meets the added limitation of claim 15. RX-0072 at 16:34-40, 18:20-25; JDX-0001C ¶¶3041-44. The combination would yield the predictable result of a wrist-worn device that acquires additional motion information for a shorter third period immediately after the second period and activates biometric sensing based on that information. JDX-0001C ¶3044. Applying a known wear detection technique to takes another motion measurement for a third period would allow for a multi-period evaluation within Utter II's framework.

b) *Graham* Factor 4

Samsung has chosen to present its alleged secondary considerations of nonobviousness in a single argument common to the '359, '677, and '675 Patents. Samsung fails to separate out the alleged inventions of the '359, '677, and '675 Patents and thereby fails its burden to demonstrate a nexus to the claimed inventions, as was its burden to do. As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '675 Patent. *See also* JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

C. Claims 13 and 15 of the '675 Patent Are Ineligible Under 35 U.S.C. § 101

Claims 13 and 15 of the '675 Patent are patent ineligible under 35 U.S.C. § 101 for claiming patent ineligible subject matter. *See Alice Corp. Pty. Ltd. v. CLS Bank Int'l.*, 573 U.S. 208, 216-17 (2014) (setting forth the well-known two-step *Alice* framework for assessing patent eligibility). These claims are directed to the abstract idea of using motion to activate and deactivate sensing of a biological signal, and as set forth herein, the particular claim elements set forth nothing more than the application of this abstract idea in the context of a generic wearable device.

1. *Alice Step 1: Each claim is directed to the abstract idea of using motion to activate and deactivate sensing of a biological signal*

Claims 13 and 15 of the '675 Patent are directed to the abstract idea of using motion to activate and deactivate sensing of a biological signal, and as explained further below, there is nothing patent eligible about such generic and conceptual claiming.

At *Alice* step one, courts determine whether the claims at issue are directed to a patent-ineligible concept by “looking at the ‘focus’ of the claims [and] their ‘character as a whole.’” *Elec. Power Grp., LLC v. Alstom S.A.*, 830 F.3d 1350, 1353 (Fed. Cir. 2016). “In the context of software-based inventions, *Alice/Mayo* step one ‘often turns on whether the claims focus on the specific asserted improvement in computer capabilities or, instead, on a process that qualifies as an abstract idea for which computers are invoked merely as a tool.’” *Trinity Info Media, LLC v. Covalent, Inc.*, 72 F.4th 1355, 1363 (Fed. Cir. 2023) (quoting *In re Killian*, 45 F.4th 1373, 1382 (Fed. Cir. 2022)).

Upon reviewing claims 13 and 15, it is evident that they broadly require only three aspects: (1) using motion data to determine if the device is being worn, (2) obtaining motion data if the device is worn, and (3) activating and deactivating a second sensor when the data from the first sensor is below or above a user movement threshold. An analysis of the claim language of claim 13 shows this:

A method comprising:

determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device;

acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body

activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;

detecting a biometric information of the user using the activated at least one second sensor;

during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and

in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.

Use motion data to determine if the device is being worn

Obtain motion data if the device is worn

Activating and deactivating a second sensor when the data from the first sensor is below or above a user movement threshold

JX-0003 at cl. 13 (annotated). And even though motion data continues to be monitored while the second sensor is active in order to assess whether the second sensor should be deactivated, this is just an extension of the abstract idea, namely using motion to activate and deactivate sensing of a biological signal.

Claim 15 does not change the abstract idea to which independent claim 13 is directed, nor does Samsung claim that it does. Rather claim 15 merely sets forth a third period to obtain motion data in the performance of the same abstract idea. Thus, it adds no patentable eligible subject matter to claim 13, nor changes the abstract idea to which the claims are directed, and claim 13 of the '675 Patent may be treated as representative. *See Berkheimer v. HP Inc.*, 881 F.3d 1360, 1365 (Fed. Cir. 2018) (“Courts may treat a claim as representative in certain situations, such as if the patentee does not present any meaningful argument for the distinctive significance of any claim limitations not found in the representative claim”); *Mortg. Grader, Inc. v. First Choice Loan*

Servs., 811 F.3d 1314, 1324 n.6 (Fed. Cir. 2016) (a claim may be representative if the claims do not “differ in any manner that is material to the patent eligibility inquiry”).

Each individual claim element of claims 13 and 15 requires only routine aspects of wearable components. Such claims do not set forth any particular “improvement in computer capabilities”—indeed, the claim elements are far too generic to do so—and instead set forth the generic idea of using motion to activate and deactivate sensing of a biological signal and merely place this abstract idea in the context of method performed by a generic wearable device. For example, there is no particular manner of motion detection or biometric information detection required: all that is claimed is a broad causal and temporal relationship between sensors of a wearable device. This is an abstract idea. See *GoTV Streaming, LLC v. Netflix, Inc.*, 166 F.4th 1053, 1064 (Fed. Cir. 2026) (“receiving inputs, storing and retrieving, processing, outputting (including displaying), and transmitting” are all examples of “ordinary functions” of computers and networks); *Intellectual Ventures I LLC v. Capital One Fin. Corp.*, 850 F.3d 1332, 1341 (Fed. Cir. 2017) (routine computer activity does not change the *Alice* step one analysis); *Elec. Power Grp.*, 830 F.3d at 1353 (“[t]he claims in this case do not even require a new source or type of information, or new techniques for analyzing it”); *SAP Am., Inc. v. InvestPic, LLC*, 898 F.3d 1161, 1168 (Fed. Cir. 2018) (“even if a process of collecting and analyzing information is ‘limited to particular content’ or a particular ‘source,’ that limitation does not make the collection and analysis other than abstract”); *AGI SureTrack LLC v. Farmers Edge Inc.*, 176 F.4th 1373, 1378 (Fed. Cir. 2026) (“The fact that the claims here are limited to a particular type of information—farming data—does not remove them from the realm of the abstract... As we have previously emphasized, ‘an abstract idea remains an abstract idea even when narrowed—e.g., by subject matter—to a particular use or environment.’”); *In re Killian*, 45 F.4th 1373, 1380 (Fed. Cir. 2022) (“We have

distinguished between claims ‘directed to an improvement in the functioning of a computer,’ versus those [] that simply recite ‘generalized steps to be performed on a computer using conventional computer activity.’”).

Samsung has contended that “the ’675 Patent’s claims recite novel methods and wearable devices that “use a multitude of sensors to collect data and optimize when biometric information is measured, including based on motion signals.” JDX-0048C at 225. But even if correct (and it is not, for the reasons described above), “optimizing when biometric information is measured, including based on motion signals” *is* an abstract idea, as it just the concept of optimization based on known and generic inputs. *See Constellation Designs, LLC v. LG Elecs. Inc.*, 174 F.4th 888, 903 (Fed. Cir. 2026) (“the claims nonetheless recite nothing more than the broad abstract idea of ‘optimizing’ a constellation for PD capacity”).

Furthermore, the Federal Circuit has held that “a motion sensor system that evaluates and communicates the relative movement of a body using static and dynamic acceleration information collected from sensors” is ineligible because “is not focused on a specific means or method to improve motion sensor systems, nor is it directed to a specific physical configuration of sensors.” *iLife Techs., Inc. v. Nintendo of Am., Inc.*, 839 Fed. Appx. 534, 537 (Fed. Cir. 2021). The same rationale applies here.

2. *Alice Step 2: Each claim merely embodies the abstract idea without adding any inventive concept*

At *Alice* step two, courts remove the abstract idea from the claim and search the remainder of the claim for an “‘inventive concept’—*i.e.*, an element or combination of elements that is ‘sufficient to ensure that the patent in practice amounts to significantly more than a patent upon the [ineligible concept] itself.’” *Alice*, 573 U.S. at 217-18. *Alice* step two requires that the claim limitations “involve more than performance of ‘well-understood, routine, [and] conventional

activities previously known to the industry.” *Content Extraction & Transmission LLC v. Wells Fargo Bank, Nat’l Ass’n*, 776 F.3d 1343, 1347-48 (Fed. Cir. 2014) (citing *Alice*, 573 U.S. at 225).

There is nothing in the claims that could supply an inventive concept. To begin with, the abstract idea itself cannot form the inventive concept at *Alice* step two. *BSG Tech LLC v. Buyseasons, Inc.*, 899 F.3d 1281, 1291 (Fed. Cir. 2018) (“Here, the only alleged unconventional feature of BSG Tech’s claims is the requirement that users are guided by summary comparison usage information or relative historical usage information. But this simply restates what we have already determined is an abstract idea.”).

Once separated from the abstract idea (using motion to activate and deactivate sensing of a biological signal), there is no inventive concept claimed. There are only generic functions of motion and biometric signal detection, and unspecified and generic calculations or comparisons to an unspecified “threshold” to control the operations of the second sensor. These elements, alone or in combination, do not convey an inventive concept. *See Killian*, 45 F.4th at 1380 (“the claims here do not detail how the computer should go about determining eligibility for benefits, beyond saying that the computer should determine eligibility based ‘on the identified information’ and ‘current SSDI benefit legal requirements’”).

A wearable device comprising a motion sensor, a biological sensor, and a controller (the claimed hardware environment) was well-understood, routine, and conventional at the time of the ’675 Patent’s alleged invention. The ’675 Patent even acknowledges that obtaining movement and biometric data was conventional and recognizes that conventional wearable sensors were known. JX-0003 at 1:22-41. A commercially available example of such a wearable device would have included the Basis B1 Band, which is “a wearable device that measures signals such as user movement and biological signals.” JDX-0001C ¶105; *see also id.* ¶1284 (“[t]he Basis B1 Band

watch would be an example of a commercially available wearable device with a motion sensor, a biological sensor, and a controller”). The Basis B1 Band includes “an internal accelerometer that allows it to measure steps’ along with ‘the optical and skin temperature sensors’ wherein the optical sensor performs PPG measurements by ‘measuring the reflection of light emitted by the green sensor’ which discerns the heart rate.” JDX-0001C ¶1284 (quoting RX-0009.0013, 0017 (July 25, 2013 review of the Basis B1 Band)).

Both Hong and Utter II (discussed in relation to invalidity Grounds 2-5) are further evidence that the field of art would have found the claimed determinations to be performed by the device were well-understood, routine, and conventional, even in combination, as of the claimed priority date of the ’675 Patent. Mr. Christophy and Mr. Larson’s expected testimony regarding the Oura Gen. 1 ring will be further evidence of what was well-understood, routine, and conventional with regard to the claimed determinations of sensor activity based on other sensor activity.

Samsung has made no allegations of any particular inventive concept, and accordingly, the claims of the ’675 Patent fail at both *Alice* steps one and two, and are therefore patent ineligible under 35 U.S.C. § 101.

V. ’677 PATENT

A. Oura’s Products Do Not Infringe the Asserted Claims

As set forth in Oura’s opposition to Samsung’s motion for summary determination (EDIS Doc. ID 889195), Oura contends that Samsung will not be able to carry its burden to show infringement of the ’677 Patent by any product at issue (Oura’s Gen. 4, Ring 5, and [REDACTED] rings).

1. Oura’s Gen. 4 Ring Does Not Infringe the Asserted Claims

First, Samsung will not be able to show that the Oura Gen. 4 ring “refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy

the specified condition,” as required by claim element [1.g], or “refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range,” as required by claim element [9.f]. As Dr. Welch must admit upon cross-examination, the state transition alleged by Samsung to meet these limitations [REDACTED] [REDACTED]

[REDACTED]. Thus, as a result of this state transition, the optical sensor does *not* “inactivat[e] the at least one second sensor.”

Second, Samsung’s expert, Dr. Welch, cannot provide credible opinions sufficient to sustain Samsung’s burden of proof because he has contradictory opinions with regard to invalidity in view of Oura’s Gen. 1 ring, which has substantially similar state transitions as are accused. Specifically, Dr. Welch opines that the Oura’s Gen. 1 ring’s transition [REDACTED] (or potentially [REDACTED] state) does *not* meet the requirements of claim limitations [1.g] and [9.f], whereas the Gen. 4 ring’s transition from the [REDACTED] state (or potentially [REDACTED] state) does—a contradictory opinion, given that, as Dr. Mortazavi and Mr. Larson will testify, the code is [REDACTED]. JDX-0007C ¶¶147, 151, 195, 202. Dr. Welch’s invalidity opinions undermine his infringement opinions, and he will not be able to help Samsung carry its burden of proof.

2. Ring 5 and [REDACTED] Do Not Infringe the Asserted Claims

Samsung has put forth no contention or evidence of infringement by these rings and accordingly cannot carry its burden of proof. *See Certain Crafting Machines*, Inv. No. 337-TA-1426, ID at 40. Dr. Mortazavi and Mr. Larson will testify that although Ring 5 and [REDACTED] have broadly similar functionality to Oura’s Gen. 4 ring, the software and hardware functionality is not

the same; given this, and given the lack of any theory of representativeness, Samsung will not be able to carry its burden of proof for these reasons as well.

B. Claims 1-3 and 9 of the '677 Patent Are Invalid in View of the Prior Art

There are four asserted grounds of prior art-based invalidity that render invalid claims 1-3 and 9 of the '677 Patent:

Invalidity Ground	Basis	Applicable Claims
Ground 1	Anticipation in view of Oura Ring Gen. 1	1-3, 9
Ground 2	Anticipation in view of Ahmed486	1-3, 9
Ground 3	Anticipation in view of Presura	1, 9
Ground 4	Obviousness in view of Presura and Patel	2, 3

These four grounds of invalidity are explained in detail herein.

1. Overview of Prior Art References

The Oura Ring Gen. 1 was Oura's first generation of smart ring, and was a wearable device worn on a user's finger. JDX-0001C ¶¶1350-1353. This smart ring used physiological data signals to calculate deep sleep, light sleep, REM sleep, and awake states, including by using an optical sensor, an accelerometer, and a temperature sensor. RX-0023. As established in Section IV.B.2, *supra*, the Oura Ring Gen. 1 was in public use and/or available to the public at least prior to August 12, 2015, and accordingly qualifies as prior art to the '677 Patent under at least 35 U.S.C. § 102(a).

U.S. Patent Application Publication No. 2014/0073486 to Ahmed ("Ahmed486") (RX-0016) was filed on September 4, 2013 and published on March 13, 2014, and therefore qualifies as prior art to the '677 Patent under at least 35 U.S.C. § 102(a) and/or (b). Ahmed486 discloses a "wearable physiological measurement system" that includes "a plurality of light emitters for emitting light toward the user's skin, a plurality of light detectors for receiving light reflected from the user's skin, a motion sensor for detecting a motion of the user, and a processing module

configured to determine a motion status of the user based on data received from the motion sensor, and based on the motion status of the user, automatically and selectively activate one or more of the light emitters to determine a heart rate of the user.” RX-0016 ¶[0011].

International Publication No. WO 2015/139980 A1 to Presura (“Presura”) (RX-0015) was filed on March 9, 2015 and published on September 24, 2015, and therefore qualifies as prior art to the ’677 Patent under at least 35 U.S.C. § 102(a) and/or (b). Presura discloses a wearable heart rate monitor system in which a “primary sensor” implemented as an optical heart rate sensor is activated and deactivated based on motion information supplied by a low-power “secondary sensor” such as an accelerometer, for the purpose of reducing power consumption. JDX-0001C ¶¶1843-46 (citing RX-0015 at 5:1-2, 8:16-19).

U.S. Patent App. Pub. No. 2015/0371028 to Patel (“Patel”) (RX0014) was filed on August 21, 2014 and published December 24, 2015, and therefore qualifies as prior art to the ’677 Patent under at least 35 U.S.C. § 102(a) and/or (b). Patel discloses a “pulse sensor [that] includes a light source (*e.g.*, a light-emitting diode) and a photo sensor,” and is directed to a wearable device with optical heart rate sensors. JDX-0001C ¶¶2027-28, 2032, 2034 (citing RX-0014 at Abstract, [0012], [0016]).

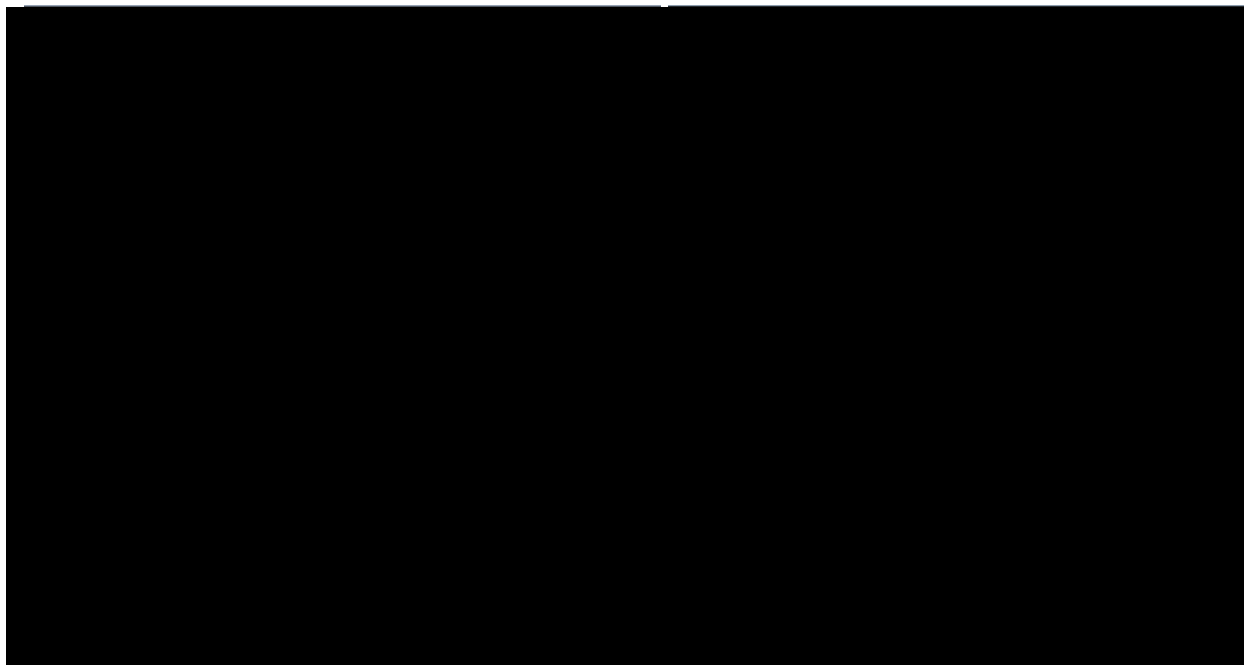
2. Ground 1: Anticipation in view of Oura Ring Gen. 1

Claims 1-3 and 9 of the ’677 Patent are invalid as anticipated by the Oura Ring Gen. 1, which meets each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi. This is especially clear should the opinions of Dr. Welch regarding alleged infringement by the Oura Gen. 4 ring be afforded weight, as claims should be construed in the same for purposes of both anticipation and infringement.

i. Claim elements [1.pre]-[1.c]

Claim elements [1.pre]-[1.c] are each met by the Oura Ring Gen. 1. Samsung does not contest elements [1.pre]-[1.b]. CSUCE at 13.⁵

The Oura Ring Gen. 1 was a smart ring worn on a user's finger. JDX-0001C ¶¶1350-1353.



RX-0029C (Oura Ring Gen. 1 CAD file)



RX-0054 at OURA1478_00017075, -83; RX-0055; RX-0056 (pictures from Quantified Self Expo 2015, uploaded to Flickr.com in June 2015).

⁵ Statements of “Uncontested” with regard to the Oura Ring Gen. 1 are made while acknowledging that Samsung challenges the prior art status of the Oura Ring Gen. 1. *See* CSUCE at 13.

The Oura Ring Gen. 1 includes one or more accelerometers that detects movement and is therefore a “motion sensor.” JDX-0001C ¶¶1354-1359; RX-0069; RX-0028C; RX-0034C.

The Oura Ring Gen. 1 also includes “Infrared (IR) LEDs” and a “phototransistor,” which together function as an optical sensor that includes a light emitting unit and a light receiving unit. JDX-0001C ¶¶1360-1363; RX-0033C at OURA1478_00001496-500; RX-0069; RX-0023; RX-0034C.

The Oura Ring Gen. 1 also includes at least one processor, for example a CPU. JDX-0001C ¶¶1364-67; RX-0026C (Oura Ring Gen. 1 BOM [REDACTED]).

Further evidence that the Oura Ring Gen. 1 was a wearable electronic device comprising: a motion sensor; one or more optical sensors includes a light emitting unit and a light receiving unit; and at least one processor will be provided by the testimony of Mr. Christophy.

ii. [Uncontested] Claim element [1.d]

Claim element [1.d] is met by the Oura Ring Gen. 1.

In the Oura Ring Gen. 1, [REDACTED]
[REDACTED]
[REDACTED]. RX-0033C at OURA1478_00001496-500; JDX-0001C ¶¶1368-71.

For instance, [REDACTED]
[REDACTED]
[REDACTED]:

[REDACTED]

[REDACTED]

RX-0033C at OURA1478_00001495-501

[REDACTED]

[REDACTED]

[REDACTED]. For example,

[REDACTED]

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]. Thus, the Oura Ring Gen. 1 is configured to obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors.

iii. Claim element [1.e]

Claim element [1.e] is met by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 was configured such that [REDACTED]
[REDACTED]. JDX-0001C ¶¶1370-76 (citing RX-0033C; RX-0034C). As one example of a specified condition, the Oura Ring Gen. 1 [REDACTED]
[REDACTED]

[REDACTED]. As another example, [REDACTED]
[REDACTED]
[REDACTED]. See JDX-0001C ¶¶1370, 1375 (citing RX-0034C ([REDACTED]
[REDACTED])). The device [REDACTED]
[REDACTED]. *Id.*; RX-0033C.

In [REDACTED]
[REDACTED]. JDX-0001C ¶1375 (citing RX-0033C at OURA1478_00001496).

iv. *[Uncontested] Claim element [1.f]*

Claim element [1.f] is met by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 enables an IR LED and phototransistor for measuring a biological signal (e.g., heart rate) [REDACTED]. JDX-0001C ¶¶1379-82 (citing RX-0069; RX-0023; RX-0033C). The Oura Ring Gen. 1 [REDACTED]

[REDACTED]. See, e.g., JDX-0001C ¶1381 (citing RX-0034C regarding [REDACTED]). Given the orientation of the light emitting unit and light receiving unit, the fact that the device is to be worn in direct contact with a user's skin, and the fact that code [REDACTED] [REDACTED], it follows that the light receiving unit receives the light emitted from the light emitting unit via reflection off of the user's skin. JDX-0001C ¶1381.

v. *[Uncontested] Claim element [1.g]*

Claim element [1.g] is met by the Oura Ring Gen. 1.

During the obtaining of the biological signal (e.g., [REDACTED] [REDACTED]), the Oura Ring Gen. 1 [REDACTED] [REDACTED]. JDX-0001C ¶1384 (citing RX-0033C; RX-0034C).

The [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]. For instance, [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. *Id.*

vi. Claim element [1.h]

Claim element [1.g] is met by the Oura Ring Gen. 1.

For example, the Oura Ring Gen. 1 t [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. JDX-0001C ¶1386 (citing RX-0034C). Thus,

according to the movement of the Oura Ring Gen. 1 [REDACTED]

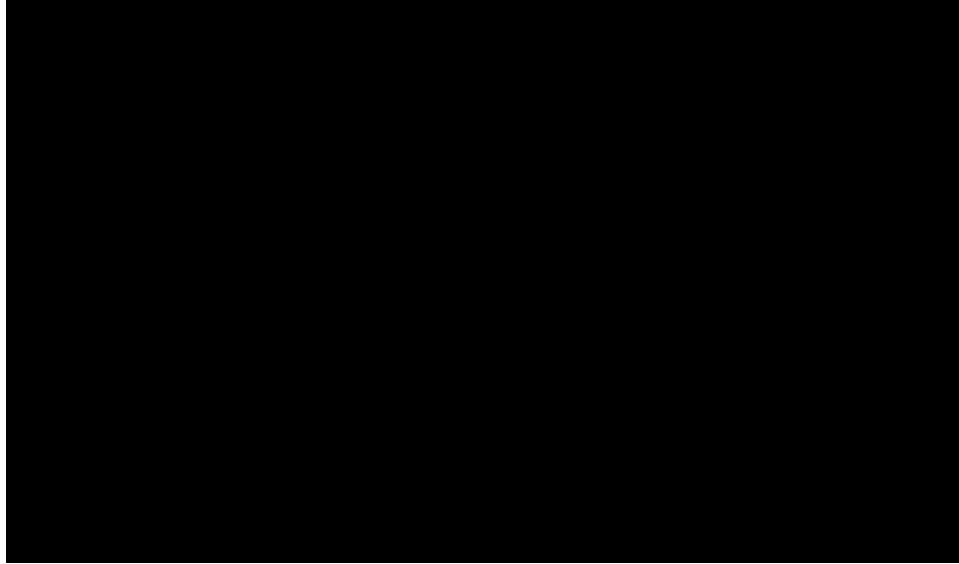
[REDACTED]

[REDACTED].

vii. Claim element [2.a]

Claim element [2.a] is met by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 determines whether the wearable electronic device is attached to the user (*i.e.*, being worn on a user's finger) using its optical sensor:



JDX-0001C ¶1390 (citing RX-0033C at OURA147800001495). Specifically, the Oura Ring Gen.

1 [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED].

viii. Claim element [2.b]

Claim element [2.b] is met by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 [REDACTED]. The device
will not t [REDACTED]

[REDACTED]. JDX-0001C ¶¶1395-96 (citing OURA1478_00001496-1500; RX-0034C). The source code
establishes that [REDACTED]

[REDACTED]
[REDACTED]
[REDACTED]. *Id.*

ix. Claim element [3]

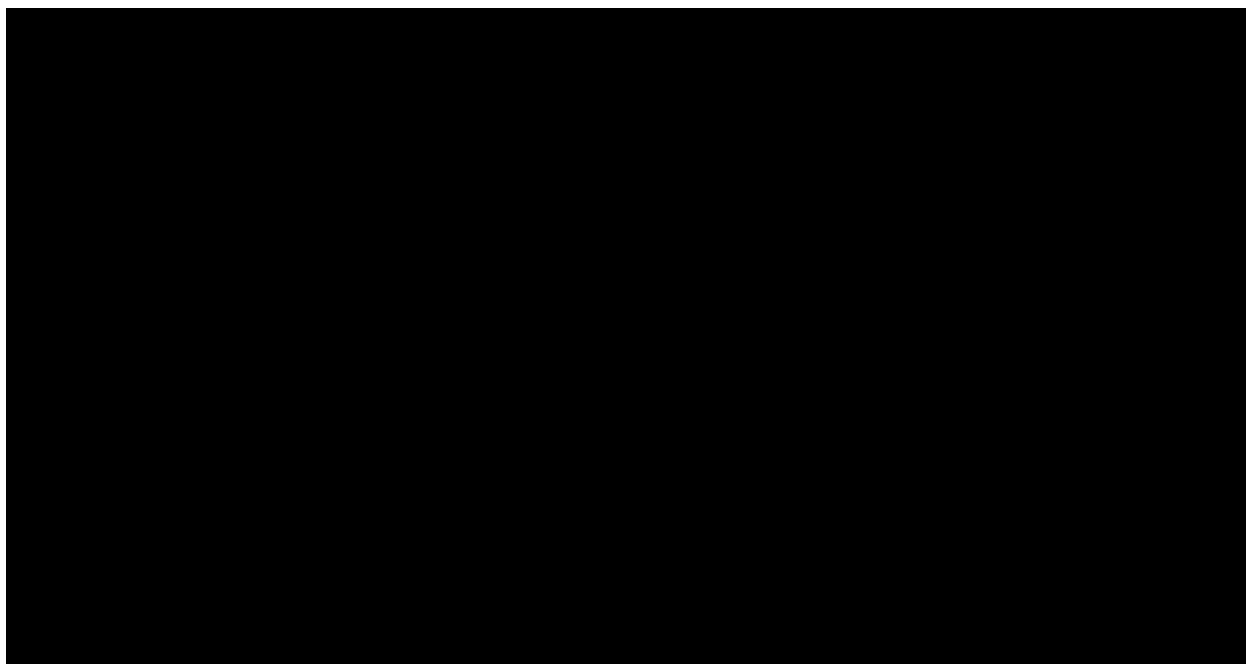
Claim 3 requires the wearable electronic device of claim 2, wherein the processor is further configured to “perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.”

The Oura Ring Gen. 1 cannot [REDACTED]
[REDACTED], as explained above for [1.e]. Thus, the
Oura Ring Gen. 1 [REDACTED]
[REDACTED]
[REDACTED]. RX-0033C;
JDX-0001C ¶1401. As one example, [REDACTED]
[REDACTED]. JDX-
0001C ¶¶1401-02 (citing RX-0034C).

x. Claim elements [9.pre]-[9.c]

Claim elements [9.pre]-[9.c] are each is met by the Oura Ring Gen. 1. Samsung does not dispute claim elements [9.pre] – [9.b]. CSUCE at 13.

The Oura Ring Gen. 1 was a smart ring worn on a user’s finger. JDX-0001C ¶¶1424-26.



RX-0029C (Oura Ring Gen. 1 CAD file)



RX-0054 at OURA1478_00017075, -83; RX-0055; RX-0056 (pictures from Quantified Self Expo 2015, uploaded to Flickr.com in June 2015).

The Oura Ring Gen. 1 includes one or more accelerometers that detects movement and is therefore a “first sensor.” JDX-0001C ¶¶1428-33; RX-0069; RX-0033C at OURA147800001496-1500; RX-0034C.

The Oura Ring Gen. 1 also includes “Infrared (IR) LEDs” and a “phototransistor,” which together function as a “second sensor.” JDX-0001C ¶¶1434-37; RX-0033C at OURA1478_00001496-500; RX-0069; RX-0023; RX-0034C.

The Oura Ring Gen. 1 also includes at least one processor, for example a CPU. JDX-0001C ¶¶1438-41; RX-0026C (Oura Ring Gen. 1 BOM detailing [REDACTED] [REDACTED]).

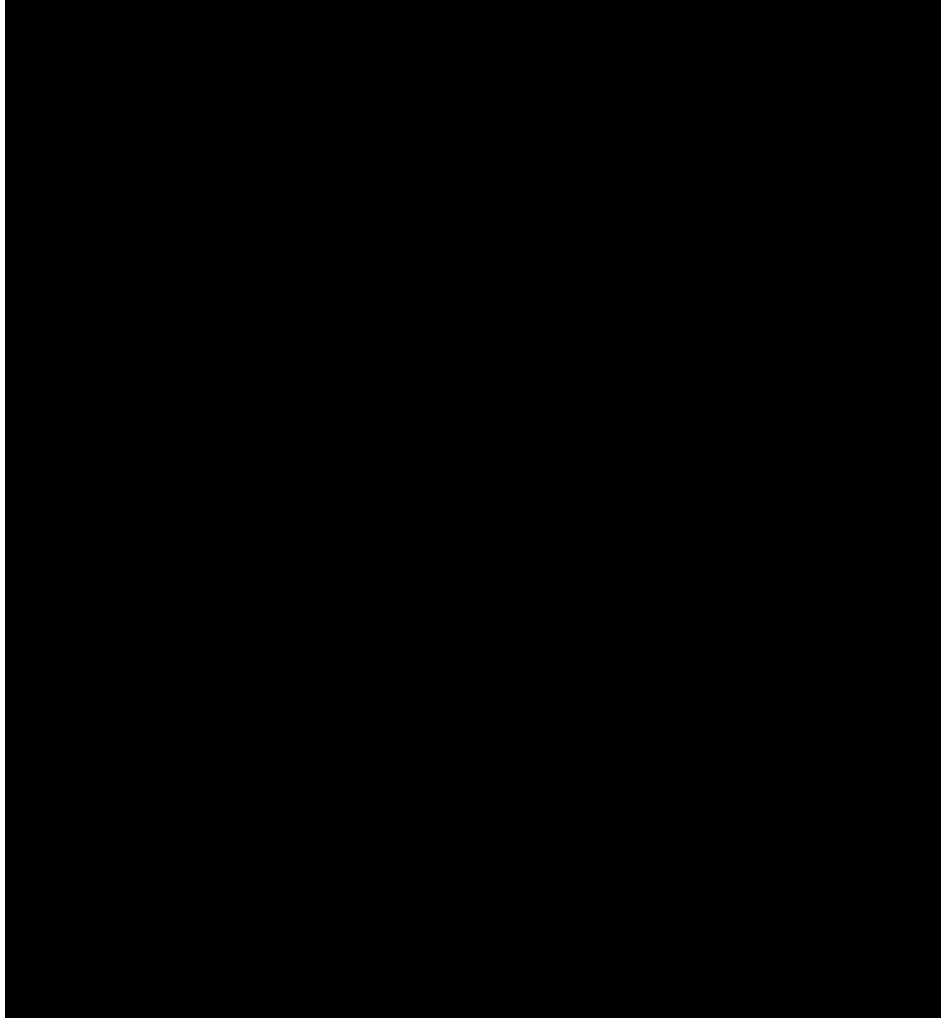
Further evidence that the Oura Ring Gen. 1 was a wearable electronic device comprising a first sensor, a second sensor, and at least one processor will be provided by the testimony of Mr. Christophy.

xi. [Uncontested] Claim element [9.d]

Claim element [9.d] is met by the Oura Ring Gen. 1.

In the Oura Ring Gen. 1, the accelerometer (*i.e.*, first sensor) is used to detect motion (*i.e.*, obtain first information corresponding to movement of the wearable electronic device). RX-0033C; JDX-0001C ¶¶1442-48.

For instance, t [REDACTED]
[REDACTED]
[REDACTED]:
[REDACTED]



RX-0033C at OURA147800001495-1501

In [REDACTED]

[REDACTED]. For example, [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

See JDX-0001C ¶1370 (citing RX-0034C ([REDACTED])

[REDACTED])).

xii. Claim element [9.e]

Claim element [9.e] is met by the Oura Ring Gen. 1.

The Oura Ring Gen. 1 was configured such that LEDs (*i.e.*, light emitting unit) [REDACTED]
[REDACTED]. See JDX-0001C ¶¶1449-53 (citing RX-0033C at
OURA147800001495-1501; RX-0034C). For example, [REDACTED]
[REDACTED]
[REDACTED]. JDX-0001C ¶1452 (citing RX-0034C).
As another example, [REDACTED]
[REDACTED]
[REDACTED]. See JDX-0001C ¶¶1451 (citing RX-0034C (f [REDACTED]
[REDACTED])), and
[REDACTED]. The device n [REDACTED]
[REDACTED]. *Id.*; RX-0033C at OURA1478_00001496.

In [REDACTED]
[REDACTED]. JDX-0001C ¶1451 (citing RX-0033C at
OURA1478_00001496).

xiii. [Uncontested] Claim element [9.f]

Claim element [9.f] is met by the Oura Ring Gen. 1.

During t [REDACTED]
[REDACTED], the Oura Ring Gen. 1 [REDACTED]
[REDACTED]. JDX-0001C ¶¶1457-58 (citing RX-0033C; RX-0034C). The
Oura Ring Gen. 1 [REDACTED]
[REDACTED]

See, e.g., RX-0034C ([REDACTED]); JDX-0001C ¶¶1458.

xiv. Claim element [9.g]

Claim element [9.g] is met by the Oura Ring Gen. 1.

When the Oura Ring Gen. 1 [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. JDX-0001C ¶¶1462 (citing RX-0034C). Thus, based at least in part on a [REDACTED]

[REDACTED]

[REDACTED]. *Id.*

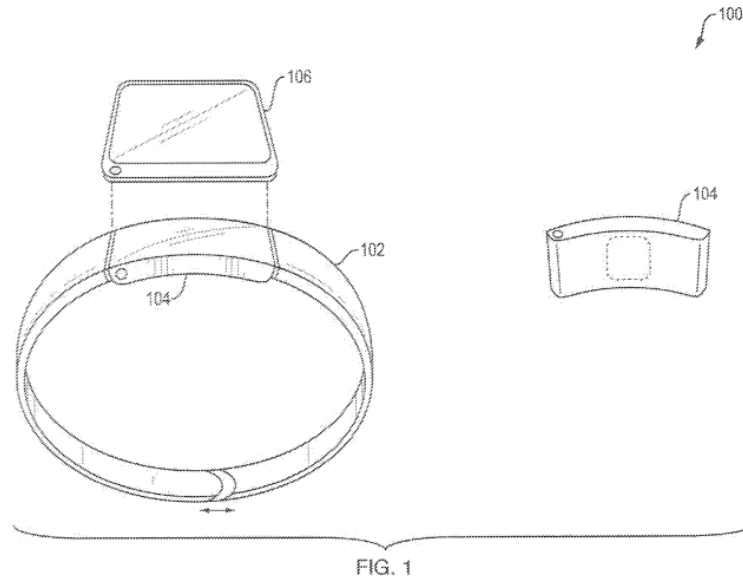
3. Ground 2: Anticipation in view of Ahmed486

Claims 1-3 and 9 of the '677 Patent are invalid as anticipated by the teachings of Ahmed486, which meet each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi. Samsung does not dispute claim elements [1.pre] – [1.b]. CSUCE at 26.

i. Claim elements [1.pre]-[1.c]

Claim elements [1.pre]-[1.c] are each is taught by Ahmed486.

Ahmed486 discloses a “lightweight wearable system” that is a “wearable electronic device.” RX-0016 ¶¶[0004]; JDX-0001C ¶¶2086-89.



Ahmed486 at Fig. 1

Ahmed486 discloses that the lightweight wearable system includes “a motion sensor for detecting a motion of the user,” which “may receive data on the motion of the user using, for example, an accelerometer,” and is accordingly a motion sensor. JDX-0001C ¶¶2090-93 (citing RX-0016 ¶¶[0011], [0014], [0124]).

Ahmed486 further discloses that the lightweight wearable system includes “one or more optical sensors includes a light emitting unit and a light receiving unit,” specifically “a plurality of light emitters for emitting light toward the user’s skin, a plurality of light detectors for receiving light reflected from the user’s skin,” which Ahmed486 describes as “an optical sensor” and is accordingly one or more optical sensors includes a light emitting unit and a light receiving unit. JDX-0001C ¶¶2094-98 (citing RX-0016 ¶¶[0011], [0062], [0064], [0124], [0130]).

Ahmed486 also discloses that the lightweight wearable system includes a processor or processing module and accordingly includes at least one processor. JDX-0001C ¶¶2099-103 (citing RX-0016 ¶¶[0080], [0087], [0203], [0216]).

ii. [Uncontested] Claim element [1.d]

Claim element [1.d] is taught by Ahmed486.

The processing module of Ahmed486 obtains a motion signal from the user by using the motion sensor prior to measuring heart rate. JDX-0001C ¶¶2104-08 (citing RX-0016 ¶¶[0011], [0124], [0130]) (“the processing module may receive data on the motion of the user using, for example, an accelerometer”). Specifically, “processing module may process the motion data to determine a motion status of the user [and] may adjust the duty cycle of one or more light emitters and the corresponding sampling rate of the one or more light detectors based on the motion status.” RX-0016 ¶[0124]. Thus, because the light emitters are selectively activated based on the motion status of the user, the motion of the user is acquired prior to the heart rate being measured. RX-0016 ¶[0011] (“based on the motion status of the user, automatically and selectively activate one or more of the light emitters to determine a heart rate of the user”).

This disclosure meets claim element [1.d]. The processor obtains a first motion signal (*i.e.*, motion data from the accelerometer) corresponding to the wearable electronic device prior to a biological signal being obtained (*i.e.*, prior to obtaining a biological signal used to determine heart rate). JDX-0001C ¶¶2104-08.

iii. Claim element [1.e]

Claim element [1.e] is taught by Ahmed486.

The processing module of Ahmed486 determines a motion status and, based on that determination, activates the light emitters. JDX-0001C ¶¶2109-14 (citing RX-0016 ¶¶[0011], [0124], [0130]). Specifically, “upon determining that the motion status indicates that the user is at

a first higher level of motion, the processing module may activate the light emitters at a first higher duty cycle and sample the reflected light using light detectors sampling at a first higher sampling rate.” RX-0016 ¶[0124]; RX-0016 ¶[0011] (“based on the motion status of the user, automatically and selectively activate one or more of the light emitters to determine a heart rate of the user”); RX-0016 ¶[0130] (“upon determining that the motion status indicates that the user is at a first higher level of motion (e.g., exercising), one or more light emitters may be activated to emit light at a first wavelength”).

This disclosure meets claim element [1.e]. The processor emits light using the light emitting unit (*i.e.*, activating the light emitters) based at least in part on a determination that the first motion signal satisfies a specified condition (*i.e.*, when the motion signal is sufficient to indicate a higher level of motion of the user). JDX-0001C ¶¶2109-14.

iv. [Uncontested] Claim element [1.f]

Claim element [1.f] is taught by Ahmed486.

Ahmed486 discloses that biological signals for determining heart rate are obtained using “light emitters for emitting light toward the user’s skin” and “a plurality of light detectors for receiving light reflected from the user’s skin.” JDX-0001C ¶¶2115-19 (citing RX-0016 ¶[0011], [0014], [0084], [0124]). Specifically, Ahmed486 teaches the use of “a plurality of electronic components configured for analyzing data corresponding to the reflected light to automatically and continually determine a heart rate of the user.” RX-0016 ¶[0084]. Furthermore, the “processing module [is] configured to determine a heart rate of the user based on light received by one or more of the light detectors.” RX-0016 ¶[0014]. Thus, light reflected on the user is obtained using the light sensors and is used to determine the heart rate of the user.

This disclosure meets claim element [1.f]. The biological signal (*i.e.*, the reflected light corresponding to heart rate) is obtained using the light receiving unit (*i.e.*, the light detectors) based at least in part on light reflected from the user. JDX-0001C ¶¶2115-19.

v. *[Uncontested] Claim element [1.g]*

Claim element [1.g] is taught by Ahmed486.

Ahmed486 discloses that the duty cycle of the light emitters and the corresponding sampling rate of the light detectors “may be adjusted in a graduated or continuous manner based on the motion status or level of motion of the user.” JDX-0001C ¶¶2120-24 (citing RX-0016 ¶¶[0124], [0130]). Specifically, the “processing module may adjust the duty cycle of one or more light emitters and the corresponding sampling rate of the one or more light detectors based on the motion status.” RX-0016 ¶[0124]. Because the motion status is continuously monitored and used to adjust the duty cycle during receiving the biological signal used for the heart rate measurement, a second motion signal is obtained by the motion sensor during the obtaining of the biological signal.

vi. *Claim element [1.h]*

Claim element [1.h] is taught by Ahmed486.

Ahmed486 discloses that when the level of motion is at a second lower level and not a higher level of motion, “the processing module may activate the light emitters at a second lower duty cycle and sample the reflected light using light detectors sampling at a second lower sampling rate.” RX-0016 ¶[0124]. At this lower duty cycle, the light emitters refrain from emitting light for a greater percentage of the time: for example, a “[w]hen the user is at rest, the light detectors may be activated for about a 1% duty cycle—10 milliseconds each second (*i.e.*, 1% of the time).” RX-0016 ¶[0126]. As a result of this teaching, the light emitters refrain from emitting light some

percentage of the time “to minimize consumption of power by the wearable physiological measurement system.” RX-0016 ¶[0014].

This disclosure meets claim element [1.h]. The processor refrains from emitting light (*i.e.*, emits light only at a lower duty cycle) based at least in part on a determination that the second motion signal does not satisfy the specified condition (*i.e.*, if the motion level of the user is not a higher level of motion). JDX-0001C ¶¶2127-31.

vii. Claim element [2.a]

Claim element [2.a] is taught by Ahmed486.

Ahmed486 discloses that “the same sensor used for measuring heart rate may be used to indicate whether the user is wearing the wearable system or not.” JDX-0001C ¶¶2132-36 (citing RX-0016 ¶¶[0069], [0085]). Specifically, the “identity of the portion of the user’s body to which the wearable system is attached may be determined based on one or more parameters including, but not limited to, absorbance level of light as returned from the user’s skin, reflectance level of light as returned from the user’s skin.” RX-0016 ¶[0085]. Thus, the optical sensors are used to determine whether the device is attached to the user.

viii. Claim element [2.b]

Claim element [2.b] is taught by Ahmed486.

Ahmed486 discloses that when the wearable system is not being worn, “power to the one or more LEDs may be cut off as soon as this situation is detected, and reset once the user has put the wearable system back on their wrist.” RX-0016 ¶¶[0069], [0086] (“[i]f the wearable system is determined to be taken off from the user’s body, the processing module is configured to deactivate the light emitters and the light detectors and cease monitoring of the heart rate of the user to conserve power.”). Thus, when the wearable system is back on a user’s wrist, the light emitters are

activated. JDX-0001C ¶¶2137-41. Thus, light is emitted based on a determination that the wearable device is attached to the user.

ix. Claim element [3]

Claim 3 requires the wearable electronic device of claim 2, wherein the processor is further configured to “perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.” This is taught by Ahmed486.

Ahmed486 discloses that part of the determination of whether the device is being worn (*i.e.*, “attached to a user”) includes detecting motion using a motion sensor. JDX-0001C ¶¶2142-46 (citing RX-0016 ¶[0085]). Specifically, the “identity of the portion of the user’s body to which the wearable system is attached may be determined based on one or more parameters including [] motion sensor data (e.g., accelerometer and/or gyroscope).” RX-0016 ¶[0085]

x. Claim elements [9.pre]-[9.c]

Claim elements [9.pre]-[9.c] are each taught by Ahmed486. Samsung does not dispute claim elements [9.pre] – [9.b]. CSUCE at 27.

Ahmed486 discloses a “lightweight wearable system” that is a “wearable electronic device.” RX-0016 ¶[0004]; JDX-0001C ¶¶2166-69.

Ahmed486 discloses that the lightweight wearable system includes “a motion sensor for detecting a motion of the user,” which “may receive data on the motion of the user using, for example, an accelerometer,” and is accordingly a first sensor. JDX-0001C ¶¶2170-73 (citing RX-0016 ¶¶[0011], [0014], [0124]).

Ahmed486 further discloses that the lightweight wearable system includes “one or more optical sensors includes a light emitting unit and a light receiving unit,” specifically “a plurality of light emitters for emitting light toward the user’s skin, a plurality of light detectors for receiving light reflected from the user’s skin,” which Ahmed486 describes as “an optical sensor” and is

accordingly a second sensor. JDX-0001C ¶¶2174-78 (citing RX-0016 ¶¶[0011], [0062], [0064], [0124], [0130]).

Ahmed486 also discloses that the lightweight wearable system includes a processor or processing module and accordingly includes at least one processor. JDX-0001C ¶¶2179-83 (citing RX-0016 ¶¶[0080], [0087], [0203], [0216]).

xi. [Uncontested] Claim element [9.d]

Claim element [9.d] is taught by Ahmed486.

The processing module of Ahmed486 obtains a motion signal from the user by using the motion sensor. JDX-0001C ¶¶2184-88 (citing RX-0016 ¶¶[0011], [0124], [0130]). Specifically, “the processing module may receive data on the motion of the user using, for example, an accelerometer.” RX-0016 ¶[0124]. This disclosure meets claim element [9.d]. The processor obtains, using the first sensor (*i.e.*, the motion sensor or accelerometer), first information corresponding to movement of the wearable electronic device (*i.e.*, the motion data indicating the user’s level of motion).

xii. Claim element [9.e]

Claim element [9.e] is taught by Ahmed486.

The processing module of Ahmed486 activates the optical sensor (*i.e.*, second sensor) based on the motion status of the user. JDX-0001C ¶¶2189-93 (citing RX-0016 ¶¶[0011], [0014], [0124], [0130]). Specifically, “processing module [is] configured to determine a motion status of the user based on data received from the motion sensor, and based on the motion status of the user, automatically and selectively activate one or more of the light emitters to determine a heart rate of the user.” RX-0016 ¶[0011].

xiii. [Uncontested] Claim element [9.f]

Claim element [9.f] is taught by Ahmed486.

The processing module of Ahmed486 obtains a biological signal corresponding to heart rate using the activated optical sensor while continuing to monitor the user's motion. JDX-0001C ¶¶2194-99 (citing RX-0016 ¶¶[0011], [0014], [0124]). Specifically, Ahmed486 discloses that “based on the motion status of the user, automatically and selectively activate one or more of the light emitters to determine a heart rate of the user” (RX-0016 ¶[0011]) and the “processing module may adjust the duty cycle of one or more light emitters and the corresponding sampling rate of the one or more light detectors based on the motion status.” RX-0016 ¶[0124]. Thus, processing module obtains, using the activated second sensor (*i.e.*, the optical sensor), second information corresponding to a biological signal related to the user (*i.e.*, the reflected light used to determine heart rate) during the obtaining of the first information (*i.e.*, while the motion data continues to be obtained).

xiv. Claim element [9.g]

Claim element [9.g] is taught by Ahmed486.

Ahmed486 discloses that when the level of motion is at a second lower level (when the “motion status indicates that the user is at a second lower level of motion”) and not a higher level of motion, “the processing module may activate the light emitters at a second lower duty cycle and sample the reflected light using light detectors sampling at a second lower sampling rate.” JDX-0001C ¶¶2200-06 (citing RX-0016 ¶¶[0011], [0014], [0124], [0130]). At this lower duty cycle, the light emitters refrain from emitting light for a greater percentage of the time: for example, a “[w]hen the user is at rest, the light detectors may be activated for about a 1% duty cycle—10 milliseconds each second (*i.e.*, 1% of the time).” RX-0016 ¶[0126]. As a result of this teaching, the light emitters refrain from emitting light some percentage of the time “to minimize consumption of power by the wearable physiological measurement system.” RX-0016 ¶[0014].

This disclosure meets claim element [9.g]. The processor refrains from obtaining the second information using the activated second sensor (*i.e.*, the light detectors cease or reduce sampling of reflected light) based at least in part on a determination that the first information satisfies a specified range (*i.e.*, when the motion status indicates a second lower level of motion). JDX-0001C ¶¶2200-06.

4. Ground 3: Anticipation in view of Presura

Claims 1 and 9 of the '677 Patent are invalid as anticipated by the teachings of Presura, which meet each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi.

i. *Claim elements [1.pre]-[1.c]*

Claim elements [1.pre]-[1.c] are each taught by Presura. Samsung does not contest elements [1.pre]-[1.b]. CSUCE at 21.

Presura discloses a “heart rate monitor system 100” that is a “wearable electronic device.” JDX-0001C ¶¶1843-46 (citing RX-0015 at 5:1-2, 8:16-19). Specifically, Presura discloses that “the heart rate monitor system 100 can be arranged at a wrist or a forearm or behind an ear of a user” and that the “heart rate monitor system can be arranged as a wrist device.” RX-0015 at 5:1-2, 8:16.

The heart rate monitor system of Presura includes a “secondary sensor 120” that is a “motion sensor.” JDX-0001C ¶¶1847-51 (citing RX-0015 at 4:22-31, 6:22-28, 8:7-8, 11:1-2). Specifically, Presura discloses that “the secondary sensor 120 is a motion sensor such as an accelerometer,” and claims “[a] heart rate monitor device (100) ... wherein the at least one secondary sensor (110) is a motion sensor, for example an accelerometer.” RX-0015 at 8:7-8, 11:1-2.

The heart rate monitor system of Presura also includes a “primary sensor 110” that is an optical sensor having both a light emitting unit and a light receiving unit. JDX-0001C ¶¶1852-56 (citing RX-0015 at 1:9-10, 5:3-9, 8:20, 10:31-32). Specifically, Presura discloses that “[t]he primary sensor 110 can be implemented as an optical sensor which comprises a light source for example in form of an LED for producing artificial light,” and that the emitted light “can be reflected back to a photo detector which may also be part of the optical sensor 110.” RX-0015 at 5:3-9. Presura claims “[a] heart rate monitor device (100) ... wherein the at least one primary sensor (110) is an optical sensor.” RX-0015 at 10:31-32 (claim 4), 8:20 (“The primary sensor 110 can be embodied as an optical sensor”).

The heart rate monitor system of Presura further includes microcontroller with a control unit and processing unit that therefore includes “at least one processor.” JDX-0001C ¶¶1857-60 (citing RX-0015 at 4:26-29, 5:15-24). Specifically, Presura discloses that “the control unit 131, the processing unit 132 and the power management unit 160 can be implemented by a [microcontroller] 130” and that the “heart rate monitor system 100 furthermore comprises optionally a control unit 131 for controlling the operation of the at least one primary sensor 110” and that “a processing unit 132 is provided for processing the output of the at least one primary sensor 110 to determine the heart rate of a user.” RX-0015 at 4:26-29, 5:21-22.

ii. Claim element [1.d]

Claim element [1.d] is taught by Presura.

The processor of Presura obtains motion information from the secondary (motion) sensor and uses that information to control whether to activate the primary (optical) sensor. JDX-0001C ¶¶1861-65 (citing RX-0015 at 5:25-31, 6:1-7). Specifically, Presura teaches that “the secondary sensors can be used to control the activation and deactivation (on/off switching) of the primary

sensor.” RX-0015 at 5:25-31, at 6:24-28 (primary sensor is the heart rate sensor; secondary sensor is the motion sensor).

This disclosure meets claim element [1.d]. The processor obtains, using the motion sensor (*i.e.*, the accelerometer), a first motion signal corresponding to the wearable electronic device prior to a biological signal being obtained using the optical sensor (*i.e.*, before the primary sensor is switched on to measure heart rate). JDX-0001C ¶¶1861-65.

iii. Claim element [1.e]

Claim element [1.e] is taught by Presura.

The processor of Presura activates the primary optical sensor, and therefore its LED light source, when the secondary sensor detects a change in the user’s motion pattern. JDX-0001C ¶¶1866-71 (citing RX-0015 at 5:3-9, 5:25-31, 6:22-28). Specifically, “[a]s soon as the secondary sensors 120 detect a change in the motion or motion pattern of the user, the primary heart rate sensor 100 can be activated again to measure the actual heart [rate] of the user.” RX-0015 at 6:22-28. Because the primary sensor “comprises a light source for example in form of an LED” that “emits the artificial light onto the skin of a user,” activating the primary sensor necessarily causes light to be emitted. RX-0015 at 5:3-5.

This disclosure meets claim element [1.e]. The processor emits light using the light emitting unit (*i.e.*, the LED light source of the primary sensor) based at least in part on a determination that the first motion signal satisfies a specified condition (*i.e.*, that the secondary sensors have detected a change in the “motion pattern of the user”). JDX-0001C ¶¶1866-71.

iv. [Uncontested] Claim element [1.f]

Claim element [1.f] is taught by Presura.

The photo detector of Presura’s primary sensor receives light reflected from the user’s skin and generates the output signal from which heart rate is determined. JDX-0001C ¶¶1872-76 (citing

RX-0015 at 1:9-13, 4:12-21, 5:3-9). Specifically, the “light source emits the artificial light onto the skin of a user [and] the artificial light can be scattered throughout the skin and can be reflected back to a photo detector which may also be part of the optical sensor 110. The photo detector is detecting the reflected light through the skin of the user and is generating an output signal.” RX-0015 at 5:4-9. Presura confirms that “[b]ased on the received signals from the sensor, the heart rate of a user can be determined,” and that the “processing unit 132 is provided for processing the output of the at least one primary sensor 110 to determine the heart rate of a user.” RX-0015 at 1:11-13, 4:28-29.

v. *Claim element [1.g]*

Claim element [1.g] is taught by Presura.

The secondary sensor of Presura continues to supply motion information while the primary optical sensor is measuring heart rate, and that motion information is used to adjust the primary sensor’s operation. JDX-0001C ¶¶1877-81 (citing RX-0015 at 5:27-29, 6:8-10, 6:22-28). Specifically, “additional information from the secondary sensors can be used to control the activation and deactivation (on/off switching) of the primary sensor.” RX-0015 at 5:27-29. Deactivation of the optical sensor based on information from the motion sensors is only possible if the motion signal continues to be obtained while heart rate is being measured.

This disclosure meets claim element [1.g]. The processor obtains, using the motion sensor, a second motion signal corresponding to the wearable electronic device (*i.e.*, the continuing accelerometer output) during the obtaining of the biological signal (*i.e.*, while the primary sensor measures heart rate). JDX-0001C ¶¶1877-81.

vi. *Claim element [1.h]*

Claim element [1.h] is taught by Presura.

When the processor determines that there is no change to the motion pattern of an individual, Presura deactivates the primary optical sensor and estimates heart rate from the secondary sensor data instead, so that no light is emitted. JDX-0001C ¶¶1882-86 (citing RX-0015 at 5:3-9, 5:25-31, 6:8-17, 6:22-28). Specifically, “if the information from the secondary sensors is sufficient to estimate or predict the heart rate of the user... the primary sensor 110 can be activated and deactivated at least for a duration during which the heart rate can be estimated accurately enough.” RX-0015 at 6:8-14. This can occur if “it can be determined that the user is in a constant motion or is not moving.” RX-0015 at 6:24-25. Because the LED light source is part of the primary sensor, deactivating that sensor necessarily means that the sensor refrains from emitting light. RX-0015 at 5:3-9.

This disclosure meets claim element [1.h]. The processor refrains from emitting light (*i.e.*, the primary sensor and its LED are deactivated) based at least in part on a determination that the second motion signal does not satisfy the specified condition (*i.e.*, that the accelerometer shows that there is no change in the “motion pattern of the user”). JDX-0001C ¶¶1882-86.

vii. Claim elements [9.pre]-[9.c]

Claim elements [9.pre]-[9.c] are each taught by Presura. Samsung does not dispute claim elements [9.pre] – [9.b]. CSUCE at 22.

Presura discloses a “heart rate monitor system 100” that is a “wearable electronic device,” which “can be arranged at a wrist or a forearm or behind an ear of a user” and “can be arranged as a wrist device.” JDX-0001C ¶¶1902-04 (citing RX-0015 at 5:1-2, 8:17-19).

The heart rate monitor system of Presura includes a “secondary sensor 120” that is “a first sensor.” JDX-0001C ¶¶1905-09 (citing RX-0015 at 4:22-31, 6:22-28, 8:7-8, 11:1-2). Specifically, Presura discloses that “the secondary sensor 120 is a motion sensor such as an accelerometer,” and

claims “[a] heart rate monitor device (100) ... wherein the at least one secondary sensor (110) is a motion sensor, for example an accelerometer.” RX-0015 at 8:7-8, 11:1-2.

The heart rate monitor system of Presura also includes a “primary sensor 110” that is “a second sensor.” JDX-0001C ¶¶1910-14 (citing RX-0015 at 1:9-10, 5:3-9, 8:20, 10:31-32). Specifically, Presura discloses that “[t]he primary sensor 110 can be implemented as an optical sensor which comprises a light source for example in form of an LED for producing artificial light,” and that the emitted light “can be reflected back to a photo detector which may also be part of the optical sensor 110.” RX-0015 at 5:3-9. Presura claims “[a] heart rate monitor device (100) ... wherein the at least one primary sensor (110) is an optical sensor.” RX-0015 at 10:31-32 (claim 4), 8:20 (“The primary sensor 110 can be embodied as an optical sensor”).

The heart rate monitor system of Presura further includes microcontroller with a control unit and processing unit that therefore includes “at least one processor.” JDX-0001C ¶¶1915-18 (citing RX-0015 at 4:26-29, 5:15-24). Specifically, Presura discloses that “the control unit 131, the processing unit 132 and the power management unit 160 can be implemented by a [microcontroller] 130” and that the “heart rate monitor system 100 furthermore comprises optionally a control unit 131 for controlling the operation of the at least one primary sensor 110” and that “a processing unit 132 is provided for processing the output of the at least one primary sensor 110 to determine the heart rate of a user.” RX-0015 at 4:26-29, 5:21-22.

viii. [Uncontested] Claim element [9.d]

Claim element [9.d] is taught by Presura.

The processor of Presura obtains motion measurements (first information corresponding to movement of the wearable electronic device) from the first sensor (the accelerometer). JDX-0001C ¶¶1919-23 (citing RX-0015 at 6:22-28, 8:7-8). Specifically, an accelerometer is used to determine “that the user is in a constant motion or is not moving.” RX-0015 at 6:22-28. Determining whether

“the user is in a constant motion or is not moving” requires the processor to obtain motion measurements from the accelerometer that correspond to movement of the wearable electronic device.

ix. Claim element [9.e]

Claim element [9.e] is taught by Presura.

The processor of Presura activates the primary (optical) sensor (the second sensor) when the accelerometer detects a change in the user’s motion. JDX-0001C ¶¶1924-28 (citing RX-0015 at 5:25-31, 6:22-28). Specifically, “[a]s soon as the secondary sensors 120 detect a change in the motion or motion pattern of the user, the primary heart rate sensor 100 can be activated again to measure the actual heart [rate] of the user.” RX-0015 at 6:22-28, *id.* at 5:25-31 (“the addition information from the secondary sensors can be used to control the activation and deactivation (on/off switching) of the primary sensor”).

x. [Uncontested] Claim element [9.f]

Claim element [9.f] is taught by Presura.

Once activated, the second sensor (the primary optical sensor) obtains an output signal from the received light that corresponds to the user’s heart rate while the first sensor (the accelerometer) continues to obtain motion measurements. JDX-0001C ¶¶1929-34 (citing RX-0015 at 4:12-21, 5:3-9, 5:12-14, 5:27-29, 6:8-10, 6:22-28). Specifically, “the artificial light can be scattered throughout the skin and can be reflected back to a photo detector which may also be part of the optical sensor 110 [and the] photo detector is detecting the reflected light through the skin of the user and is generating an output signal.” RX-0015 at 5:3-9. Using the output signal from the photo detector, a “processing unit 132 receives the signal from the optical sensor 110 and determines a pulse rate or a heart rate of the user.” RX-0015 at 5:12-14.

Furthermore, “additional information from the secondary sensors can be used to control the activation and deactivation (on/off switching) of the primary sensor.” RX-0015 at 5:27-29. Thus, while the primary sensor is active, the secondary sensor continues to obtain measurements, such as motion, to determine whether to keep the primary heart rate sensor activated or whether to deactivate the primary heart rate sensor.

This disclosure meets claim element [9.f]. The processor obtains, using the activated second sensor (*i.e.*, the primary optical sensor), second information corresponding to a biological signal related to the user (*i.e.*, output signal from the reflected light, from which heart rate is determined) during the obtaining of the first information (*i.e.*, while the accelerometer continues to report motion). JDX-0001C ¶¶1929-34.

xi. Claim element [9.g]

Claim element [9.g] is taught by Presura.

When the processor determines that there is no change to the motion pattern of an individual, Presura deactivates the primary optical sensor and estimates heart rate from the secondary sensor data instead, so that no light is emitted. JDX-0001C ¶¶1935-44 (citing RX-0015 at 5:3-9, 5:25-31, 6:8-17, 6:22-28). Specifically, “if the information from the secondary sensors is sufficient to estimate or predict the heart rate of the user... the primary sensor 110 can be activated and deactivated at least for a duration during which the heart rate can be estimated accurately enough.” RX-0015 at 6:8-14. This can occur if “it can be determined that the user is in a constant motion or is not moving.” RX-0015 at 6:24-25. Because the LED light source is part of the primary sensor, deactivating that sensor necessarily means that the sensor refrains from emitting light. RX-0015 at 5:3-9.

This disclosure meets claim element [9.g]. The processor refrains from obtaining the second information using the activated second sensor (*i.e.*, the primary optical sensor is deactivated

and no longer detects reflected light to generate an output signal) based at least in part on a determination that the first information satisfies a specified range (*i.e.*, that the accelerometer shows the user “is in a constant motion or is not moving”). JDX-0001C ¶¶1935-40.

5. Ground 4: Obviousness in view of Presura and Patel

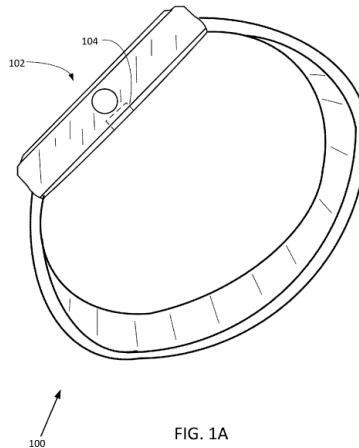
Claims 2 and 3 of the ’677 Patent are invalid as obvious by the teachings of Presura in view of Patel, which together meet or render obvious each limitation of these claims as explained herein and as will be shown by the testimony of Dr. Mortazavi.

a) *Graham* Factors 1-3

As set forth above, Presura discloses a heart rate monitor system with a primary optical sensor and a secondary accelerometer for power management, in which the accelerometer controls activation and deactivation of the optical sensor based on motion patterns. RX-0015 at 4:12-21, 5:25-31, 6:22-28; JDX-0001C ¶¶2026, 1843-60. As addressed above for Ground 3, Presura discloses all elements of claim 1.

But while Presura does not expressly disclose using the optical sensor to determine whether the device is attached to the user (and therefore does not teach the added limitations of claims 2 and 3), these additional limitations are rendered obvious by the teachings of Patel.

Patel discloses a “pulse sensor [that] includes a light source (*e.g.*, a light-emitting diode) and a photo sensor,” and is directed to a wearable device with optical heart rate sensors. JDX-0001C ¶¶2027-28, 2032, 2034 (citing RX-0014 at Abstract, [0012], [0016]).



Patel at Fig. 1A

Furthermore, Patel discloses that the optical sensor can be used to “accurately determine whether it is being worn by a user.” RX-0014 ¶[0012]. Patel therefore expressly describes using an optical sensor of a wearable heart rate detector to determine whether the device is being worn. JDX-0001C ¶2027.

A POSITA would have found it obvious to incorporate Patel’s optical sensor-based process for determining whether the device is being worn by a user into the heart rate monitor system of Presura.

First, Patel teaches that using optical detection of a pulse to confirm on-body status “allows for greater certainty than, for example, using proximity detection alone, which may be triggered by mere proximity of an object, such as the surface of a table.” JDX-0001C ¶2028 (citing RX-0014 ¶[0012]). Incorporating Patel’s on-body detection functionality would provide an additional power-saving feature to Presura’s system by preventing activation of the heart rate sensor when the device is not actually worn, complementing Presura’s motion-based power management with an additional power-saving check using the same hardware already present in the device. JDX-0001C ¶¶2028, 2033-34; RX-0015 at 2:14-3:2.

Furthermore, Presura and Patel are both directed to wearable heart rate monitoring devices employing optical sensors and motion sensors for power management. JDX-0001C ¶2028 (citing Presura at 2:1-8; Patel at Abstract). Their combination would yield the predictable result of improved power saving capability by adding an on-body detection check before or during the heart rate sensor activation logic that Presura already discloses. JDX-0001C ¶2034.

The incorporation of Patel’s optical sensor-based process for determining whether the device is being worn by a user into the heart rate monitor system of Presura would meet all added requirements of claims 2 and 3 of the ’677 Patent:

i. Claim element [2.a]

Claim element [2.a] is met by the combined teachings of Presura and Patel.

By incorporating Patel’s optical sensor-based process for determining whether the device is being worn into the heart rate monitor system of Presura (which otherwise meets all requirements of claim 1), the processor of the resulting combination would be configured to determine whether the wearable electronic device is attached to the user using the one or more optical sensors. JDX-0001C ¶¶2029-35 (citing RX-0014 at Abstract, [0012], [0016], [0020]; RX-0015 at 1:28-30, 2:14-3:2, 4:22-31). Specifically, the optical sensor of the combination would be configured to “accurately determine whether it is being worn by a user.” RX-0014 ¶[0012].

ii. Claim element [2.b]

Claim element [2.b] is met by the combined teachings of Presura and Patel.

Patel teaches that the light source should only be turned on after a determination that the device is being worn by the user. RX-0014 at [0020] (“If, at block 404, the wearable device 100 determines that there is a pulse present, the wearable device 100 obtains a baseline reflection profile for the user at block 408. For example, the processor 202 turns on the light source 214.”); JDX-0001C ¶¶2036-41. Using the determination that the device is being worn as a prerequisite for

heart rate detection would be consistent with, and improve, the power saving functionality of Presura. JDX-0001C ¶2038 (citing RX-0015 at 1:28-30).

iii. Claim element [3]

Claim 3 requires the wearable electronic device of claim 2, wherein the processor is further configured to “perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.” This is met by the combined teachings of Presura and Patel. JDX-0001C ¶¶2036-41 (citing RX-0015 at 6:22-28, RX-0014 at [0014]).

As discussed above, incorporating Patel’s optical sensor-based process for determining whether the device is being worn into the heart rate monitor system of Presura results in an additional power saving feature in the combined system. But Presura also teaches that movement should be used to determine whether to activate the optical sensors: “[a]s soon as the secondary sensors 120 detect a change in the motion or motion pattern of the user, the primary heart rate sensor 100 can be activated again to measure the actual heart [rate] of the user.” RX-0015 at 6:26-28).

Patel suggests that this functionality should be implemented alongside the optical sensor-based process for determining whether the device is being worn: “if, for example, the wearable device detects motion, then it checks for a pulse.” JDX-0001C ¶¶2045-46 (citing RX-0014 at [0014]).

Thus, the combined teachings of Presura and Patel meet the requirements of claim 3, as they teach a processor that is further configured to perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition (*i.e.*, there is a detected a change in the motion or motion pattern of the user).

b) *Graham* Factor 4

Samsung has chosen to present its alleged secondary considerations of nonobviousness in a single argument common to the '359, '677, and '675 Patents. Samsung fails to separate out the alleged inventions of the '359, '677, and '675 Patents and thereby fails its burden to demonstrate a nexus to the claimed inventions, as was its burden to do. As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '677 Patent. *See also* JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

C. Claims 1-3 and 9 of the '677 Patent Are Ineligible Under 35 U.S.C. § 101

Claims 1-3 and 9 of the '677 Patent are patent ineligible under 35 U.S.C. § 101 for claiming patent ineligible subject matter. These claims are directed to the abstract idea of using motion to activate sensing of a biological signal, and as set forth herein, the particular claim elements set forth nothing more than the application of this abstract idea in the context of a generic wearable device.

1. *Alice Step 1: Each claim is directed to the abstract idea of using motion to activate sensing of a biological signal*

Claims 1-3 and 9 of the '677 Patent are directed to the abstract idea of using motion to activate sensing of a biological signal, and as explained further below, there is nothing patent eligible about such generic and conceptual claiming.

Using independent claim 1 of the '677 Patent as exemplary, the key elements are simple: (1) activating an optical biological sensor if a first motion signal meets a condition, and (2) detecting a biological signal from the optical sensor. The remaining claim language concerns only the a generic environment of a wearable sensing device, broad causal and temporal relationships

between sensors which remains in the realm of abstraction, or else the step of refraining from emitting light, which is just the converse of the first concept. An annotation of claim 1 may help to illustrate this:

A wearable electronic device comprising:

a motion sensor;

one or more optical sensors includes a light emitting unit and a light receiving unit;
and

at least one processor configured to:

obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,

emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition,

obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user,

obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and

refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.

Generic
environment of
wearable sensing
device

Activate a
biological sensor
if motion meets a
condition

Use biological
sensor

Refrain from
emitting light if
motion does not
meet a condition

JX-0002 cl. 1 (annotated).

Claims 2-3 and 9 do not change the abstract idea to which independent claim 1 is directed, nor does Samsung claim that it does. Rather:

- Dependent claim 2 differs from claim 1 just by adding the threshold condition of determining that the device is being worn by the user using the optical sensor in an unspecified manner, which is a generic computer-based technique being used in a conventional manner.
- Dependent claim 3 differs from claim 1 just by adding the trivial requirement that the first motion signal must satisfy an unspecified condition, which is a generic computer-based technique being used in a conventional manner.
- Independent claim 9 differs from claim 1 principally by not requiring that the second sensor is optical. Claim 9 does not differ from claim 1 in the nature of the abstract idea that the claim is directed to, nor does it set forth an inventive concept.

Thus, claim 1 of the '677 Patent may be treated as representative. *See Berkheimer*, 881 F.3d at 1365; *Mortg. Grader*, 811 F.3d at 1324 n.6.

Such claims require only routine aspects of wearable components and do not set forth any particular “improvement in computer capabilities”—indeed, the claim elements are far too generic to do so—and instead set forth the generic idea of using motion to activate sensing of a biological signal and merely place this abstract idea in the context of method performed by a generic wearable device. For example, there is no particular manner of motion detection or biometric information detection required: all that is claimed is a broad causal and temporal relationship between sensors of a wearable device. This is an abstract idea. *See GoTV*, 166 F.4th at 1064; *Intellectual Ventures I*, 850 F.3d at 1341; *Elec. Power Grp.*, 830 F.3d at 1353 (“[t]he claims in this case do not even require a new source or type of information, or new techniques for analyzing it”); *SAP Am.*, 898 F.3d at 1168; *AGI SureTrack*, 176 F.4th at 1378; *Killian*, 45 F.4th at 1380.

It also matters little that the claims set forth a “motion sensor,” “biological sensor,” and “controller,” as no particular structure for these generic components of a wearable device are claimed; instead, the various generic computer functionality in claim 1 are just the tools used to accomplish the claimed abstract idea. Where such generic computer processes are claimed, patent eligibility “requires more than result-focused functional language and more than just using those functions in the context of specifically identified content.” *GoTV*, 166 F.4th 1053, 1064; *see also PersonalWeb Techs. LLC v. Google LLC*, 8 F.4th 1310, 1316 (Fed. Cir. 2021) (“The claims do this in a computer environment, but that doesn’t transfigure an idea out of the realm of abstraction”); *In re Killian*, 45 F.4th 1373, 1380 (Fed. Cir. 2022) (“We have distinguished between claims ‘directed to an improvement in the functioning of a computer,’ versus those [] that simply recite ‘generalized steps to be performed on a computer using conventional computer activity.’”).

Samsung contends that the claims “recite novel methods and wearable devices that use a multitude of sensors to collect data and optimize when biometric information is measured, including based on motion signals.” JDX-0048C at 340. But even if correct (and it is not, for the reasons described above), “optimizing when biometric information is measured, including based on motion signals” is an abstract idea, as it just the concept of optimization based on known and generic inputs. *See, e.g., Constellation Designs*, 174 F.4th at 903.

Furthermore, the Federal Circuit has held that “a motion sensor system that evaluates and communicates the relative movement of a body using static and dynamic acceleration information collected from sensors” is ineligible because “is not focused on a specific means or method to improve motion sensor systems, nor is it directed to a specific physical configuration of sensors.” *iLife Techs.*, 839 Fed. Appx. at 537. The same rationale applies here.

2. *Alice Step 2: Each claim merely embodies the abstract idea without adding any inventive concept*

There is nothing in the claims that could supply an inventive concept. The abstract idea itself cannot form the inventive concept at *Alice* step two, and once separated from the abstract idea (using motion to activate sensing of a biological signal), there is no inventive concept claimed. There are only generic functions of motion and biometric signal detection, and unspecified and generic calculations using a “condition” to control the operations of the optical sensor. These elements, alone or in combination, do not convey an inventive concept. *See Killian*, 45 F.4th at 1380 (“the claims here do not detail how the computer should go about determining eligibility for benefits, beyond saying that the computer should determine eligibility based ‘on the identified information’ and ‘current SSDI benefit legal requirements’”).

A wearable device comprising a motion sensor, an optical sensor, and a controller (the claimed hardware environment) was well-understood, routine, and conventional at the time of

the '677 Patent's alleged invention. The '677 Patent even acknowledges that obtaining movement and biometric data was conventional and recognizes that conventional wearable sensors were known. *See* JX-0002 at 1:26-43. A commercially available example of such a wearable device would have included the Basis B1 Band, which is “a wearable device that measures signals such as user movement and biological signals.” JDX-0001C ¶105; *see also id.* ¶1284 (“[t]he Basis B1 Band watch would be an example of a commercially available wearable device with a motion sensor, a biological sensor, and a controller”). The Basis B1 Band includes “an internal accelerometer that allows it to measure steps’ along with ‘the optical and skin temperature sensors’ wherein the optical sensor performs PPG measurements by ‘measuring the reflection of light emitted by the green sensor’ which discerns the heart rate.” JDX-0001C ¶1284 (quoting RX-0009.0013, 0017 (July 25, 2013 review of the Basis B1 Band)).

Ahmed, Presura, and Patel (discussed in relation to invalidity Grounds 2-4) are further evidence that the field of art would have found the claimed determinations to be performed by the device were well-understood, routine, and conventional, even in combination, as of the claimed priority date of the '677 Patent. Mr. Christophy and Mr. Larson's expected testimony regarding the Oura Gen. 1 ring will be further evidence of what was well-understood, routine, and conventional with regard to the claimed determinations of sensor activity based on other sensor activity.

Samsung has made no allegations of any particular inventive concept, and accordingly, the claims of the '677 Patent fail at both *Alice* steps one and two, and are therefore patent ineligible under 35 U.S.C. § 101.

VI. '359 PATENT

A. Oura's Products Do Not Infringe the Asserted Claims

Samsung will not be able to carry its burden to show infringement of the '359 Patent by any product at issue (Oura's Gen. 4, Ring 5, and [REDACTED] rings).

1. Oura's Gen. 4 Ring Does Not Infringe the Asserted Claims

Oura's Gen. 4 ring does not infringe either claims 1 or 4 of the '359 Patent for the reasons explained herein.

- a) Oura's Gen. 4 ring does not meet the limitations of claim element [1.f]

The evidence will show that the Oura Ring 4 does not infringe limitation [1.f], which requires "a select portion of the biological signal."

The applicants for the '359 Patent amended this limitation to include the "select" modifier to the "select portion" element to make clear that the claim does not include using the full biological signal obtained during a select period. RDX-0032C, ¶¶73-74. Specifically, after the Examiner rejected the claim with this language the applicants amended the claim limitation to state "perform the specified function using a **select portion** of the biological signal obtained during a select period of the movement signal, determined to correspond to the specified condition":

1. (Currently amended) A wearable device comprising:
a motion sensor;
a biological sensor; and
a controller configured to:
 obtain a movement signal of an object using the motion sensor, and a biological signal of
the object using the biological sensor, and
 determine, based on an analysis of the movement signal, whether to perform a specified
function of the wearable device using the biological signal, and
 perform ~~a~~ the specified function using ~~at least a~~ select portion of the biological signal
obtained during a ~~specified select period, that of~~ the movement signal, determined to corresponds
to a specified condition.

JX-0005 (SAMS382-0002902). On this basis, Dr. Mortazavi will explain that a POSITA would understand that the plain and ordinary meaning of a “portion” to mean part of a whole or a specific part and, specific to limitation [1.f], not all of or the full biological signal obtained during the select period. RDX-0032C, ¶76.

Indeed, Dr. Mortazavi’s understanding of the plain and ordinary meaning of a “portion” is also consistent with what the co-inventor of the ’359 Patent, Dr. Changmok Choi, understood the word to mean:

[REDACTED]

JX-0020, 70:21-71:3 (objections omitted).

Moreover, Dr. Mortazavi will explain that the ’359 Patent’s use of “average” does not mean using a select portion of the biological signal obtained during the select period. *See, e.g.*, RDX-0032C, ¶78; JX-0001, 6:10-15 (“The first threshold value may be a reference value used to

determine the intentional motion of the target 105. The first threshold value may be determined as an **average value of magnitudes of first signals** obtained when the user moves a muscle for an actual operation.”), 6:46-49 (“The second threshold value may be set as **an averaged minimum value obtained based on data obtained** by simulations in which users moved their muscles to initiate an actual operation.”), 11:56-58 (“The first threshold value may be determined as **an average value of magnitudes of the first signals** obtained while a user moves a muscle for an actual operation.”), 12:23-26 (“The second threshold value may be determined as an averaged minimum value obtained a user moves a muscle for an actual operation.”) (emphases added). Rather, the ’359 Patent refers to averages as a POSITA would understand them—using all of the signal or data in the context of a calculation, and not a portion of signal or data. RDX-0032C, ¶78.

First, in his analysis to support his assertion that the Oura Ring 4 meets this limitation, Dr. Welch repeatedly cites to the function [REDACTED]. RDX-0035C, ¶¶251-253. However, Dr. Mortazavi and Mr. Larson—an Oura engineer responsible for firmware implementation—will testify that the function [REDACTED] is *not* implemented in the Oura Ring 4. RDX-0032C, ¶79. In fact, Mr. Larson will testify that the function [REDACTED] [REDACTED]. The following source code illustrates that the function [REDACTED] is neither activated nor implemented by the Oura Ring 4:

[REDACTED]

RX-0147C (OURA_1478_SC_0000203-206 at -205), [REDACTED] at lines 174-184. Since the [REDACTED] for the Oura Ring 4. RDX-0032C, ¶79.

In view of the fact that this code is not used in Oura Ring 4, Dr. Welch’s citation of this code to support infringement does not support a conclusion that the Oura Ring 4 infringes claim 1 of the ’359 Patent. RDX-0032C, ¶80.

Second, separate from Dr. Welch’s reliance on firmware code that is not used in the Oura Ring 4 (*i.e.*, the S [REDACTED]), the [REDACTED] in the Oura Ring 4 is not configured to “perform the specified function using a **select portion** of the biological signal obtained during a select period, of the movement signal, determined to correspond to the specified condition.” RDX-0032C, ¶82. Dr. Welch appears to assert that the claim requirement to use only a select portion of the biological signal is met by the fact that [REDACTED] [REDACTED]:

[REDACTED]

Id.

But, as will be explained by Dr. Mortazavi and confirmed by Mr. Larson as a factual matter, this section of the firmware [REDACTED] [REDACTED]. RDX-0032C, ¶83. Dr. Mortazavi will further testify that this is consistent with a POSITA’s understanding of what it means to obtain an average. *Id.* That is, when obtaining the average of a collection of data, [REDACTED]

use only a portion of the data (portion of the biological signal as claimed). *Id.* Instead, Dr. Mortazavi will explain, and Mr. Larson will confirm, that the Oura Ring 4 [REDACTED]

[REDACTED]. *Id.*

Further, Dr. Welch improperly relies on source code and/or functionality performed on devices other than the Oura Ring 4 to show that the [REDACTED] in the Oura Ring 4 is configured to perform the specified function, as defined by Dr. Welch—s [REDACTED] [REDACTED] (RDX-0035C, ¶239)—as claimed in limitation [1.f]. RDX-0032C, ¶85. For example, Dr. Welch acknowledges that “the **application software** for the Accused Product [REDACTED] [REDACTED] [REDACTED]” *Id.*; RDX-0032C, ¶85 (citing RDX-0035C, ¶254) (emphasis added).

- b) Oura’s Gen. 4 ring does not meet the limitations of dependent claim 4

Oura’s Gen. 4 ring also does not meet the added limitation of dependent claim 4. RDX-0032C, ¶¶89-90.

Specifically, Oura will demonstrate that the dynamic sensing feature of Oura Ring 4 does not satisfy the limitation “wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.” RDX-0035C, ¶¶279-280, 282 (with ¶282 citing [REDACTED] at RX-0147C (OURA_1478_SC_0000437-444)); RDX-0032C, ¶91.

First, Mr. Larson will testify that the Oura firmware does not include “[REDACTED] [REDACTED],” which

precludes infringement. JX-0027, 259:5-260:9; JX-0025, 132:4-14; 135:25-136:8; RDX-0032C, ¶92.

Second, the '359 Patent explains that a device may compare the biological signal threshold value that is based upon many operations performed by the target to determine whether the second signal is valuable or invalid. RDX-0032C, ¶93 (citing JX-0001, 12:17-28). If the second signal does not meet the second signal threshold value, the wearable device may use that to determine whether the target moved intentionally. *Id.* at 12:29-39. The wearable device uses this comparison to determine whether it should transition to active mode in order to control an external device. *Id.* at 12:40-57.

Dr. Welch states: “Oura Ring 4 [REDACTED]
[REDACTED]
[REDACTED]” but cites no evidence to support this position. RDX-0032C, ¶94 (citing RDX-0035C, ¶279).

Dr. Welch discusses the testimony of Ketan Patel about dynamic sensing to support his position by explaining that the function [REDACTED]
[REDACTED] RDX-0032C, ¶95 (citing RDX-0035C, ¶280). However, the cited testimony from Mr. Patel explains that, i [REDACTED]
[REDACTED]
[REDACTED]. *Id.*

The Oura Ring 4 [REDACTED] implementation of dynamic sensing does not show that the controller is configured “to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.” RDX-0032C, ¶97. Rather, the [REDACTED]

[REDACTED]. *Id.* In fact, t [REDACTED]
[REDACTED]
[REDACTED]. *Id.*

Thus, Dr. Welch’s opinions relating to dynamic sensing in the Oura Ring 4 do not meet the elements of claim 4. RDX-0032C, ¶¶98, 99 (opining that Dr. Welch’s reliance on Simon Ratner’s testimony on the application software, which runs on, for example, a user’s mobile phone, does not provide evidence that the Oura Ring 4 infringes claim 4).

2. Ring 5 and [REDACTED] Do Not Infringe the Asserted Claims

Samsung has put forth no contention or evidence of infringement by these rings and accordingly cannot carry its burden of proof. *See Certain Crafting Machines*, Inv. No. 337-TA-1426, ID at 40. Dr. Mortazavi and Mr. Larson will testify that although Ring 5 and [REDACTED] have broadly similar functionality to Oura’s Gen. 4 ring, the software and hardware functionality is not the same; given this, and given the lack of any theory of representativeness, Samsung will not be able to carry its burden of proof for these reasons as well.

B. The Technical Prong of the Domestic Industry Inquiry is Not Met

Samsung’s alleged DI products do not practice either claims 1 or 4 of the ’359 Patent for the reasons explained herein.

- a) The Asserted DI Products do not meet the limitations of claim element [1.f]

The Asserted DI Products do not practice limitation [1.f], which requires “a select portion of the biological signal.”

As discussed above, the applicants for the ’359 Patent amended this limitation to include the “select” modifier to the “select portion” element to make clear that the claim does not include using the full biological signal obtained during a select period. JX-0005 at SAMS382-0002902;

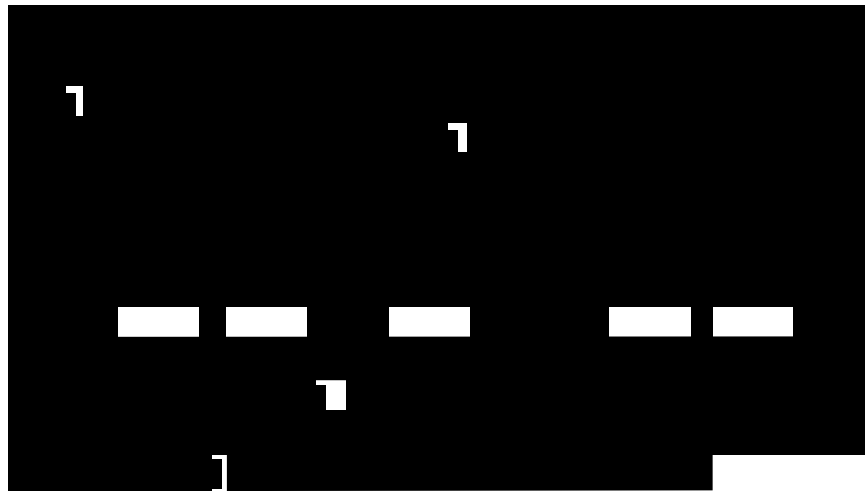
RDX-0033C ¶94. Thus, a POSITA would understand that the plain and ordinary meaning of a “portion” to mean part of a whole or a specific part and, specific to limitation [1.f], not all of or the full biological signal obtained during the select period. RDX-0033C, ¶97; JX-0020 (Choi), 70:21-71:3.

Dr. Welch’s opinion that the Asserted DI Products practice limitation [1.f] first relies on the assertion that “‘there’s practically almost no difference’ or ‘almost no difference’” in the implementation of the SpO₂ algorithm in the Samsung Galaxy Watch and Ring products. RDX-0035C ¶719 (quoting JX-0013, 21:3-25:14). However, Dr. Welch provides no analysis of the alleged similarities across the Asserted DI Products and only relies on source code for a single Asserted DI Product, the Samsung Galaxy Watch8. RDX-0035C ¶717 (citing RX-0150C at SAMS382-SC-0000099-132); RDX-0033C ¶99.

Based on Dr. Welch’s analysis, including a lack of identification of source code files, Samsung will not be able to carry its burden that each of the Asserted DI Products practice this limitation. *Compare* Final Technical Domestic Industry Contentions (JX-0049), Exhibit 1 (served May 27, 2026) at 89-101 (failing to identify any source code related to Samsung’s assertion that the Asserted DI Products practice limitation [1-e]) *with* RDX-0035C, ¶¶709-720 (only identifying source code for the Samsung Galaxy Watch8 in support of the assertion that the Asserted DI Products practice limitation [1-e]). Thus, Dr. Mercier will testify that Dr. Welch has not demonstrated all Asserted DI Products rely on purported similar implementations of the SpO₂ algorithm. RDX-0032C, ¶100.

Additionally, Dr. Mercier will testify that, contrary to Dr. Welch’s opinions, the Samsung source code (RX-0150C) does not illustrate the computing and storing of “a select portion of the SpO₂ data.” RDX-0032C ¶101; RDX-0035C ¶17. Dr. Welch appears to assert that [REDACTED]

██████████ is equivalent to “using a select portion of the biological signal obtained during a select period”:



RDX-0035C, ¶717 (emphasis added).

But Dr. Welch’s analysis ignores the distinction between ██████████
██████████ in a calculation. RDX-0032C, ¶102. A POSITA would understand that if data is collected for purposes of a calculation, the full (not a select portion) collected dataset remains part of the calculation even if some entries are later assigned a value of zero. *Id.* Setting the value of a data entry to zero does not remove the data entry from a dataset, rather it changes the value of the data entry within the dataset. *Id.* Dr. Mercier will further testify that ██████████ may still affect other calculations like totals, averages, and weighting. *Id.*

Dr. Mercier will show that Dr. Welch has not demonstrated, in the firmware that he relies on, that any of the collected data is discarded or not used in the context of performing the specified function of storing, transmitting, computing, and/or displaying SpO₂ levels. RDX-0035C, ¶700 (defining the specified function). Rather, the collected data—██████████
██████████ and, thus, failing to meet limitation [1-e]. RDX-0032C, ¶103.

Dr. Mercier will further testify that Samsung’s technical documents and engineer testimony cited by Dr. Welch provide only a high level descriptions of SpO₂ determination or display and do not demonstrate that the code meets the “select portion” requirement of limitation [1-e]. *Id.* ¶¶711-716, 718; RDX-0032C, ¶104.

- b) The Asserted DI Products do not meet the limitations of dependent claim 4

The Asserted DI Products also does not meet the added limitation of dependent claim 4. RDX-0032C, ¶¶109-110.

Specifically, Dr. Welch’s argument with regard to this element (RDX-0035C, ¶735) ignores that the [REDACTED]:

[REDACTED]

[REDACTED] RX-0150C (lines 1140-1153, SAMS382-SC-0000099-132 at -117); RDX-0032C, ¶111.

In other words, [REDACTED]

[REDACTED]

[REDACTED] RDX-0032C, ¶112.

Thus, the functionality of Samsung’s asserted DI products does not meet this claim limitation.

Additionally, Dr. Welch has failed to provide any analysis related to the required “second threshold value.” RDX-0032C, ¶113. For example, Dr. Welch appears to allege that the second threshold value is related to the “[REDACTED].” *Id.* However, besides generically citing lines of source code, Dr. Welch provides no additional analysis related to how [REDACTED]

[REDACTED]

RDX-0035C, ¶735 (citing RX-0150C ([REDACTED], lines 1442-1457, SAMS382-SC-0000099-132 at -122)).

C. Claims 1 and 4 of the ’359 Patent Are Invalid in View of the Prior Art

There are four asserted grounds of prior art-based invalidity that render invalid claims 1 and 4 of the ’359 Patent:

Invalidity Ground	Basis	Applicable Claims
Ground 1	Anticipation in View of Kuroda II	1
Ground 2	Obviousness in View of Kuroda II and Ahmed066	4
Ground 3	Obviousness in View of Kuroda II and Dixon	4
Ground 4	Anticipation in View of Suzuki	1

These four grounds of invalidity are explained in detail herein.

1. Overview of Prior Art References Relied Upon

U.S. Patent App. Pub. No. 2008/0139952 to Kuroda (“Kuroda II”) (RX-0003) was filed on December 6, 2007, and published June 12, 2008, and therefore qualifies as prior art to the ’359 patent under at least 35 U.S.C. 102(a)(1). Kuroda II discloses “a biometric information processing device” with a “heart rate detection unit” to measure resting heart rate, wherein the user’s resting state is determined by “an activity level determination unit” comprising an “activity sensor” such as “an acceleration detector.” JDX-0001C ¶¶92-95 (citing RX-0003 ¶¶[0017], [0034], [0064]).

U.S. Patent App. Pub. No. 2012/0078066 to Ahmed (“Ahmed066”) (RX-0008) was published March 29, 2012, and therefore qualifies as prior art to the ’359 patent under at least 35 U.S.C. 102(a)(1). Ahmed is directed to a wearable device that measures signals such as user movement and biological signals, wherein a specified function is carried out using the biological signal, depending on user movement. JDX-0001C ¶104 (citing RX-0008 ¶¶[0006], [0007], [0022]-[0025]).

Compressed Sensing System Considerations for ECG and EMG Wireless Biosensors, by Anna M. R. Dixon et al. (“Dixon”) (RX-0006) was an article published by *IEEE Transactions on Biomedical Circuits and Systems*, Vol. 6, No. 2 on April 16, 2012, and therefore qualifies as prior art to the ’359 patent under at least 35 U.S.C. 102(a)(1). Dixon is directed to a wearable device that measures signals such as user movement and biosignals, and discloses a device wherein a function is carried out using the biosignal, depending on user movement. JDX-0001C pages 28-29 (citing RX-0006 at Abstract, Section I, Figs. 1, 3, 7).

U.S. Patent No. 7,351,206 to Suzuki (“Suzuki”) (RX-0005) was filed on January 31, 2005 and issued on April 1, 2008, and therefore qualifies as prior art to the ’359 Patent under at least 35 U.S.C. § 102(a)(1). Suzuki discloses “a biotic sleep state determining apparatus that determines a sleep state of a subject based on a series of pulse interval data” and “body movement data that indicates a body movement of the subject.” RX-0005 at Abstract; JDX-0001C ¶¶96-99.

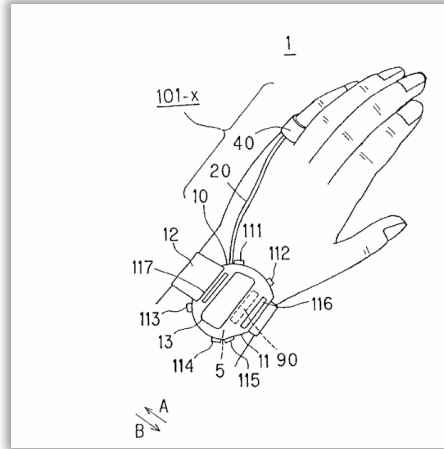
2. *Ground I: Anticipation in View of Kuroda II*

Claim 1 of the ’359 Patent is invalid as anticipated by the teachings of Kuroda II, which meet each limitation of this claim as explained herein and as will be shown by the testimony of Dr. Mortazavi.

i. Claim elements [1.pre]-[1.c]

Claim elements [1.pre]-[1.c] are each taught by Kuroda II. Samsung does not contest claim elements [1.a]-[1.c]. CSUCE at 2.

Kuroda II teaches “biometric information processing device 1” that is a “wearable device.” JDX-0001C ¶¶116-19 (citing RX-0003 ¶¶[0055]-[0056]). Specifically, Kuroda II discloses that the “biometric information processing device 1... includes a wristwatch-type main unit 10, a cable 20 that is connected to this main unit 10, and a pulse sensor 30 that is disposed to the distal end of the cable 20.” RX-0003 ¶¶[0055]-[0056].



Kuroda II at FIG. 1

The “biometric information processing device 1” of Kuroda II includes an “activity sensor 302” that is a “motion sensor.” JDX-0001C ¶¶121-23 (citing RX-0003 ¶¶[0064], [0087], [0134], Fig. 6). Specifically, Kuroda II discloses that “an acceleration detector, for example, can be used as the activity sensor 302,” and that “[t]he activity sensor 302 detects user activity and outputs the detected activity signal to the activity signal amplifier circuit 304.” RX-0003 ¶¶[0064], [0087], [0134], Fig. 6. The activity sensor 302 is disclosed as part of the watch case 11, which in turn is a part of the main unit 10 of the biometric information processing device 1. RX-0003 ¶¶[0062], [0064].

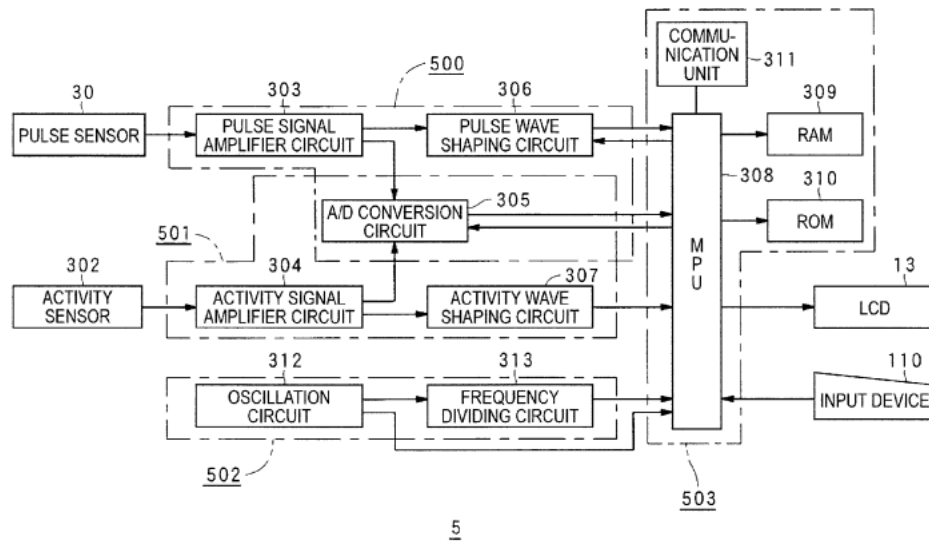
The “biometric information processing device 1” of Kuroda II also includes a “pulse sensor 30” that is a “biological sensor.” JDX-0001C ¶¶127-30 (citing RX-0003 ¶¶[0056], [0058], [0059], [0066], [0080], Fig. 5). Specifically, Kuroda II discloses that the pulse sensor 30 outputs a “pulse wave signal,” which is used to calculate “heart rate.” RX-0003 ¶[0066].

The “biometric information processing device 1” of Kuroda II also includes a “control unit 5” that is a “controller.” JDX-0001C ¶¶134-37 (citing RX-0003 ¶¶[0065], [0088], [0100]-[0101], Fig. 6). Specifically, Kuroda II discloses that “[t]he control unit 5 includes a pulse data processing unit 500 for determining the heart rate based on input from the pulse sensor 30, a pitch data processing unit 501 for determining the pitch based on input from the activity sensor 302, a clock generator 502 for generating an operating clock signal, and a control section 503 for controlling the control unit 5,” and that “the control section 503 includes an MPU 308, RAM 309, ROM 310, and a communication unit 311,” wherein “the MPU 308 controls the control unit 5, and therefore the biometric information processing device 1, based on a control program stored in the ROM.” RX-0003 ¶¶[0088], [0100]-[0101], Fig. 6. The control unit 5 is disclosed as part of the watch case 11, which in turn is a part of the main unit 10 of the biometric information processing device 1. RX-0003 ¶¶[0062], [0065].

ii. Claim element [1.d]

Claim element [1.d] is taught by Kuroda II.

The “control unit 5” of Kuroda II obtains a movement signal of an object (moreover, one that is external to the wearable device) using the motion sensor (“activity sensor 302”). JDX-0001C ¶¶140-44 (citing RX-0003 ¶¶[0088], [0133], [0134], Figure 6). Specifically, “[t]he control unit 5 includes... a pitch data processing unit 501 for determining the pitch based on input from the activity sensor 302,” and the “activity sensor 302 detects user activity,” which is then processed and “output[] to the MPU 308” as “activity data.” RX-0003 ¶¶[0088], [0133], [0134].



Kuroda II at FIG. 6

Because the “activity sensor 302 (see FIG. 6) for determining the pace is disposed inside the watch case 11” (RX-0003 ¶[0064]), the motion being detected is that of the user, which is an object external to the wearable device. RX-0003 at Fig. 1.

The “control unit 5” of Kuroda II also obtains a biological signal of the object (the user) using the biological sensor (“pulse sensor 30”). JDX-0001C ¶¶140-44 (citing RX-0003 ¶¶[0066], [0088], [0100]-[0101], [0133], [0136], Figure 6). Specifically, “[t]he control unit 5 includes a pulse data processing unit 500 for determining the heart rate based on input from the pulse sensor 30,” the “the pulse sensor 30 detects the pulse of the user,” which is then processed as a “pulse signal” and “output[] to the MPU 308” as “pulse data.” RX-0003 ¶¶[0088], [0133], [0136], Figure 6.

iii. *Claim element [1.e]*

Claim element [1.e] is taught by Kuroda II.

The “control unit 5” of Kuroda II performs this analysis as set forth by the claim language of element [1.e]. JDX-0001C ¶¶147-52 (citing RX-0003 ¶¶[0088], [0137]-[0139], [0142], [0143],

[0150], Figure 8). Specifically, Kuroda II teaches the following process for which the user activity is used to determine whether to calculate heart rate:

Measuring the heart rate HR is described in detail first.

[...]

The MPU 308 then determines if the activity component is greater than a prescribed threshold value for determining if the heart rate can be calculated.

If the activity component is greater than the prescribed threshold value for determining if the heart rate can be calculated, the activity component is too large to calculate the heart rate, and the current heart rate cannot be measured.

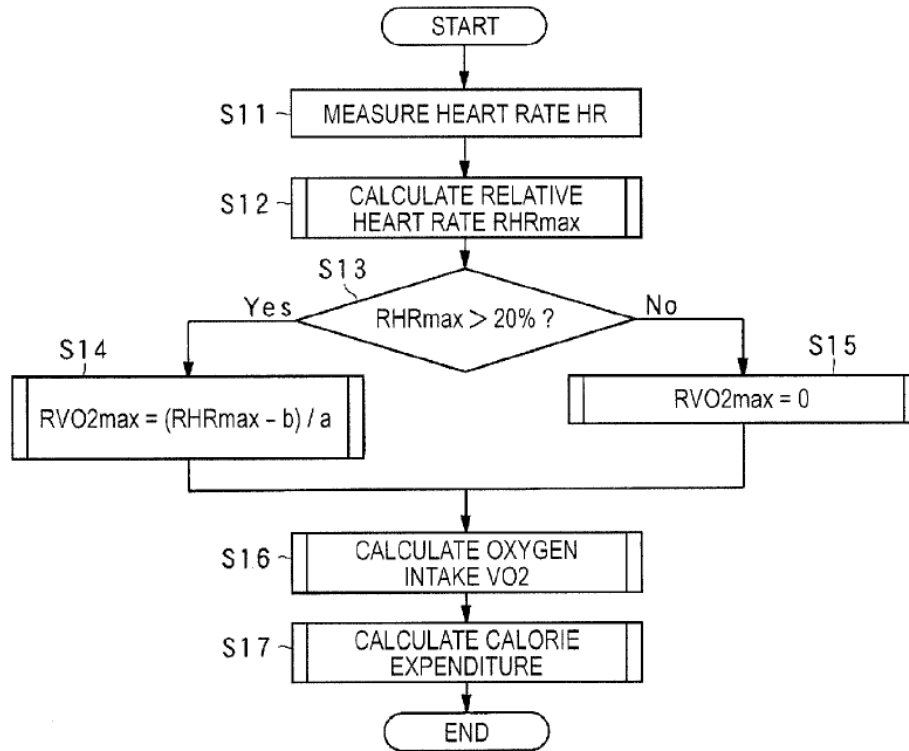
However, if the activity component is less than or equal to the prescribed threshold value for determining if the heart rate can be calculated, the activity component is subtracted from the heart rate component.

[...]

Based on the frequency of the extracted heart rate spectrum, the MPU 308 then calculates the heart rate HR.

RX-0003 ¶¶[0131], [0137]-[0139], [0142]. Thus, Kuroda II teaches that when the activity is determined to be below a threshold, “the activity component is subtracted from the heart rate component” in determining the user’s heart rate. *Id.*

Furthermore, Kuroda II discloses that once heart rate is calculated, “the MPU 308 then calculates the relative heart rate RHRmax” using this heart rate, which is used to calculate “the relative oxygen intake RVO2max,” which is used to calculate oxygen intake VO2 and, ultimately, calorie expenditure C. RX-0003 ¶¶[0143], [0145], [0147], [0150]. Furthermore, relative oxygen intake RVO2 “is calculated differently when the relative heart rate RHRmax is greater than or equal to 20% and when the relative heart rate RHRmax is less than 20%.” *Id.* at [0116]. These steps are depicted in Figure 8:



Kuroda II at FIG. 8

This disclosure meets claim element [1.e]. The controller determines, based on an analysis of the movement signal (*i.e.*, based an analysis of whether “the activity component is less than or equal to the prescribed threshold value”) whether to perform a specified function of the wearable device using the biological signal (*i.e.*, whether to calculate oxygen intake VO₂ and, ultimately, calorie expenditure C). JDX-0001C ¶¶147-152.

iv. Claim element [1.f]

Claim element [1.f] is taught by Kuroda II.

The “control unit 5” of Kuroda II performs this performance step as set forth by the claim language of element [1.f]. JDX-0001C ¶¶155-60 (citing RX-0003 ¶¶[0035], [0116], [0137]-[0142], [0143], [0150], Figure 8). Specifically, Kuroda II teaches the following process for which the user activity is used to determine whether to calculate heart rate:

Measuring the heart rate HR is described in detail first.

[...]

The MPU 308 then determines if the activity component is greater than a prescribed threshold value for determining if the heart rate can be calculated... if the activity component is less than or equal to the prescribed threshold value for determining if the heart rate can be calculated, the activity component is subtracted from the heart rate component. More specifically,

$$F_m = F_m - F_t.$$

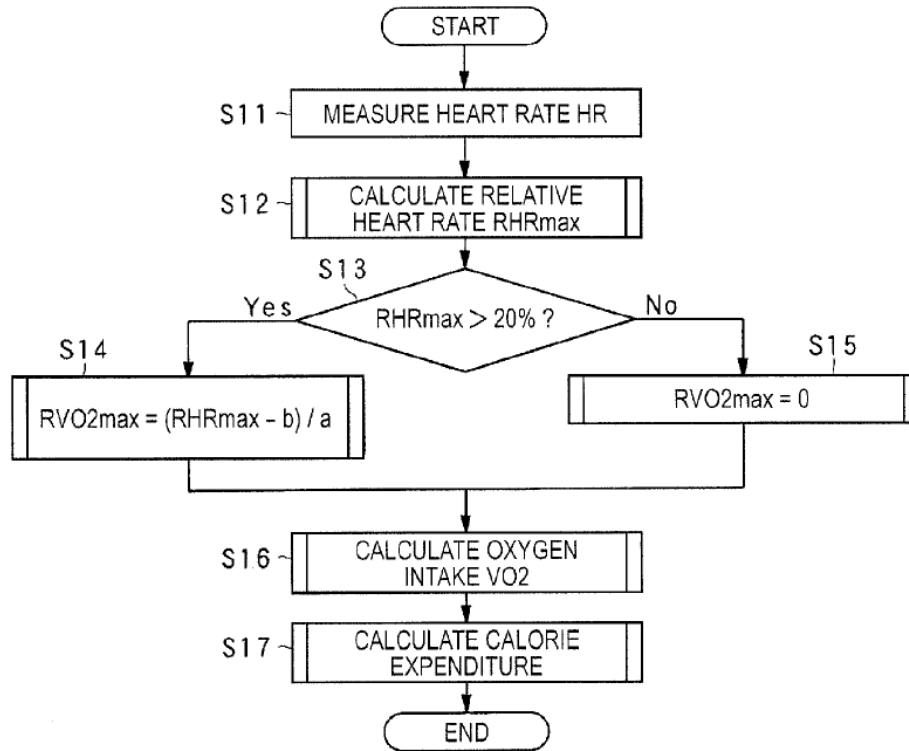
That is, the frequency component that is present only in the pulse signal is extracted.

[...]

Based on the frequency of the extracted heart rate spectrum, the MPU 308 then calculates the heart rate HR.

RX-0003 ¶¶[0131], [0137]-[0140], [0142]. Furthermore, Kuroda II teaches that once heart rate is calculated, “the MPU 308 then calculates the relative heart rate RHRmax” using this heart rate, which is used to calculate “the relative oxygen intake RVO2max,” which is used to calculate oxygen intake VO2 and, ultimately, calorie expenditure. RX-0003 ¶¶[0143], [0145], [0147], [0150].

Furthermore, Kuroda II also discloses that “when the relative heart rate is less than or equal to the prescribed level (such as when relative heart rate RHRmax<20%), the heart rate can fluctuate easily due to psychological effects, meals, and physical condition,” and “heart rate values in this range can therefore be omitted from the calorie expenditure calculation” to obtain a valid signal. RX-0003 ¶[0035]. Specifically, Kuroda II discloses that “the relative oxygen intake RVO2,” which is used to calculate oxygen intake VO2, “is calculated differently when the relative heart rate RHRmax is greater than or equal to 20% and when the relative heart rate RHRmax is less than 20%.” *Id.* at [0116]. These steps are depicted in Figure 8:



Kuroda II at FIG. 8

This disclosure meets claim element [1.f]. The controller performs the specified function (e.g., calculating oxygen intake VO₂ and, ultimately, calorie expenditure) using a select portion of the biological signal obtained during a select period of the movement signal determined to correspond to a specified condition (i.e., using the heart rate signal when “the activity component is less than or equal to the prescribed threshold value”). JDX-0001C ¶¶155-60.

3. Ground 2: Obviousness in View of Kuroda II and Ahmed066

Claim 4 of the '359 Patent is invalid as obvious by the teachings of Kuroda II in view of Ahmed066, which together meet or render obvious each limitation of this claim as explained herein and as will be shown by the testimony of Dr. Mortazavi.

a) *Graham* Factors 1-3

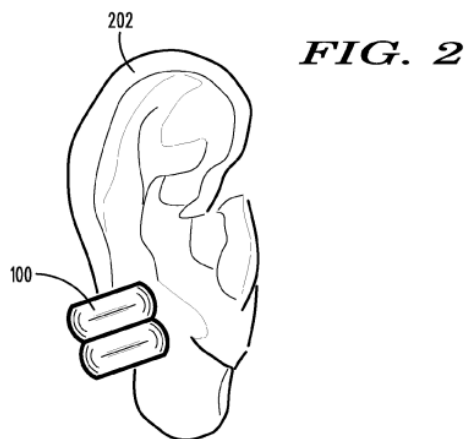
Kuroda II discloses “a biometric information processing device” with a “heart rate detection unit” to measure resting heart rate, wherein the user’s resting state is determined by “an

activity level determination unit” comprising an “activity sensor” such as “an acceleration detector.” RX-0003 ¶¶[0017], [0034], [0064]; JDX-0001C ¶¶376-78. As addressed above for Ground 1, Kuroda II discloses all elements of claim 1.

Kuroda II also teaches that the biological signal (*i.e.*, the “input from the pulse sensor 30”) is not used for calculating oxygen intake VO₂ under certain conditions. Specifically, Kuroda II teaches that “when the relative heart rate is less than or equal to the prescribed level (such as when relative heart rate RHR_{max}<20%), the heart rate can fluctuate easily due to psychological effects, meals, and physical condition,” and “heart rate values in this range can therefore be omitted from the calorie expenditure calculation.” RX-0003 ¶¶[0035], [0116], Fig. 8; JDX-0001C ¶¶386-91.

But to the extent that Kuroda II does not disclose the added requirement of claim 4 (“wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value”), this is rendered obvious by the teachings of Ahmed066.

Like Kuroda II, Ahmed066 teaches a wearable device that measures signals such as user movement and biological signals. *See, e.g.*, JDX-0001C ¶379; RX-0008 at FIG. 1, FIG. 2, [0006], [0007], [0022]-[0025].

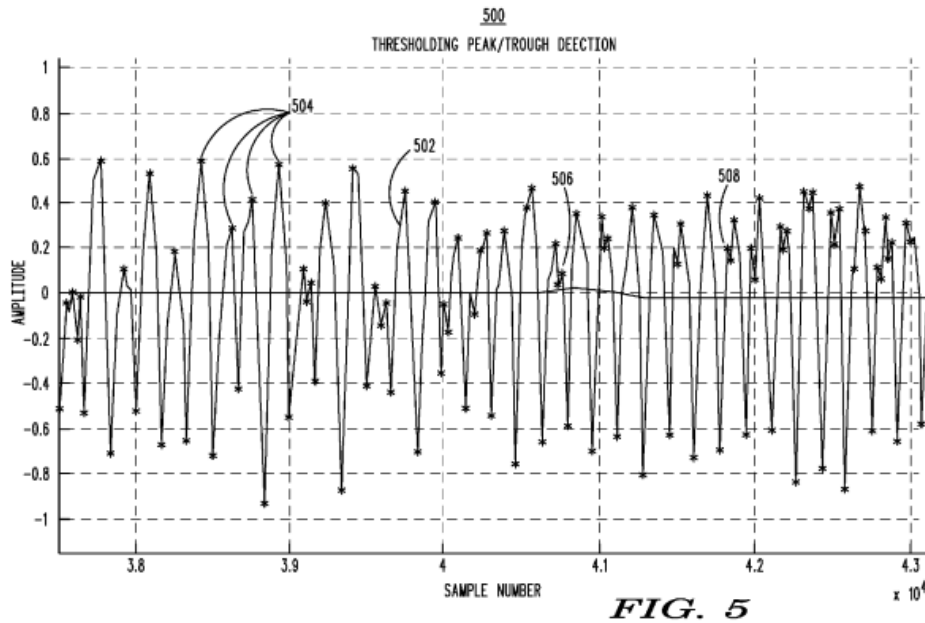


Ahmed066 also discloses a device wherein a specified function is carried out using the biological signal, depending on user movement. *See, e.g.*, JDX-0001C ¶379; RX-0008 at FIG. 3, [0021], [0023]-[0026], [0046]. Ahmed066 discloses the use of sensors, including motion and bio signal sensors that detect user heart rate. *See, e.g.*, JDX-0001C ¶379; RX-0008 at FIG. 1, FIG. 2, [0006], [0007], [0022]-[0025]. Specifically, Ahmed066 discloses an optical sensor used to measure user heart rate. *See, e.g.*, JDX-0001C ¶379; RX-0008 at FIG. 1, FIG. 2, [0006], [0007], [0022]-[0023].

In this similar setting to Kuroda II, Ahmed066 discloses a biological signal validation process that labels valid or nonvalid values based on peak amplitude in reference to specific threshold ranges. *See, e.g.*, RX-0008 at FIG. 5, FIG. 8, [0010], [0026]-[0027], [0029]-[0032], [0038]-[0039]; JDX-0001C ¶379.

More specifically, Ahmed066 discloses an adaptive noise canceller that is coupled to a bandpass filter designed to filter out frequency components that fall outside the range of normal heartbeats, for example outside the range of 0.5 to 4.0 Hertz equivalent to a heart rate range of 30 to 240 beats per minute. JDX-0001C ¶394 (citing RX-0008 ¶[0026]). Using the bandpass filter helps clean up the signal by further reducing noise and facilitate heart beat detection. *Id.*

Ahmed066 also discloses a first peak detector that scans each window to identify peak signal samples, *i.e.*, samples greater than both preceding and succeeding samples ($S_{k-1} < S_k > S_{k+1}$). JDX-0001C ¶395 (citing RX-0008 ¶[0029]). This condition is necessary but not sufficient. The first peak detector may include comparators to evaluate this inequality. To qualify as a valid peak, the sample S_k must also exceed a threshold proportional to the window energy $|S_k| > T \cdot \text{RMS}$ where T is a preprogrammed threshold factor and RMS represents signal energy. *Id.* For normalized signals (-1 to 1), the threshold factor T is suitably in the range of 0.0 to 0.3 , preventing low-amplitude noise from being detected as peaks. *Id.*



RX-0008 ¶[0030] (“FIG. 5 is a graph 500 including a plot of a filtered Photoplethysmography signal 502 along with points 504 (only a few of which are numbered to avoid crowding) identifying peaks associated with individual heart beats that have been detected by a peak detector.”).

The biological signal validation process taught by Ahmed066 thus determines that the biological signal is invalid in response to the magnitude of the biological signal not satisfying a threshold value different from the first threshold value from claim 1, and irrespective of the movement signal. JDX-0001C ¶396. This meets the added limitation of claim 4.

A POSITA would have found it obvious to incorporate the biological signal validation process taught by Ahmed066 into the heart rate detection unit of Kuroda II.

First, a POSITA would recognize that use of validation and filtering techniques is well-known and built into devices with sensor circuitry so as to avoid redundant sensor measurements or to have an ability to remove outlier data samples such that the sensing analysis is not skewed in view of anomalies, erroneous data, and/or to ensure reliable results. JDX-0001C ¶383. Ahmed066 teaches such a validation and filtering technique. Incorporation of the biological signal validation

process taught by Ahmed066 into the heart rate detection unit of Kuroda II would achieve this benefit. *Id.*

Second, both Kuroda II and Ahmed066 disclose wearable devices with motion and biological sensors such as accelerometers and heart rate sensors within the wearable device, wherein a specified function is carried out using the biological signal, depending on user movement. JDX-0001C ¶¶377-81 (citing RX-0003 ¶¶[0055], [0065]-[0066], [0087]-[0088], [0100]-[0101], [0130], [0132]-[0134], [0137]-[0142], [0150], Figs. 1, 6; RX-0008 ¶¶[0006], [0007], [0021]-[0026], [0046], [0048]-[0050], [0058]-[0060], [0082]-[0083], Figs. 1-3). The technical similarities of the teachings of wearable biological and movement signal measuring systems and methods to perform a specified function of a wearable device using a validated biological signal in response to a movement signal meeting a specified condition would render the combination obvious to try. JDX-0001C ¶383. A POSITA would have had a reasonable expectation of success in achieving the benefits disclosed in making the modification, which would involve only routine skill. *Id.*

b) *Graham* Factor 4

Samsung has chosen to present its alleged secondary considerations of nonobviousness in a single argument common to the '359, '677, and '675 Patents. Samsung fails to separate out the alleged inventions of the '359, '677, and '675 Patents and thereby fails its burden to demonstrate a nexus to the claimed inventions, as was its burden to do. As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '359 Patent. *See also* JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

4. Ground 3: Obviousness in View of Kuroda II and Dixon

Claim 4 of the '359 Patent is invalid as obvious by the teachings of Kuroda II in view of Dixon, which together meet or render obvious each limitation of this claim as explained herein and as will be shown by the testimony of Dr. Mortazavi.

a) *Graham* Factors 1-3

Kuroda II discloses “a biometric information processing device” with a “heart rate detection unit” to measure resting heart rate, wherein the user’s resting state is determined by “an activity level determination unit” comprising an “activity sensor” such as “an acceleration detector.” RX-0003 ¶¶[0017], [0034], [0064]; JDX-0001C ¶¶321-23. As addressed above for Ground 1, Kuroda II discloses all elements of claim 1.

Kuroda II also teaches that the biological signal (*i.e.*, the “input from the pulse sensor 30”) is not used for calculating oxygen intake VO₂ under certain conditions. Specifically, Kuroda II teaches that “when the relative heart rate is less than or equal to the prescribed level (such as when relative heart rate RHR_{max}<20%), the heart rate can fluctuate easily due to psychological effects, meals, and physical condition,” and “heart rate values in this range can therefore be omitted from the calorie expenditure calculation.” RX-0003 ¶¶[0035], [0116], Fig. 8; JDX-0001C ¶¶321-23, 332-37.

But to the extent that Kuroda II does not disclose the added requirement of claim 4, this is rendered obvious by the teachings of Dixon.

Like Kuroda II, Dixon teaches wearable device that measures signals, such as user movement, and biosignals, such as EMG signals. *See, e.g.*, JDX-0001C ¶324 (citing RX-0006 at Abstract, Section I, Figs. 1, 3, 7). Dixon also discloses a device wherein a function is carried out using the biosignal (*i.e.*, alteration of biosignal “sampling rate”), depending on user movement (*i.e.*, determination of high signal activity vs. low-activity period). *See, e.g.*, JDX-0001C ¶324

(citing RX-0006 at Section III, Table 1). Dixon discloses the use of sensors, including motion and bio signal sensors that detect ECG/EMG signals. *See, e.g.*, JDX-0001C ¶324 (citing RX-0006 at Abstract, Section I, Figs. 1, 3, 7).

In this similar setting to Kuroda II, Dixon specifically discloses dynamic thresholding of the biosignal, which compares collected EMG signal magnitude to a threshold level to reduce baseline drift and noise, which is set to zero if not met. *See, e.g.*, JDX-0001C ¶325 (citing RX-0006 at Section I, II.B, III, IV, Fig. 6, Table 1).

For example, Dixon discloses dynamic thresholding of an EMG/ECG signal “to reduce baseline drift and noise,” as shown below:

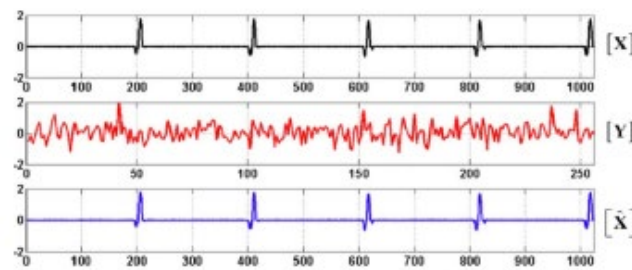


Fig. 6. A thresholded ECG signal before, $[X]$, and after CS compression, $[Y]$, and after reconstruction, $[\hat{X}]$.

Dixon at FIG. 6, Section II.B. Specifically, Dixon discloses comparison of peak-to-peak amplitude of the ECG signal against threshold value “ th ,” describing that “if the magnitude of the next input sample is $> th$, it is processed as is; if it is $< th$, it is set to zero” and not processed at all. RX-0006 at Section IV; JDX-0001C ¶¶340-42.

The determination to zero an ECG/EMG signal where the magnitude of the peak-to-peak amplitude is less than a threshold value “ th ” results in an invalidity determination of “the biological signal,” determined “irrespective of the movement signal” and “in response to a magnitude of the biological signal not satisfying a second threshold value,” thereby meeting the added limitation of claim 4. JDX-0001C ¶342.

A POSITA would have found it obvious to incorporate the dynamic thresholding of the biosignal taught by Dixon into the heart rate detection unit of Kuroda II.

First, a POSITA would recognize that use of validation and filtering techniques is well-known and built into devices with sensor circuitry so as to avoid redundant sensor measurements or to have an ability to remove outlier data samples such that the sensing analysis is not skewed in view of anomalies, erroneous data, and/or to ensure reliable results. JDX-0001C ¶329. Dixon teaches such a validation and filtering technique. JDX-0001C ¶¶325-29. Incorporation of the dynamic thresholding of the biosignal taught by Dixon into the heart rate detection unit of Kuroda II would achieve this benefit. *Id.*

Second, both Kuroda II and Dixon disclose wearable devices with motion and biological sensors such as accelerometers and heart rate sensors within the wearable device, wherein a specified function is carried out using the biological signal, depending on user movement. JDX-0001C ¶¶321-23, 332-37 (citing RX-0003 ¶¶[0055], [0065]-[0066], [0087]-[0088], [0100]-[0101], [0130], [0132]-[0134], [0137]-[0142], [0150], Figs. 1, 6; JDX-0001C ¶¶321-23, 332-37 (citing RX-0006 at Abstract, Sections I-IV, Table 1, Table 1, Figs. 1, 3, 6, 7). The technical similarities of the teachings of wearable biological and movement signal measuring systems and methods to perform a specified function of a wearable device using a validated biological signal in response to a movement signal meeting a specified condition would render the combination obvious to try. JDX-0001C ¶329. A POSITA would have had a reasonable expectation of success in achieving the benefits disclosed in making the modification, which would involve only routine skill. *Id.*

b) *Graham* Factor 4

Samsung has chosen to present its alleged secondary considerations of nonobviousness in a single argument common to the '359, '677, and '675 Patents. Samsung fails to separate out the alleged inventions of the '359, '677, and '675 Patents and thereby fails its burden to demonstrate

a nexus to the claimed inventions, as was its burden to do. As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '359 Patent. *See also* JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

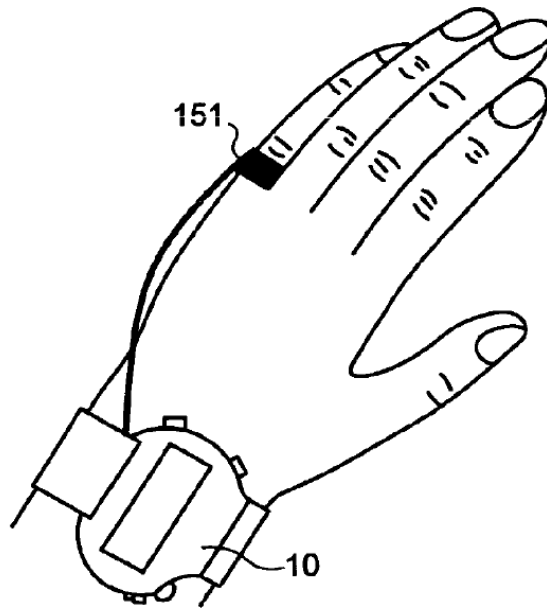
5. *Ground 4: Anticipation in View of Suzuki*

Claim 1 of the '359 Patent is invalid as anticipated by the teachings of Suzuki, which meet each limitation of this claim as explained herein and as will be shown by the testimony of Dr. Mortazavi.

i. *[Uncontested] Claim elements [1.pre]-[1.c]*

Claim elements [1.pre]-[1.c] are each taught by Suzuki.

Suzuki discloses a “sleep state determining apparatus 10” in combination with a “pulse wave sensor 151” that constitutes a “wearable device.” JDX-0001C ¶¶429-33. Specifically, Suzuki states that “a pulse wave sensor 151 is attached to a finger and the sleep state determining apparatus 10 is attached to a wrist like a wristwatch.” RX-0005 at 3:61-63. Suzuki alternatively discloses that “the pulse wave sensor 151 may be attached to the palm of the hand,” or can be integrated with the sleep state determining apparatus “and attached onto an artery of the wrist.” RX-0005 at 3:65-4:02.



Suzuki at FIG. 2

Suzuki's sleep state determining apparatus includes an "acceleration sensor 141" that is a "motion sensor." JDX-0001C ¶¶434-38. Specifically, Suzuki discloses that "[t]he body movement measuring unit 14 includ[ing] an acceleration sensor 141," wherein "the acceleration sensor 141 measures accelerations of -2 g to 2 g in three-axis directions, and is mounted in the sleep state determining apparatus 10." RX-0005 at 4:14-18.

Suzuki's sleep state determining apparatus also includes a "pulse wave measuring unit 15 [that] includes pulse wave sensor 151" that is a "biological sensor." JDX-0001C ¶¶439-45. Specifically, Suzuki discloses that "[t]he pulse wave sensor 151, which is composed by a blue LED and a photodiode, irradiates a light onto a surface of a finger skin, detects a fluctuation in a reflected light that changes according to a bloodstream fluctuation in a capillary using the photodiode, thereby measuring a pulse wave." RX-0005 at 4:27-31.

Suzuki's sleep state determining apparatus further includes a "controller 25" that is a "controller." JDX-0001C ¶¶446-51. Specifically, Suzuki discloses that "the controller 25 controls

entirety of the sleep state determining apparatus 10,” specifically “control[ing] a processing request and a data flow for each processor.” RX-0005 at 9:1-5.

ii. [Uncontested] Claim element [1.d]

Claim element [1.d] is taught by Suzuki.

The “controller 25” of Suzuki obtains a movement signal of an object using the motion sensor (“acceleration sensor 141”). JDX-0001C ¶¶452-60. Specifically, Suzuki discloses a “sleep state determining apparatus 10 includ[ing] ... a body movement measuring unit 14 ... and a controller 25,” wherein “[t]he body movement measuring unit 14 adjust[ing] a gain and an offset of analog data measured by the acceleration sensor 141” and “input[ting] the digital quantity to the controller 25.” RX-0005 at 3:48-58, 4:18-23. Suzuki defines this data as “body movement data.” RX-0005 at 4:12-13.

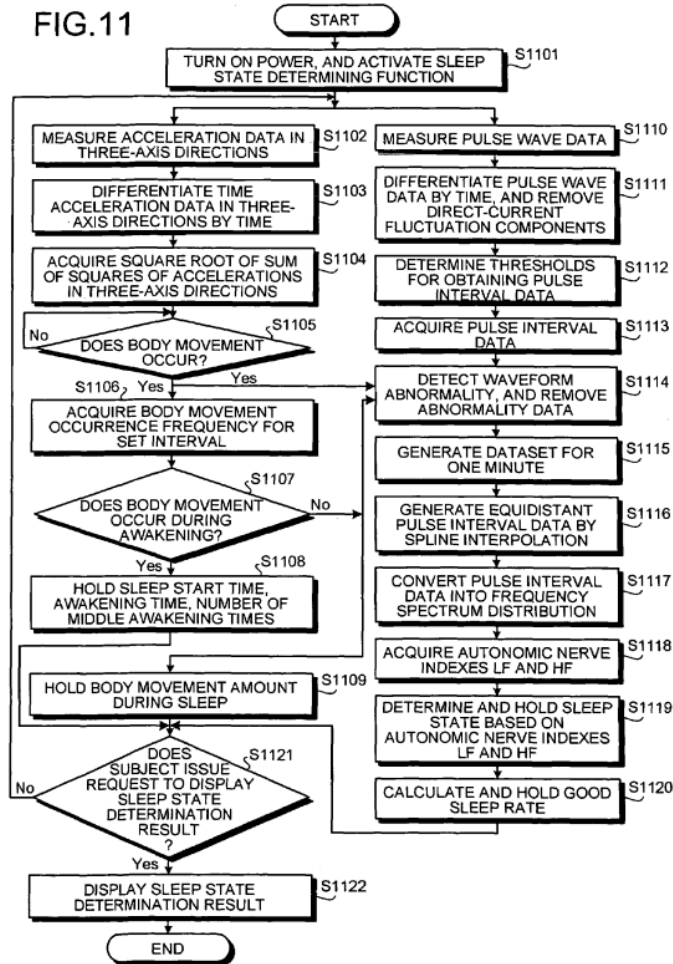
The “controller 25” of Suzuki also obtains a biological signal of the object (the user) using the biological sensor (“pulse wave sensor 151”). JDX-0001C ¶¶476. Specifically, Suzuki discloses that “[t]he pulse wave measuring unit 15 convert[ing] an output current from the photodiode of the pulse sensor 151 ... into a digital quantity using a 10-bit A/D converter, and inputs the digital quantity to the controller 25.” RX-0005 at 4:31-40. Suzuki defines data measured from the pulse wave sensor 151 as “pulse wave data.” RX-0005 at 4:24-25.

iii. Claim element [1.e]

Claim element [1.e] is taught by Suzuki.

Suzuki discloses a determination of the body movement data as being above or equal to/less than a predetermined threshold, which is a teaching of an “analysis of the movement signal.” JDX-0001C ¶¶462-69 (citing RX-0005 at 4:51-5:12, 9:62-10:08, 10:29-11:09, Fig. 11). Specifically, Suzuki discloses an “awakening/sleep determining unit 19” that determines whether a subject is awake or asleep based on body movement data by comparing the frequency of body movement to

a “predetermined threshold.” RX-0005 at 4:51-5:12. This analysis is performed at step S1107 of Figure 11. JDX-0001C ¶¶463-65; RX-0005 at 4:51-5:12, 9:62-10:08, 10:29-11:09, Fig. 11.



Suzuki at Fig. 11

If, based on this analysis of the body movement data, the awakening/sleep determining unit 19” determines that the subject is asleep (“No” at Step S1107), Suzuki teaches that the controller will perform data processing functions on the biological signal. JDX-0001C ¶¶465-68 (citing RX-0005 at 9:62-10:08, 10:29-11:09, Fig. 11). Specifically, Suzuki teaches that if the subject is determined to be asleep, the controller proceeds to detect and remove abnormal pulse wave data steps S1114 and S1115. *Id.* Suzuki further teaches that the controller proceeds through steps

S1116-1120 of Fig. 11 to further process this data, and ultimately display a sleep state determination result. *Id.* This process includes determining “the sleep state of the subject based on the autonomic nerve indexes LF . . . and HF” using “the series of pulse interval data processed by the pulse interval data processor 22” at steps S1118-1120 *Id.* The “pulse wave data” or “pulse interval data” is the biological signal obtained using “pulse sensor 151.” JDX-0001C ¶468.

This disclosure meets claim element [1.e]. The controller determines, based on an analysis of the movement signal (*i.e.*, whether the frequency of body movement is less than a predetermined threshold), whether to perform a specified function of the wearable device using the biological signal (*i.e.*, determining the sleep state of the subject using pulse interval data). JDX-0001C ¶¶461-70.

iv. Claim element [1.f]

Claim element [1.f] is taught by Suzuki.

The “specified function” of Suzuki (*i.e.*, determining the sleep state of the subject using pulse interval data) is performed using a portion of the biological signal obtained during a select period (*i.e.*, the portion of the pulse interval data that was not removed at step S1114 and was obtained over the “one minute”). JDX-0001C ¶¶472-81 (citing RX-0005 at 4:66-5:03, 5:10-5:12, 9:62-10:08, 10:29-11:09, Fig. 11).

Specifically, Suzuki teaches that the controller detects and removes abnormal pulse wave data step S1114 and S1115 to create a “data set for one minute.” RX-0005 at 10:29-50, Fig. 11; JDX-0001C ¶¶477-78. It is on this one minute data set from which the abnormal pulse waves were removed that steps 1116-1120 are performed, which ultimately determine the sleep state of the subject using pulse interval data (which is the “specified function”). RX-0005 at 10:57-11:09, Fig. 11; JDX-0001C ¶478. This select portion of the movement signal corresponds to the condition of the subject being determined to be asleep. RX-0005 at 4:66-5:12.

This disclosure meets claim element [1.f]. The controller performs the specified function (e.g., determining the sleep state of the subject using pulse interval data) using a select portion of the biological signal obtained during a select period of the movement signal determined to correspond to a specified condition (*i.e.*, the non-removed portion of the pulse rate data obtained when body movement frequency is less than the second predetermined threshold for a given minute). JDX-0001C ¶¶471-82.

D. Claims 1 and 4 of the '359 Patent Are Ineligible Under 35 U.S.C. § 101

Claims 1 and 4 of the '359 Patent are patent ineligible under 35 U.S.C. § 101 for claiming patent ineligible subject matter. These claims are directed to the abstract idea of performing a function based on an amount of movement, and as set forth herein, the particular claim elements set forth nothing more than the application of this abstract idea in the context of a generic wearable device.

1. Alice Step 1: Each claim is directed to the abstract idea of performing a function based on an amount of movement

Claims 1 and 4 of the '359 Patent are directed to the abstract idea of performing a function based on an amount of movement, and as explained further below, there is nothing patent eligible about such generic and conceptual claiming.

An analysis of claim 1 lays bare that the claimed invention merely sets forth, in a generic environment of a wearable sensing device, the conceptual steps of obtaining signals from sensors, and then, using the signal from the motion sensor, deciding whether to perform a specified function of the wearable device that involves the signal from the biological signal (and then performing that signal)—this is the abstract idea of performing a function based on an amount of movement:

1. A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement signal of an object using the motion sensor,
and a biological signal of the object using the biological
sensor,

determine, based on an analysis of the movement signal,
whether to perform a specified function of the wearable device
using the biological signal, and

perform the specified function using a select portion of the
biological signal obtained during a select period, of the
movement signal, determined to correspond to a specified
condition.

Generic
environment of
wearable sensing
device

Obtain signals

Use motion signal to
determine whether to
execute a function that
uses the biological
signal; then perform it

JX-0001 at cl. 1 (annotated).

Such a claim does not set forth any particular “improvement in computer capabilities”—indeed, the claim elements are far too generic to do so—and instead sets forth the generic idea of performing a function based on an amount of movement, and merely places this abstract idea in the context of a generic wearable device. For example, there is no claimed manner in which the controller “determine[s] whether to perform a specified function of the wearable device.” JX-0001 cl. 1. At most, this it is “based on an analysis of the movement signal,” but there is no specificity as to what that analysis is. This is an abstract idea. *See GoTV*, 166 F.4th at 1064; *Intellectual Ventures I*, 850 F.3d at 1341; *Elec. Power Grp.*, 830 F.3d at 1353 (“[t]he claims in this case do not even require a new source or type of information, or new techniques for analyzing it”); *SAP Am.*, 898 F.3d at 1168; *AGI SureTrack*, 176 F.4th at 1378; *Killian*, 45 F.4th at 1380.

It also matters little that the claims set forth a “motion sensor,” “biological sensor,” and “controller,” as no particular structure for these generic components of a wearable device are claimed; instead, the various generic computer functionality in claim 1 are just the tools used to accomplish the claimed abstract idea. Where such generic computer processes are claimed, patent

eligibility “requires more than result-focused functional language and more than just using those functions in the context of specifically identified content.” *GoTV*, 166 F.4th 1053, 1064; *PersonalWeb Techs.*, 8 F.4th at 1316 (Fed. Cir. 2021) (“The claims do this in a computer environment, but that doesn’t transfigure an idea out of the realm of abstraction”); *In re Killian*, 45 F.4th 1373, 1380 (Fed. Cir. 2022) (“We have distinguished between claims ‘directed to an improvement in the functioning of a computer,’ versus those [] that simply recite ‘generalized steps to be performed on a computer using conventional computer activity.’”).

Samsung’s contention that the claims are directed to “a number of intricate and novel processing functions using data collected from the sensors” (JDX-0048C at 86) is belied by the language of the claims, which set forth absolutely no “intricate and novel processing functions.” *Hawk Tech. Sys., LLC v. Castle Retail, LLC*, 60 F.4th 1349, 1357 (Fed. Cir. 2023) (“[t]he analysis at step one ‘must focus on’ the claim language”); *ChargePoint, Inc. v. SemaConnect, Inc.*, 920 F.3d 759, 769 (Fed. Cir. 2019) (“Even a specification full of technical details about a physical invention may nonetheless conclude with claims that claim nothing more than the broad law or abstract idea underlying the claims”); *Accenture Global Servs. v. Guidewire Software, Inc.*, 728 F.3d 1336, 1345 (Fed. Cir. 2013) (“the system claims themselves only contain generalized software components arranged to implement an abstract concept on a computer”). Likewise, Samsung’s contention that the claims improved prior art devices by “recogniz[ing] an intention of a user with a greater precision” also has no basis in the claims, which set forth no improvement in precision. JDX-0048C at 87 (citing JX-0001 at 1:43-45).

Dependent claim 4 differs from claim 1 just by requiring the use of a threshold value to filter invalid biological measurement signals, which is a generic computer-based technique being used in a conventional manner (*i.e.*, signal filtering for an incoming signal). As explained by Dr.

Mortazavi, the use of validation and filtering techniques is well-known and built into devices with sensor circuitry so as to avoid redundant sensor measurements or to have an ability to remove outlier data samples such that the sensing analysis is not skewed in view of anomalies, erroneous data, and/or to ensure reliable results, and is exemplified by the teaches of Ahmed066 and Dixon. JDX-0001C ¶¶325-29, 379, 383. Thus, it adds no patentable eligible subject matter to claim 1, nor changes the abstract idea to which the claims are directed, and claim 1 of the '359 Patent may be treated as representative. *See Berkheimer*, 881 F.3d at 1365; *Mortg. Grader*, 811 F.3d at 1324 n.6.

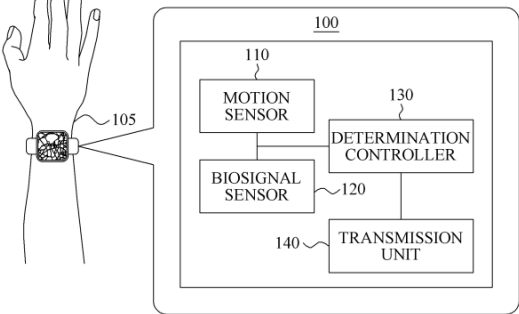
Ultimately, there is nothing non-abstract about these claims. There is no particular “analysis” required; there is no particular “function” to be performed; there is no particular “portion of the biological signal” needed to be used in the function; and there is no particular “condition” that must be used. Rather, the claims just set forth the abstract idea of performing a function based on an amount of movement. Thus, the claims are directed to an abstract idea at *Alice* step one.

2. *Alice Step 2: Each claim merely embodies the abstract idea without adding any inventive concept*

There is nothing in the claims that could supply an inventive concept. The abstract idea itself cannot form the inventive concept at *Alice* step two, and once separated from the abstract idea (performing a function based on an amount of movement), there is no inventive concept claimed. All that is left is (1) a generic hardware environment of a wearable device, (2) generic functions to be accomplished by the wearable device in the performance of the abstract idea, and (3) a generic filtering step (in the case of claim 4). These elements, alone or in combination, do not convey an inventive concept.

A wearable device comprising a motion sensor, a biological sensor, and a controller (the claimed hardware environment) was well-understood, routine, and conventional at the time of

the '359 Patent's alleged invention. To begin with, this is acknowledged by the '359 Patent itself: “wearable-type mobile devices are emerging at an accelerated rate amid the proliferation of smartphones... [a] gesture recognized by an accelerometer or a biosignal sensor may be used to operate the wearable mobile device.” JX-0001 at 1:28:-35. A commercially available example of such a wearable device would have included the Basis B1 Band, which is “a wearable device that measures signals such as user movement and biological signals.” JDX-0001C ¶105; *id.* ¶1284 (“[t]he Basis B1 Band watch would be an example of a commercially available wearable device with a motion sensor, a biological sensor, and a controller”).

'359 Patent	Basis B1 Band
FIG. 1  JX-0001 at Fig. 1.	 RX-0009.0009

The Basis B1 Band includes ““an internal accelerometer that allows it to measure steps’ along with ‘the optical and skin temperature sensors’ wherein the optical sensor performs PPG measurements by ‘measuring the reflection of light emitted by the green sensor’ which discerns the heart rate.” JDX-0001C ¶1284 (quoting RX-0009.0013, 0017 (July 25, 2013 review of the Basis B1 Band)). Further evidence that such a hardware environment was well-understood, routine, and conventional includes Kuroda II, a U.S. Patent Application published on June 12, 2008 (over five

years prior to the claimed priority of the '359 Patent), which discloses “a biometric information processing device” with a “heart rate detection unit” to measure resting heart rate, wherein the user’s resting state is determined by “an activity level determination unit” comprising an “activity sensor” such as “an acceleration detector.” JDX-0001C ¶93 (quoting RX-0003 at [0017], [0034], [0064]). Indeed, as Suzuki disclosed at least as of its filing date (January 1, 2005), “[t]here is conventionally known a biotic sleep state determining apparatus that determines a sleep state of a subject based on pulse interval data which is data on one cycle of a pulse wave of the subject, and on body movement data which indicates a body movement of the subject.” Suzuki at 1:18-22, 1:23-58 (disclosing exemplary conventional capabilities of such devices); JDX-0001C ¶1284 (discussing Suzuki).

Furthermore, the remaining limitations of the asserted claims “concern the programming of the controller of such a well-understood, routine, and conventional device” and “there are no recited functions for the controllers of any of the Asserted Claims that would not have been able to be easily programmed by a POSITA.” JDX-0001C ¶1286. That the relevant programming of the accused devices was conventional may also be attested to, for example, by Mr. Larson. Indeed, the claims of the '359 Patent do not set forth any particular coding technique, and the '359 Patent itself notes that, conventionally, “[a] gesture recognized by an accelerometer or a biosignal sensor may be used to operate the wearable mobile device” and that the object of the invention was “a wearable mobile device model that recognizes an intention of a user with *a greater precision* is desirable to improve user experience.” JX-0001 at 1:28-35, 1:43-45. But of course, precision is not set forth by the claims. Further evidencing the fact that the claimed software limitations were well-understood, routine, and conventional is the teachings of Kuroda II, Suzuki, Ahmed066, and Dixon, discussed in the context of invalidity grounds 1-4, *supra*, and showing that the field of art

would have found the claimed software limitations well-understood, routine, and conventional, even in combination, as of the claimed priority date of the '359 Patent. JDX-0001C ¶¶1283-87.

Finally, merely implementing the claimed capabilities within the environment of a generic wearable device also does not convey an inventive concept. *See Killian*, 45 F.4th at 1380 (“the claims here do not detail how the computer should go about determining eligibility for benefits, beyond saying that the computer should determine eligibility based ‘on the identified information’ and ‘current SSDI benefit legal requirements’”); *Elec. Power Group*, 830 F.3d at 1355 (“the claims’ invocation of computers, networks, and displays does not transform the claimed subject matter into patent-eligible applications”); *United Servs. Auto. Ass’n v. PNC Bank N.A.*, 139 F.4th 1332, 1339 (Fed. Cir. 2025) (“Our precedent is clear that computer-mediated implementation of routine or conventional activity is not enough to provide an inventive concept.”).

Samsung has made no allegations of any particular inventive concept, and accordingly, the claims of the '359 Patent fail at both *Alice* steps one and two, and are therefore patent ineligible under 35 U.S.C. § 101.

VII. '789 PATENT

A. Oura’s Products Do Not Infringe the Asserted Claims

Samsung will not be able to carry its burden to show infringement of the '789 Patent by any product at issue (Oura’s Gen. 4, Ring 5, and [REDACTED] rings).

1. Oura’s Gen. 4 Ring Does Not Infringe the Asserted Claims

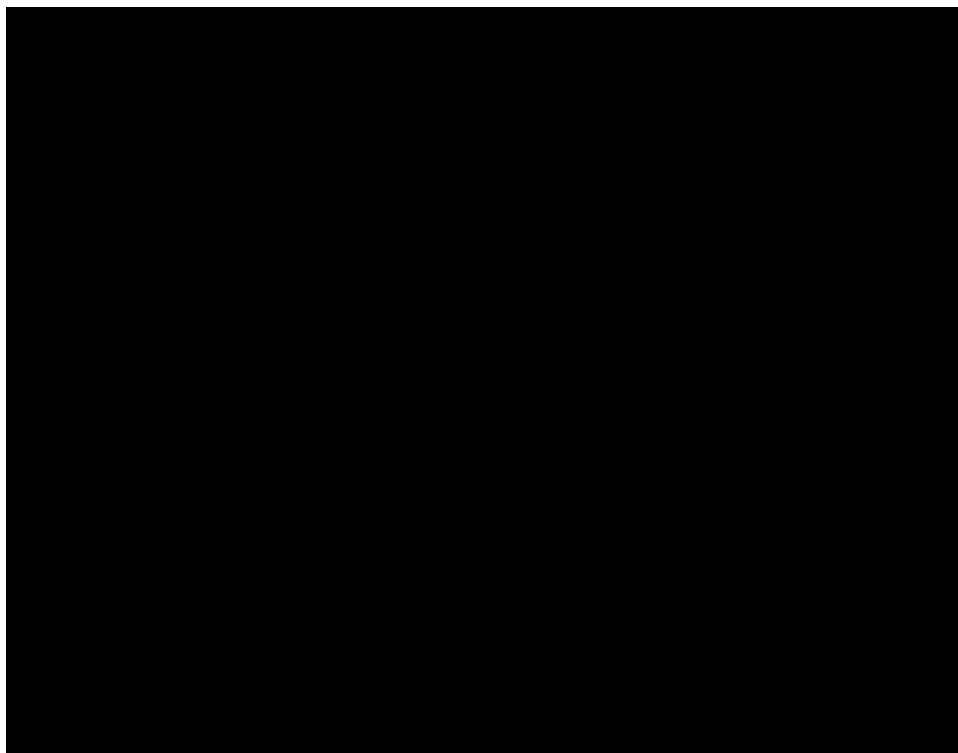
Oura’s Gen. 4 ring does not infringe either claims 7 or 9 of the '789 Patent for the reasons explained herein.

- a) Oura's Gen. 4 ring does not meet the limitations of claim element [7.j]

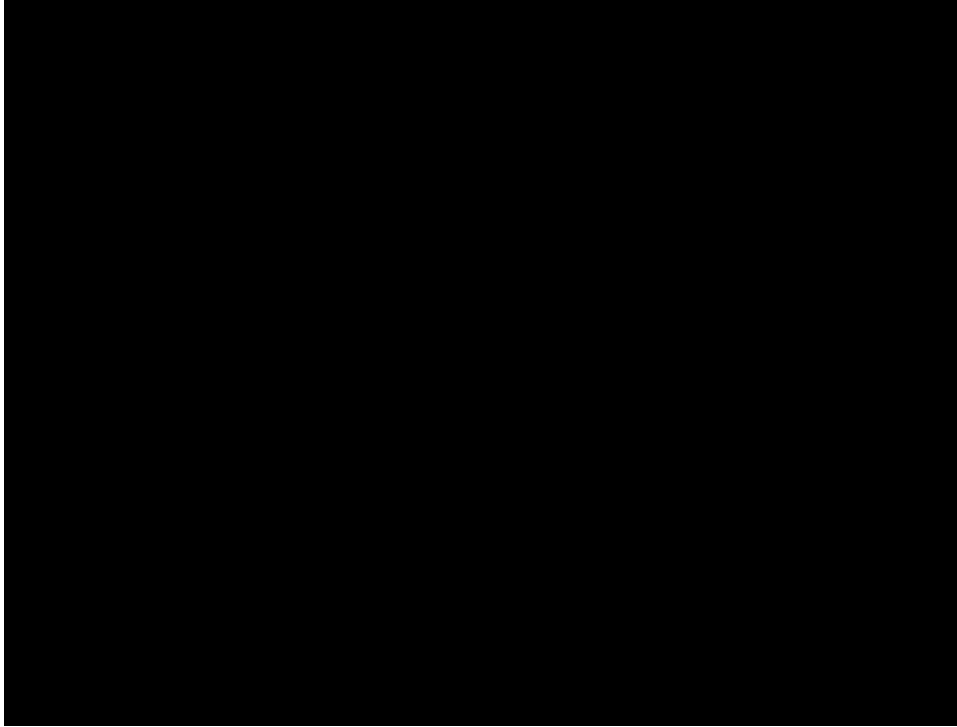
Dr. Mortazavi will explain that each of Samsung's three "alternative mappings" fail to include a "second insulating layer" as required by claim element [7.j]. RDX-0032C, ¶¶597-99.

Specifically, as illustrated below, Dr. Mortazavi will explain that in each of the three alternative mappings, the "second insulating layer" is identified as a [REDACTED]

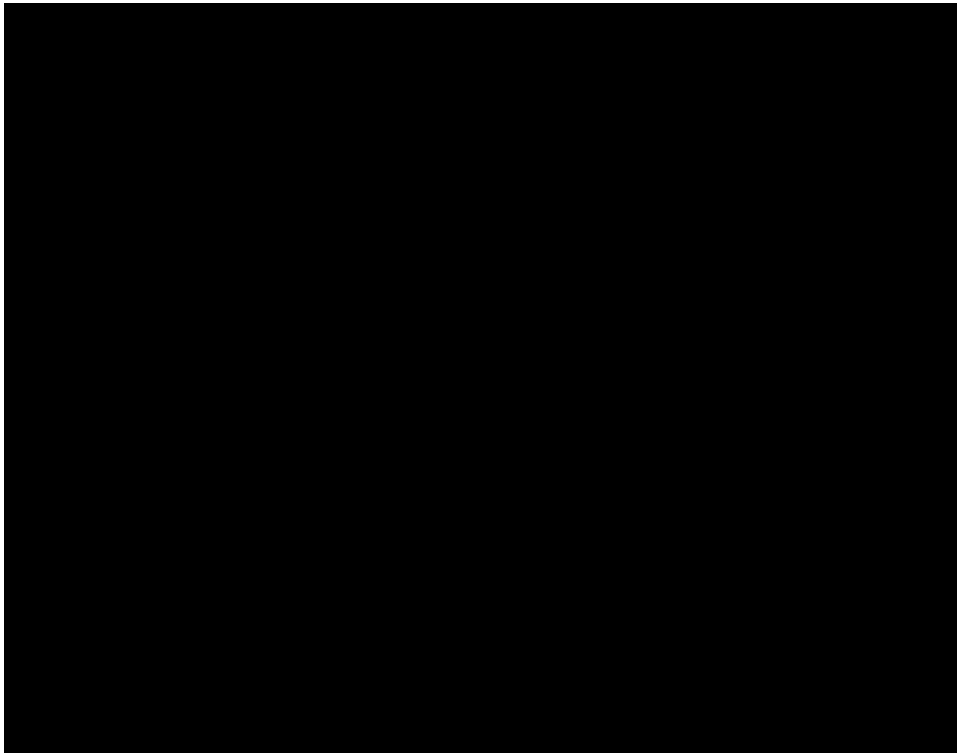
[REDACTED]. Compare RDX-0032C, ¶598 with RDX-0036C, ¶162.



First Alternative Mapping (RDX-0036C, ¶161 (citing RX-0255C (annotated)))



Second Alternative Mapping (RDX-0036C, ¶171 (citing RX-0255C (annotated)))



Third Alternative Mapping (RDX-0036C, ¶181 (citing RX-0255C (annotated)))

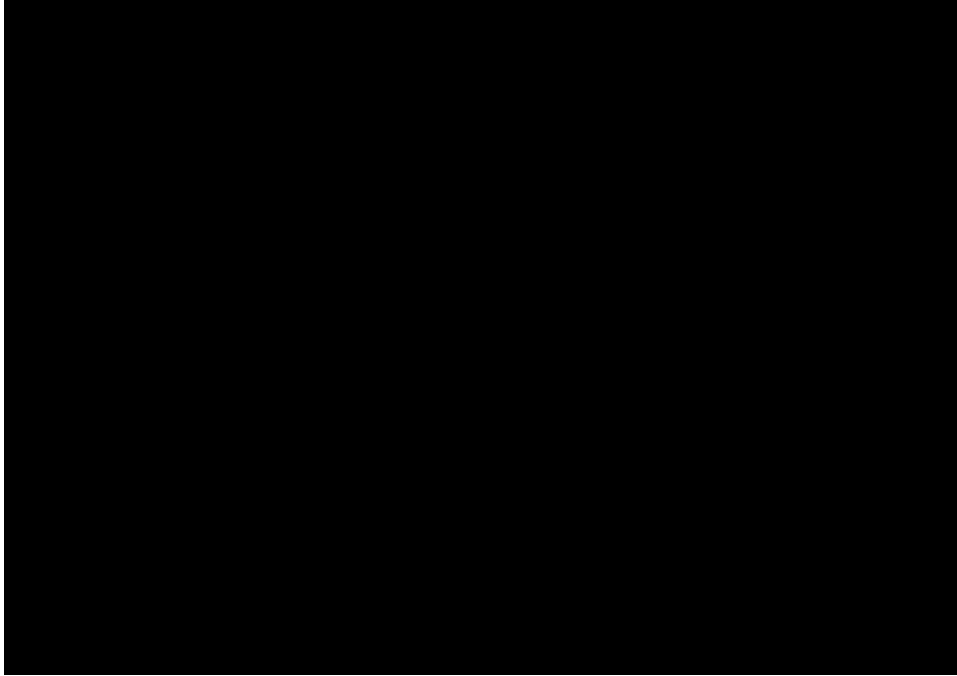
While [REDACTED] a POSITA may recognize as an insulating material, Dr. Mortazavi will testify that a POSITA would not recognize a b [REDACTED] [REDACTED] as forming a single insulating layer. RDX-0032C, ¶599. Nor does the intrinsic evidence disclose that the second insulating layer may include a plurality of layers. *Id.* (citing JX-0004, 5:52-56).

Thus, there is no “second insulating layer,” and Samsung cannot demonstrate that the Oura Ring 4 infringes limitation [7.j].

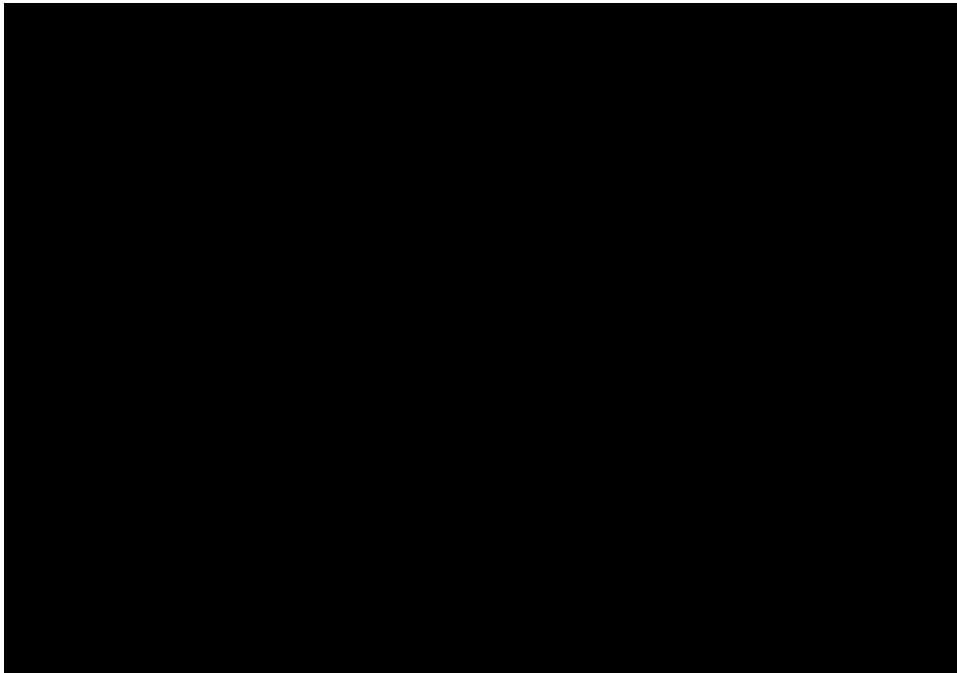
- b) Oura’s Gen. 4 ring does not meet the limitations of claim element [7.o]

Dr. Mortazavi will explain that each of Samsung’s three “alternative mappings” fail to include a “second insulating layer . . . having a second thickness thinner than the first thickness” as required by claim element [7.o]. RDX-0032C, ¶¶617-618.

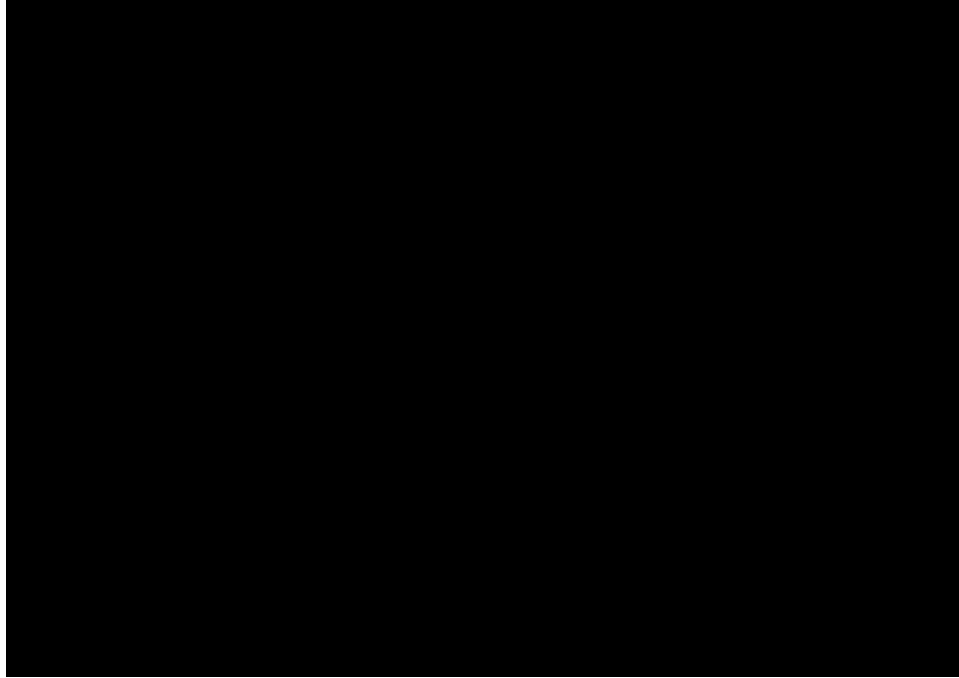
Dr. Warren alleges that the thickness of the second insulating layer in the flexible PCB portion is [REDACTED] (*i.e.*, the alleged second thickness) and that the thickness of the second insulating layer in the rigid PCB portion is [REDACTED] (*i.e.*, the alleged first thickness). RDX-0036C, ¶246. But Dr. Warren improperly identifies [REDACTED] as forming the alleged second insulating layer in the rigid PCB portion. RDX-0032C, ¶619. Specifically, [REDACTED] a POSITA may recognize as an insulating material, a POSITA would not recognize [REDACTED] [REDACTED]. *Id.* At best, the stack-up diagrams relied on by Dr. Warren show that the thickness of the alleged second insulating layer in the flexible PCB portion and the thickness of the second insulating layer in the rigid PCB portion are the same, not different as the claim requires (*i.e.*, both are 1 [REDACTED]). RDX-0032FC, ¶619.



First Alternative Mapping (RDX-0036C, ¶245 (citing RX-0255C (annotated)))



Second Alternative Mapping (RDX-0036C, ¶252 (citing RX-0255C (annotated)))



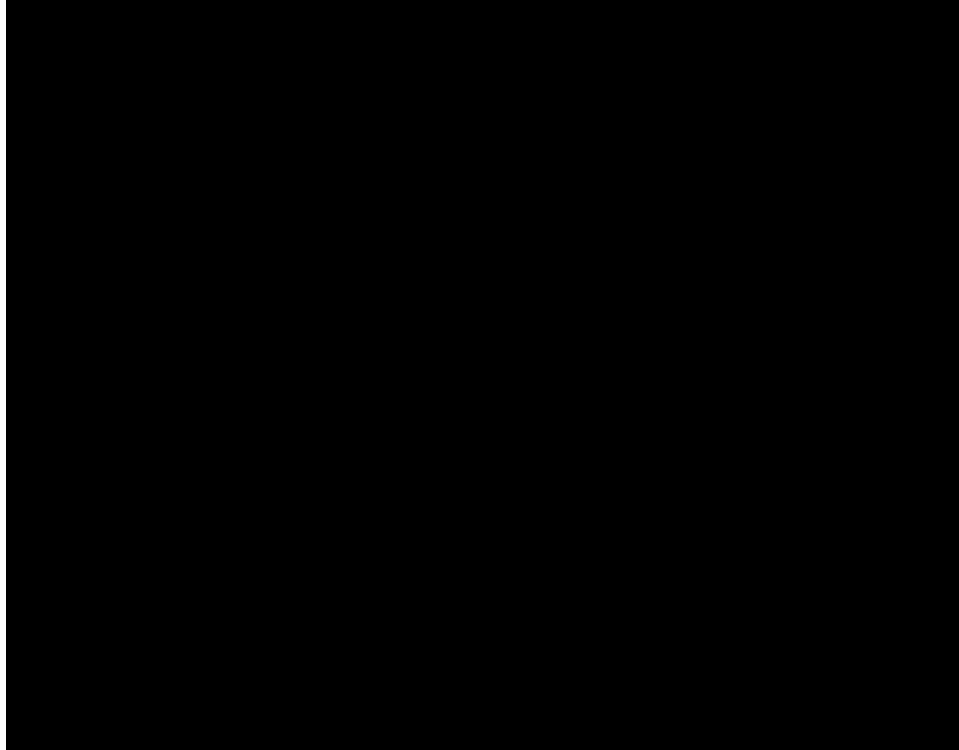
Third Alternative Mapping (RDX-0036C, ¶259 (citing RX-0255C (annotated)))

Accordingly, the evidence will show that Samsung cannot demonstrate that the Oura Ring 4 infringes limitation [7.o].

- c) Oura's Gen. 4 ring does not meet the limitations of dependent claim 9

The evidence will show that the Oura Ring 4 does not infringe dependent claim 9 for at least the same reasons that claim 7 is not infringed. RDX-0032C, ¶¶635-636.

Moreover, Dr. Warren has failed to show a fourth insulating layer as required by claim 9. *Id.* ¶¶637-644. Specifically, Dr. Mortazavi will explain—as illustrated below—that Dr. Warren improperly identifies [REDACTED] as forming the alleged fourth insulating layer. *Compare* RDX-0032C, ¶639 *with* RDX-0036C, ¶¶303-304.



Mapping (RDX-0036C, ¶303 (citing RX-0255C (annotated by Dr. Warren)))

While [REDACTED] a POSITA may recognize as an insulating material, a POSITA would not recognize [REDACTED] [REDACTED]. RDX-0032C, ¶640. Nor does the intrinsic evidence disclose that the fourth insulating layer may include a plurality of layers. *Id.* (citing JX-0004, 5:52-56).

Accordingly, Samsung has failed to demonstrate that the Oura Ring 4 infringes dependent claim 9.

2. Ring 5 and [REDACTED] Do Not Infringe the Asserted Claims

Oura's arguments with regard to Oura's Gen. 4 ring also apply to the Ring 5 and [REDACTED] but additionally arguments apply for these particular products, as was set forth in Oura's motion for summary determination. *See* EDIS Doc. ID 887851.

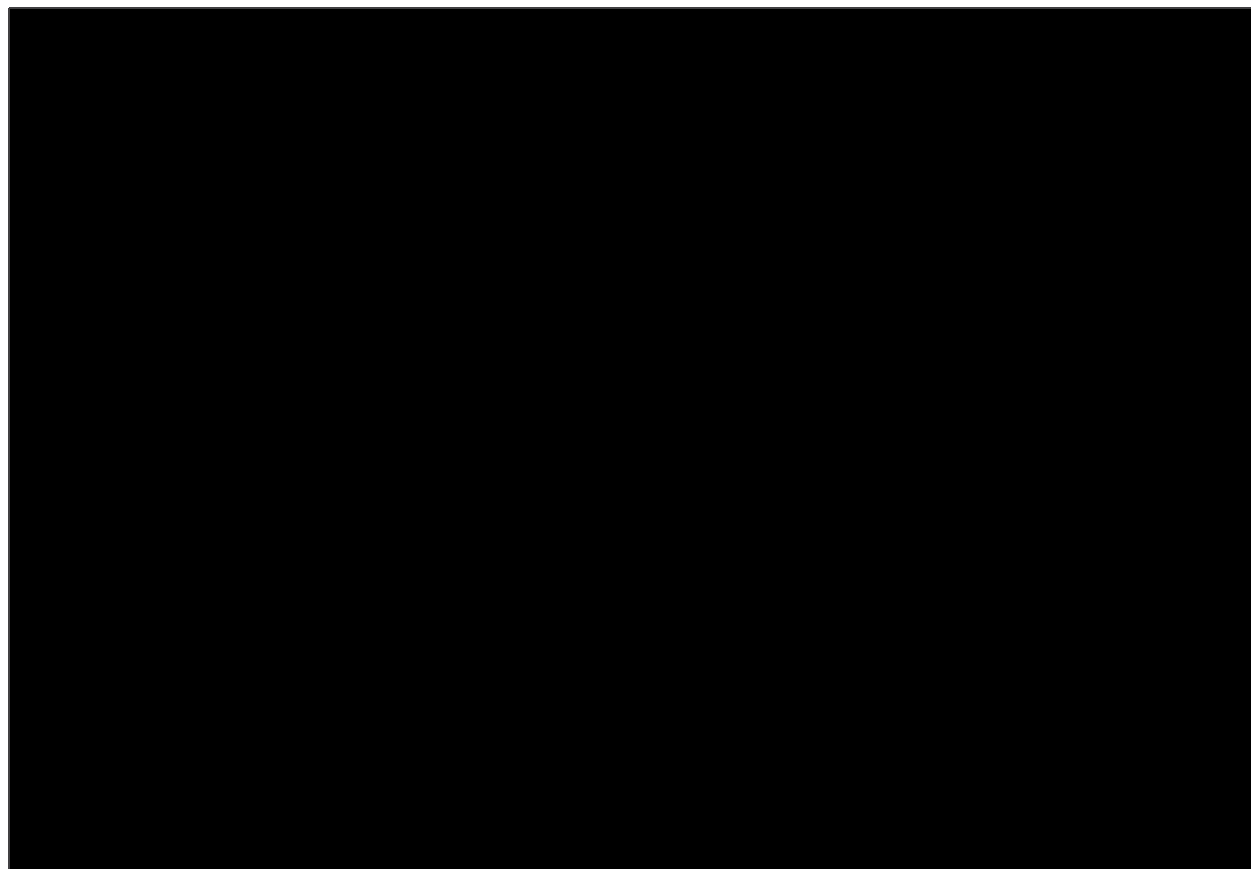
- a) Samsung has put forth no infringement contentions or evidence of infringement

Samsung has put forth no contention or evidence of infringement by Ring 5 or [REDACTED] and this basis alone it cannot carry its burden of proof on infringement for either product. *See Certain Crafting Machines*, Inv. No. 337-TA-1426, ID at 40.

- b) Ring 5 does not infringe the asserted claims

As Dr. Mortazavi will testify, there are at least two reasons why the Ring 5 does not infringe claims 7 or 9 of the '789 Patent.

First, the Ring 5 does not meet claim element [7.o], which requires that the second insulating layer for the flexible PCB portions have a “second thickness thinner than the first thickness” of the second insulating layer for the rigid PCB portions. Rather, [REDACTED] [REDACTED]. JDX-0007C ¶¶662-64. This is shown by an annotated version of the [REDACTED] PCB stackup diagram:

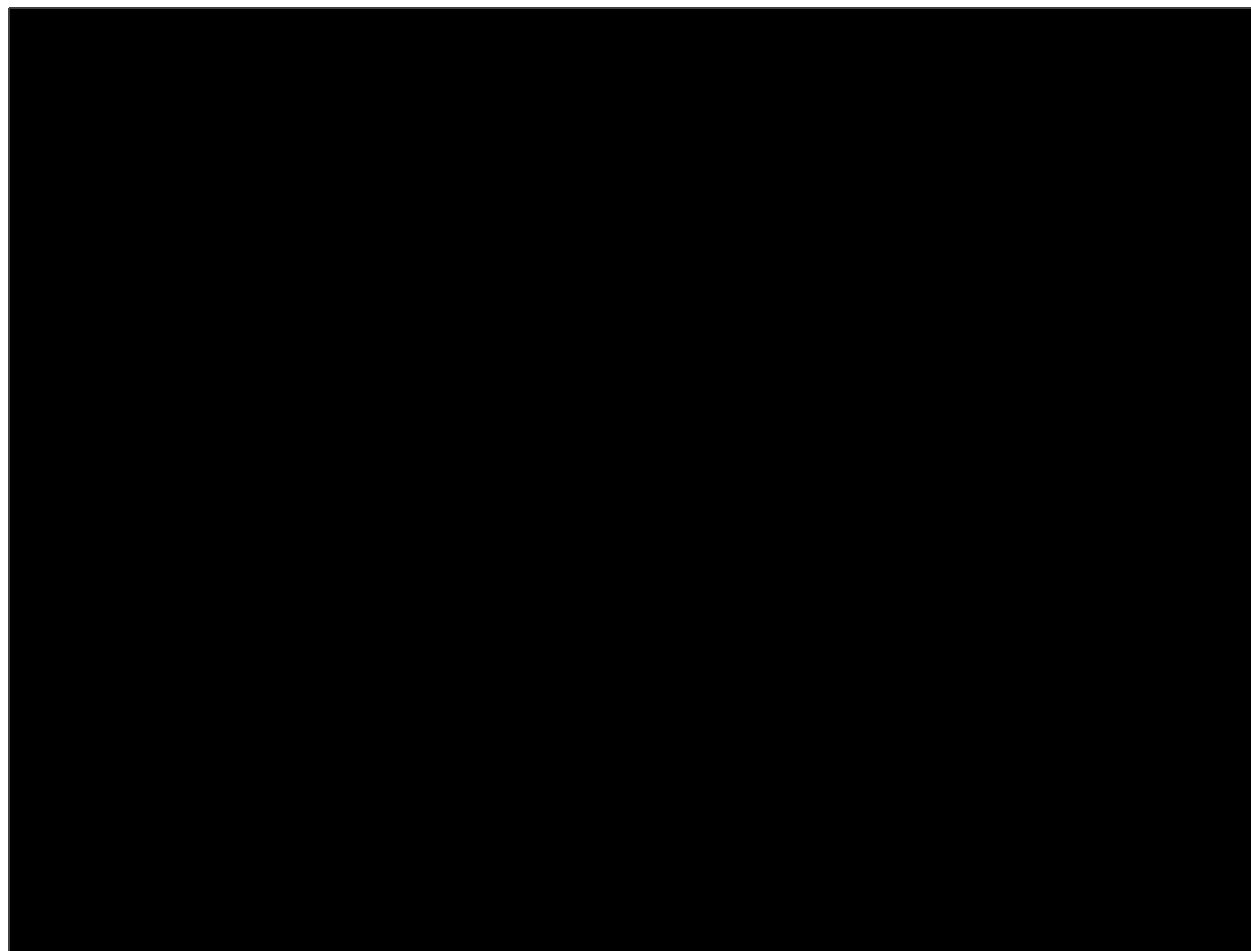


Id. ¶¶663 (annotating a color version of RX-0258C to show [REDACTED]); RX-0169 at OURA1478_00014259 (showing [REDACTED]).

First, the Ring 5 does not meet claim element [7.q], which requires a “second material” in the third insulating layer of the flexible PCB portions that is different from the “first material” in the third insulating layer of the rigid PCB portions. Dr. Mortazavi showed that the [REDACTED]

[REDACTED] JDX-0007C

¶¶667-70. This is shown by an annotated version of the [REDACTED] PCB stackup diagram:



Id. ¶669 (annotating a color version of OURA1478_00006316).

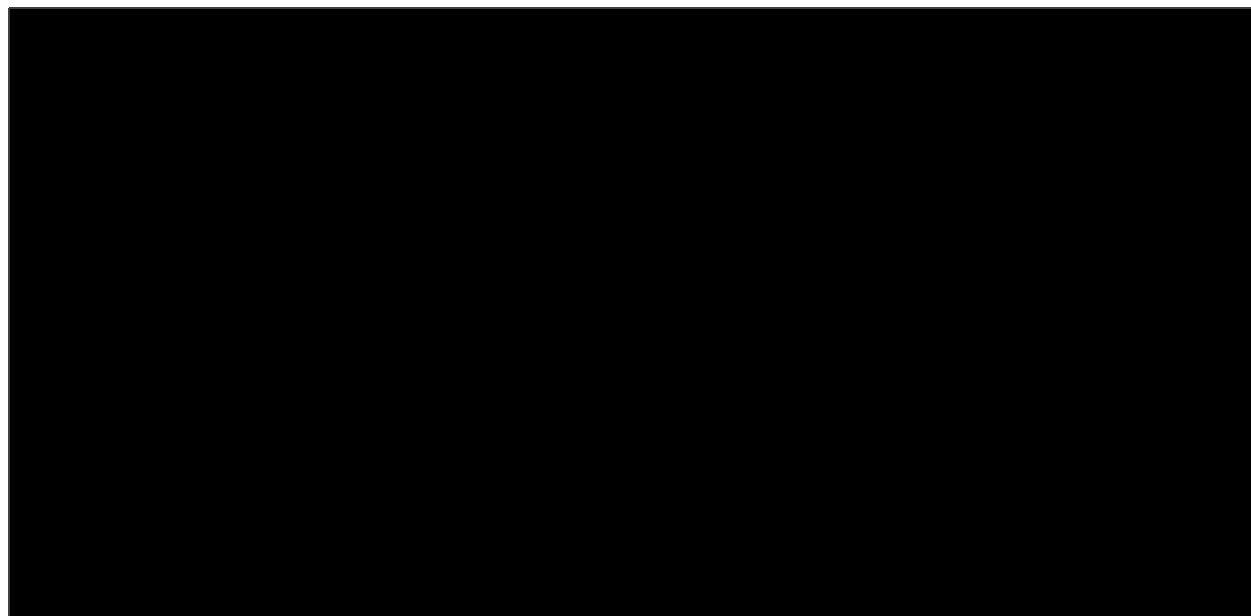
c) [REDACTED] does not infringe the asserted claims

[REDACTED] also does not infringe claims 7 or 9 of the '789 Patent, as it does not meet the requirement of claim element [7.o]. Dr. Mortazavi will show that the [REDACTED]

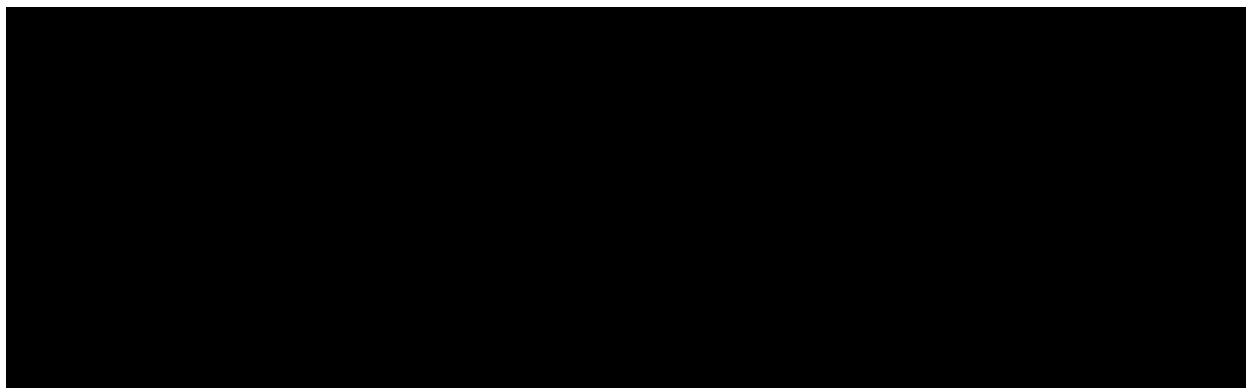
[REDACTED]

[REDACTED]. JDX-0007C ¶¶677-79. This is shown by an annotated version of the [REDACTED]

PCB stackup diagram:



Id. ¶678 (annotating RX-0151C). The [REDACTED] can also be seen in the scanning electron microscope images of the [REDACTED] PCB:



RX-0259C.

B. The Technical Prong of the Domestic Industry Inquiry is Not Met

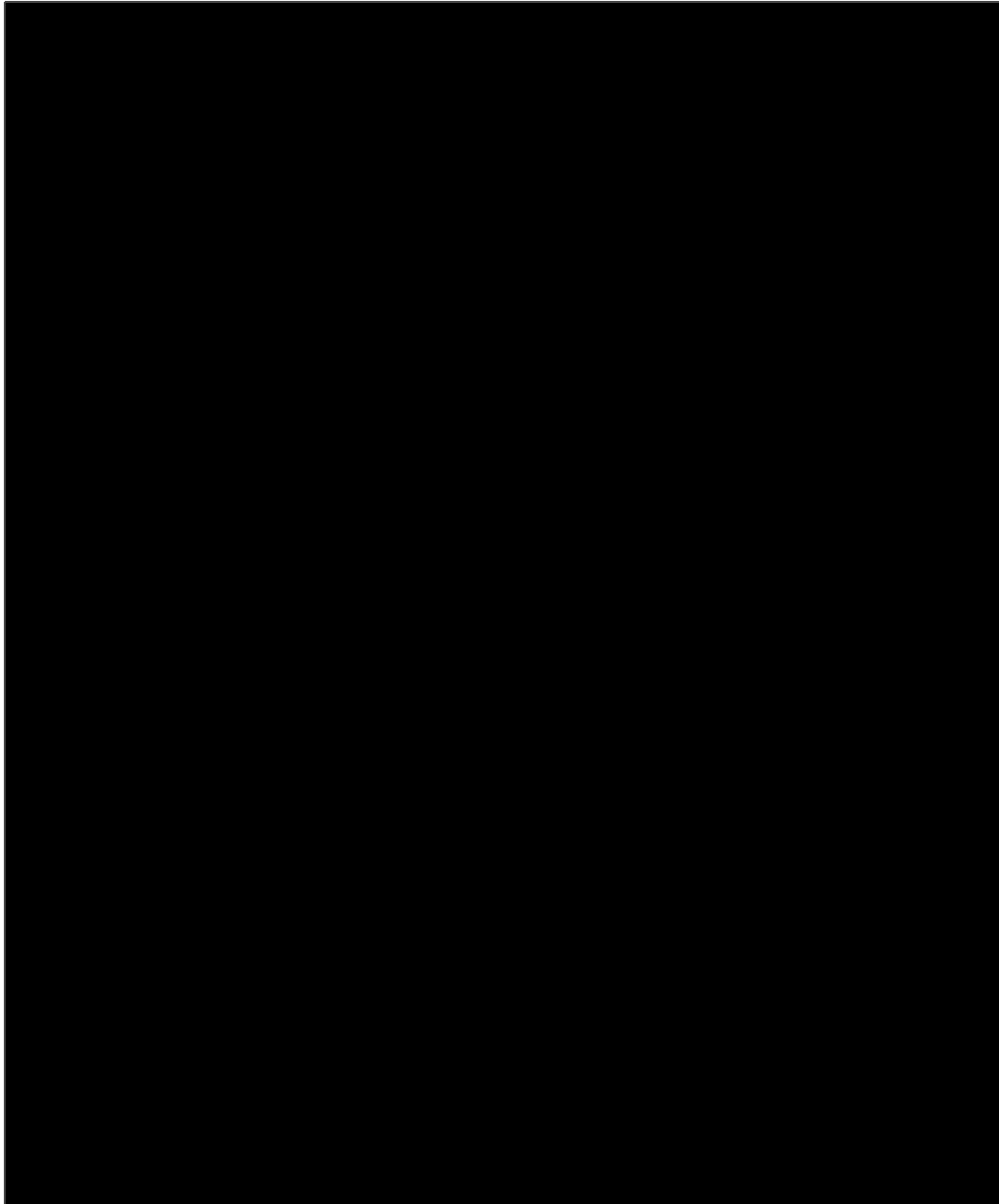
Samsung's alleged DI products do not practice claim 6 of the '789 Patent for the reasons explained herein. Claim 6 depends on independent claim 1, so the non-practice of elements of claim 1 shows non-practice of claim 6.

PUBLIC VERSION

- a) The Asserted DI Products do not meet the limitations of claim element [1.c]

The Asserted DI Products do not practice limitation [1.c], which requires “a first printed circuit board (PCB).”

Samsung argues that each of the Asserted DI Products “includes a first PCB, in the form of the a [REDACTED],” and identifies the “[REDACTED]



RDX-0036C, ¶333 (citing CX-0119C (S25 Basic) (annotated)).

However, Dr. Mercier will testify that the “[REDACTED]” is nothing more than [REDACTED]
[REDACTED]. RDX-0033C, ¶166. While [REDACTED]
[REDACTED] may have similarities, they are different components in an electronic system. RDX-0033C,
¶166; RDX-0039C, 101:1-13. Specifically, Dr. Mercier will further testify that an [REDACTED]
[REDACTED]
[REDACTED]. RDX-0033C, ¶166; RDX-
0039C, 101:1-13. In contrast, a [REDACTED]
[REDACTED]
[REDACTED]. RDX-0033C, ¶166; RDX-0039C, 101:1-13.

Accordingly, Dr. Mercier will demonstrate that the Asserted DI Products do not practice limitation [1.c].

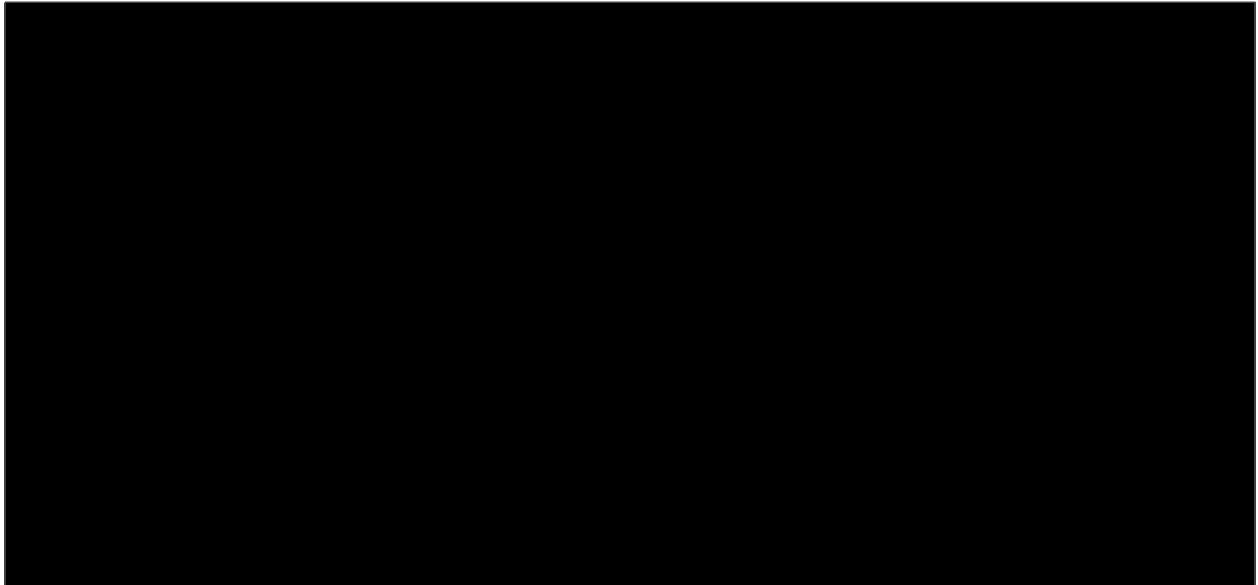
- b) The Asserted DI Products do not meet the limitations of claim element [1.h]

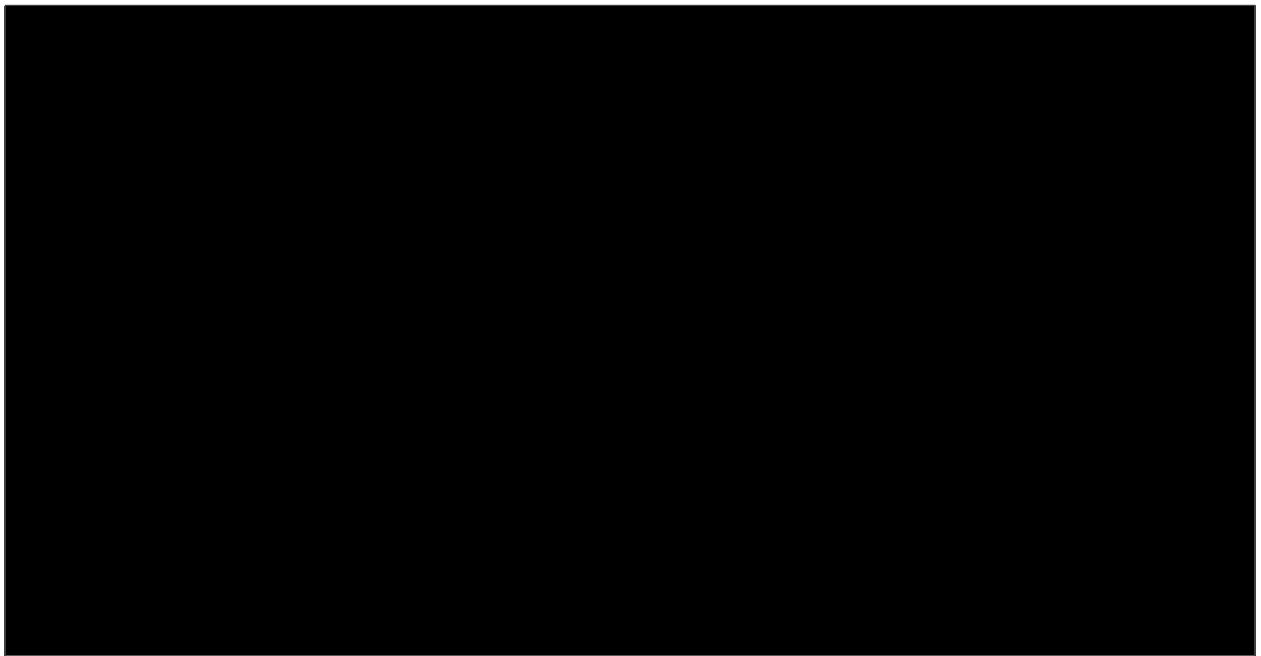
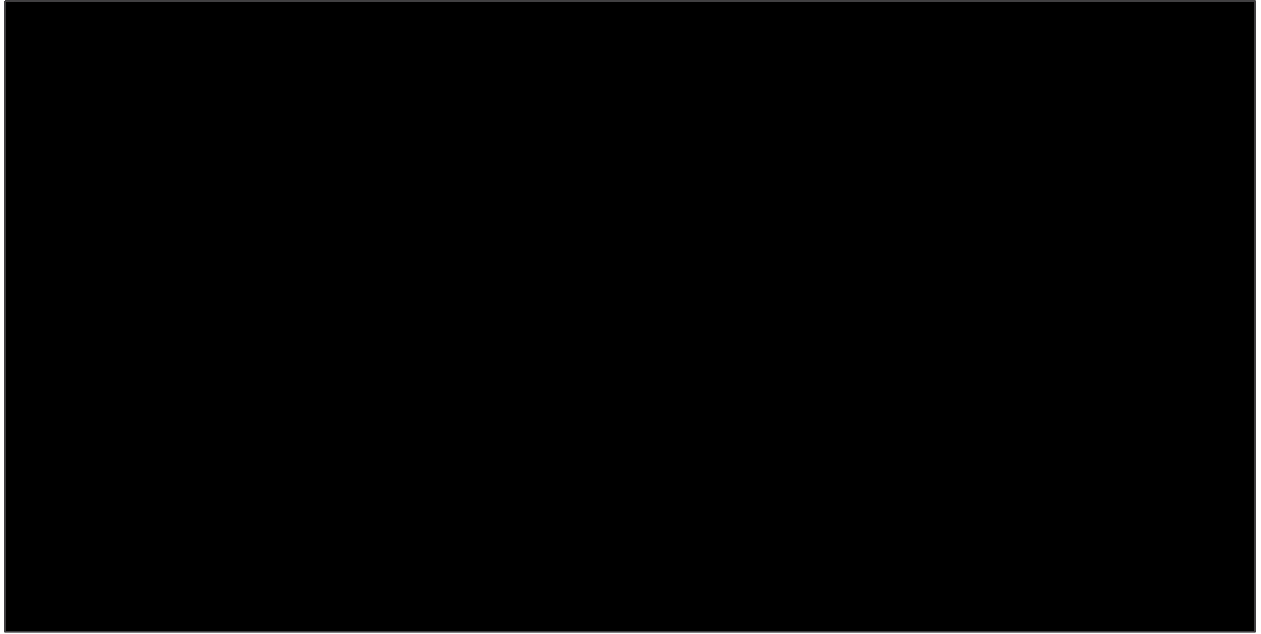
The Asserted DI Products do not practice limitation [1.h], which requires “a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line.”

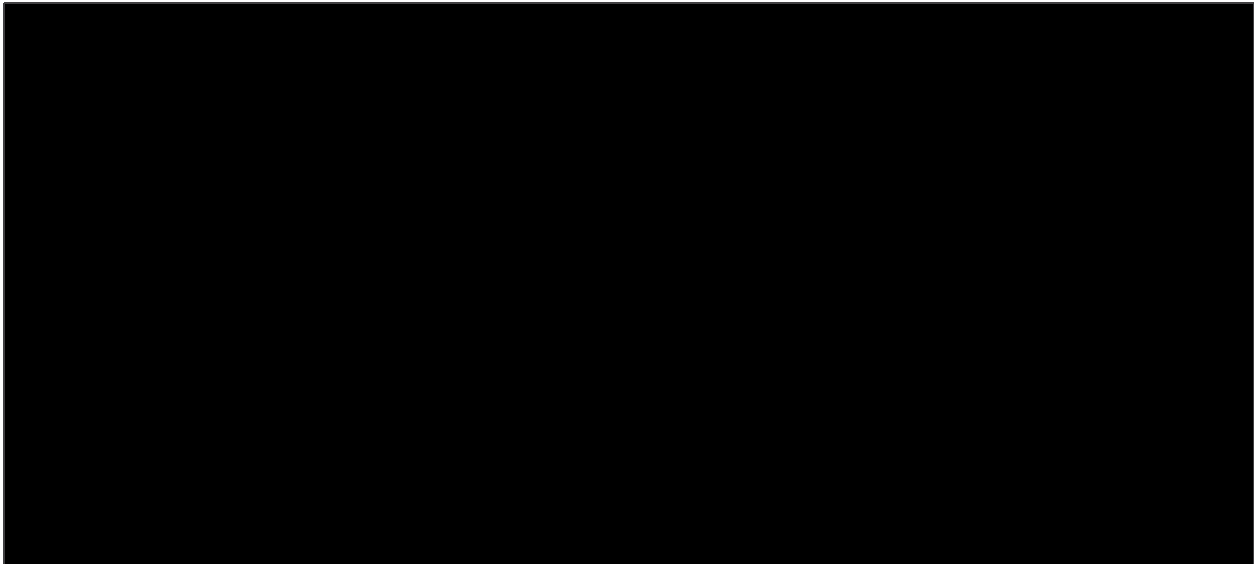
To begin, Dr. Mercier will testify that a ground line—as understood by a POSITA—is typically a physical copper path or wire on a PCB that connects various components to the system’s electrical ground. RDX-0033C, ¶173; *id.* ¶177. In one application, a signal lines and ground lines may run next to each other to minimize the impact of noise created by other signal lines and/or electronic components. RDX-0033C, ¶178 (citing JDX-0018C (Cho), 97:22-98:1). In contrast, a ground portion (sometimes referred to as a ground area, ground plane, or ground plate) is a large section of ground extending in both the x and y directions that exists in one or more layers or an

entire layer of, for example, a PCB, and has significantly less inductance than a ground line. RDX-0039C (Mercier), 132:5-134:3; JX-0004, 31:33-44, 32:62-33:6 (claiming a “ground portion” as opposed to a “ground line”), 6:22-46 (distinguishing between ground lines and ground portions or areas); RDX-0033C, ¶¶173, 177-180; RX-0193 (OURA1478_00017839-18700 at -18254 (“a ground plane has significantly less inductance than a trace”); RX-0194 (OURA1478_00018701-18707).

Dr. Mercier will further explain that Samsung has failed to identify the required ground lines in the Asserted DI Products, but has rather identified [REDACTED] [REDACTED]. RDX-0033C, ¶¶173-175. For example, Dr. Warren identifies the following images and alleges that the “[REDACTED] [REDACTED].” RDX-0036C, ¶353.

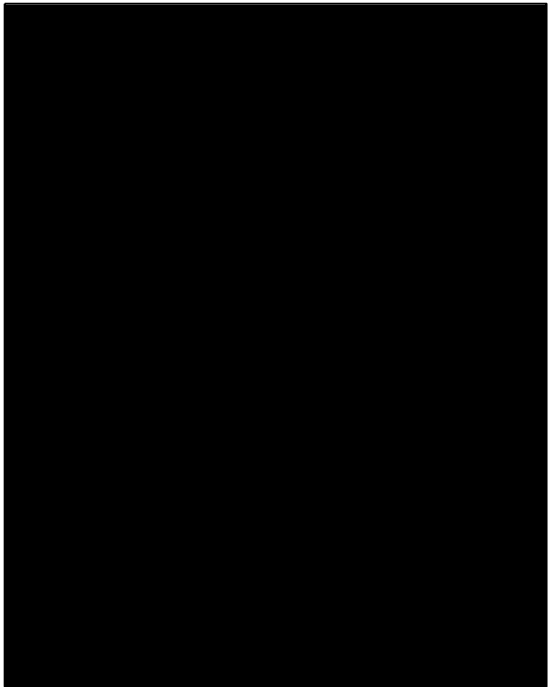
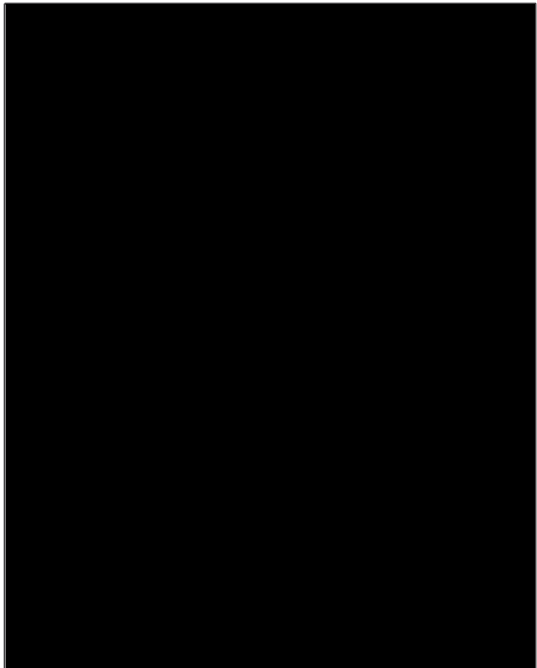


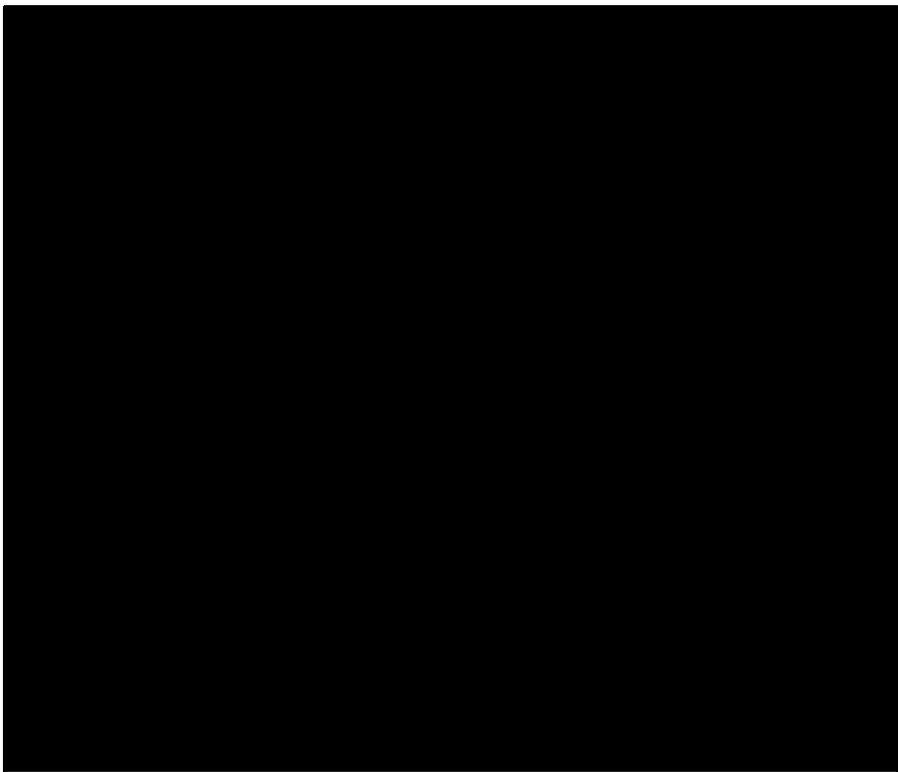
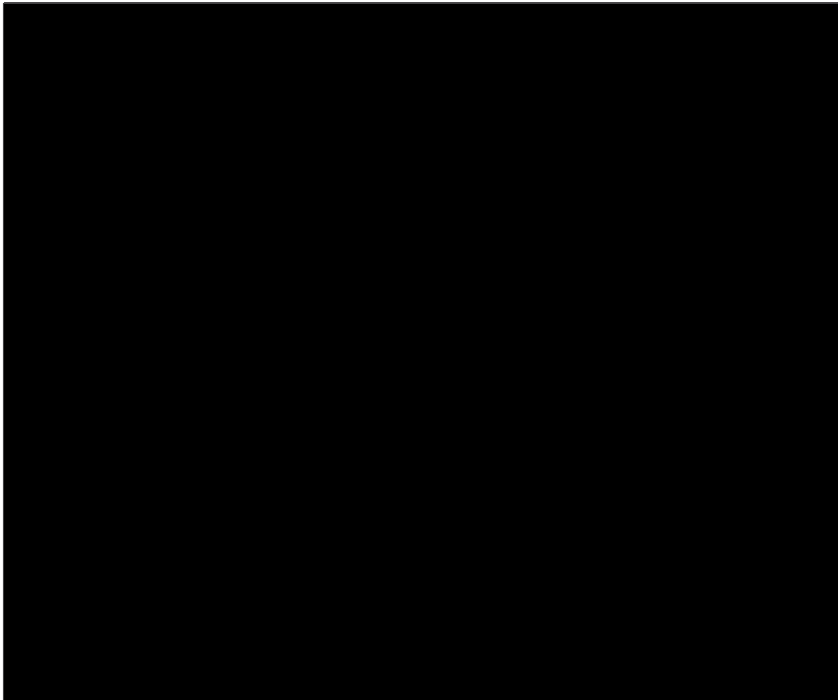


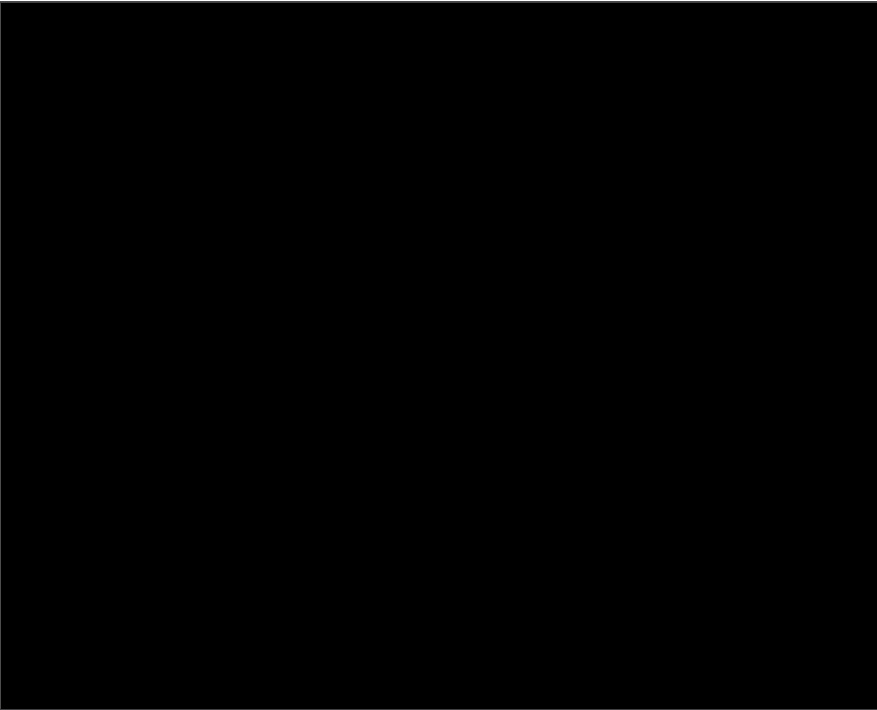


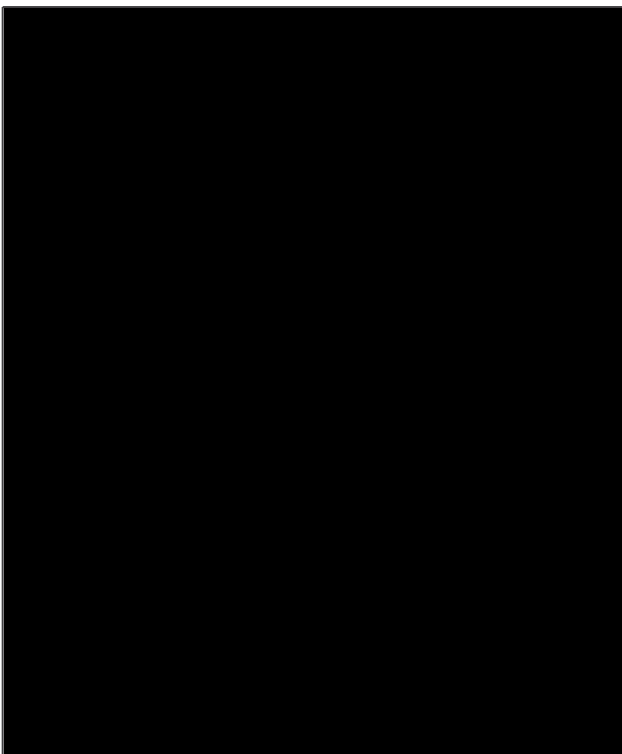
RDX-0036C, ¶353 (citing CX-0671C (S25 design files annotated by Complainant)), 354 (citing additional design files for other Asserted DI Products).

However, Dr. Warren has not identified the required first and second ground lines that are partially spaced apart. RDX-0033C, ¶175. Rather, Dr. Warren only identifies [REDACTED]
[REDACTED]. *Id.* For reference, Dr. Warren's [REDACTED]
[REDACTED]:

Asserted DI Product Gerber File(s) Cited By Samsung RDX-0036C, ¶353 (citing CX-0671C (S25 design files annotated by Complainant))	
Asserted DI Product Gerber File(s) Cited By Samsung and Further Annotated by Oura CX-0671C (S25 design files annotated by Complainant and further annotated by Respondent)	

Asserted DI Product Gerber File(s) Cited By Samsung RDX-0036C, ¶353 (citing CX-0671C (S25 design files annotated by Complainant))	
Asserted DI Product Gerber File(s) Cited By Samsung and Further Annotated by Oura CX-0671C (S25 design files annotated by Complainant and further annotated by Respondent)	

Alleged DI Product Gerber File(s) Cited By Dr. Warren CX-0671C (S25 design files annotated by Complainant and further annotated by Respondent)	
Alleged DI Product Gerber File(s) Cited By Dr. Warren and Further Annotated by Respondent CX-0671C (S25 design files annotated by Complainant and further annotated by Respondent)	

Alleged DI Product Gerber File(s) Cited By Dr. Warren CX-0671C (S25 design files annotated by Complainant and further annotated by Respondent)	
Alleged DI Product Gerber File(s) Cited By Dr. Warren and Further Annotated by Respondent CX-0671C (S25 design files annotated by Complainant and further annotated by Respondent)	

Dr. Mercier will explain that, [REDACTED]

[REDACTED]. RDX-0033C, ¶175. Indeed, a Samsung engineer, Mr. Kim, provided testimony supporting Dr. Mercier's conclusion, namely that the [REDACTED]

[REDACTED] JX-0015 (Kim), 63:7-17, 65:2-16.

Finally, Dr. Warren has failed to identify the required "first signal line located between the first ground line and the second ground line" and the required second signal line located between the first signal line and the second ground line." Specifically, Dr. Mercier will explain that Dr. Warren relies on [REDACTED]. RDX-0033C, ¶¶181-182. And, as Dr. Warren [REDACTED] in other Asserted DI Products, he cannot demonstrate that these products practice this limitation for the same reasons. *Id.* ¶183.

- c) The Asserted DI Products do not meet the limitations of claim element [6.b]

Samsung cannot prove the Asserted DI products practice dependent claim 6 for at least the same reasons as explained for claim 1. RDX-0033C, ¶203.

Further, with respect to limitation [6.b], Dr. Warren has provided no opinions related to whether "the first ground line and the second ground line are electrically connected with a ground member located in the second PCB." RDX-0036C, ¶¶391-393. Dr. Warren failed to identify the required "first ground line" and the required "second ground line" let alone any ground lines electrically connected with a ground member. RDX-0033C, ¶205. Thus, Samsung has failed to demonstrate that the Asserted DI Products practice limitation [6.b]. *Id.* ¶206.

C. Claims 6, 7, and 9 of the '789 Patent Are Invalid in View of the Prior Art

There are seven asserted grounds of prior art-based invalidity that render invalid claims 6,⁶ 7, and 9 of the '789 Patent:

Invalidity Ground	Basis	Applicable Claims
Ground 1	Anticipation in view of Oura Ring Gen. 1	7
Ground 2	Obviousness in view of Williams	7, 9
Ground 3	Obviousness in View of Williams and Aoyama	7, 9
Ground 4	Obviousness in View of Williams and Naganuma	7, 9
Ground 5	Obviousness in View of Aoyama	7, 9
Ground 6	Obviousness in View of Aoyama and Williams	7, 9
Ground 7	Obviousness in View of Galaxy S5, Mow, and Kim	6

These seven grounds of invalidity are explained in detail herein.

1. Overview of Prior Art References Relied Upon

The Oura Ring Gen. 1 was Oura's first generation of smart ring, and was a wearable device worn on a user's finger. JDX-0001C ¶¶1350-1353. This smart ring used physiological data signals to calculate deep sleep, light sleep, REM sleep, and awake states, including by using an optical sensor, an accelerometer, and a temperature sensor. RX-0023. As established in Section IV.B.2, *supra*, the Oura Ring Gen. 1 was in public use and available to the public at least prior to August 12, 2015, and as Mr. Christophy will testify, was on sale by August 18, 2015, and accordingly qualifies as prior art to the '789 Patent under at least 35 U.S.C. § 102(a).

U.S. Patent No. 10,064,276 B2 ("Williams") (RX-0076), is titled "3D Bendable Printed Circuit Board with Redundant Interconnections," was issued on August 28, 2018. Williams was

⁶ Claim 6 is only relevant to alleged domestic industry and is not an asserted claim.

filed on October 21, 2015, published on April 27, 2017, and issued on August 28, 2018, and therefore qualifies as prior art to the '789 patent under at least 35 U.S.C. § 102.

U.S. Patent No. 8,383,948 B2 (“Aoyama”) (RX-0074) is titled “Flex-Rigid Wiring Board and Method for Manufacturing the Same,” Aoyama claims priority to U.S. provisional application No. 61/243,576 (filed on September 18, 2009), was filed on June 4, 2010, published on March 24, 2011, and issued on February 26, 2013, and therefore qualifies as prior art to the '789 Patent under at least 35 U.S.C. § 102.

U.S. Patent Application Publication No. 2012/0325524 A1 (“Naganuma”) (RX-0075), titled “Flex-Rigid Wiring Board and Method for Manufacturing the Same.” Naganuma claims priority to provisional application No. 61/500,361 (filed on June 23, 2011), was filed on May 29, 2012, and published on December 27, 2012, and therefore qualifies as prior art to the '789 patent under at least 35 U.S.C. § 102.

The Samsung Galaxy S5 (“S5”) is a smartphone released in April 2014 and qualifies as prior art to the '789 Patent under at least 35 U.S.C. § 102(a). JDX-0001C ¶¶ 3138-40; RX-0077-87; RX-0142-0145; RX-0939C, RFAs 6-9.

U.S. Patent Publication No. 2015/0280771 (“Mow”) (RX-0110) was filed on June 11, 2015, published on October 1, 2015, and has a priority date of July 6, 2010, and therefore qualifies as prior art to the '789 patent under at least 35 U.S.C. § 102.

U.S. Patent Application Publication No. 2019/0045630 A1 (“Kim”) (RX-0112) was filed on February 6, 2017, published on February 7, 2019, and claims priority to Korean application No. 10-2016-0022978, filed on February 26, 2016, and therefore qualifies as prior art to the '789 patent under at least 35 U.S.C. § 102.

2. Ground 1: Anticipation in View of Oura Gen. 1 Ring

Claim 7 of the '789 Patent is invalid as anticipated by the Oura Ring Gen. 1, which meets each limitation of this claim as explained herein and as will be shown by the testimony of Dr. Mortazavi. This is especially clear should the opinions of Dr. Warren regarding alleged infringement by the Oura Gen. 4 ring be afforded weight, as claims should be construed in the same for purposes of both anticipation and infringement.

a) [Uncontested]⁷ Claim Elements [7.pre]-[7.c]

The Oura Ring Gen. 1 meets requirement of claim elements [7.pre]-[7.c]. JDX-0001C ¶¶3342-3379.

The Oura Ring Gen. 1 is a ring that uses Bluetooth wireless circuitry to transmit biometric data to a paired mobile phone and is therefore a “portable communication device.” JDX-0001C ¶¶3342-3355 (citing JX-0026C at 45:1-6, 45:9-46:6; RX-0029C; RX-0023; RPX-0001; RPX-0005).

The Oura Ring Gen. 1 includes a battery that is a “first electronic component.” JDX-0001C ¶¶3356-3362 (citing RX-0023; RX-0026C; RX-0028C; RX-0032; RX-0029; RX-0077C; JX-0026 at 45:1-6, 45:9-46:21, 127:16-17, 127:22-23, 128:8-23; RPX-0001; RPX-0005).

BREAKTHROUGH MINIATURIZATION



⁷ Samsung challenges the prior art status of the Oura Ring Gen. 1. *See* CSUCE at 40.

RX-0023

The Oura Ring Gen. 1 includes two IR LED emitters, a photodetector, a microcontroller, flash memory, and charging pads, and therefore includes at least a “second electronic component.” JDX-0001C ¶¶3363-3371 (citing RX-0023; RX-0026C; RX-0028C; RX-0029C; RX-0032; RX-0077C; JX-0026C at 45:9-46:21, 127:16-17, 127:22-23, 128:8-23; RPX-0001; RPX-0005).

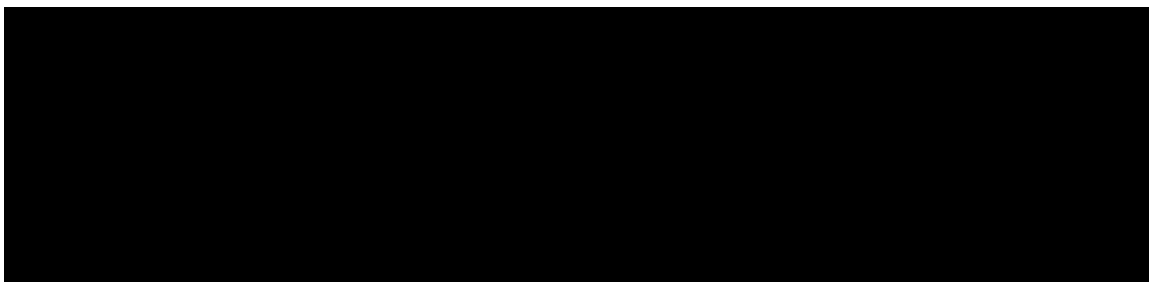
The Oura Ring Gen. 1 includes “a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component.” JDX-0001C ¶¶3372-3379 (citing RX-0031C; RX-0023; RX-0028C; RX-0029C; JX-0026C at 45:6-46:21, 53:17-19, 153:19-25, 159:14-160:1, 166:7-9, 166:23-167:17, 168:6-169:18; RX-0032; RPX-0001; RPX-0005). The PWB Stackup Specification (RX-0031C) confirms [REDACTED]

[REDACTED]. Mr. Christophy is also expected to testify that the Oura Ring Gen. 1 PCB [REDACTED]
[REDACTED]
[REDACTED].

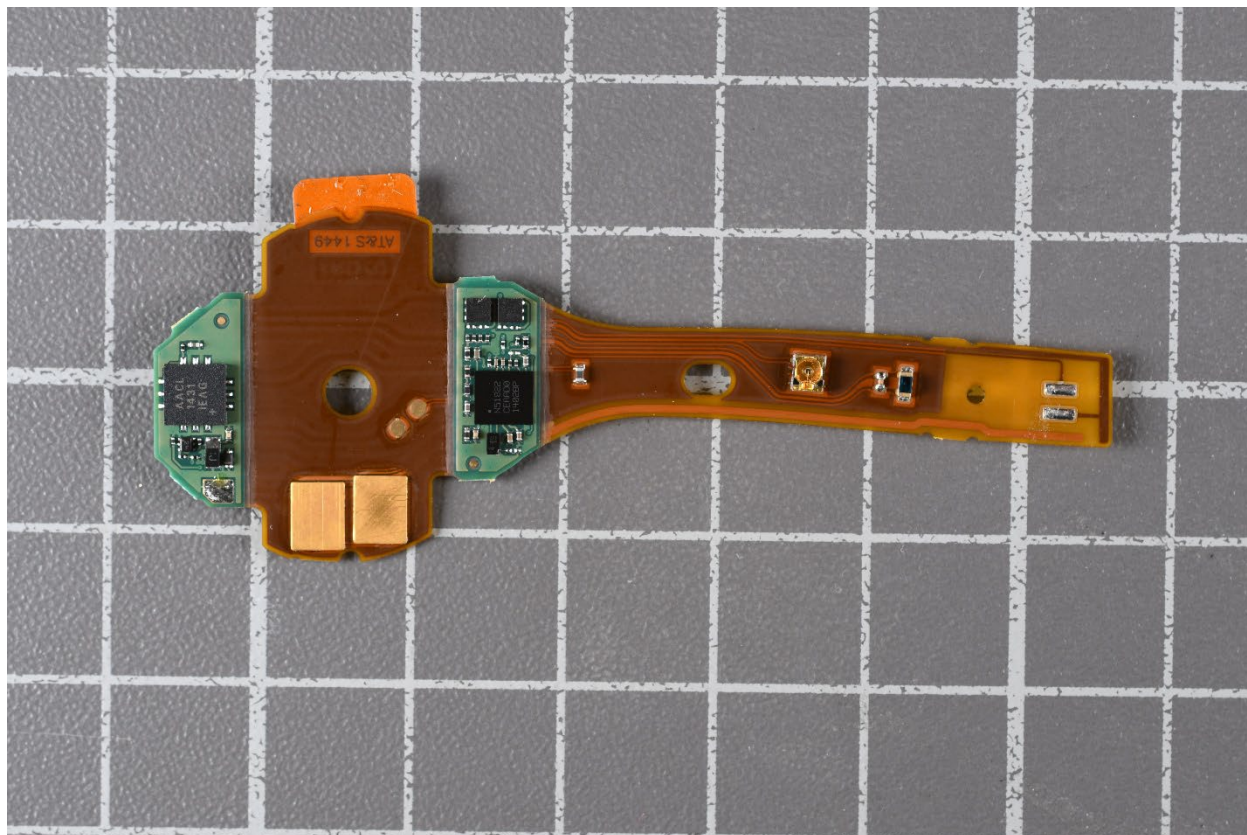
b) Claim Elements [7.d]-[7.g]

Claim elements [7.d]-[7.g] are met by the Oura Ring Gen. 1.

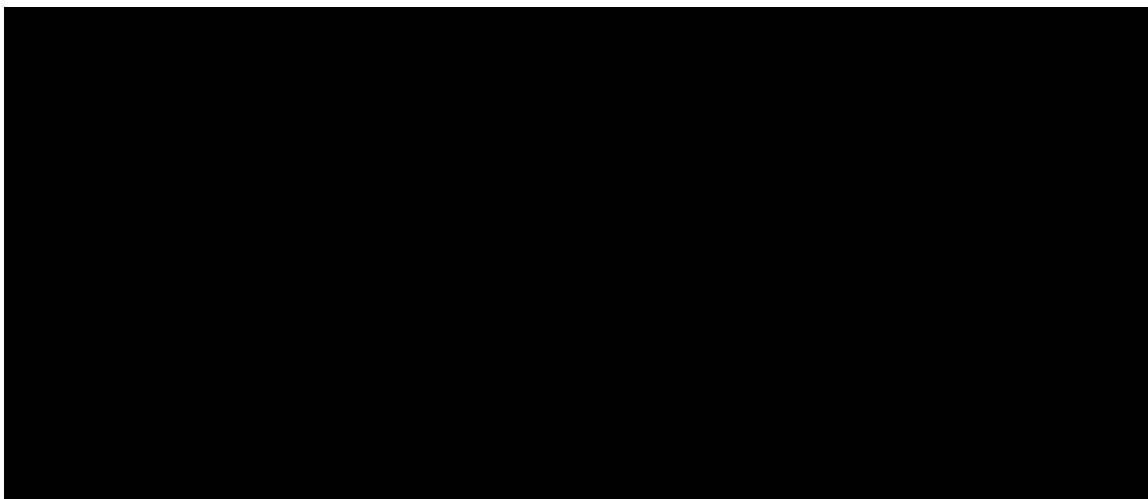
Specifically, as Dr. Mortazavi will explain, the PCB of the Oura Ring Gen. 1 features [REDACTED]
[REDACTED] around the ring’s circumference: the [REDACTED]
[REDACTED]
[REDACTED]. JDX-0001C ¶¶3380-3410 (citing RX-0030C; RX-0031C; RX-0032; RX-0276C-RX-0286C; JX-0026C at 53:17-54:11, 129:15-131:19, 140:1-144:22, 146:19-147:5, 147:12-149:16, 153:19-25, 158:7-160:1, 163:14-17, 165:14-169:18; RPX-0001; RPX-0005).



RX-0030C



RX-0032



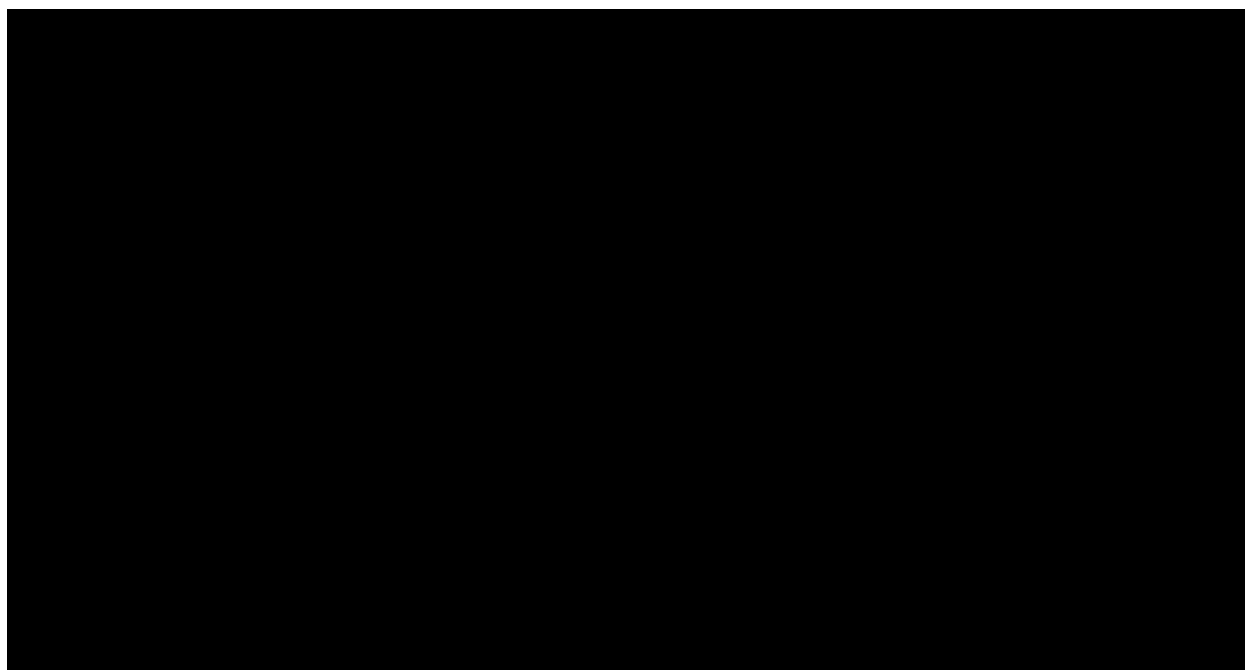
RX-0031C

Mr. Christophy is also expected to testify that the Oura Ring Gen. 1 PCB is [REDACTED]
[REDACTED]
[REDACTED]. Because [REDACTED]
[REDACTED] as required by claim elements
[7.d]-[7.g].

c) Claim Element [7.h]

Claim element [7.h] is disclosed by the Oura Ring Gen. 1.

Specifically, the Oura Ring Gen. 1 PCB includes a [REDACTED]
[REDACTED]
[REDACTED]. RX-0031C; JDX-0001C ¶¶3411-3420 (citing RX-0030C; RX-0031C; JX-
0026 at 147:12-149:16, 163:14-17; RPX-0001; RPX-0005; RX-0276C-RX-0286C).



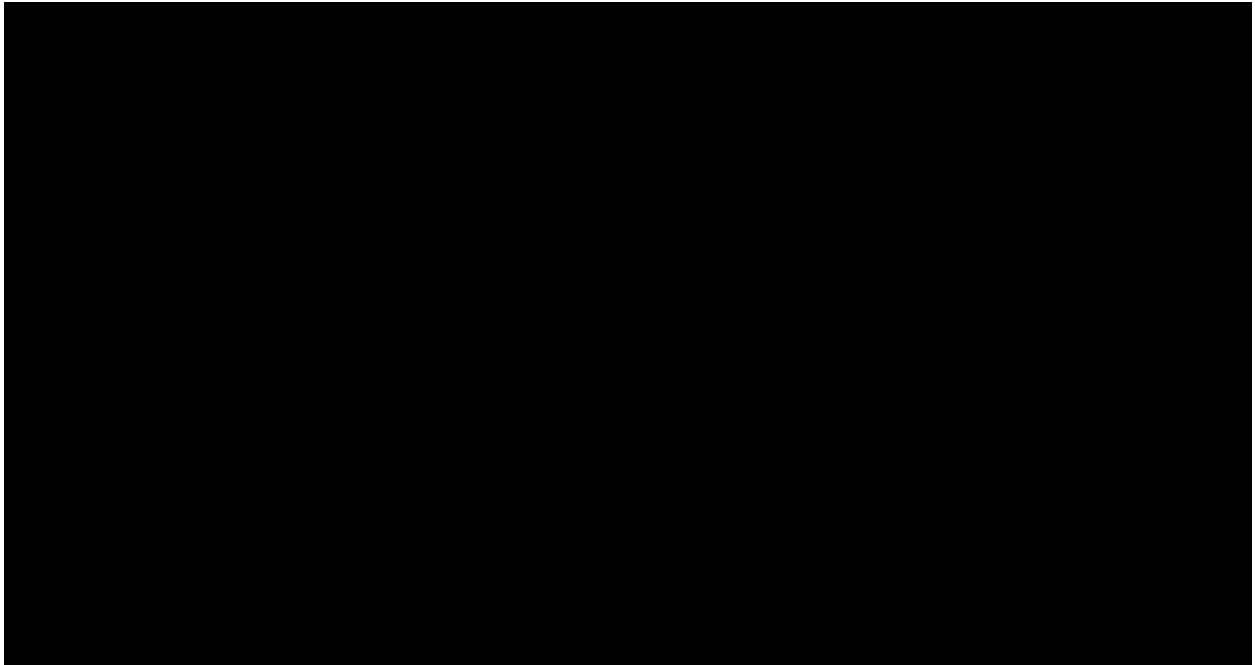
RX-0031C (annotated)

Mr. Christophy is also expected to testify that the [REDACTED]
[REDACTED]. A POSITA would understand [REDACTED]
[REDACTED] forming the claimed “first insulating layer.” JDX-
0001C ¶¶3411-3420.

d) Claim Element [7.i]

Claim element [7.i] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm, [REDACTED]
[REDACTED] as the first insulating
layer for [7.h]. JDX-0001C ¶¶3421-3428 (citing RX-0030C; RX-0031C; JX-0026C at 142:18-25,
147:12-149:16, 163:14-17); RPX-0001; RPX-0005; RX-0276C-RX-0286C). [REDACTED]
[REDACTED]. *Id.*

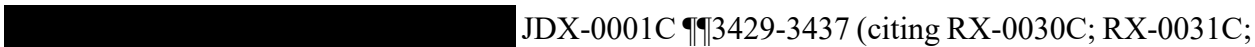
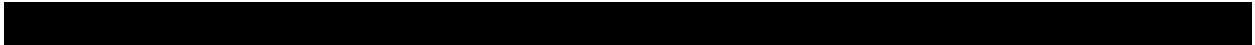


RX-0031C (annotated)

e) Claim Element [7.j]

Claim element [7.j] is met by the Oura Ring Gen. 1.

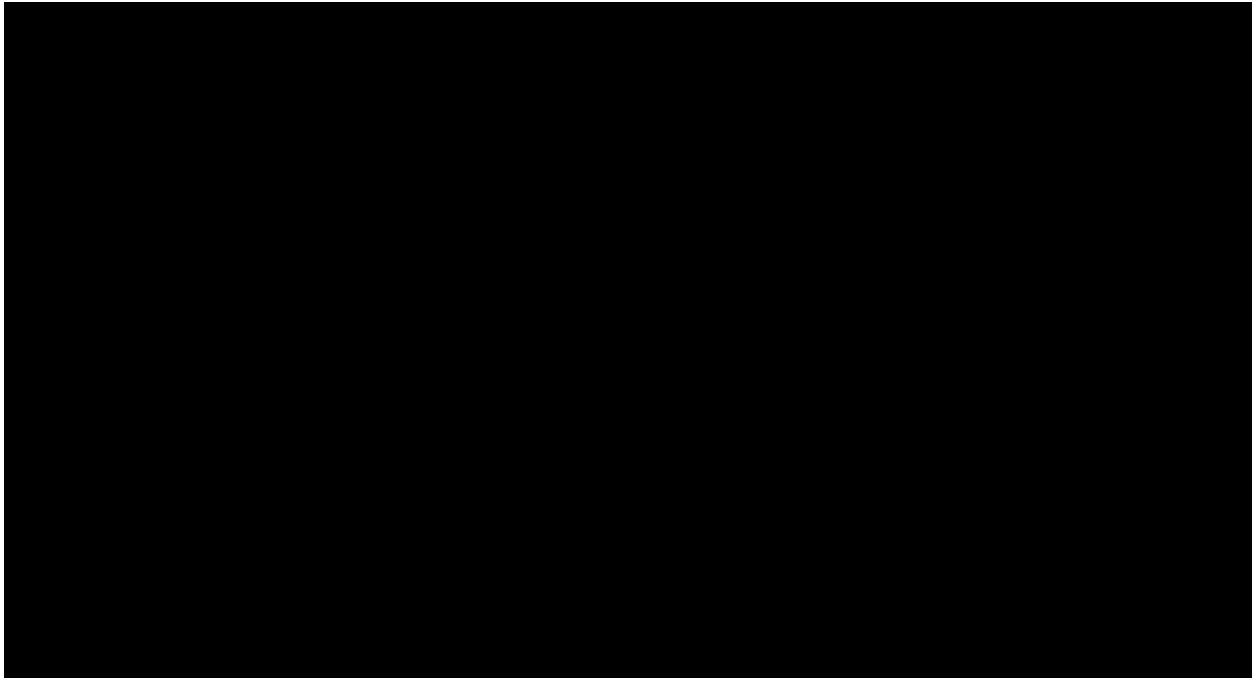
Specifically, as Mr. Christophy is expected to confirm,



JDX-0001C ¶¶3429-3437 (citing RX-0030C; RX-0031C; JX-0026C at 141:5-6, 142:18-25, 144:2-22, 147:12-149:16; RPX-0001; RPX-0005; RX-0276C-RX-0286C).



mapped as the first conduction layer.



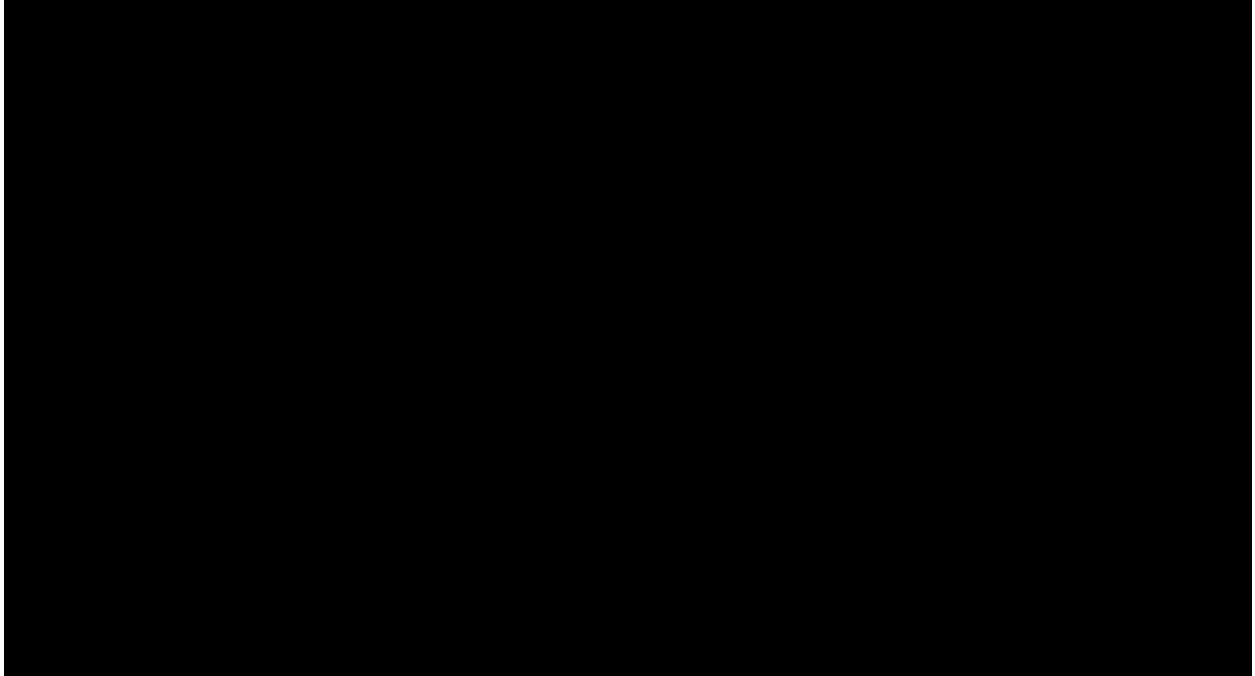
RX-0031C (annotated).

f) Claim Element [7.k]

Claim element [7.k] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm,

[REDACTED]. JDX-0001C ¶¶3438-3446 (citing RX-0030C; RX-0031C; JX-0026C at 102:5-6, 102:23-56, 142:22-25, 146:16-17, 147:12-149:16, 163:14-17; RPX-0001; RPX-0005; RX-0276C-RX-0286C).

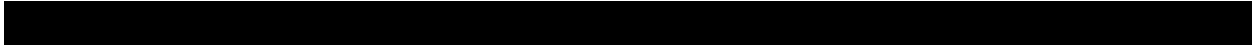


RX-0031C (annotated).

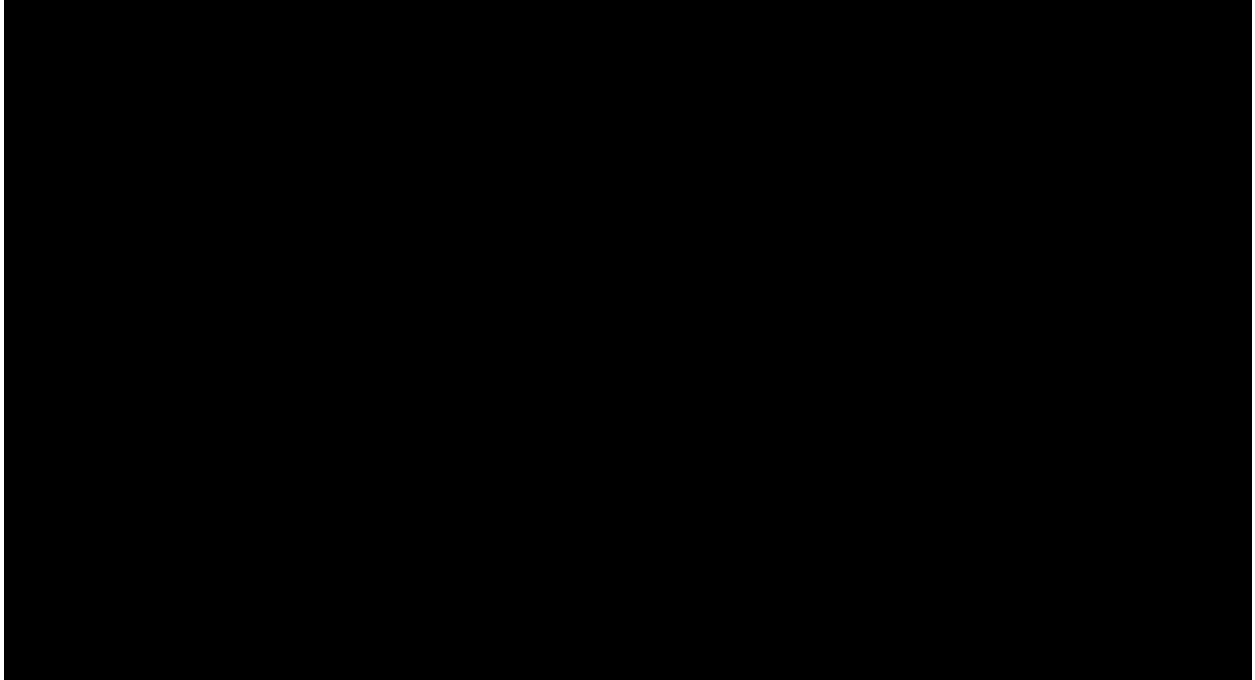
g) Claim Element [7.1]

Claim element [7.1] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm,



. JDX-0001C ¶¶3447-3455 (citing RX-0030C; RX-0031C; JX-0026C at 147:12-149:16, 163:14-17).



RX-0031C (annotated).

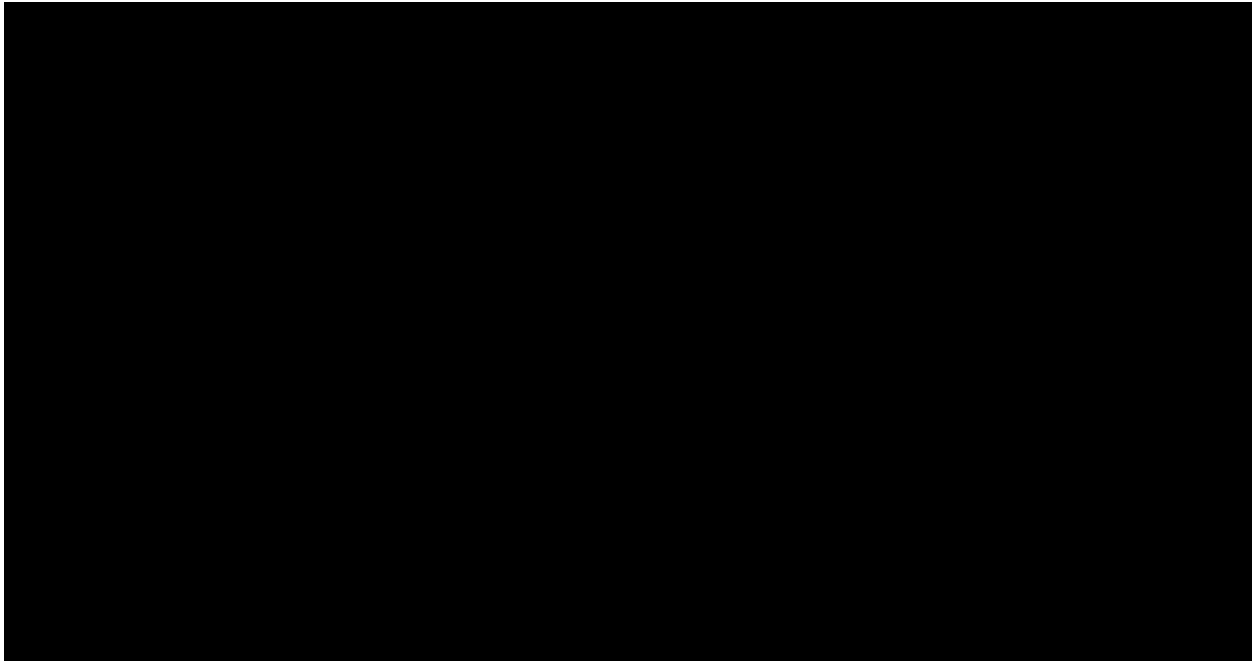
h) Claim Element [7.m]

Claim element [7.m] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm,



JDX-0001C ¶¶3456-3461 (citing RX-0030C; RX-0031C; JX-0026C at 147:12-149:16, 163:11-17).



RX-0031C (annotated).

The [REDACTED]

[REDACTED] and Dr. Mortazavi will testify this disclosure is sufficient to demonstrate a different material under the meaning of the '789 Patent. RX-0031C. The '789 Patent fails to recite any particularities regarding what comprises a “different” material and states that “solder resist” and polyimide are different materials. JX-0008 at 7:5-8, 7:40-54, 8:4-9, cl. 12; JX-0008 at cls. 4, 19.

i) Claim Element [7.n]

Claim element [7.n] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm, [REDACTED]

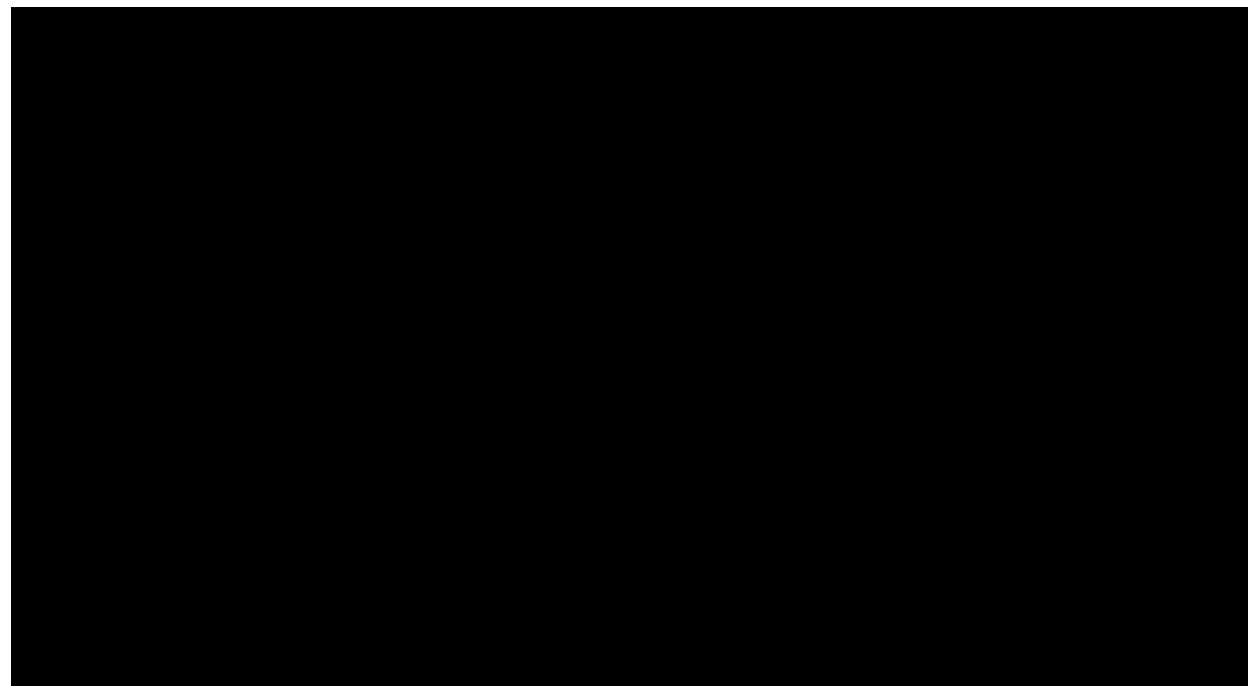


[REDACTED]. JDX-0001C

¶¶3462-3470 (citing RX-0030C; RX-0031C; JX-0026C at 116:22-25, 147:12-149:16 153:19-25, 159:14-160:1. 163:11-17, 168:6-169:18; RPX-0001; RPX-0005; RX-0276C-RX-0286C). Mr.

Christophy is expected to confirm that [REDACTED]

[REDACTED] in the Oura Ring Gen. 1.



RX-0031C (annotated).

j) Claim Element [7.o]

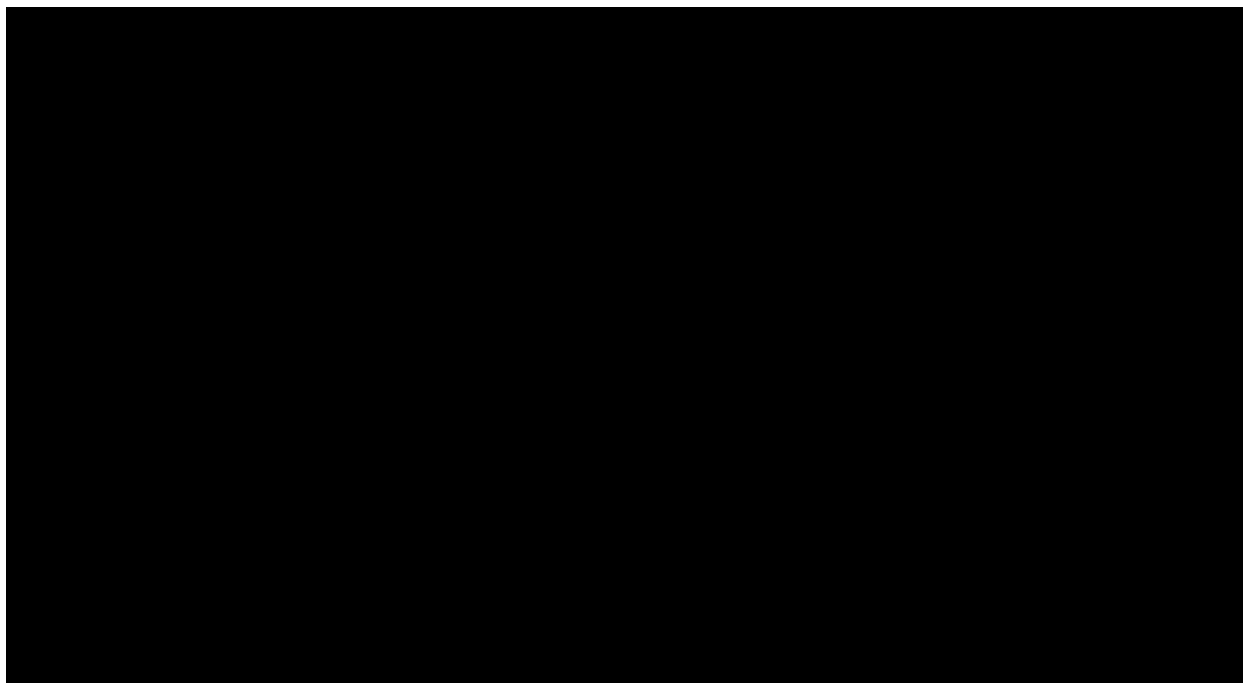
Claim element [7.o] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm, [REDACTED]



[REDACTED]. JDX-0001C

¶¶3472-3478 (citing RX-0030C; RX-0031C; JX-0026C at 107:19-23, 141:5-6, 141:18-142:10, 147:12-149:16, 150:20, 163:11-17, 168:6-169:18; RPX-0001; RPX-0005; RX-0276C-RX-0286C).



RX-0031C (annotated).

k) Claim Element [7.p]

Claim element [7.p] is met by the Oura Ring Gen. 1.

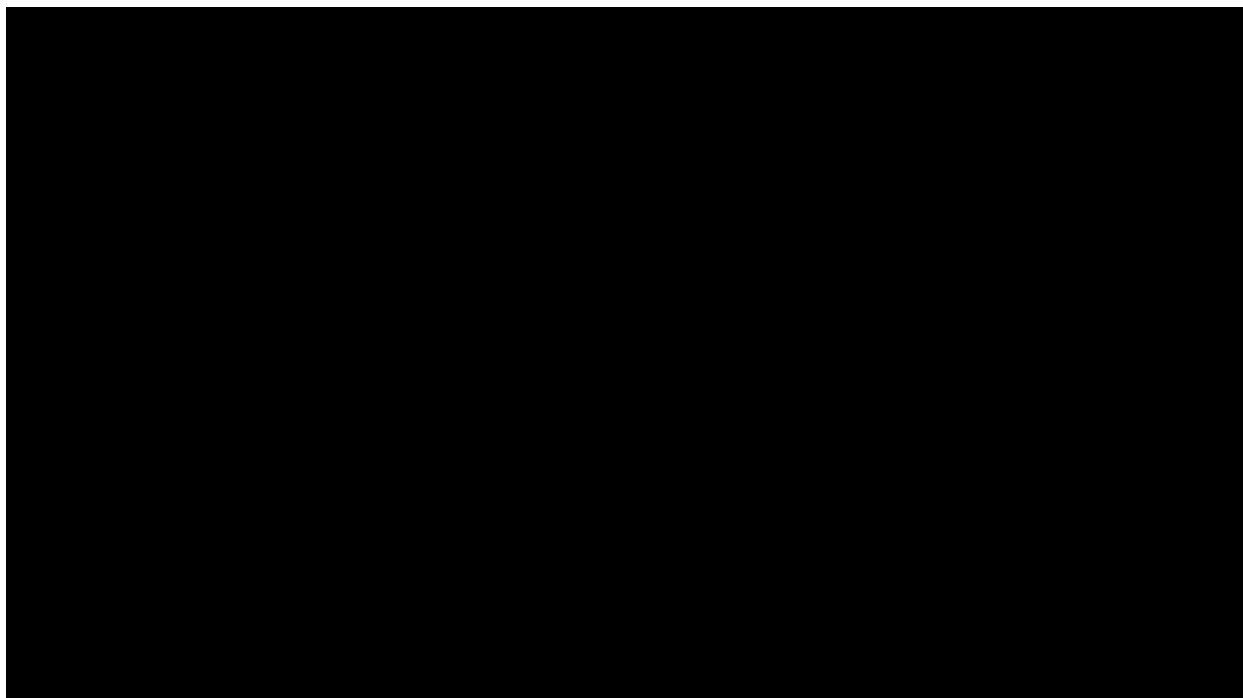
Specifically, as Mr. Christophy is expected to confirm, [REDACTED]



[REDACTED]. JDX-0001C

¶¶3479-3488 (citing RX-0030C; RX-0031C; JX-0026C at 102:5-6, 116:22-25, 146:16-17, 147:12-149:16, 153:19-25, 159:14-160:1, 163:11-17, 168:6-169:18; RPX-0001; RPX-0005; RX-0276C-RX-0286C). Mr. Chrisophy is also expected to testify that [REDACTED]





RX-0031C (annotated).

l) Claim Element [7.q]

Claim element [7.q] is met by the Oura Ring Gen. 1.

Specifically, as Mr. Christophy is expected to confirm,

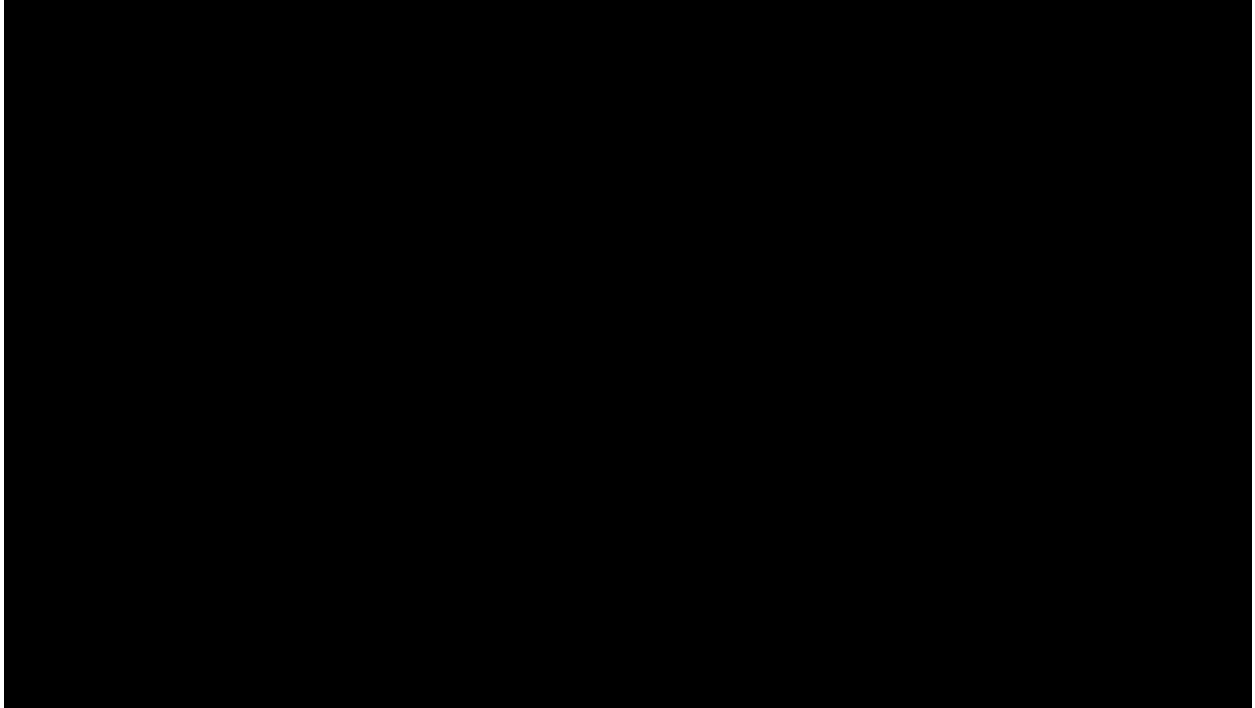


. JDX-0001C ¶¶3489-3496

(citing RX-0030C; RX-0031C; JDX-0025C at 147:12-149:16, 148:6-9, 163:11-17; RPX-0001;

RPX-0005; RX-0276C-RX-0286C). Mr. Christophy is also expected to testify that





RX-0031C (annotated).

3. Ground 2: Obviousness in view of Williams

The teachings of Williams either explicitly disclose or render obvious each and every claim element of claim 7 and 9 of the '789 Patent, and accordingly claims 7 and 9 of the '789 Patent are rendered obvious by Williams in view of the skill and general knowledge of a POSITA. JDX-0001C ¶¶3895-3900.

a) *Graham* Factors 1-3

Williams is directed to 3D bendable printed circuit boards and discloses rigid-flex PCB architectures for portable communication devices (*e.g.*, including “a variety of wearable electronics and medical devices,” including “wrist bands”), including rigid PCB portions with insulating exterior layers and metal conduction layers, flexible PCB portions with different, thinner insulating layers and metal conduction layers, electrical interconnection through conductive vias and continuous metal layers, and multi-layer flexible PCB constructions. JDX-0001C, ¶¶3507-3509 (citing RX-0076). Williams falls within the same field of endeavor as the '789 Patent—the

structure of rigid-flex printed circuit boards for portable communication devices and/or antenna and communication design in portable communication devices—and is reasonably pertinent to the particular problem the ’789 Patent addresses. *Id.* ¶¶3164-3165, 3170.

As explained in detail in the element-by-element analysis below, Williams discloses or renders obvious each and every limitation of claims 7 and 9 of the ’789 Patent, such that there is no material difference between the teachings of Williams and the claimed invention. JDX-0001C ¶¶3506-3521. However, to the extent that Williams does not provide its teachings within the context of a single embodiment (and specifically, does not provide a specific teaching that the embodiment of Figure 69E and its associated disclosures has specific layer properties or electrical connections), the modifications discussed herein would have been obvious to a POSITA.

Specifically, Williams itself teaches that “[t]he electrical interconnection of the various metal layers within a given rigid PCB, between rigid PCBs, and within flex PCB’s can be accomplished without the need for wires, connectors or solder joints, using conductive vias.” RX-00076 at 72:31-41. Given a POSITA’s motivation to connect conductive layers of a PCB to allow functioning of the overall PCB (indeed, this is what conductive layers are for), a POSITA would be motivated to modify the embodiment depicted in Figure 69E to include electrical interconnection elements to connect the claimed conductive layers relevant to elements [7.n] and [7.p].

Furthermore, the specific layer arrangements and thicknesses recited in claims 7 and 9 (and discussed in greater detail in the context of certain claim elements below) represent nothing more than the predictable application of known rigid-flex PCB construction techniques to the architecture disclosed in Williams. JDX-0001C, ¶¶3895-3900. A POSITA would have had a reasonable expectation of success in implementing the claimed PCB architecture because rigid-

flex PCBs with the specific layer structures recited in claims 7 and 9—including alternating conduction and insulating layers, exterior insulating layers made of different materials in the rigid and flexible portions, and multi-layer flexible constructions—were well-established in the art at the time of the invention. *Id.* ¶¶3895-3900, 3152-3322, 3326-3330.

Accordingly, obvious modifications to the embodiment of Williams depicted in Figure 69E, including modifications suggested by Williams itself, would teach all elements of claims 7 and 9. The detailed scope and content of Williams’s teachings applicable to claims 7 and 9 are set forth immediately below.

i. Claim Element [7.pre]

Williams meets claim element [7.pre]. JDX-0001C, ¶¶3522-3534 (citing RX-0076).

Specifically, Williams discloses that its rigid-flex PCB invention is applicable to “a variety of wearable electronics and medical devices,” including “wrist bands,” “cuffs, arm bands,” and communication circuitry (*e.g.*, radios, antennas) can be used with its invention and, therefore, teaches a portable device with wireless communication capabilities. JDX-0001C, ¶¶3524-3528; RX-0076 at 1:12-31; 8:39-13:12, 88:1-27, Figs. 33-35, 45.



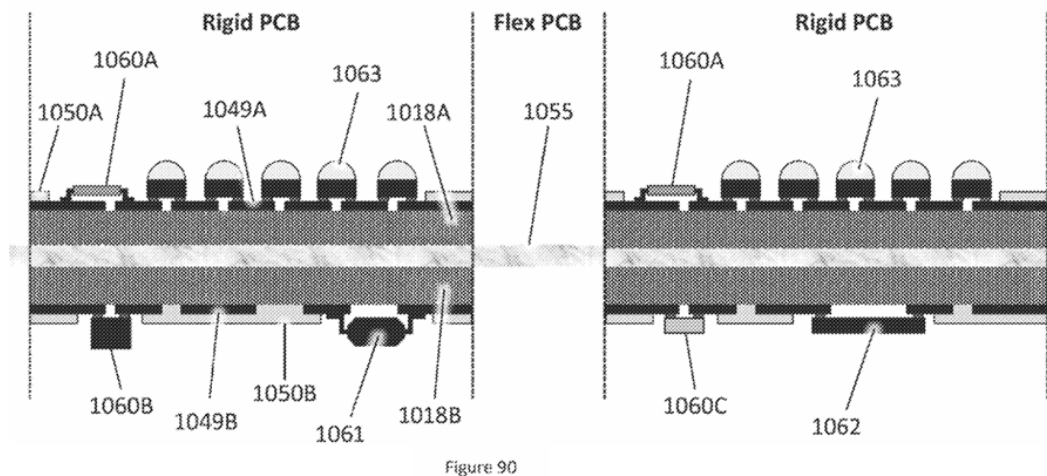
Dr. Mortazavi will further testify that Williams further teaches that its invention (*e.g.*, array of rigid PCBs interconnected with flexible PCBs) overcomes issues such as allowing electronics to conform to arbitrary shape and allowing PCBs to be “squeezed into enclosures otherwise too small to accommodate required PCB surface areas” and allow PCBs to be fit into, for example, “motor casings, watch enclosures, miniaturized surveillance cameras” and “wearable electronics for sports applications.” JDX-0001C, ¶3530; RX-0076 at 8:39-9:62, 1:12-31, 8:66-13:12, 88:1-27, Figs. 33-35, 45.

Thus, Williams discloses its invention in the context of a portable communication device such as a cell phone, camera, tablet computer, notebook, consumer device, or wearable electronics. JDX-0001C, ¶¶3532, 3526.

ii. Claim Elements [7.a]-[7.c]

The evidence will show that Williams meets claim element [7.a]-[7.c]. JDX-0001C, ¶¶3535-3563 (citing RX-0076). Samsung does not dispute that Williams discloses elements [7.a] and [7.b]. CSUCE at 41.

Specifically, Williams discloses multiple electronic components such as integrated circuits, LEDs, sensors or emitters, which meet the “first electronic component” (7.a) and “second electronic component” (7.b) limitations. JDX-0001C, ¶¶3536-3538 (7.a), 3546-3549 (7.b); RX-0076 at 1:12-55, 87:19-31, Fig. 90. Dr. Mortazavi will further testify that the purpose of printed circuit boards is to physically mount electronic components and provide electrical connections with electronic components. JDX-0001C, ¶¶3539, 3550; RX-0076 at 1:12-19, 86:60-67. Figure 90 depicts a PCB electrically connected with multiple electronic components. RX-0076 at 86:60-67, 87:19-31, Fig. 90; JDX-0001C, ¶¶3540-3542, 3551-3553.



RX-0076, Fig. 90

Regarding element [7.c], Williams further discloses its array of multiple rigid-flex PCB islands interconnected by a flex PCB used to mount and electrically connect electronic components, including battery, sensors, radio, and other electronic components—indeed, the entirety of its disclosure is directed to aspects of aspects of PCBs, such as the composition, application, manufacturing, and/or the like of PCBs. JDX-0001C, ¶3557; RX-0076 at 1:6-8, 1:12-20, 1:32-37, 1:11-55, 1:56-6:23 (discussing background of rigid PCBs), 6:24-8:38, 9:1-13:12, 23:06-24, 23:25-61, 23:62-67, 24:1-5, 71:41-56, 72, 11-30, 73:36-55, 74:12-23, 78:25-35, 86:60-67, 87:19-31, Figs. 1, 2, 11, 11b, 13a, 13b, 15-17, 33, 45, 61-90.

The use of PCBs is to mechanically support and provide electrical connection between “electric components” and that Williams does, in fact, disclose a PCB electrically connected with the first electronic component and the second electronic component. JDX-0001C, ¶¶3558-3561; RX-0076 at 1:12-55. For example, Figures 33 and 45, with supporting disclosure in Figure 15, demonstrate how Williams discloses a PCB electrically connected with the first electronic component and the second electronic component (JDX-0001C, ¶3562):

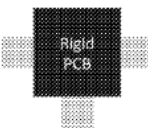
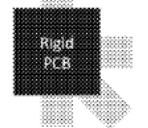
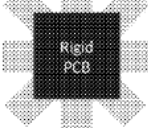
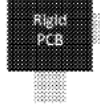
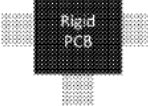
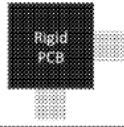
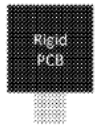
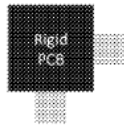
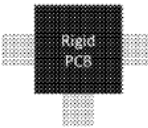
!!!	Circuit Function	Minimal	Good	Superior
Critical	• Primary External Connector, ESD • Power Supply, Regulator, Battery • Logic, DSP, ASP, Clock, A/D • MCU, BIOS, NVM, Firmware • Analog / Digital / USB Control, RF • Safety, OTSD, OCSD, OVP	RF ≥ 2 	RF ≥ 3 	RF ≥ 7 
Important	• Unique Sensors, Drivers, LEDs • Block level control and processing • RW Data / Scratch Pad Memory • Secondary External connectors • Antenna for RF Links	RF ≥ 1 	RF ≥ 2 	RF ≥ 4 
Basic	• 1-of-n Sensor Array Element • 1-of-n Driver Array Element • 1-of-n LED Array Element • POL Regulator, Local Function • Storage Cap, Interconnect Links	RF ≥ 0 	RF ≥ 1 	RF ≥ 3 
Ancillary	• Supplemental Sensors • Monitors, Use Tracking Functions • Indicators • Convenience Features • Tertiary Connectors	RF ≥ 0 	RF ≥ 1 	RF ≥ 2 

Figure 33

RX-0076, Fig. 33

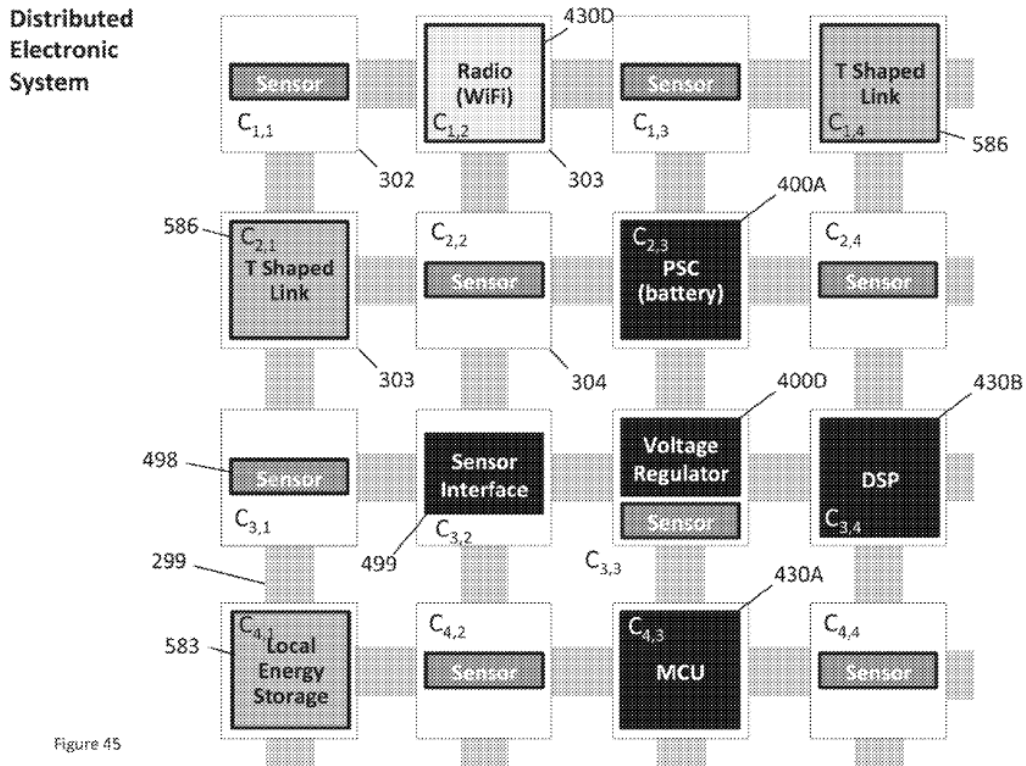


Figure 45

RX-0076, Fig. 45

iii. Claim Elements [7.d]-[7.g]

Williams meets claim elements Claim elements [7.d]-[7.g]. JDX-0001C, ¶¶3564-3587 (citing RX-0076).

Specifically, Williams discloses rigid-flex PCBs with multiple rigid and flexible portions in an alternating sequence (*i.e.*, PCBs having at least two rigid portions and two flexible portions as claimed), including in Figure 15 (rigid-flex PCB array with multiple rigid and flexible portions), examples of multiple rigid PCBs arranged in arrays and connected by flex connectors (*i.e.*, flexible PCBs), advantages of a rigid-flex PCBs (such as better reliability through redundancy and better fit), and discloses that this arrangement was well-known in the art. JDX-0001C, ¶¶3566-3568 (7.d, first rigid portion), 3571-3573 (7.e, first flexible portion), 3577-3578 (7.f, second rigid portion),

and 3582-3584 (7.g, second flexible portion); RX-0076 at Abstract, 1:6-8, 1:12-20, 1:32-37, 1:47-51, Figs. 2, 11A, 11B, 13A, 13B, 15.

Williams also discloses that the first flexible portion is in contact with the first rigid portion, the second rigid portion is in contact with the first flexible portion, and the second flexible portion is in contact with the second rigid portion. *Id.*

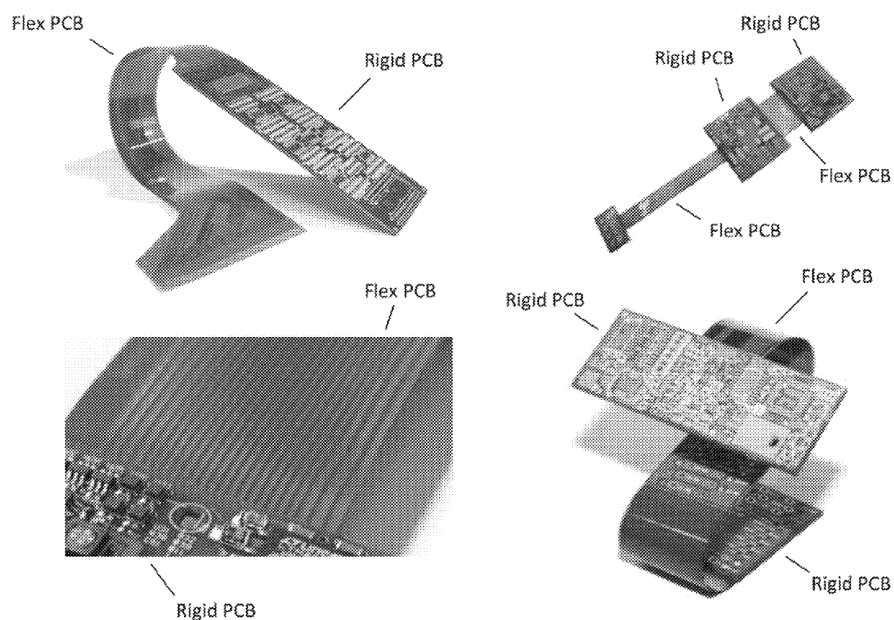


Figure 11A

RX-0076, Fig. 11A (demonstrating PCBs in the right-hand side images with multiple rigid portions (labeled as “Rigid PCB”) and flexible portions (labeled as “Flex PCB”) in contact with one another as claimed)

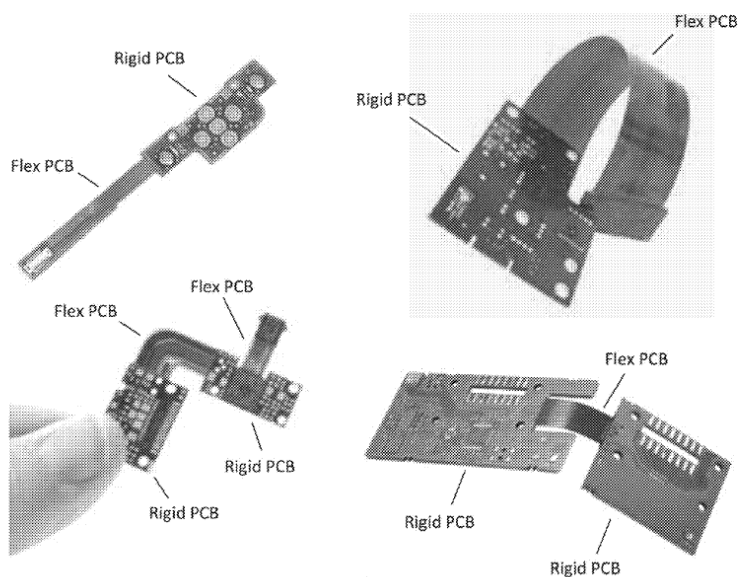


Figure 11B

RX-0076, Fig. 11B (demonstrating PCBs in the bottom two images with multiple rigid portions (labeled as “Rigid PCB”) and flexible portions (labeled as “Flex PCB”) in contact with another as claimed)

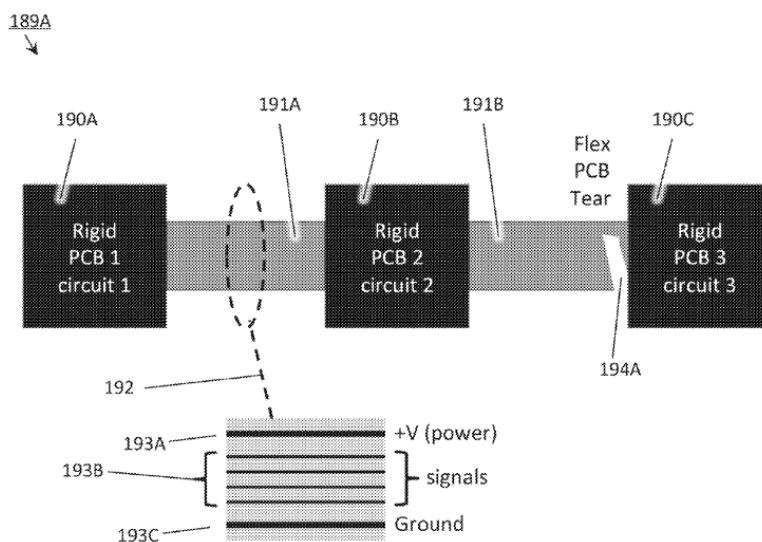
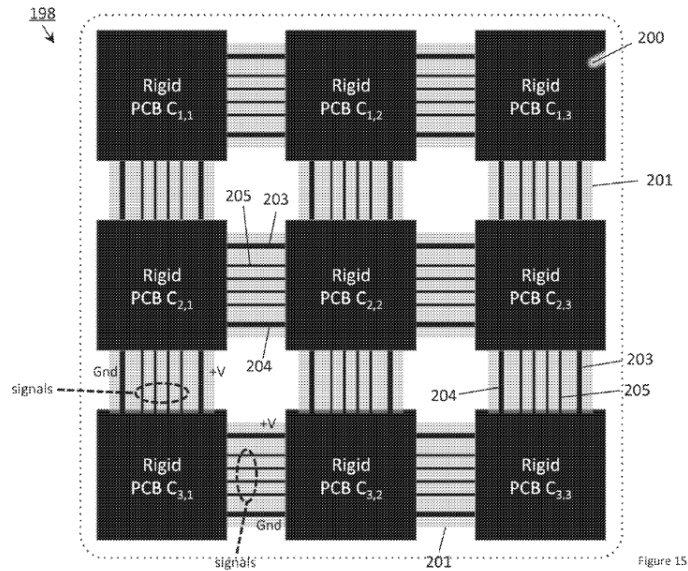


Figure 13A

RX-0076, Fig. 13A (showing rigid-flex PCB with multiple rigid portions 190A, 190B, 190C and multiple flexible portions 191A and 191B)



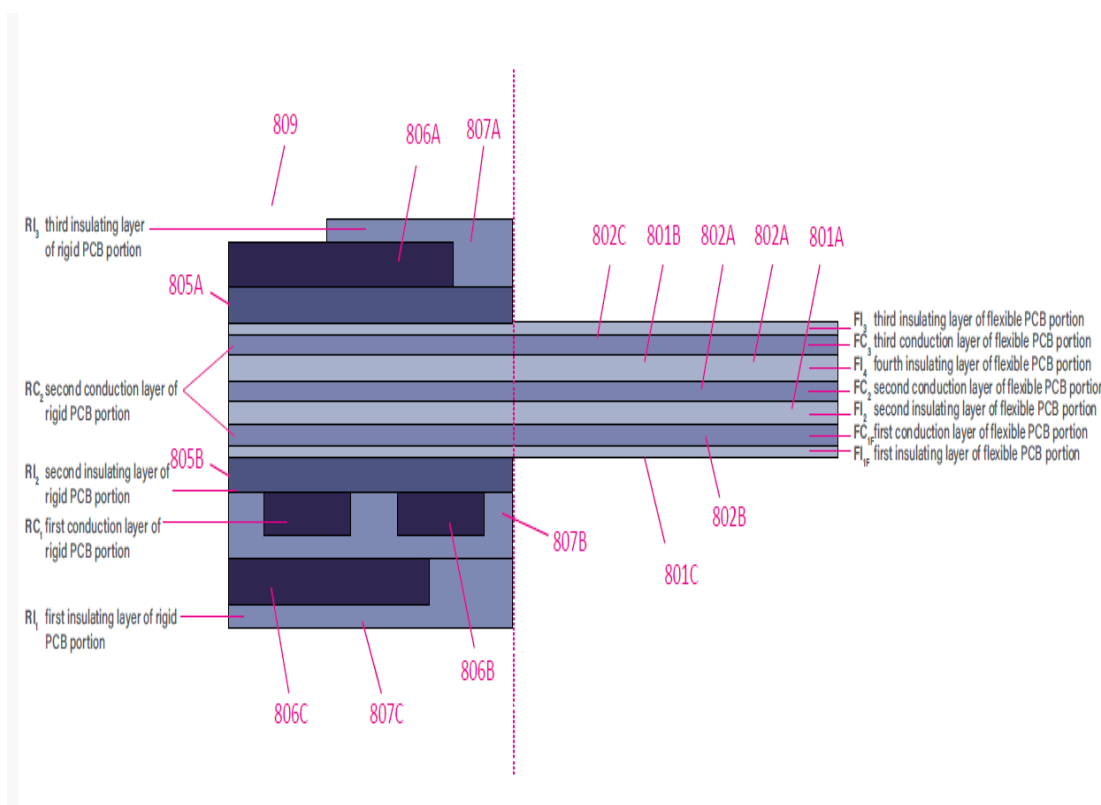
RX-0076, Fig. 15 (demonstrating a PCB with multiple rigid and flexible portions in contact with another as claimed)

iv. Claim Element [7.h]

The evidence will show that Williams meets claim element [7.h]. JDX-0001C, ¶¶3588-3598 (citing RX-0076).

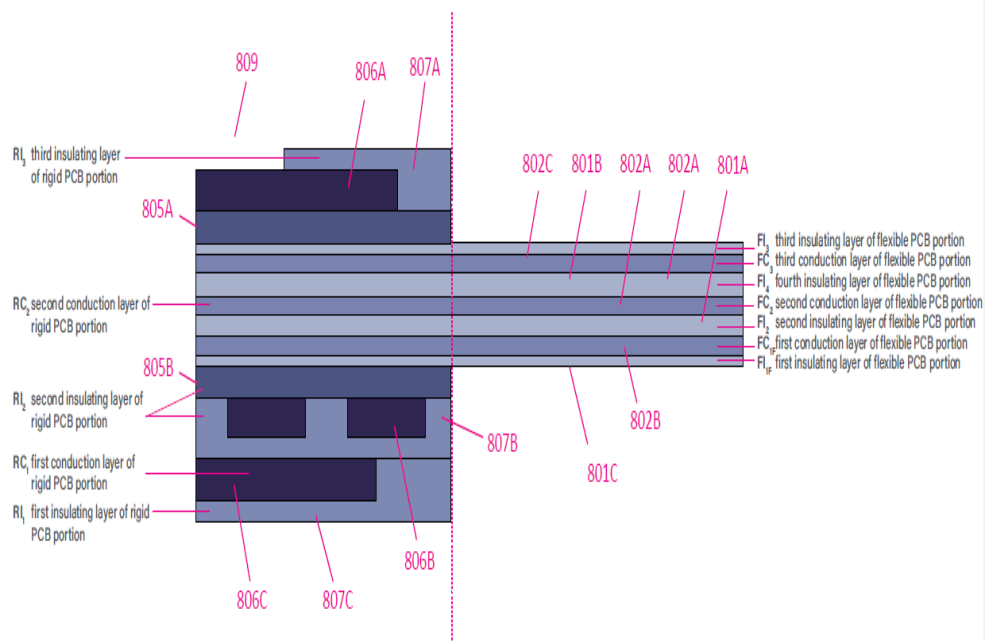
Specifically, Williams discloses rigid-flex PCBs specifically with rigid portions with a first insulating layer (e.g., 807C), such as solder mask, forming an exterior surface (as shown in annotated Fig. 69E mappings below). JDX-0001C, ¶¶3591-3595; RX-0076 at Fig. 69E, 71:41-56, 74:12-23, 75:52-59, 84:48-60, 86:4-17. Williams discloses that the rigid portions can be comprised of “any PCB material more rigid and ‘less flexible’ than the flex portions.” JDX-0001C, ¶¶3595, 3597; RX-0076 at 71:41-56. Indeed, the first insulating surface must be made of a first material, such as solder mask, as it needs to be made of something. JDX-0001C, ¶¶3596.

Dr. Mortazavi will explain these key teachings (a) of Williams using an annotated version of Figure 69E (JDX-0001C, ¶¶3588-3598):

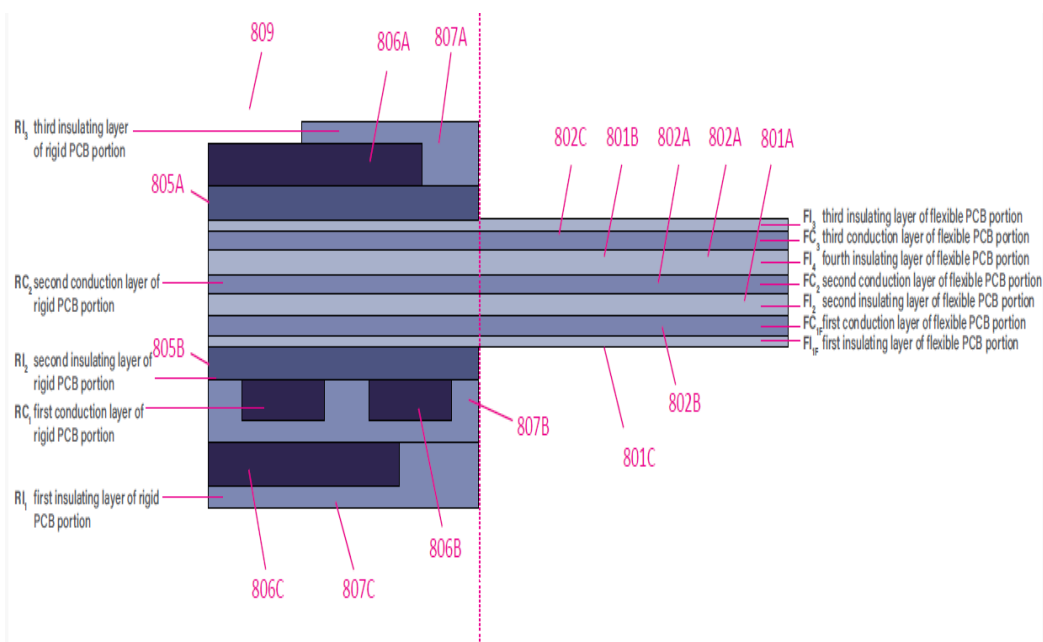


RX-0076, Fig. 69E (annotated)

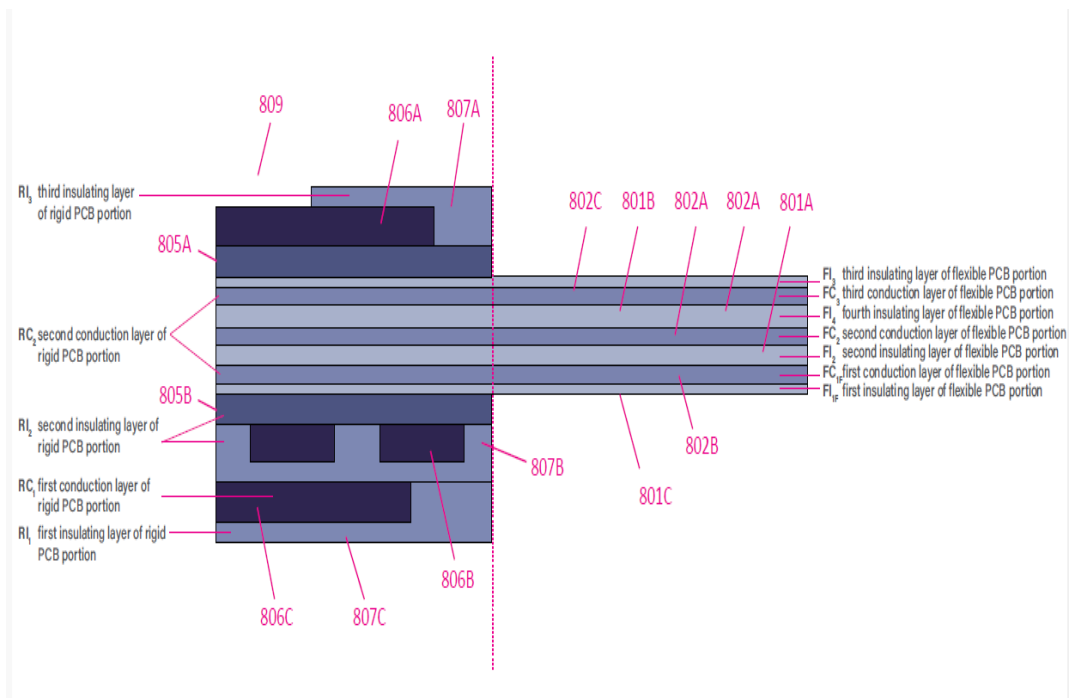
As necessary, Dr. Mortazavi may explain that there are alternative mappings onto the layers depicted in Figure 69E also match the claim requirements under the same logic (JDX-0001C, ¶¶3588-3598):



RX-0076, Fig. 69E (annotated)



RX-0076, Fig. 69E (annotated)



RX-0076, Fig. 69E (annotated)

Williams discloses that PCBs “comprise one or more layers of conductors, typically copper, separated by insulating layers such as glass, epoxy, or polyimide on which electronic components are physically mounted, providing mechanical support for electronic circuitry.” JDX-0001C, ¶3272; RX-0076 at 1:12-16, 84:48-60 (describing application of solder mask on a portion of a rigid PCB portion). Dr. Mortazavi will further testify that a POSITA would have been motivated to use one or more insulating layers as an exterior surface, because this was a basic purpose of insulating PCBs: for example, to protect the underlying conductive traces from environmental damage, to prevent unintended electrical contact of the conductive traces, provide structural support, resist thermal stress, control electromagnetic environments, prevent short circuits during assembly and use, and/or define solder mask openings for component attachment. JDX-0001C, ¶¶3272, 3275; RX-0076 at 2:30-48, 4:19-21, 6:51-55, 7:24-26, 8:17-32.

v. *Claim Element [7.i]*

The evidence will show that Williams meets claim element [7.i]. JDX-0001C, ¶¶3599-3606 (citing RX-0076).

Williams discloses rigid-flex PCBs wherein each rigid portion can include a conduction layer (*e.g.*, 806B, 806C), such as metal layers or conductive traces, located on or above the first insulating layer 807C, which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3602-3605; RX-0076 at Fig. 69E, 75:52-59.

vi. *Claim Element [7.j]*

The evidence will show that Williams meets claim element [7.j]. JDX-0001C, ¶¶3607-3618 (citing RX-0076).

Williams discloses rigid portions with a second insulating layer (*e.g.*, insulating layer 805B or 807B) located on or above the first conduction layer, which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3611, 3616; RX-0076 at Fig. 69E, 72:1-11, 72:24-26.

William also teaches that the second insulating layer is of a particular thickness. *See, e.g.*, JDX-0001C, ¶¶3612, 3617, 3281-3283; RX-0076 at 6:38-50, 12:54-13:3, 29:53-30:30, 71:41-56, 86:18-52, Figs. 62, 66A-66B, 69A-69E, 87A-88, 89A-89E.

vii. *Claim Element [7.k]*

The evidence will show that Williams meets claim element [7.k]. JDX-0001C, ¶¶3619-3627 (citing RX-0076).

Specifically, Williams discloses that each rigid portions can include a second conduction layer (802A, 802B, or 802C) (such as metal layers or conductive traces) located on or above the second insulating layer 805B or 807B, which Dr. Mortazavi will show using the annotated versions

of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3622-3624; RX-0076 at Fig. 69E, 71:58-72:10.

viii. Claim Element [7.l]

The evidence will show that Williams meets claim element [7.l]. JDX-0001C, ¶¶3628-3640 (citing RX-0076).

Williams discloses rigid portions with a third insulating layer (*e.g.*, 807A) located on or above the second conduction layer and forming a second exterior surface, which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3632-3639; RX-0076 at Fig. 69E, 71:41-56, 84:48-60. Dr. Mortazavi will further testify that Williams also discloses solder mask forming exterior surfaces. JDX-0001C, ¶¶3633, 3639; RX-0076 at 84:48-60, 1:12-31, 2:9-28, 22:4-6, 71:41-56, 72:11-30, 82:31-83:3, 86:4-17, 87:19-31.

To the extent not expressly disclosed, a POSITA would have found it obvious to use solder mask to form exterior surfaces, including on the embodiment depicted by Figure 69E, because using the same material for both exterior insulating layers was well-known and standard practice for reliable PCB construction (*e.g.*, to protect components from environmental damage) and provides consistent mechanical properties and simplified manufacturing. RX-0076 at 84:48-60, 87:19-31, Figs. 69A-69E; JDX-0001C, ¶¶3288-3293.

ix. Claim Element [7.m]

The evidence will show that Williams meets claim element [7.m]. JDX-0001C, ¶¶3641-3656 (citing RX-0076).

Williams discloses that the each flexible portion can include a first insulating layer 801C forming a first exterior surface, using materials such as polyester, silk, polyimide, or flexible plastics (*i.e.*, a first insulating layer including a second material different from the first material in the rigid portion, solder mask, in element [7.h]), which Dr. Mortazavi will show using the

annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3644; RX-0076 at Fig. 69E, 6:42-49, 71:58-72:10. Specifically, Williams discloses a range of materials that can be used for insulating layers of flexible PCBs, including materials different from material used for the exterior insulating layers of rigid PCBs. *Id.* ¶¶3650 (citing RX-0076 at 6:42-49 (“Flex PCBs generally utilize much thinner copper layers and thinner insulating substrates than rigid PCBs. Substrates may involve polyester, silk, polyimide, semi-crystalline thermoplastics (also known as PEEK polymers), or flexible plastics and polymeric materials. Like rigid PCBs, flex PCBs may comprise, single, dual or multi-layer constructions generally with conductive vias.”); *id.* ¶¶3654; RX-0076 at 12:54-13:3, 86:5-17 (describing a manufacturing method relating to the polymeric material or polyimide compound that is different than the solder mask).

Dr. Mortazavi will further testify that Williams also explicitly discloses that its rigid portions use “any PCB material more rigid and ‘less flexible’ than the flex portions of the PCB”—*i.e.*, different material. JDX-0001C, ¶¶3652-3653; RX-0076 at 71:41-56.

To the extent not expressly disclosed, Dr. Mortazavi will testify that a POSITA would have found it obvious to use insulating layers forming a first exterior surface on the flexible portions because it was well-known in the art for flexible PCB portions to use materials (*e.g.*, polyimide) for an exterior surface with different mechanical properties (*e.g.*, superior flexibility, thermal stability, chemical resistance, and dielectric properties) than rigid portions (*e.g.*, solder mask) to enable flexion. JDX-0001C, ¶¶3294-3299; RX-0076 at 6:38-50, 12:54-13:3, 86:5-17.

x. Claim Element [7.n]

The evidence will show that Williams meets claim element [7.n]. JDX-0001C, ¶¶3657-3667 (citing RX-0076).

Williams discloses that the mechanical construction of the flexible portions can include a first conduction layer (metal layer 802B) (such as metal layers or conductive traces) located on or

above the first insulating layer (insulator 801C) and being electrically connected with the first conduction layer (metal layer 806C) of the rigid PCB portions (*e.g.*, based on being continuous, being electrically connected through partial or full vias, and/or the like), which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3661-3666; RX-0076 at Fig. 69E, 71:57-75:59.

To the extent that such electrical connections are not expressly disclosed within the specific context of the embodiment of Figure 69E, Dr. Mortazavi will testify that a POSITA would have found the use of electrical connections between conduction layers obvious because electrical interconnection between conduction layers of flexible and rigid PCB portions was well-known in the art to accommodate different applications such as more layers for more complex systems and to, for example, create electrical pathways through the entire PCB assembly; indeed, such connections are essential for the PCB to function as an integrated unit. JDX-0001C, ¶¶3300-3304, 3156-61; RX-0076 at 6:48-50, 73:45-55, Figs. 66A-66B, 69C. Williams even expressly suggests using such connections: Williams discloses that “[t]he electrical interconnection of the various metal layers within a given rigid PCB, between rigid PCBs, and within flex PCB’s can be accomplished without the need for wires, connectors or solder joints, using conductive vias.” JDX-0001C, ¶3664 (citing RX-0076 at 72:31-41, Figs. 68, 69C); *id.* ¶3665 (citing RX-0076 at 73:45-55) (identifying via elements, such 831 and 832 for connecting metal layers). Interconnects between conductive layers in a PCB is even shown in Figure 66B:

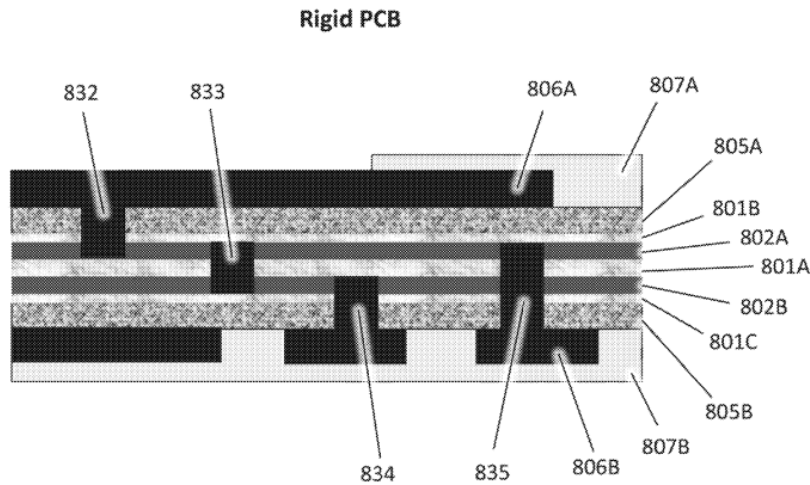


Figure 66B

RX-0076 at Fig. 66B, 73:50-55 (“in the cross section of FIG. 66B, partial via 832 connects” metal layers).

xi. Claim Element [7.o]

The evidence will show that Williams meets claim element [7.o]. JDX-0001C, ¶¶3668-3680 (citing RX-0076).

Williams discloses flexible portions with a second insulating layer (e.g., 801A) located on or above the first conduction layer and having a thickness thinner than the first thickness in the rigid portions, which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3669-3670; RX-0076 at Fig. 69E, 6:38-50, 71:41-56, 78:25-42, Figs. 62, 66A-66B, 69A-69E, 87A-88, 89A-89E. For example, Williams teaches that “[f]lex PCBs generally utilize much thinner copper layers and thinner insulating substrates than rigid PCBs” (JDX-0001C, ¶3675 (citing RX-0076 at 6:42-49)) and that “thinner layers are preferable for superior flexibility.” JDX-0001C, ¶3677 (citing RX-0076 at 78:25-35), ¶3677

(citing RX-0076 at 71:41-56), ¶3678 (citing 12:54-13:3 and 86:5-17) (discussing manufacturing methods).

To the extent not expressly disclosed, Dr. Mortazavi will testify that a POSITA would have found it obvious to use a second insulating layer thinner than the first thickness in the rigid portions because thickness differentiation between rigid and flexible insulating layers was well-known in the art and because a POSITA would be motivated to use a layer and/or material of a thickness that possesses the flexibility, thermal stability, and durability required for portions of the PCB that must bend during device assembly or use. JDX-0001C, ¶3308. The selection of a different thickness of a layer for flexible applications versus rigid applications was dictated by the physical requirements of each portion and selecting appropriate materials for flexible versus rigid PCB portions was well within the ordinary skill of a POSITA, and thus would have been at least obvious to try. *Id.* ¶¶3305-3307, 3309.

xii. Claim Element [7.p]

The evidence will show that Williams also provide teachings that meet claim element [7.p]. JDX-0001C, ¶¶3681-3690 (citing RX-0076).

Williams discloses flexible portions with a second conduction layer (e.g., 802C) (metal layers or conductive traces) located on or above the second insulating layer and electrically connected (e.g., based on being continuous, being electrically connected through partial or full vias, and/or the like) with the rigid PCB portions, which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3683-3684; RX-0076 at Figs. 69C-69E, 71:57-75:59.

To the extent that such electrical connections are not expressly disclosed within the specific context of the embodiment of Figure 69E, Dr. Mortazavi will testify that a POSITA would have found the use of electrical connections between conduction layers obvious because electrical

interconnection between conduction layers of flexible and rigid PCB portions was well-known in the art to accommodate different applications such as more layers for more complex systems and to, for example, create electrical pathways through the entire PCB assembly, and such connections are essential for the PCB to function as an integrated unit. JDX-0001C, ¶¶3310-3314, 3156-61; RX-0076 at 6:48-50, 73:45-55, Figs. 66A-66B, 69C. Williams even expressly suggests using such connections: Williams discloses that “[t]he electrical interconnection of the various metal layers within a given rigid PCB, between rigid PCBs, and within flex PCB’s can be accomplished without the need for wires, connectors or solder joints, using conductive vias.” JDX-0001C, ¶3664 (citing RX-0076 at 72:31-41, Figs. 68, 69C), ¶3665 (citing RX-0076 at 73:45-55) (identifying via elements, such as 831 and 832 for connecting metal layers). Interconnects between conductive layers in a PCB is even shown in Figure 66B:

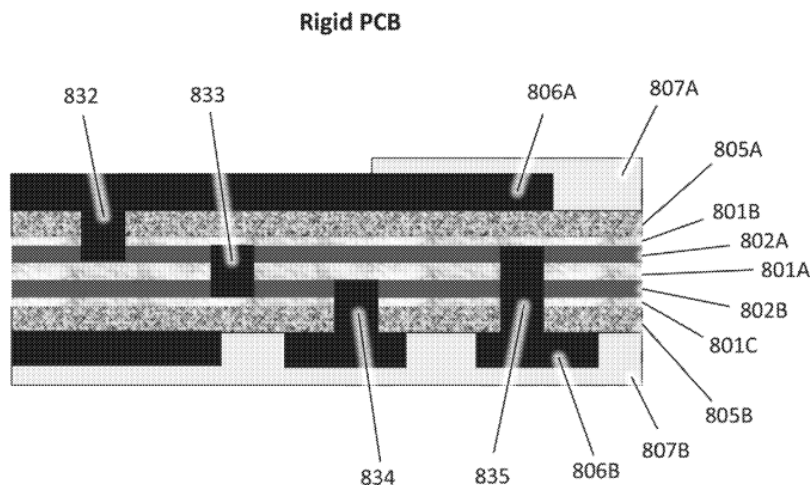


Figure 66B

RX-0076 at Fig. 66B, 73:50-55 (“in the cross section of FIG. 66B, partial via 832 connects” metal layers).

xiii. Claim Element [7.q]

The evidence will show that Williams meets claim element [7.q]. JDX-0001C, ¶¶3691-3703 (citing RX-0076).

Williams discloses flexible portions with a third insulating layer forming a second exterior surface, which Dr. Mortazavi will show using the annotated versions of Figure 69E illustrated for element [7.h]. JDX-0001C, ¶¶3692-3693; RX-0076 at Fig. 69E, 6:38-50, 71:58-72:10. For example, Williams teaches that flexible portions use materials such as “polyester, silk, polyimide, semi-crystalline thermoplastics (also known as PEEK polymers), or flexible plastics and polymeric materials” that are more flexible than rigid portion materials. JDX-0001C, ¶3697 (citing RX-0076 at 6:42-49), ¶3699 (citing RX-0076 71:41-46). Thus, Dr. Mortazavi will further testify that Williams explicitly discloses that its rigid portions use any PCB material more rigid or “less flexible” (solder mask) than the flex portions of the PCB, the polymeric material or polyimide compound (*i.e.*, different material as claimed). JDX-0001C, ¶¶3700-3702.

To the extent not expressly disclosed, a POSITA would have found it obvious to use a third insulating layer forming a second exterior surface on the flexible portion because material differentiation between rigid and flexible insulating layers was well-known in the art and been motivated to use the same material—such as polyimide or coverlay polyimide—for both the first and third insulating layers (forming the first and second exterior surfaces, respectively) of the flexible PCB portions for several well-known reasons such as uniform mechanical properties and simplification of manufacturing, and thus would have been at least obvious to try. JDX-0001C, ¶¶3315-3317.

b) Graham Factor 4

As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '789 Patent. *See also* JDX-

0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

4. Ground 3: Obviousness in View of Williams and Aoyama

Williams, in combination with Aoyama, discloses or renders obvious each element of claims 7 and 9 of the '789 Patent. JDX-0001C, ¶¶4465-4469. To the extent Samsung contends that Williams does not disclose limitations [7.l], [7.m], [7.o], or [7.q], or claim 9, the combination of Williams with Aoyama renders these limitations obvious.

a) *Graham* Factors 1-3

As set forth above for Ground 2, Williams alone discloses or renders obvious all elements of claims 7 and 9. But to the extent Samsung contends that Williams alone does not disclose or render obvious limitations [7.l], [7.m], [7.o], or [7.q], or claim 9, the combination of Williams with Aoyama renders these limitations obvious. Dr. Mortazavi will testify that both references are in the same field of endeavor as the '789 Patent (*e.g.*, both are directed to rigid-flex PCB construction for portable electronic devices). *Id.* ¶4467.

As related to elements [7.l], [7.m], [7.q], and claim 9, Dr. Mortazavi will testify that Aoyama provides express teachings on the selection of materials for flexible PCB portions, including and a POSITA would understand that materials with equal coefficients of thermal expansion avoid differential expansion during thermal cycling. *Id.* ¶4467; *see also infra* Sections VII.C.6.a.v-vii (obviousness in view of Aoyama). Thus, Aoyama provides specific teachings with regard to the specific properties of the layers to be used in practicing Williams' disclosures, and would be obvious to use in combination with Williams.

For example, Aoyama discloses insulation layers and/or solder-resist layers (*e.g.*, insulation layers 107, 108; solder-resist layers 209a, 209b). JDX-0001C, ¶4041 (citing RX-0074

at 7:44-55, Figs. 1, 2, 5, 31-36), 4042, 4048-4051 (citing RX-0074 at 5:41-50, 6:30-39, 7:44-55, 8:1-5, discussing solder resist made of resins, or the insulation made of prepreg). Further, Figure 1 of Aoyama shows solder-resist layer 209b forming a first exterior surface of each of the first rigid PCB portion 11 and second rigid PCB portion 12. JDX-0001C, ¶4043 (citing RX-0074 at 10:55-62; Fig. 1). Aoyama also discloses additional examples of insulating layers, such as insulation layer 105 or 107. For example, in the annotated Figure 32 image below, insulation layer 105 forms the exterior surface of the rigid portions of the PCB:

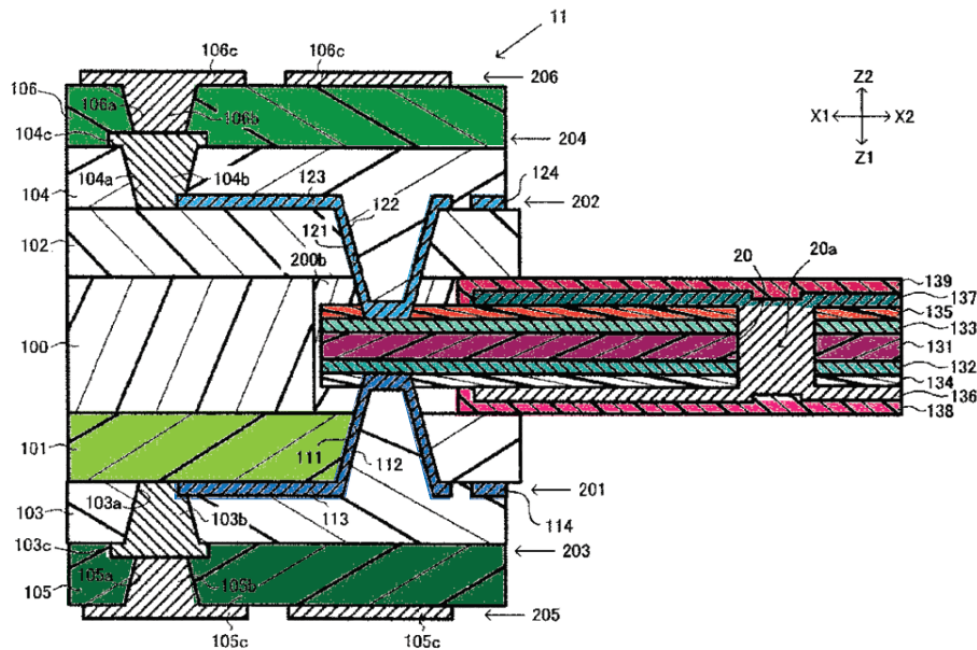


FIG. 32

- | | |
|---|--|
|  | third insulating layer of first rigid portion and second rigid portion |
|  | second conducting layer of first rigid portion and second rigid portion |
|  | second insulating layer of first rigid portion and second rigid portion |
|  | first conducting layer of first rigid portion and second rigid portion |
|  | first insulating layer of first rigid portion and second rigid portion |
| | |
|  | third insulating layer of first flexible portion and second flexible portion |
|  | third conducting layer of first flexible portion and second flexible portion |
|  | fourth insulating layer for first flexible portion and second flexible portion |
|  | second conducting layer of first flexible portion and second flexible portion |
|  | second insulating layer of first flexible portion and second flexible portion |
|  | first conducting layer of first flexible portion and second flexible portion |
|  | first insulating layer of first flexible portion and second flexible portion |

RX-0074 at Fig. 32 (annotated)

See *id.* ¶¶4047, 4059, 4081, 4052-4053.

Moreover, Aoyama discloses that the first and third insulating layers of its flexible portions include a second material different from the first material. *Id.* ¶¶4059, 4081. For example, Aoyama discloses that its coverlays and insulation layers are made of polyimide, for example. *Id.* (citing RX-0074 at 4:16-20, 4:31-36, 5:25-32). This second material differs from the first material, which

is a resin such as a photosensitive resin using acrylic epoxy resin, thermosetting resin mainly containing epoxy resin or UV-setting resin; cured prepreg; or another material such as liquid or film type thermosetting resin or thermoplastic resin or resin coated copper foil. *Id.* (citing RX-0074 at 6:58-63, 7:44-55, 8:1-5). Aoyama's disclosure of the claimed third insulating layer, including the second material, can form a second exterior surface of the first and second flexible portions as shown in the annotated Figure 32E image above. *Id.* ¶4081. And Aoyama also teaches the fourth insulating pursuant to claim 9, which is also shown on the annotated Figure 32E image. *Id.* ¶4092.

Regarding element [7.o], Aoyama provides express teachings on layer thickness considerations, including the use of thinner insulating layers in flexible PCB portions compared to rigid PCB portions, which a POSITA would have understood to be essential for achieving the required flexibility and bend radius without cracking, delamination, or mechanical failure. JDX-0001C, ¶4467; *see also infra* Section VII.C.6.a.vi (obviousness in view of Aoyama). For example, Dr. Mortazavi will further testify that Aoyama discloses a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness, including as shown in the annotated Figure 32 image above. *Id.* ¶4069 (citing RX-0074 at 4:13-36, 8:8-20; Figs. 1, 2, 5, 31-36). Aoyama discloses that the second insulating layer (*e.g.*, flexible substrate 131, insulation layer 134, 135, and/or the like) has a second thickness (*e.g.*, 20-50 μm , preferably 30 μm , or 5-15 μm), which is thinner than the first thickness of 50-100 μm . *Id.* (citing RX-0074 at 4:13-36, 6:58-63, 8:8-20); *see also id.* at Figs. 1, 2, 5, 31-36.

Dr. Mortazavi will further testify that a POSITA would have been motivated to combine Williams with Aoyama for elements [7.l], [7.m], [7.o], [7.q], and [9] to disclose the claimed insulating layers, particularly as different materials (between the first and second materials) used

for the exterior surfaces, and the claimed insulating layers of different thicknesses (*i.e.*, a second insulating layer having a second thickness thinner than the first thickness) because: (1) both references address overlapping technical problems in rigid-flex PCB construction for portable electronic devices; (2) Aoyama teaches material selection for flexible PCB portions using materials with equal coefficients of thermal expansion for improved thermal stability and reduced warping; (3) Aoyama teaches thinner insulating layers in flexible portions to achieve required flexibility and bend radius; (4) both references teach using different materials for rigid versus flexible portions—solder resist for rigid exterior surfaces and polyimide for flexible exterior surfaces; (5) Aoyama discloses using the same materials on both exterior surfaces within each region, simplifying manufacturing and ensuring consistent mechanical and protective properties; and (6) combining these teachings would yield predictable results given the maturity of rigid-flex PCB design. JDX-0001C, ¶¶4467, 3288-3293 (7.l), 3294-3299 (7.m), 3305-3309 (7.o), 3315-3322 (7.q), 3326-3330 (9); *see also* 3149-3173, 3239-3333, 4465-4469. For example, Dr. Mortazavi will testify that a POSITA would have been motivated to make this combination because a thinner interior insulating layer in the flexible PCB portions would assist with flexibility of such portions, and/or a thicker interior insulating layer in the rigid PCB portions would assist with rigidity (and therefore structural integrity) of the rigid PCB portions. *Id.* ¶4469. A POSITA would have understood the well-known aspects of rigid-flex PCB design disclosed in Williams and Aoyama and would have been able to combine their teachings using only routine skill and predictable design choices. *Id.* ¶¶ 3149-3173, 3239-3333, 4465-4469.

b) *Graham* Factor 4

As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '789 Patent. *See also* JDX-0006C ¶¶ 209-25. Even if there were some demonstrated secondary considerations of

nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

5. Ground 4: Obviousness in View of Williams and Naganuma

Williams, in combination with Naganuma, discloses or renders obvious each element of claims 7 and 9 of the '789 Patent. JDX-0001C, ¶¶4470-4474. To the extent Samsung contends that Williams does not disclose limitations [7.l], [7.m], [7.q], or claim 9, the combination of Williams with Naganuma renders these limitations obvious.

a) *Graham* Factors 1-3

As set forth above for Ground 2, Williams alone discloses or renders obvious all elements of claims 7 and 9. But to the extent Samsung contends that Williams alone does not disclose or render obvious limitations [7.l], [7.m], [7.q], or claim 9, the combination of Williams with Naganuma renders these limitations obvious. In general, Naganuma discloses rigid-flex PCB architectures with rigid portions having insulating exterior layers and conduction layers, and flexible portions with different, thinner insulating layers. *Id.* ¶¶4099-4322. Dr. Mortazavi will testify that both references are in the same field of endeavor as the '789 Patent (*e.g.*, both are directed to rigid-flex PCB construction for portable electronic devices). *Id.* ¶4472.

As to elements [7.l], [7.m], [7.q], and claim 9, Naganuma provides express teachings on the selection of materials for flexible PCB portions, including and a POSITA would understand that materials with equal coefficients of thermal expansion avoid differential expansion during thermal cycling. *Id.* ¶4472; *id.* ¶¶4237-4252 (7.l), ¶¶4253-4263 (7.m), ¶¶4292-4302 (7.q), ¶¶4308-4316 (9). Thus, Naganuma provides specific teachings with regard to the specific properties of the layers to be used in practicing Williams' disclosures, and would be obvious to use in combination with Williams.

Dr. Mortazavi will further testify that Naganuma discloses multiple insulation layers of its rigid portions (*e.g.*, solder resist layers, insulation layers, insulative substrate). JDX-0001C, ¶4240. For example, Naganuma discloses solder resist layers (*e.g.*, elements 131, 132, 231, 232) made of resin, which is an insulative material, insulation layers (*e.g.*, elements 101-108, 201-208), and insulative substrate (*e.g.*, elements 100, 200). *Id.* ¶¶4241-4243 (citing RX-0075, ¶¶ 73, 85-100, 111-113, 120, 124, 135, 146-147, 152, 155, 159, 183, 184, Figs. 3A-3B); *see also id.* at Fig. 1, 8-9, 14, 18, 20-24, 29-32. Naganuma discloses solder resists (*e.g.*, solder resist 131, 132, 231, 232) made of resin, an insulative material, forming exterior surfaces of its rigid wiring boards as shown in the mappings in the annotated Figures 2, 3A, and 3B images below. JDX-0001C, ¶¶4245-4250:

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- | | |
|--|---|
| | third insulating layer of first rigid portion and second rigid portion |
| | second conducting layer of first rigid portion and second rigid portion |
| | second insulating layer of first rigid portion and second rigid portion |
| | first conducting layer of first rigid portion and second rigid portion |
| | first insulating layer of first rigid portion and second rigid portion |

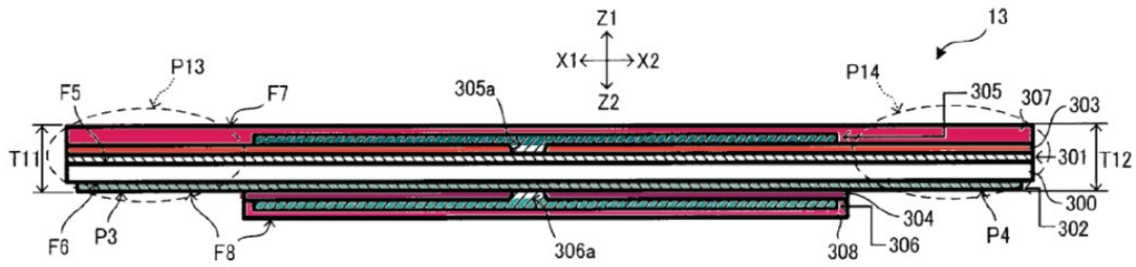


FIG.2

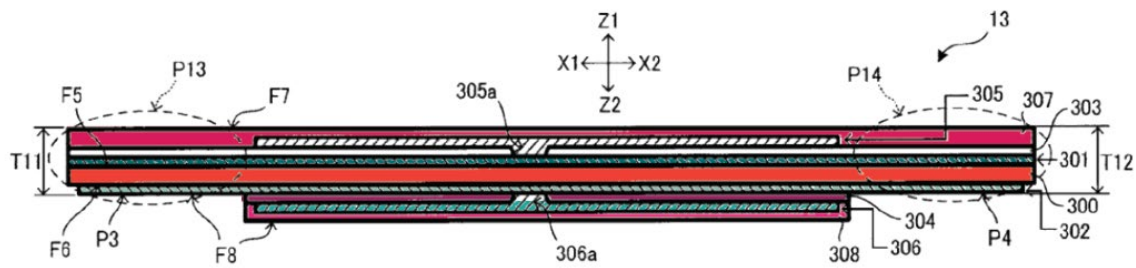


FIG.2

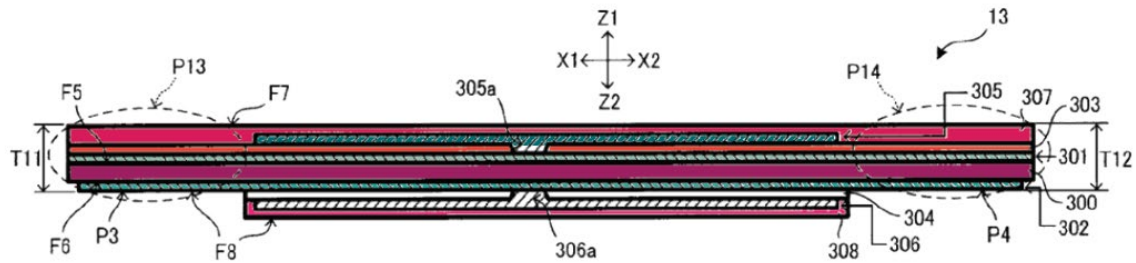


FIG.2

- third insulating layer of first flexible portion and second flexible portion
- third conducting layer of first flexible portion and second flexible portion
- fourth insulating layer for first flexible portion and second flexible portion
- second conducting layer of first flexible portion and second flexible portion
- second insulating layer of first flexible portion and second flexible portion
- first conducting layer of first flexible portion and second flexible portion
- first insulating layer of first flexible portion and second flexible portion

RX-0075 at Fig. 2 (annotated)

Moreover, Naganuma discloses that the first and third insulating layers of its flexible portions include a second material different from the first material. JDX-0001C, ¶4255. For example, Naganuma discloses that its coverlays and insulation layers are made of polyimide. *Id.*, ¶4259 (citing RX-0075, ¶¶79, 82); *id.* ¶¶4255-4257; RX-0075 ¶¶76, 77, 79-83, 129, Fig. 1, 2, 8-9, 14, 18, 20-24, 29-32. This second material differs from the first material, which is a solder resist material, or a resin, glass epoxy resin, or other suitable material such as polyester resin, bismaleimide triazine resin (BT resin), phenol resin, allyl polyphenylene ether resin (A-PPE resin). JDX-0001C, ¶4259 (citing RX-0075, ¶¶90, 94); *id.* ¶4261 (annotated Figure 2 image above).

Further, Naganuma's disclosure of the claimed third insulating layer, including the second material and located on or above the second conduction layer, can form a second exterior surface of the first and second flexible portions as shown in the annotated Figures 2 and/or 3A, 3B image above. *See id.* ¶4300, ¶¶4292-4302. Naganuma discloses multiple examples of conduction layers, such as conductive (301, 302) and shielding layers (305, 306), in its flexible wiring board. *See id.* ¶4287; RX-0075 ¶¶76, 78-83, Figs. 1, 2, 8-9, 14, 18, 20-24, 29-32. Naganuma further discloses that the conductive layers of its flexible portions are electrically connected with the conductive layers of its rigid portions by, for example, vias and pads that allow an electrical and interlayer connection within each of the rigid and flexible portions. *See id.* ¶¶4288-4289 (citing RX-0075 ¶¶78, 81, 98-103, 105-116, 137-139, 142-147, 155, 180, 183-184, Figs. 1-3, 7A-7C; *see also* Fig. 1-35, ¶¶86-93, 120, 123, 129, 148-149, 151, 153-154). And Naganuma also teaches the fourth insulating pursuant to claim 9, which is also shown on the annotated Figure 2 image above. *See id.* ¶4314, 4308-4316.

A POSITA would have been motivated to combine Williams with Naganuma for elements [7.1], [7.m], [7.q], and [9] to disclose the claimed insulating layers, particularly as different

materials (between the first and second materials) used for the exterior surfaces, and the claimed insulating layers of different thicknesses (*i.e.*, a second insulating layer having a second thickness thinner than the first thickness) because: (1) both references address overlapping technical problems in rigid-flex PCB construction for portable electronic devices; (2) Naganuma teaches material selection for flexible PCB portions using materials with equal coefficients of thermal expansion for improved thermal stability and reduced warping; (3) Naganuma teaches thinner insulating layers in flexible portions to achieve required flexibility and bend radius; (4) both references teach using different materials for rigid versus flexible portions—solder resist for rigid exterior surfaces and polyimide for flexible exterior surfaces; (5) Naganuma discloses using the same materials on both exterior surfaces within each region, simplifying manufacturing and ensuring consistent mechanical and protective properties; and (6) combining these teachings would yield predictable results given the maturity of rigid-flex PCB design. *Id.* ¶¶4472, 3288-3293 and 4251 (7.l), 3294-3299 (7.m), 3315-3322 (7.q), 3326-3330 (9); *id.* ¶¶3149-3173, 3239-3333, 4470-4474. For example, Dr. Mortazavi will testify that a POSITA would have been motivated to make this combination because a thinner interior insulating layer in the flexible PCB portions would assist with flexibility of such portions, and/or a thicker interior insulating layer in the rigid PCB portions would assist with rigidity (and therefore structural integrity) of the rigid PCB portions. *Id.* ¶¶4474. A POSITA would have understood the well-known aspects of rigid-flex PCB design disclosed in Williams and Naganuma and would have been able to combine their teachings using only routine skill and predictable design choices. *Id.* ¶¶ 3149-3173, 3239-3333, 4472-4474.

b) *Graham* Factor 4

As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '789 Patent. JDX-0006C ¶¶

209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

6. Ground 5: Obviousness in View of Aoyama

The teachings of Aoyama either explicitly disclose or render obvious each and every claim element of claim 7 and 9 of the '789 Patent, and accordingly claims 7 and 9 of the '789 Patent are rendered obvious by Aoyama in view of the skill and general knowledge of a POSITA. JDX-0001C ¶¶3919-4093.

a) *Graham* Factors 1-3

Aoyama discloses rigid-flex PCB architectures with rigid portions having insulating exterior layers (e.g., solder resist) and conduction layers separated by insulating layers (50-100 μm), and flexible portions having insulating exterior layers of a different material (e.g., polyimide coverlay) with thinner insulating layers (20-50 μm or 5-15 μm). JDX-0001C ¶¶3919-28; RX-0074 at 4:13-36, 5:25-32, 6:30-63, 6:58-63, 7:7-55, 8:1-5, 8:8-20, Figs. 1, 2, 5, 31-36. Aoyama is in the same field of endeavor as the '789 Patent. *Id.*

As explained in detail in the element-by-element analysis below, Aoyama discloses or renders obvious each and every limitation of claims 7 and 9 of the '789 Patent, such that there is no material difference between the teachings of Aoyama and the claimed invention. JDX-0001C ¶¶3919-4093. However, to the extent that Aoyama does not expressly disclose a second flexible portion with the same layer arrangement as the first flexible portion in the context of a particular embodiment, the modifications discussed herein would have been obvious to a POSITA.

For example, a POSITA would have been motivated to include a second flexible PCB portion to extend the PCB assembly and enable electrical connections to additional device regions—a common requirement in portable communication devices, which Aoyama expressly

discloses as an intended application. RX-0074 at 7:64-66. The alternating rigid-flex-rigid-flex pattern was standard feature of PCB design. RX-0073 at Fig. 29; JDX-0001C ¶¶3264-3268.

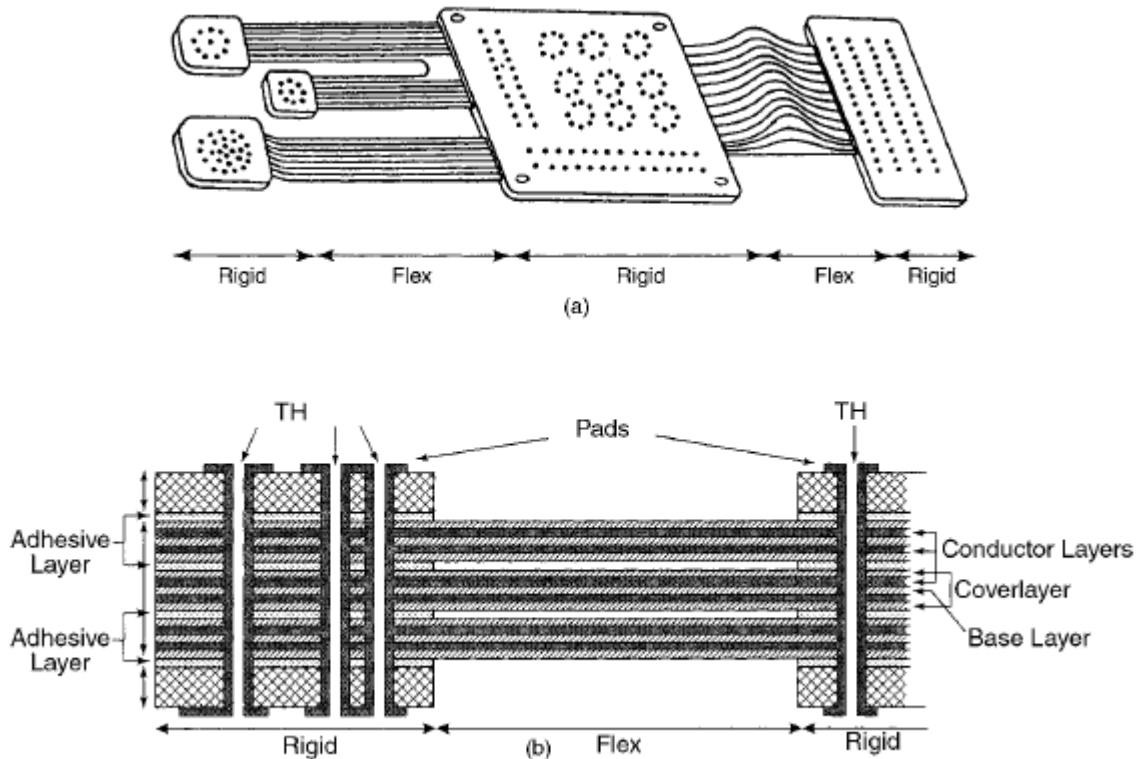


FIGURE 65.1 Basic construction of multilayer circuits (a) Plane view, (b) Cross section.

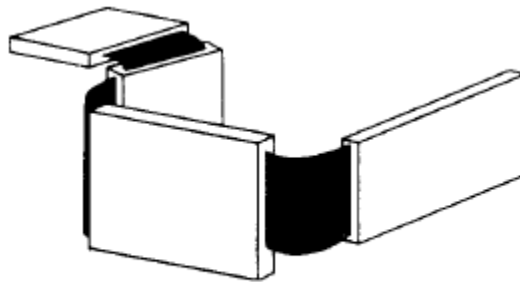


FIGURE 65.3 Folding type rigid/flexible circuit.

RX-0115 at Figs. 65.1, 65.3

Thus, a POSITA would be motivated to, and find it obvious to, include a second flexible PCB portion with the same structure as the first flexible PCB portion to practice the invention taught by Aoyama. *See generally* JDX-0001C ¶¶3149-3173, 3264-3268, 3239-3333, 3919-4098.

Accordingly, obvious modifications to Aoyama, including modifications suggested by Aoyama itself, would teach all elements of claims 7 and 9. The detailed scope and content of Aoyama's teachings applicable to claims 7 and 9 are set forth immediately below.

i. [Uncontested] Claim Element [7.pre]

Claim element [7.pre] is disclosed by Aoyama. JDX-0001C ¶¶3929-3934 (citing RX-0074).

Aoyama teaches that its "flex-rigid wiring board" is used in "electronic devices such as mobile phones" and that flexible wiring boards "are used widely" in "cellular telephones" (*i.e.*, portable communication devices). RX-0074 at 1:7-12, 1:30-40.

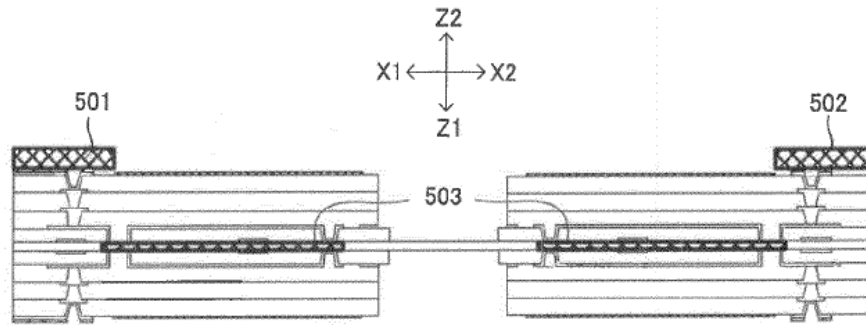
Accordingly, Aoyama discloses or renders obvious claim element [7.pre].

ii. [Uncontested] Claim Elements [7.a]-[7.c]

Claim elements [7.a]-[7.c] are taught by Aoyama. JDX-0001C ¶¶3935-3965 (citing RX-0074).

Specifically, Aoyama discloses external connection terminals "used for electrical connection with other wiring boards or electronic components" and Figure 36 showing electronic components 501 and 502 (exemplary first electronic components) mounted on the surface and component 503 (exemplary second electronic component) built into and electrically connected with flex-rigid wiring board 10. RX-0074 at 3:29-30, 7:62-67, 12:27-29, Fig. 36.

FIG.36



RX-0074 at Fig. 36

iii. *Claim Elements [7.d]-[7.f]*

Claim elements [7.d]-[7.f] are each taught by Aoyama. JDX-0001C ¶¶3966-3993 (citing RX-0074).

Specifically, Aoyama discloses flex-rigid wiring boards with multiple rigid and flexible portions in physical contact. RX-0074 at 4:1-12, 5:33-50, 6:11-16, Figs. 1, 2, 5, 31-36. Specifically, Aoyama discloses “first rigid wiring board 11” and “second rigid wiring board 12” as the rigid portions of flex-rigid wiring board 10, connected by “flexible wiring board 13.” RX-0074 at 4:1-12, Figs. 1, 2. Aoyama discloses that “[b]oth ends of flexible wiring board 13 are connected respectively to first rigid wiring board 11 and second rigid wiring board 12” and that the rigid and flexible portions “are electrically connected without using connectors.” RX-0074 at 5:33-40, 6:11-16, Figs. 1, 5. Figure 5 provides a magnified view of the “connected section of first rigid wiring board 11 and flexible wiring board 13,” confirming physical contact between these portions. RX-0074 at 5:41-50, Fig. 5. Aoyama further confirms that the connection structure between second rigid wiring board 12 and flexible wiring board 13 “is the same as the structure of the connected section between first rigid wiring board 11 and flexible wiring board 13.” RX-0074 at 5:41-50; JDX-0001C ¶¶3966-3991.

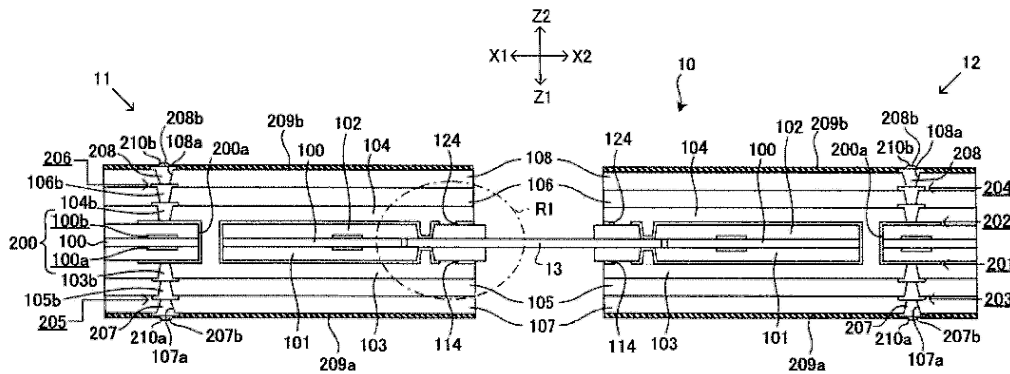


FIG.1

RX-0074 at Fig. 1

iv. Claim Element [7.g]

Claim element [7.g] is taught or rendered obvious by Aoyama. JDX-0001C ¶¶3994-4001 (citing RX-0074).

Specifically, Aoyama discloses a flex-rigid wiring board with two rigid portions (11 and 12) connected by a flexible portion (13), and teaches that “[t]he structure of the connected section between second rigid wiring board 12 and flexible wiring board 13 is the same as the structure of the connected section between first rigid wiring board 11 and flexible wiring board 13.” RX-0074 at 4:1-12, 5:33-50, Figs. 1, 2, 5, 31-36.

To the extent a second flexible portion is not expressly disclosed, a POSITA would have found it obvious. JDX-0001C ¶¶3580-3587, 3994-4001. Williams teaches rigid-flex PCBs with multiple alternating rigid and flexible portions and that this was well-known in the art. RX-0076 at 1:47-51, Figs. 11B, 13A, 15; JDX-0001C ¶¶3580-3587. A POSITA would have been motivated to include a second flexible PCB portion to extend the PCB assembly and enable electrical connections to additional device regions—a common requirement in portable communication

devices, which Aoyama expressly discloses as an intended application. RX-0074 at 7:64-66. The alternating rigid-flex-rigid-flex pattern was standard feature of PCB design. RX-0073 at Fig. 29; JDX-0001C ¶¶3264-3268.

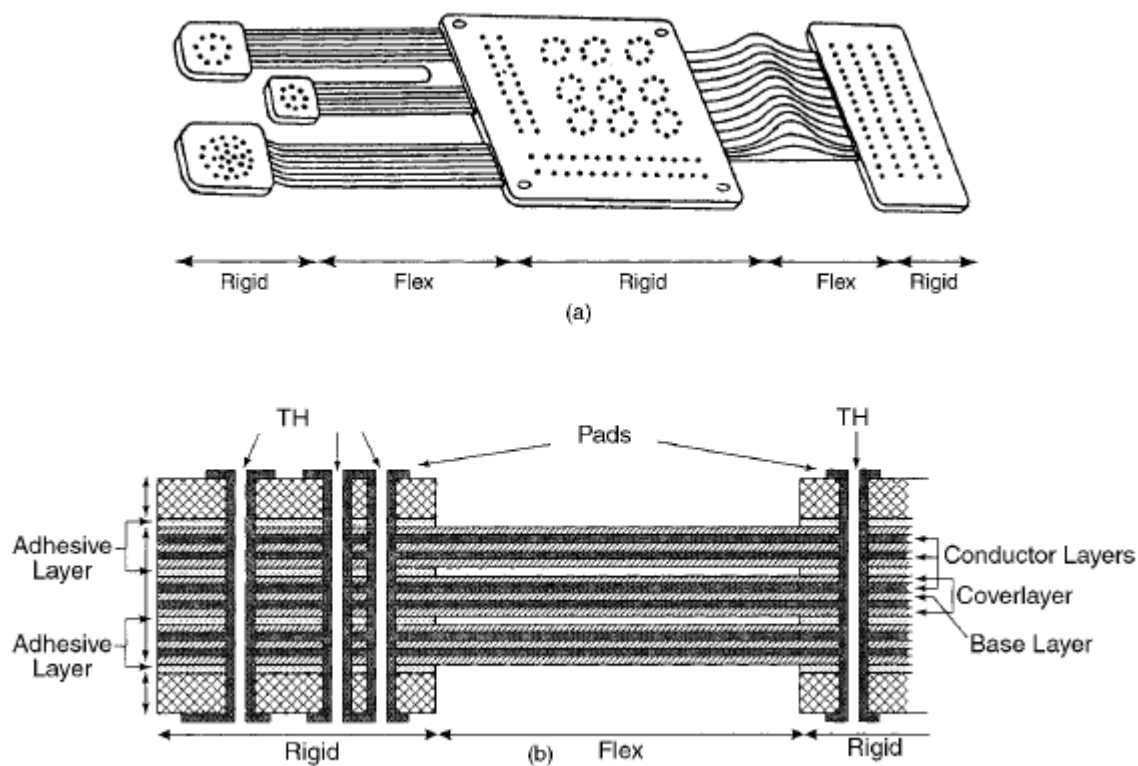


FIGURE 65.1 Basic construction of multilayer circuits (a) Plane view, (b) Cross section.

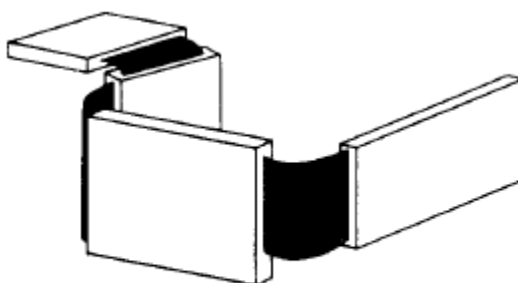


FIGURE 65.3 Folding type rigid/flexible circuit.

RX-0115 at Figs. 65.1, 65.3

v. *Claim Elements [7.h]-[7.l]*

Claim elements [7.h]-[7.l] are each taught or rendered obvious by Aoyama. JDX-0001C ¶¶4002-4054 (citing RX-0074).

Specifically, Aoyama discloses rigid-flex PCBs wherein each rigid PCB portion includes insulating layers and conduction layers in an alternating stack. RX-0074 at 6:30-63, 6:64-7:6, 7:7-55, Figs. 1, 2, 5, 31-36. For the first and third insulating layers (claim elements [7.h] and [7.l]), Aoyama discloses solder-resist layers 209a and 209b formed on the exterior surfaces of rigid PCB portions 11 and 12, both made of the same materials—“resins such as photosensitive resin using acrylic epoxy resin, thermosetting resin mainly containing epoxy resin or UV-setting resin.” RX-0074 at 6:30-39, 7:44-55, 8:1-5, 10:55-62, Figs. 1, 5, 31. For the conduction layers (claim elements [7.i] and [7.k]), Aoyama discloses wiring layers (e.g., wiring layers 201, 206) and conductive patterns located on or above insulation layers in both rigid wiring boards. RX-0074 at 6:30-39, 6:52-57, 6:64-7:6, 7:10-43, Figs. 1, 2, 5, 31-36. For the second insulating layer (claim element [7.j]), Aoyama discloses insulation layers (e.g., insulation layers 103-108) with thicknesses of 50-100 μm . RX-0074 at 6:30-63, 7:7-55, Figs. 1, 5, 31-36.

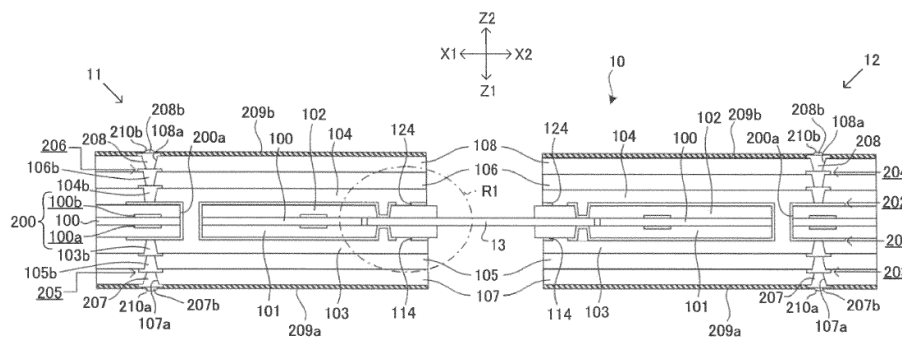


FIG. 1

RX-0074 at Fig. 1

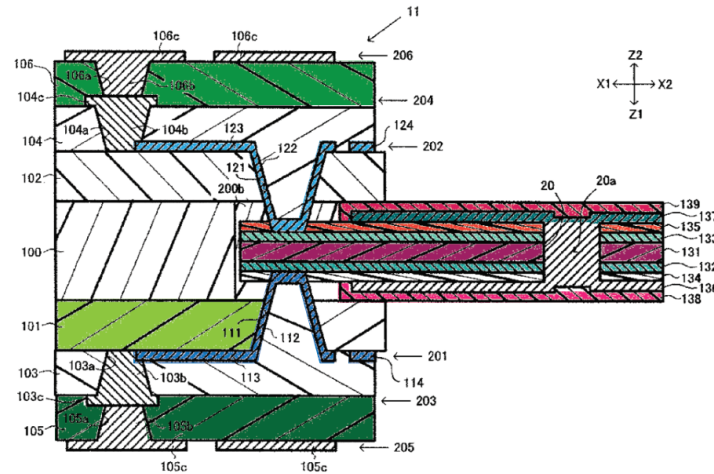
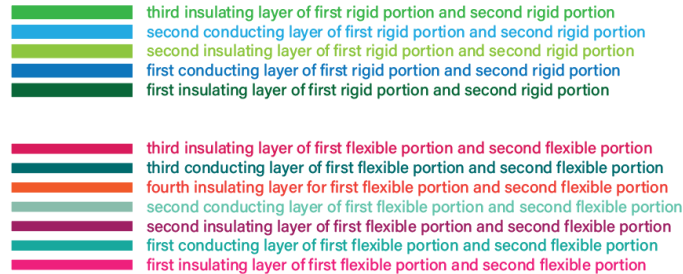


FIG.32



RX-0074 at Fig. 32 (annotated to show layers applicable to claim elements [7.h] – [7.l])

Finally, to the extent not expressly disclosed, a POSITA would have found that it is inherent or obvious that each layer, including the second insulating layer, has a thickness, because thickness is an inherent property of a physical layer. JDX-0001C ¶¶3588-3598, 4002-4054.

vi. *Claim Elements [7.m]-[7.q]*

Claim elements [7.m]-[7.q] are each taught or rendered obvious by Aoyama. JDX-0001C ¶¶4055-4100 (citing RX-0074).

Specifically, Aoyama discloses flexible wiring boards with insulation layers, coverlays, and conduction layers in an alternating stack. RX-0074 at 4:13-36, 4:37-5:24, 5:25-32, 5:51-58, 8:8-20, Figs. 1, 2, 5, 31-36. For the first and third insulating layers (claim element [7.m] and [7.q]), Aoyama discloses coverlays and insulation layers made of polyimide forming exterior surfaces,

which differs from the first material (solder-resist resins or cured prepreg) used in the rigid portions. RX-0074 at 4:16-20, 4:31-36, 5:25-32, 6:58-63, 7:44-55, 8:1-5, Figs. 1, 2, 5. For the conduction layers (claim elements [7.n] and [7.p]), Aoyama discloses conductive layers and shield layers electrically connected between the flexible and rigid portions. RX-0074 at 4:1-5:24, 4:13-16, 4:21-30, 5:33-40, 6:1-23, 6:52-57, 8:8-20, Figs. 1, 2, 5, 31-36. For the second insulating layer (claim element [7.o]), Aoyama discloses insulating layers in the flexible portions with thicknesses of 20-50 μm (preferably 30 μm) or 5-15 μm , which are thinner than the 50-100 μm thickness in the rigid portions. RX-0074 at 4:13-36, 5:25-32, 6:30-63, 6:58-63, 8:8-20.

As described above, a POSITA would have found it obvious to implement the same layer structure as in the first flexible portion in the second flexible portion. JDX-0001C ¶¶3301-3320, 3639-3678, 4055-4100. Williams discloses rigid-flex PCBs with multiple alternating rigid and flexible portions, including configurations where the second flexible portion has the same layer structure as the first flexible portion. RX-0076 at 1:47-51, 6:38-50, 74:12-23, 78:25-35, Figs. 11B, 13A, 15, 69A-69E; JDX-0001C ¶¶3580-3587, 3639-3678. *Id.* A POSITA would have been motivated to implement the same layer structure in both flexible portions to simplify manufacturing, ensure uniform mechanical properties, and provide consistent signal routing and flexibility across the PCB assembly. JDX-0001C ¶¶3301-3320.

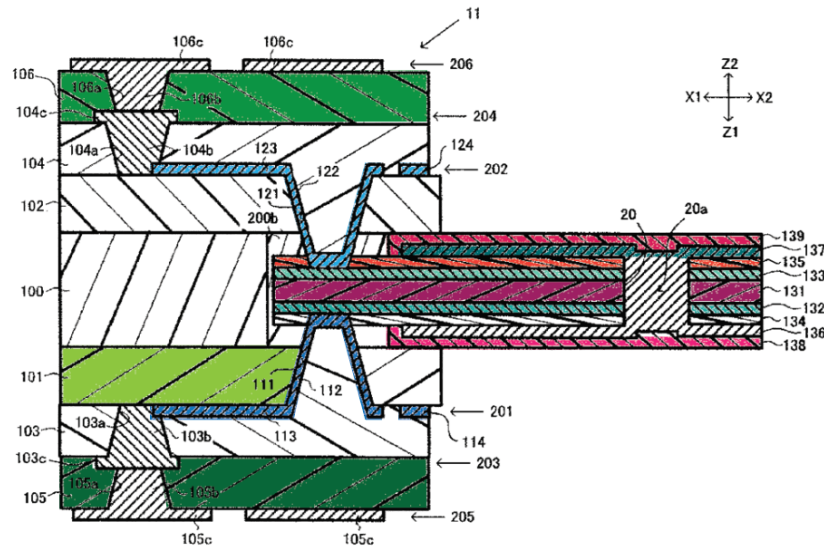


FIG.32

- | | |
|---|--|
|  | third insulating layer of first rigid portion and second rigid portion |
|  | second conducting layer of first rigid portion and second rigid portion |
|  | second insulating layer of first rigid portion and second rigid portion |
|  | first conducting layer of first rigid portion and second rigid portion |
|  | first insulating layer of first rigid portion and second rigid portion |
|
 | |
|  | third insulating layer of first flexible portion and second flexible portion |
|  | third conducting layer of first flexible portion and second flexible portion |
|  | fourth insulating layer for first flexible portion and second flexible portion |
|  | second conducting layer of first flexible portion and second flexible portion |
|  | second insulating layer of first flexible portion and second flexible portion |
|  | first conducting layer of first flexible portion and second flexible portion |
|  | first insulating layer of first flexible portion and second flexible portion |

RX-0076 at Fig. 32 (annotated)

vii. *Claim 9*

The additional requirement of Claim 9 is taught or rendered obvious by Aoyama. JDX-0001C ¶¶4089-4093.

Specifically, Aoyama discloses flexible PCB portions with multiple insulating and conduction layers, including a fourth insulating layer between the second conduction layer and third insulating layer, and conduction layers (*e.g.*, conductive layers and shield layers) between the third and fourth insulating layers. RX-0074 at 4:13-36, 4:21-30, 4:37-5:24, 8:8-20, Figs. 1, 2, 5, 31-36. As discussed, to the extent a second flexible portion has the same layer structure as the first

flexible portion is not expressly disclosed, a POSITA would have found it obvious to implement the same layer structure in the second flexible portion. JDX-0001C ¶¶3301-3320, 3639-3678, 3713-3722, 4055-4100. A POSITA would have been motivated to implement the same layer structure in both flexible portions to simplify manufacturing, ensure uniform mechanical properties, and provide consistent signal routing and flexibility across the PCB assembly. JDX-0001C ¶¶3301-3325.

b) *Graham* Factor 4

As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '789 Patent. *See also* JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

7. *Ground 6: Obviousness in View of Aoyama and Williams*

Aoyama, in combination with Williams, discloses or renders obvious each element of claims 7 and 9 of the '789 Patent. JDX-0001C ¶4482. To the extent Complainants contend that Aoyama does not disclose limitation [7.g] (a second flexible PCB portion) or any limitation reliant on it (broadly speaking, those elements concerning second flexible PCB portion and satisfied by a second flexible PCB portion with a structure the same as the first flexible PCB portion), the combination with Williams renders these limitations obvious.

a) *Graham* Factors 1-3

The scope and content of Aoyama is set forth above in the context of Ground 5. JDX-0001C ¶¶3919-3921. The scope and content of Williams is set forth in the context of Ground 2, and it too discloses rigid-flex PCBs with multiple alternating rigid and flexible portions. RX-0076

at 1:47-51, Figs. 11B, 13A, 15; JDX-0001C ¶¶3507-3509. Both references are in the same field of endeavor as the '789 Patent. *Id.* ¶¶3170, 4480-82.

To the extent Aoyama does not expressly disclose limitation [7.g] (a second flexible PCB portion) or any limitations setting forth the structure of the second flexible PCB portion, a POSITA would have turned to Williams's teachings regarding a second flexible portion with the same layer structure. *Id.*

A POSITA would have been motivated to combine Aoyama with Williams because: (1) both references address overlapping technical problems in rigid-flex PCB construction; (2) Williams teaches rigid-flex PCBs with multiple alternating rigid and flexible portions, including configurations where the second flexible portion has the same layer structure as the first; (3) Aoyama teaches materials and layer configurations including solder-resist layers on rigid exterior surfaces and polyimide for flexible portions; and (4) combining these teachings would yield predictable results given the maturity of rigid-flex PCB design. JDX-0001C ¶4467. Moreover, a POSITA would have been motivated to include a second flexible PCB portion to extend the PCB assembly and enable electrical connections to additional device regions—a common requirement in portable communication devices, which Aoyama expressly discloses as an intended application. RX-0074 at 7:64-66. The alternating rigid-flex-rigid-flex pattern was standard feature of PCB design. RX-0073 at Fig. 29; JDX-0001C ¶¶3264-3268.

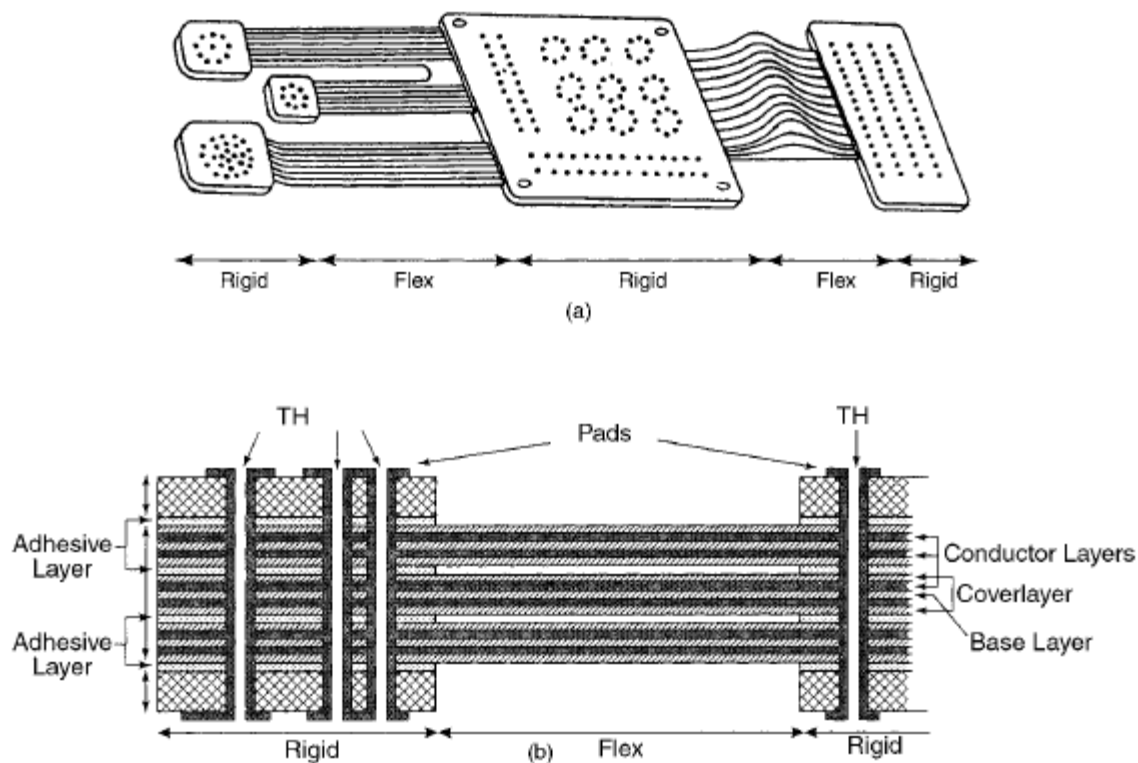


FIGURE 65.1 Basic construction of multilayer circuits (a) Plane view, (b) Cross section.

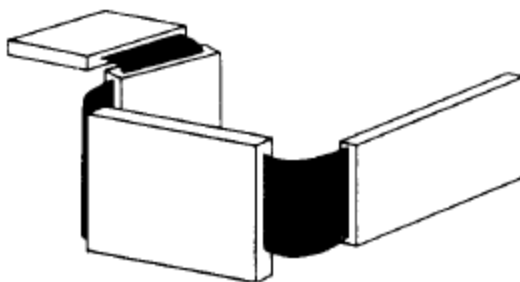


FIGURE 65.3 Folding type rigid/flexible circuit.

RX-0115 at Figs. 65.1, 65.3

A POSITA would have understood the well-known aspects of rigid-flex PCB design disclosed in Aoyama and Williams and would have been able to implement or adapt their teachings to arrive at the claimed layer structures using only routine skill. JDX-0001C ¶¶3161, 3169, 4467.

b) *Graham* Factor 4

As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '789 Patent. *See also* JDX-0006C ¶¶209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

8. *Ground 7: Obviousness in View of Galaxy S5, Mow, and Kim*

The Samsung Galaxy S5, in combination with Mow and Kim, discloses or renders obvious each element of claim 6 of the '789 Patent. JDX-0001C, ¶¶4470-4474.

a) *Graham* Factors 1-3

The S5 is a smartphone that includes a housing, at least two antennas, wireless communication circuitry, and at least two PCBs. JDX-0001C, ¶4487; JX-0070C, RFA Nos. 11-14; RX-0086.8-9; RX-0078.

Mow discloses a portable electronic device with a housing including conductive portions forming side surfaces that operate as antennas, wireless communication circuitry, and PCB arrangements with rigid and flexible portions. JDX-0001C, ¶4488; RX-0110 at Abstract, ¶¶2-6, 8, 25-32, 34-37, 39, 41-43, 46-56, 65-71, 82, 85-89, 92-99, Figs. 1-3, 6, 15-16, 20.

Kim discloses a flexible PCB for high-frequency transmission with signal and ground line arrangements that minimize signal interference and loss. JDX-0001C, ¶4489; RX-0112 at Abstract, Figs. 1, 3, 5, 8.

Accordingly, each reference is in the same field of endeavor as the '789 Patent (*e.g.*, printed circuit boards and/or the structure of portable communication devices, including signal and/or ground line configuration). JDX-0001C, ¶¶3127, 3170.

As explained in detail in the element-by-element analysis below, the S5 provides a base smartphone platform; Mow sets forth specific housing and antenna configurations that would be obvious to implement on such a smartphone platform; and Kim sets forth specific PCB signal and ground line arrangements that would be obvious to implement on such a smartphone platform. JDX-0001C, ¶¶4491, 4487-4497. Thus, the combination of the S5, Mow, and Kim discloses or renders obvious each limitation of claims 1 and 6. JDX-0001C, ¶4498.

A POSITA would have been motivated to combine these references in the manner alleged for several reasons. JDX-0001C, ¶¶4490-4493. First, using conductive portions of a device housing as antenna elements, as taught by Mow, eliminates the need for separate antenna structures, reducing device size—a known technique and objective before the priority date. JDX-0001C, ¶4492; RX-0110 at 7:38-52. Second, Kim’s signal and ground line arrangement is “advantageous for minimizing signal interference and signal loss”—critical for high-frequency wireless communication devices and therefore obvious to implement with the S5 platform. JDX-0001C, ¶4493; RX-0112 at Abstract, 2:30-45.

A POSITA would have understood the well-known aspects of housing-integrated antennas, wireless communication circuitry, and rigid-flex PCB design disclosed in these references and would have been able to combine their teachings using routine skill. JDX-0001C, ¶¶3161, 3169, 4490-4498. Thus, a POSITA would have had a reasonable expectation of success because rigid-flex PCB fabrication techniques and transmission line principles were well-established in the art and the combination would have yielded predictable results. JDX-0001C, ¶¶4494-4498.

i. [Uncontested] Claim Element [1.pre]

The S5/Mow/Kim combination discloses or renders obvious the preamble of claim 1 [1.pre]. For example the S5 is a smartphone and therefore a portable communication device. JDX-

0001C, ¶4502; JX-0070C, RFA 10. Samsung does not contest that the combination meets [1.pre]. CSUCE at 47.

ii. Claim Element [1.a]

The S5/Mow/Kim combination discloses or renders obvious claim element [1.a]. JDX-0001C, ¶¶4504-4516. Samsung admits the S5 includes a housing and at least two antennas. *Id.* ¶¶4506-4507; JX-0070C, RFA-11, RFA-12.

Mow discloses a housing with conduction portions forming side surfaces that operate as antennas. JDX-0001C, ¶¶4509-4516; RX-0110 ¶¶2-6, 8, 25-32, 34-37, 39, 41-43, 46-56, 65-71, 82, 85-89, 92-99, Figs. 1-3, 6, 15-16, 20. For example, Mow discloses a “device 10 may have antennas in regions of device 10 such as upper region 22 and lower region 20.” *Id.* ¶4510 (citing RX-0110, ¶52). Mow further discloses that antennas can be formed of conductive structures, such as parts of an electronic device housing conductive components. *Id.* ¶4511 (citing RX-0110, ¶67). In addition, Mow states that “Conductive peripheral member 180 may surround substantially all of the rectangular periphery of device 10. Conductive peripheral member 180 may have the shape of a display bezel for display 14 or a trim structure for device 10 (i.e., a bezel or trim piece that runs around the upper rim on the front of device 10) or may be implemented using a flat or curved member that forms housing sidewalls that cover substantially all of the sides of housing 12 (as examples).” JDX-0001C, ¶4513 (citing RX-0110, ¶82), ¶4514 (citing RX-0110, ¶93). In sum, Mow discloses conduction portions forming portions of a side surface, each operating as an antenna, which can be seen in Figure 15 below (JDX-0001C, ¶¶4509-4516):

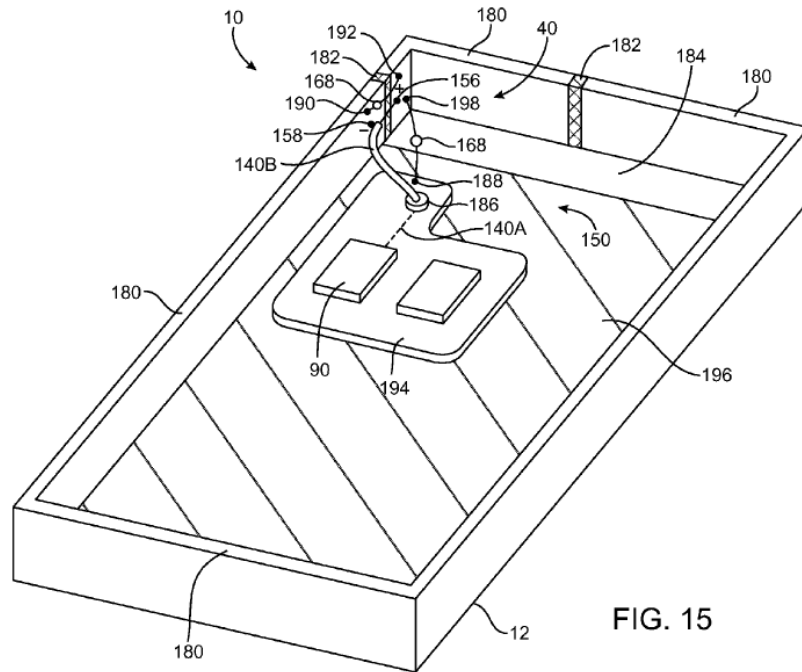


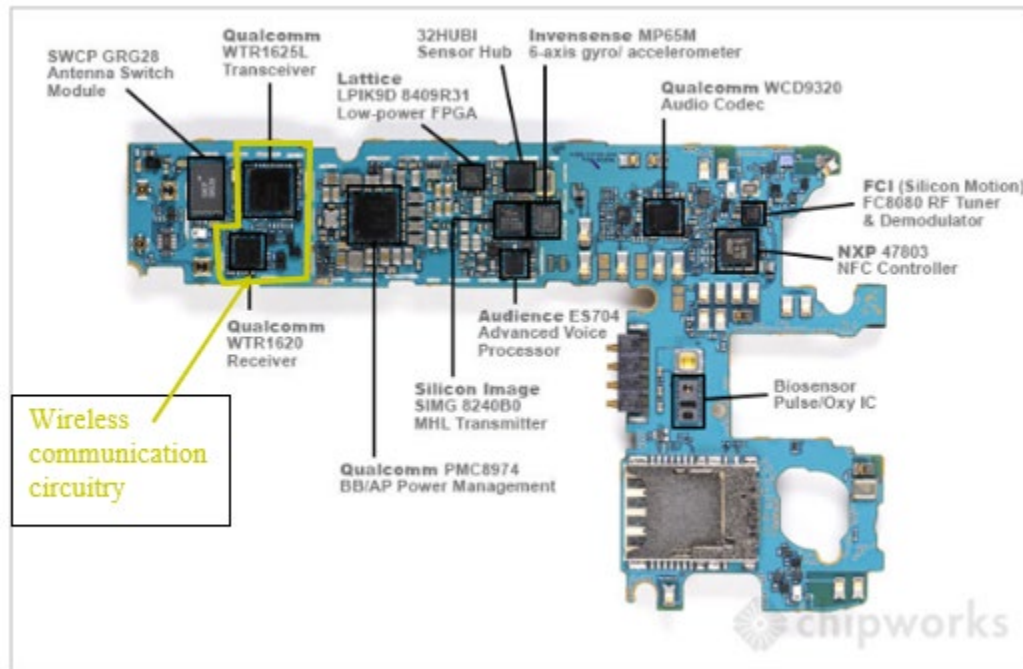
FIG. 15

Dr. Mortazavi will testify that a POSITA would have been obvious to modify S5 with the Mow to disclose [1.a] for several reasons. For example, a POSITA would have been motivated to use conductive portions of a device housing as antenna elements because doing so eliminates the need for separate, dedicated antenna structures, thereby reducing device size and improving design. JDX-0001C, ¶4515. Using metal bezels or housing portions for antennas was a known technique for maximizing antenna performance while maintaining compact device form factors. *Id.* Implementing such antenna configurations was well within the ordinary skill of a POSITA, as the integration of antenna elements into device housings was an established practice in the wireless communication device industry. *Id.*

iii. [Uncontested] Claim Element [1.b]

The S5/Mow/Kim combination discloses or renders obvious claim element [1.b]. JDX-0001C, ¶¶4517-4523. Samsung does not contest that the combination meets [1.b]. CSUCE at 47.

Samsung admits that the S5 includes wireless communication circuitry. JDX-0001C, ¶4520; JX-0070C, RFA-13. The S5 includes wireless communication circuitry such as its Qualcomm receiver and transceiver. JDX-0001C, ¶4521; RX-0079 (teardown); RX-0145 (S5 teardown/unboxing images).



- QFE1100 (Envelope Tracking Power Amplifier PMIC),
- WFR1620 (RF Receiver),
- WTR1625L (RF Transceiver),
- MSM8974AC (Snapdragon 801 (BB/AP SoC)),
- WCD9320 (Audio Codec) and
- PMC8974 (BB/AP Power Management).

Wireless communication circuitry

RX-0079.2 (color version annotated)

Mow also discloses wireless communication circuitry. *See* JDX-0001C, ¶4522; RX-0110 at Abstract, ¶¶2-6, 34, 39, 46-53, 93, Figs. 1-3, 6, 15, 20.

iv. Claim Element [1.c]

The S5/Mow/Kim combination discloses or renders obvious claim element [1.c]. JDX-0001C, ¶¶4524-4529.

Samsung admits that the S5 includes at least two PCBs. *Id.*, ¶4527; JX-0070C, RFA-14. The S5 includes a main PCB and a daughterboard (sub PCB). JDX-0001C, ¶4258; RX-0079; RX-0086.32, -.38-41 (S5 Service Manual); RX-0145 (S5 teardown/unboxing images).



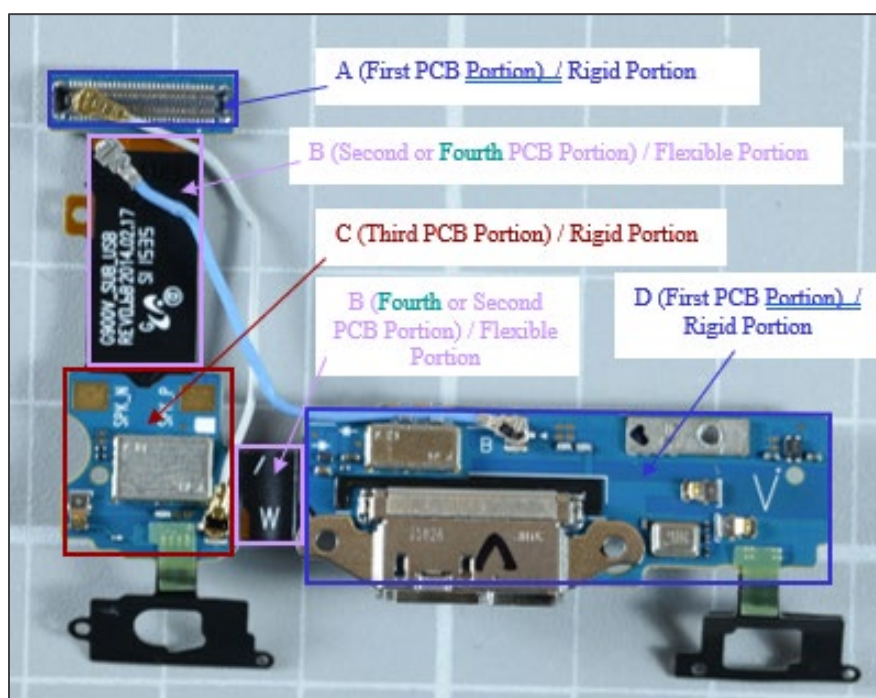
RX-0079.1 (annotated colored version of teardown)

v. Claim Elements [1.d]-[1.g]

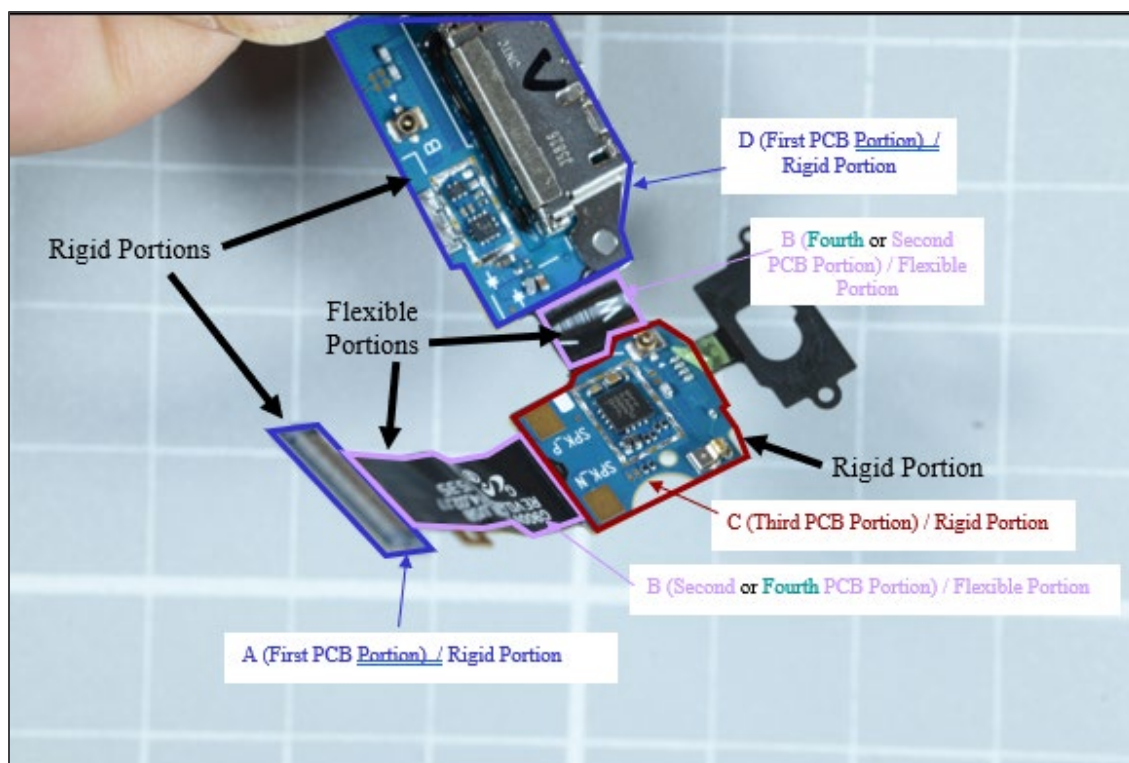
The S5/Mow/Kim combination discloses or renders obvious claim element [1.d]-[1.g]. JDX-0001C, ¶¶4530-4561; RX-0145; RX-0086C; RX-0079.

The S5 includes a PCB (e.g., daughterboard) with first, second, third, and fourth PCB portions in an alternating rigid-flex-rigid-flex configuration. JDX-0001C, ¶¶4532 (1.d), 4536-4538 (1.e), 4542-4544 (1.f), 4548-4550 (1.g); RX-0145; RX-0086C.8-.9.

As shown in the annotated S5 teardown images below, the first and third PCB portions are rigid, while the second and fourth PCB portions are “bendable”—and therefore demonstrably more flexible than the adjacent rigid portions. JDX-0001C ¶¶4532 (1.d), 4536-4538 (1.e), 4542-4544 (1.f), 4548-4550 (1.g).



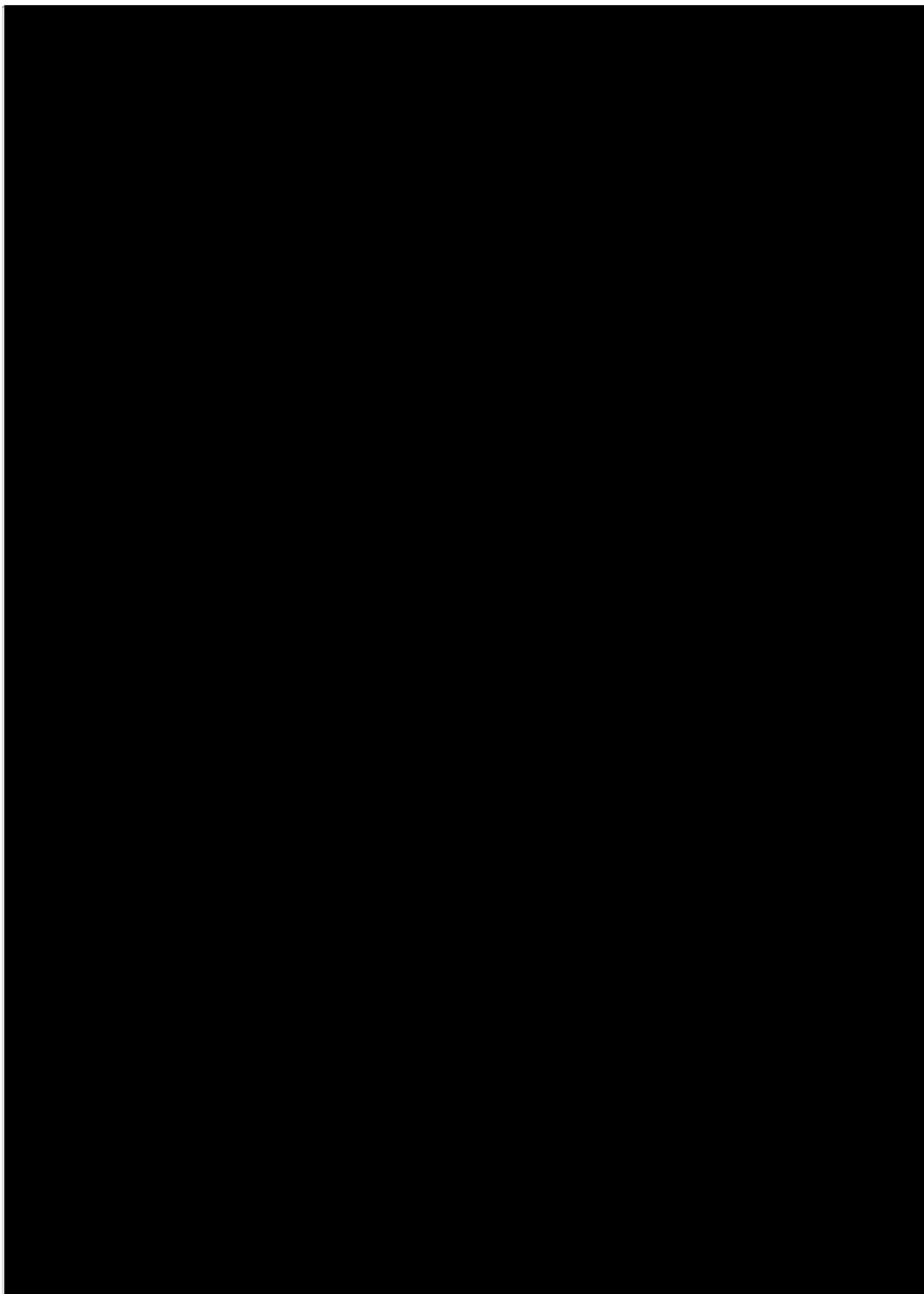
RX-0145.81 (annotated)



RX-0145.95 (annotated)

Dr. Mortazavi will further testify that the stackup diagram of the S5 confirm this flexibility differential: [REDACTED]

[REDACTED] .JDX-0001C, ¶¶4537-4538 (1.e), 4543-4544 (1.f), 4549-4550 (1.g); RX-0085C.100-104 (S5 technical specification).

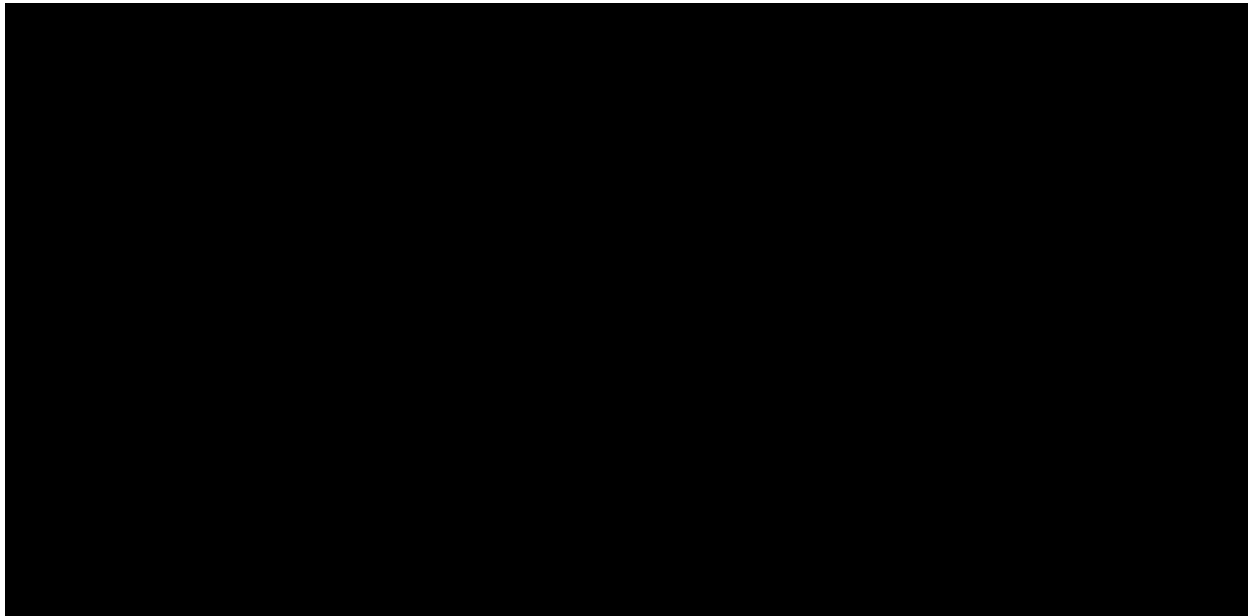


RX-0085C.104

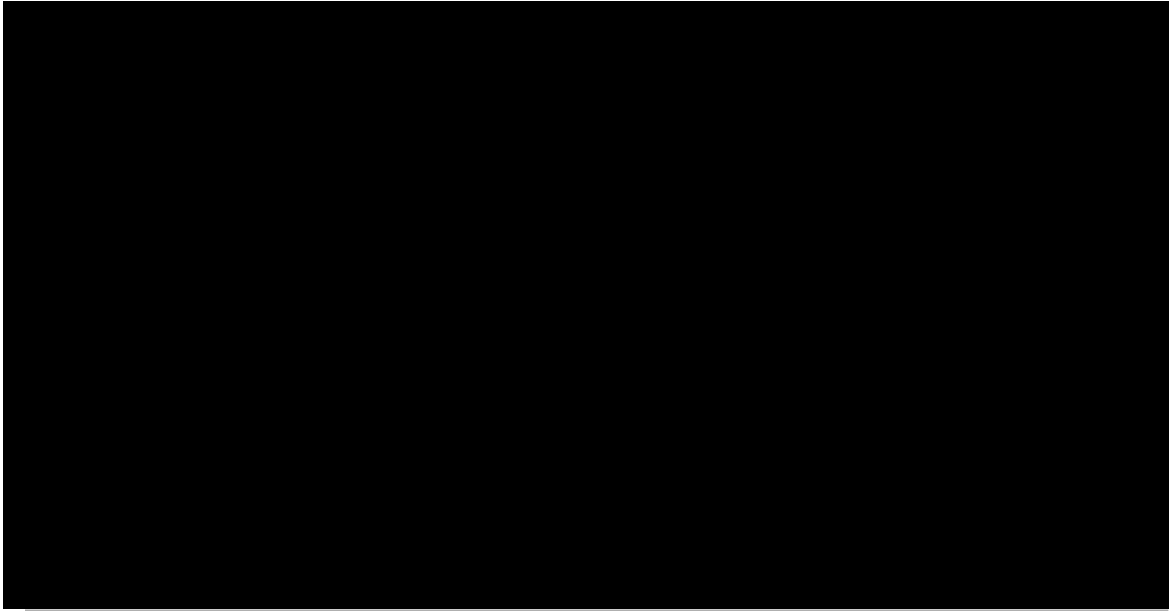
vi. *Claim Element [1.h]*

The S5/Mow/Kim combination discloses or renders obvious claim element [1.h]. JDX-0001C, ¶¶4552-4561; RX-0085C.

The S5 includes a rigid-flex PCB with a plurality of signal lines extending through its rigid and flexible portions (*i.e.*, [REDACTED]) as shown in the S5 technical documents below (JDX-0001C, ¶4554):



RX-0085C.99



RX-0086C.55

It would have been obvious to a POSITA to modify the S5 to incorporate the disclosure of Kim that teaches the specific signal line and ground line arrangement recited in claim element [1.h]. JDX-0001C, ¶4555.

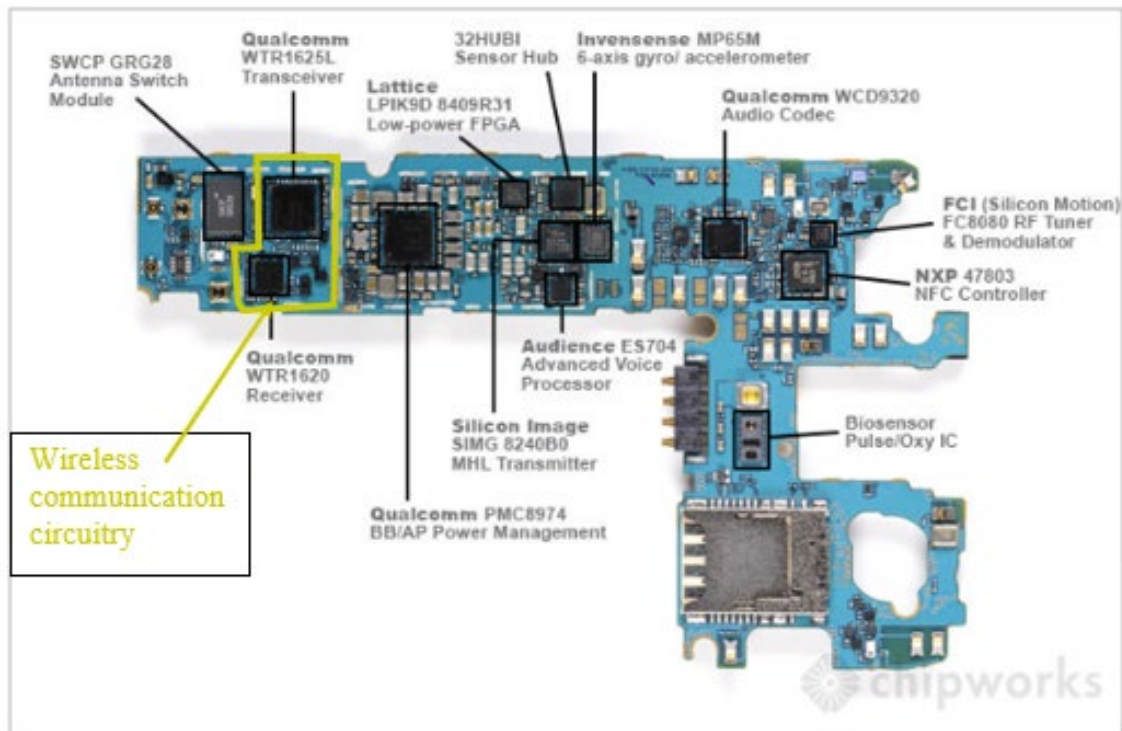
For example, Kim discloses “a first substrate part including a first signal line” and “a second substrate part including a second signal line parallel to the first signal line,” with “a shielding part therebetween,” and ground lines including “a pair of first side grounds” and “a pair of second side grounds” with “a first central ground disposed between the pair of first side grounds.” *Id.* ¶4556 (citing RX-0112, ¶¶[0008]-[0009]). In other words, Kim recites a plurality of lines (a first signal line, a second signal line, a pair of first side grounds or a pair of second side grounds, a first central ground, and a second central ground). *Id.* ¶4557. Kim further discloses signal lines electrically connected with the antenna and wireless communication circuitry (Fig. 8) to minimize “signal interference” and “signal loss” while achieving thin thickness. *Id.* ¶¶4559-4560 (citing RX-0112 at Fig. 8, ¶7).

Dr. Mortazavi will further testify that a POSITA would have been motivated to implement Kim's arrangement in the S5's PCB to [REDACTED], for example, enable signal transmission between antennas and wireless communication circuitry, which was a standard design approach for portable communication devices supporting multiple frequency bands or MIMO operations, all of which was well-known and within the ordinary skill of a POSITA. JDX-0001C, ¶¶3202, 4487-4498.

vii. *Claim Element [6.a]*

The S5/Mow/Kim combination discloses or renders obvious claim element [6.a]. JDX-0001C, ¶¶4616-4620.

For example, the S5 includes wireless communication circuitry, such as its Qualcomm receiver and transceiver, on a second PCB (e.g., motherboard). *Id.* ¶4618



PUBLIC VERSION

- QFE1100 (Envelope Tracking Power Amplifier PMIC),
- WFR1620 (RF Receiver),
- WTR1625L (RF Transceiver),
- MSM8974AC (Snapdragon 801 (BB/AP SoC)),
- WCD9320 (Audio Codec) and
- PMC8974 (BB/AP Power Management).

Wireless
communication
circuitry

RX-0079.2 (annotated color version)

As shown in the S5 teardown images, the wireless communication circuitry is located on the motherboard, which is different from the first PCB (e.g., the daughterboard) (JDX-0001C, ¶4619):





RX-0079.1; RX-0145.55-61, -.69-70, -.77-81, -.86-96 (additional teardown images in color).

viii. Claim Element [6.b]

The S5/Mow/Kim combination discloses or renders obvious claim element [6.b]. JDX-0001C, ¶¶4591-4600.

For the same reasons that combination discloses or renders obvious claim 1, the combination discloses [6.b]. *Id.*, ¶4622 (incorporating opinions for claims 1-3 into [6.b]). For example, as set forth above for claim element [1.h], the combination discloses the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry. *Id.*, ¶¶4552-4561, 3202, 4487-4498

Moreover, the first and second ground lines are electrically connected (e.g., via conductive vias) with a ground member (e.g., ground plate or ground layer) in the first PCB portion. *Id.*, ¶4622 (incorporating opinions for claims 1-3 into [6.b]), 4562-4587 (opinions for claims 2-3).

For example, Kim discloses an insulating layer (second dielectric layer E2) on which the ground lines and signal lines are located, with a ground plate (second ground layer G2) laminated on the bottom surface. *Id.* ¶¶4575-4576 (citing RX-0112 ¶¶16, 82, Fig. 1). Kim further discloses conduction vias (via holes V1-V4) electrically connecting the ground layers (G1, G2) with the ground lines (side grounds S1, S2 and central grounds C1, C2). JDX-0001C, ¶¶4584-4586 (citing RX-0112, ¶¶18-20, 81-91, Figs. 1-7). The vias pass through the dielectric layers to electrically connect the ground layers with the ground lines. *Id.* ¶¶4585-4586 (citing RX-0112, ¶¶84-85).

A POSITA would have been motivated to implement Kim's ground line and ground member (plate) configuration in the S5's PCB because Kim teaches this arrangement minimizes signal interference and signal loss. *Id.* ¶4587 (citing RX-0112, ¶7). A POSITA would have been motivated to use vias to connect the ground lines on the signal routing layer to the ground plate below the insulating layer as a matter of routine design practice, as doing so would have provided the well-known benefits such as improved grounding, reduced ground loop inductance, and enhanced electromagnetic shielding for the signal lines, and such design choices were well within the ordinary skill of a POSITA. *Id.*

b) *Graham* Factor 4

As explained in Section VIII, *supra*, there is no demonstrated nexus between any of the alleged secondary considerations and the alleged inventions of the '789 Patent. *See also* JDX-0006C ¶¶ 209-25. Even if there were some demonstrated secondary considerations of nonobviousness, they would not overcome the strong showing of obviousness as demonstrated above.

VIII. ALLEGED SECONDARY CONSIDERATIONS OF NONOBVIOUSNESS

Samsung makes essentially the same arguments for secondary considerations of nonobviousness for all asserted patents despite the patents belonging to three separate families and

capturing different inventions. *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1377 (Fed. Cir. 2019) (“Where a product embodies claims from two patents, a presumption of nexus can be appropriate only if the claims of both patents generally cover the same invention.”). No alleged secondary considerations have a nexus to the claimed inventions, as is Samsung’s burden to demonstrate. Nor has Samsung shown that the DI products are coextensive with the claimed invention to be entitled to a presumption of nexus. *Id.* Samsung’s arguments regarding secondary considerations are essentially the same for all four asserted patents: the asserted DI products practice claims of the asserted patents, and because the DI products were allegedly commercially successful, they therefore demonstrate commercial success, praise by others, and long-felt but unmet need. There are several fundamental flaws in these arguments.

First, merely selling a product meeting the requirements of the elements of the claims does not demonstrate a nexus to the alleged secondary consideration. *Id.* (patentee did not establish nexus merely because the claims covered a commercially successful product; the secondary-consideration evidence needs to be the “direct result of the unique characteristics of the claimed invention”); *Campbell Soup Co. v. Gamon Plus, Inc.*, 10 F.4th 1268, 1276–79 (Fed. Cir. 2021) (noting an “absence of evidence tying any commercial success or praise to the claimed unique characteristics” of products allegedly practicing the claimed invention). Samsung does not demonstrate a nexus from the claimed inventions, past the asserted DI products, to the actual asserted secondary considerations of nonobviousness (commercial success, praise by others, and long-felt but unmet need). Therefore, at a threshold level, Samsung fails its burden to demonstrate a nexus.

Taking next the ’675, ’677, and ’359 Patents, which Samsung does not distinguish for purposes of secondary considerations (thus illustrating a lack of nexus to the actual claimed

inventions for each patent), one of the so-called instances of praise cited by Samsung notes that success of the Galaxy Ring could be due to Samsung’s “no-subscription model” (as opposed to a monthly fee for access to certain information) and “brand recognition and financial strength.” *See* JDX-0006C ¶¶215-16. Praise and sales may have many causes, and Samsung does not demonstrate that any are attributable to the alleged inventions of the ’675, ’677, and ’359 Patents. Indeed, it stretches credulity that any sales of Samsung’s Ring (released 2024) or even Samsung’s Watch (released 2018) could demonstrate the nonobviousness of alleged inventions with asserted priority dates in 2015 and 2013. And as for broad allegations of long-felt but unmet needs, Samsung discusses such problems as “gesture recognition,” but this is not a problem even alleged to be solved by the claims themselves; the ’359 Patent claims just to “determine, based on an analysis of the movement signal, whether to perform a specified function,” but does not claim any solution for how to recognize a gesture.

For the ’789 Patent, Samsung only alleges that the asserted DI products practice claim 6, which is a different claim than the claims asserted for infringement (7 and 9); there is no nexus to claims 7 and 9 in Samsung’s arguments. Furthermore, as Dr. Mercier will testify, the majority of the articles of alleged praise of Samsung is by Samsung itself in press releases, not praise by “others.” JDX-0006C ¶¶217-23. Samsung has also shown no connection between alleged commercial success by Samsung’s smartphone products and the alleged invention of claim 6 (let alone claims 7 and 9). As for an alleged long-felt by unmet need for the alleged invention, Samsung relies just upon the say-so of the ’789 Patent and, at most, its own supposed need, which is not an indication that others in the industry needed such an invention but could not obtain it because it was not obvious (*i.e.*, the logic behind this secondary consideration).

Thus, Samsung has not demonstrated any nexus between any of its alleged secondary considerations of non-obviousness and any of the asserted claims in this Investigation.

IX. LACK OF ECONOMIC DOMESTIC INDUSTRY

Samsung has not proven an economic domestic industry (“EDI”) relating to the DI Products. 19 U.S.C. § 1337(a)(2). It has not properly allocated its investments on a patent-by-patent basis, which the Commission requires to preclude investments toward a product that practices one patent from counting toward investments in a patent which that product does not practice. *Zircon Corp. v. Int’l Trade Comm’n*, 101 F.4th 817, 822 (Fed. Cir. 2024); *see also id.* at 826-27 (noting that Section 337 relief is provided only on the patents for which EDI is proven). Samsung has framed its EDI investments in terms of the following DI Product categories:

- The **’359, ’677, and ’675 Patents**: DI Ring and DI Watches
- The **’789 Patent**: DI Phones

Samsung claims an EDI for these categories based on only the investments made by SEA and under only subsections (A) (plant and equipment) and (B) (labor and capital), not (C) (exploitation). Samsung served a report on EDI by Julia Rowe (JDX-0002C), but she fails to show an EDI through “credible and reliable evidence.” *Certain Artificial Eyelash Extension Sys., Prod. & Components Thereof*, Inv. No. 337-TA-1226, Comm’n Op. at 48 (Oct. 24, 2022). There is insufficient evidence to properly assess Samsung’s U.S. investments related to the DI Products, and Mr. Jarosz will testify that the available evidence does not show a qualifying, significant EDI.

A. Samsung Relies on Expenses Incurred by [REDACTED] Subsidiary SEA

There is a fundamental disconnect between the SEA investments Complainants rely on in attempting to satisfy economic domestic industry and the rights (or lack thereof) SEA has to practice the Asserted Patents. Samsung relies solely on expenses incurred by SEC’s subsidiary SEA. But [REDACTED] ownership of the Asserted Patents. JX-0049C (No. 13).

Despite repeated opportunities to do so throughout this Investigation, Samsung has never asserted SEA's rights in the Asserted Patents. None of the Complaint, the Amended Complaint, or the exhibits thereto claim SEA's rights in the Asserted Patents. Am. Compl. ¶5 (EDIS ID 867832); Compl. Exs. 10C (EDIS ATT ID 2463877), 11C (EDIS ATT ID 2463877). Throughout discovery, Oura sought complete information about any rights to the patents, but Samsung again failed to claim that SEA has any rights. JX-0049C (No. 13); JX-0047C (No. 21). Samsung's corporate witness on licensing could confirm only that SEA is a wholly owned subsidiary of SEC. JX-0018C (M. Kim) at 69:8-15. The Commission "consistently exclude[s]" expenses related to a patent-protected product incurred by a complainant that does not have rights to that patent. *Certain Power Semiconductors*, Inv. No. 337-TA-1308, ID at 116-17 (collecting cases) (May 18, 2023) (EDIS Doc. ID 798314), *aff'd* by Comm'n Notice at 2 (Jul. 17, 2023) (EDIS Doc. ID 800437).

Samsung has offered no facts establishing SEA has any rights to the Asserted Patents, whether express or implied. Any belated attempt to claim an implied license should be rejected because Oura would be substantially prejudiced by such a late claim. Had Samsung made such a claim during discovery Oura could have explored *if*, for example, whether (1) SEA's claimed investments were undertaken pursuant to an implied license with sublicensing rights, (2) SEC has a practice of granting implied, broad patent rights to subsidiaries, and (3) other Samsung subsidiaries, such as non-Complainant Samsung Research America ("SRA") has rights to the Asserted Patents. If SRA or other subsidiaries have those broad rights, then SEC and SEA do not have "all substantial rights" to the patents required for statutory standing here. *Lone Star Silicon Innovations LLC v. Nanya Tech. Corp.*, 925 F.3d 1225, 1228 (Fed. Cir. 2019); *see also Certain Active Matrix Organic Light-Emitting Diode Display Panels & Modules for Mobile Devices, &*

Components Thereof, 337-TA-1351, Comm’n Op. at 14 (May 15, 2024) (EDIS ID 821542) (“*Certain Active Matrix Panels*”).

There is no evidence from Samsung that SEA has any rights to the Asserted Patents. Samsung’s discovery responses and exhibit to the complaint that purport to disclose full information about licensees, none of which identifies SEA, justify a finding that SEA’s claimed investments were not made pursuant to a proper license. *See* Rule 210.33(b)(1). Samsung has not responded to these points in its contentions, waiving its position. *See* JX-0049C; JX-0047C.

B. Samsung’s Sales-Based Allocation Method Is Unreliable

Samsung uses a sales-based allocation to create its claimed investments, which is based on a blanket application of that allocation to all expenses within [REDACTED] that are assigned to Samsung’s “[REDACTED].” This allocation is improper, as set out below.

1. Samsung’s Accounting Methodology: Expenses

SEA’s expenses, which are solely for U.S. commercialization, are organized under three nested frameworks, described below. RX-0959.7-12.

First, at the highest level is the [REDACTED] labeled “[REDACTED].” JX-0047.10. These [REDACTED] are [REDACTED]. CX-1469C ([REDACTED]). Samsung provided definitions for some: [REDACTED] [REDACTED]. JX-0016C (Martinez) at 98:1-10. [REDACTED] JX-0016C (Martinez) at 97:5-25; RX-0959.9-10. The mobile products covered [REDACTED] include at least phones, tablets, watches, and earbuds, and includes the three types of DI Products (S Phones, Watches, and Ring). RX-0378C (Sheppard) at 147:18-148:4; JX-0016C (Martinez) at 61:11-23. [REDACTED] JDX-0053C (Rowe) at 132:6-10. [REDACTED]

[REDACTED].” JX-0047C.9-10. However, [REDACTED]
[REDACTED]. [REDACTED]
[REDACTED]. JX-0016C (Martinez) at 97:5-25; CX-1469C ([REDACTED]
[REDACTED], such as (example only) [REDACTED],
which Samsung did not identify as one of the [REDACTED] purportedly mobile-related cost centers)).
[REDACTED]. JDX-0002C (Rowe) ¶96; CX-1469C ([REDACTED]
[REDACTED], similarly showing expenses for cost centers not identified
by Samsung as being related to mobile products).

Second, SEA has created [REDACTED]. [REDACTED]. CX-1469 (“[REDACTED]”). These correspond to the [REDACTED] presented in § III.B of Samsung’s EDI contentions and the [REDACTED] investment categories or “departments” presented by Ms. Rowe. RX-0959C; JDX-0002C (Rowe) ¶¶16, 97. Ms. Rowe did not know how these [REDACTED] groups were assigned and “assume[d],” without asking or seeing evidence, that Mr. Martinez had spoken to the person who performed this selection. JDX-0053C (Rowe) at 136:11-137:21.

Third, SEA identifies [REDACTED]
[REDACTED]. JX-0047C.10 Each [REDACTED]. CX-1469C ([REDACTED]).

The source of Samsung's claimed investments is a single spreadsheet that it represents as containing only SEA expenses. CX-1469C; *see also* JX-0047C.10-12 (describing CX-1469C as "[redacted]"). The "[redacted]"

[REDACTED] JX-0047C.11. However, Ms. Rowe clarifies that CX-1469C contains information outside of SEA's mobile division. JDX-0053C (Rowe) at 109:18-110:3, 116:16-22. [REDACTED]

██████████. JX-0047C.12. This data is represented in the pivot tables in tabs “██████████”
██████████” *Id.*

2. *Samsung's Flawed Allocation*

DI investments must be reasonably allocated to patent-practicing products so that the Commission can determine whether there is a significant qualifying domestic industry for each article that (allegedly) practices the patent and whether, accordingly, there is a significant domestic industry for each Asserted Patent. *Zircon*, 101 F.4th 817 at 822, 826-27. Mr. Jarosz will testify that Samsung's use of a blanket allocation across unassociated expense categories does not satisfy the Section 337 standards.

Samsung and Ms. Rowe rely entirely on investments made within any [REDACTED] group, which they allocated to the DI Products using a blanket U.S.-sales-based allocation. JDX-0046C.20-21, .25-27. Ms. Rowe used a revenue-based allocation, resulting in allocation factors slightly lower than Samsung's. JDX-0002C (Rowe) ¶101. She admitted that Samsung may [REDACTED] but would not identify which. JDX-0053C (Rowe) at 33:16-42:5.

	2022	2023	2024	2025	Total
DI Phone (Rowe - revenue)					
DI Watch (Rowe - revenue)					
DI Ring (Rowe - revenue)					

a) Sales-Based Allocation Results in Improper Aggregation

Although a sales-based allocation may be a *convenient* way for Samsung to attribute expenses incurred [REDACTED] (only one of which is specific to mobile products) to the DI Products, it is not sufficient to satisfy its Section 337 burden. As Mr. Jarosz will testify, Samsung's sales-based allocation of investments in the [REDACTED] is too flawed to be reliable because it improperly sweeps in investments not associated with the DI Products and lacks "sufficient[] ex[planation] or support." *Components for Certain Environmentally-Protected LCD Digital Displays & Prods. Containing the Same*, 337-TA-1349, ID at 291-92 (Apr. 16, 2024) (EDIS Doc. ID 821125) ("*Certain Environmentally-Protected LCD Displays*") (rejecting Samsung's EDI contentions in that investigation); *see also Certain Replacement Automotive Lamps*, Inv. No. 337-TA-1291, Comm'n Op. at 27 (Mar. 7, 2024) (EDIS Doc. ID 816597) ("*Certain Replacement Automotive Lamps*") ("[A]ggregating investments in articles that *are* protected by a specific patent with investments in articles that *are not* protected by that patent precludes meaningful consideration of the relevant investments."); *Digital Video-Capable Device & Components Thereof*, Inv. No. 337-TA-1224, ID at 140-41 (Oct. 21, 2021) (EDIS Doc. ID 758903) ("*Digital Video-Capable Device*") (critiquing revenue-based allocation because it did not exclude investments in non-qualifying activities).

Samsung relies on expenses (categorized through black-box [REDACTED] [REDACTED]) without showing how they relate to the DI Products. More evidence is required to allow the Commission to cull unrelated or marginally related expense groupings. *See, e.g.*, JDX-0005C at Tab 6; JX-0016C (Martinez) at 35:11-13 ("[REDACTED] [REDACTED]"), 58:1-13 ([REDACTED] [REDACTED]); JX-0022C (D. Kim) at 66:4-12 ([REDACTED] [REDACTED])

[REDACTED]). The little information about these black-box categories that Samsung did produce indicate that many should be excluded, as the examples below show and as Mr. Jarosz will testify. Samsung made no attempt to provide good-faith surveys from employees or time-based allocations within those groups, internal documentation, or the removal of suspect categories altogether. Ms. Rowe failed to consider *any* difference among the products when allocating Samsung's investments to the products. JDX-0053C (Rowe) at 67:6-72:6.

For example, Samsung counts investments that are cellular- or carrier-exclusive indiscriminately toward all DI Products, even though the DI Rings and many DI Watches have no cellular functionality and no relationship with carriers. RX-0292.78 (69% of smart watches generally are "Bluetooth-only"); JX-0016C (Martinez) at 22:21-23:14, 24:9-15, 25:8-20, 27:15-17. Samsung admits that many of the investments (counted toward all products) are specific or almost entirely related to [REDACTED]: For example, Samsung [REDACTED] [REDACTED]. JX-0016C (Martinez) at 122:6-123:19 ([REDACTED] [REDACTED]). [REDACTED]. Compl., Ex. 11 (Martinez Decl.) (EDIS ID 2463878) (describing [REDACTED] [REDACTED]); JX-0016C (Martinez) at 123:22-124:17 ([REDACTED] [REDACTED]). Samsung's [REDACTED] are related to the DI Products primarily by training carriers. JX-0016C (Martinez) at 133:12-134:8, 137:5-18 ([REDACTED] [REDACTED] [REDACTED]); JDX-0053C (Rowe) at 179:19-180:1, 181:3-182:14 ([REDACTED]). Further, [REDACTED] directed to [REDACTED]. JX-0016C (Martinez) at 148:11-149:8, 153:3-154:22; JDX-

0053C (Rowe) at 196:15-197:15. By counting investments in these efforts toward the DI Ring and all DI Watch products, Samsung knowingly misallocates and overcounts—even while also knowing that many of its non-DI Products (such as other high-end smartphones) make extensive use of the [REDACTED]. See JX-0016C (Martinez) at 33:7-34:24 (discussing phones); CX-1590C (Sheppard) at 29:16-20 (similar).

Even for those [REDACTED] that may relate to at least some of the DI Products, it is improper to allocate those investments to *all* of the DI Products, which Samsung does. As an example, many [REDACTED] appear connected to functionalities, such as [REDACTED] which are unrelated to the DI Ring (which has no ability to load apps or coordination with many apps) or DI Watch (which has a different ability to load apps than the smartphone). JX-0016C (Martinez) at 103:17-104:7 ([REDACTED] [REDACTED]); 103:21-105:4 (admitting [REDACTED] [REDACTED]); 95:20-96:10 (referring to [REDACTED] [REDACTED] [REDACTED]); JDX-0053C (Rowe) at 187:16-188:5 (admitting [REDACTED] [REDACTED]), 190:4-19 ([REDACTED] [REDACTED]); 190:20-191:8 ([REDACTED] [REDACTED]), 192:7-22 (did not know or consider whether t [REDACTED] [REDACTED]) .

Separately, many [REDACTED] that Samsung relies on clearly have *no* connection with *any* of the DI Products. After all, it would be unlikely that all [REDACTED] involved in SEA's mobile products, service and repair, and content and services [REDACTED] are related to the handful of DI

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long conversation she had with Mr. Martinez after his deposition. JDX-0053C (Rowe) at 24:1-4. But she confirmed the relevance of (only) [REDACTED]. JDX-0053C (Rowe) at 164:2-15, 167:11-168:10; 164:16-165:19 (testifying she learned specific information from Mr. Martinez about [REDACTED]). At his deposition, Mr. Martinez could not identify which [REDACTED] were unrelated to the DI Products (JX-0016C at 94:23-96:18) could not identify whether the [REDACTED] relates to any DI Product (*id.* at 99:3-100:13).

Mr. Jarosz will testify that even if Samsung finds that a sales-based allocation approach is sufficient in the ordinary course of business to meet auditing requirements, those standards are different from the Section 337 standard, which requires a sufficient nexus between claimed expenses and each DI Product. He will further testify that allocations used to comply with external financial reporting standards are particularly inapplicable here because of inaccuracies for product-specific costs, and SEC's allocation is used across SEC globally for all products, resulting in errors from substantial cross-subsidization and distortions that limits its utility for measuring SEA's investments in the DI Products pursuant to Section 337. *See, e.g.*, JX-0022C (D. Kim) at 91:9-92:4, 94:18-95:2 ([REDACTED]). Moreover, Samsung does not claim that the sales-based allocation it used to apportion its claimed investments is the same as the allocation it uses in the ordinary course of business (and it does not appear to be). *Id.* at 66:20-67:7 ([REDACTED]).

b) Cherry-Picked Base

Mr. Jarosz will also testify that because the [REDACTED] are not limited to mobile products, the base of any sales-based allocation should be SEA's total, not just mobile, revenue. SEA's operations encompass diverse product areas, including mobile products, televisions,

computers, and home appliances. *See* JX-0016C (Martinez) at 98:3-10; CX-1590C (Sheppard) at 19:5-11. Mr. Martinez testified that at least some of [REDACTED] support non-mobile products. JX-0016C (Martinez) at 130:10-131:7, 131:21-132:5, 161:3-17 (admitting that the [REDACTED] [REDACTED]), 162:8-164:17 (admitting [REDACTED] [REDACTED]).

Mr. Jarosz will testify that SEA's 2025 revenue was [REDACTED], almost [REDACTED] its [REDACTED] revenue. RX-0390.66. Allocating SEA's expenses according to its full revenues, not just [REDACTED] revenues, reduces the claimed investments by nearly [REDACTED]%. JDX-0005C at Tab 13.

c) Calculation Error in DI Ring Allocation

Mr. Jarosz will testify that Ms. Rowe's allocation for the DI Ring in 2024 overstates the investments because her calculation erroneously assigned a full year of investment to six months of sales (effectively assuming twelve rather than six months of sales in 2024). A corrected allocation reduces Ms. Rowe's labor and capital investments for the DI Ring by [REDACTED]% in 2024 and [REDACTED]% overall. JDX-0005C (Jarosz) at Tabs 14-16.

C. Plant & Equipment Investments Are Overstated

Samsung identifies the unallocated rent amount for [REDACTED] facilities that "house the DI Activities." RX-0959C.20-22; *see also* JX-0016C at 156:24-157:3 (Martinez) (testifying that these [REDACTED] facilities are those that Samsung is relying on for DI). This is less than [REDACTED] of the "over [REDACTED] facilities across the U.S." operated by SEA that Samsung touted. RX-0959C.8.

And there is no evidence that these [REDACTED] facilities house qualifying activities for the DI Products or how much of those activities are performed. *Digital Video-Capable Device*, ID at 140 (rejecting allocation that included investments in facilities where no qualifying activities were performed). Samsung admitted that many facilities are primarily for [REDACTED]

[REDACTED]: for example, SEA’s headquarter office in New Jersey is primarily related to [REDACTED]. CX-1590C at 78:1-79:4; *see also* JX-0016C (Martinez) at 158:2-7 (agreeing “[REDACTED]”).

Ms. Rowe presents different investment amounts in Exhibit 2.0 of her report with allocated versions, citing CX-1469C as the source. JDX-0002C (Rowe); *see also* RX-0959C.22-25. The “Total” numbers in the summary below correspond to the expenses Ms. Rowe collected from those [REDACTED]. These are summarized below:

	2022	2023	2024	Jan.-Nov. 2025	Total
[REDACTED]	[REDACTED]				
Total					
DI Phone					
DI Watch					
DI Ring	[REDACTED]				

These appear to be from the “Raw Data” tab in CX-1469C, filtering to the [REDACTED] [REDACTED] that Ms. Rowe believed (without explanation) were related to plant and equipment. She did not identify the connected facility or the purpose of any equipment or payments. *See Digital Video-Capable Device*, ID at 139-43 (rejecting EDI contentions where allocations do not filter out unrelated locations).

D. Labor & Capital Investments Are Demonstrably Overstated

Samsung claims qualifying investments for labor and capital in Tables 18-23 of its contentions, which do not specifically identify the source of the reported data. RX-0959C.31-39. Ms. Rowe presents different investment amounts for each [REDACTED] in Exhibit 3.0 of her report with her allocations, citing CX-1469C as the source. JDX-0002C (Rowe). She also

presented “comprehensive” labor and capital investments, which are the plant and equipment investments added to the labor and capital investments. JDX-0002C at Ex. 9.0. Accordingly, where collapsing subsection (A) and (B), if Samsung fails to prove significant investments for labor and capital, it necessarily fails to prove significance for plant and equipment. Ms. Rowe further presented an alternative with these “comprehensive” data [REDACTED] JDX-0002C at Ex. 10.0. The results of these various sums are below, with the “Total” numbers corresponding to expenses Ms. Rowe reported from the [REDACTED]

	2022	2023	2024	Jan.-Nov. 2025	Total
[REDACTED] Total					
DI Phone					
DI Phone (comprehensive)					
DI Phone (comprehensive “excluding marketing”)					
DI Watch					
DI Watch (comprehensive)					
DI Watch (comprehensive “excluding marketing”)					
DI Ring					
DI Ring (comprehensive)					
DI Ring (comprehensive “excluding marketing”)					

When determining which expenses to include as labor and capital, Ms. Rowe did not apply the definition of capital, “a stock of accumulated goods,” required by the Federal Circuit. *Lelo Inc.*

v. Int'l Trade Comm'n, 786 F.3d 879, 883 (Fed. Cir. 2015). At her deposition, she could not explain what she understood “capital” to mean except “facility-related investments,” and that the non-facility costs she included in that category are labor. JDX-0053C (Rowe) at 237:12-239:13. Mr. Jarosz will testify that Ms. Rowe’s calculations improperly include expenses such as [REDACTED]

[REDACTED]

[REDACTED] without showing they are employment of labor or capital. *See Certain Collapsible Sockets for Mobile Elec. Devices & Components Thereof*, Inv. No. 337-TA-1056, Comm’n Op. at 16-19 (July 9, 2018) (EDIS ID 649819). For example, Mr. Jarosz will opine that the \$ [REDACTED] for “[REDACTED]” that Ms. Rowe included in all her analyses are not qualifying employments of labor or capital because they are liabilities. JDX-0005C at Tab 10, row 296. Similarly, marketing expenses for purchases of things such as “[REDACTED]” and “[REDACTED]” and incidentals do not qualify under subsection (B). JDX-0005C at Tab 10; *see Lashify, Inc. v. Int’l Trade Comm’n*, 130 F.4th 948, 961-62 (Fed. Cir. 2025) (holding that legislative history indicates that (B) excludes “simply purchasing advertising” that is not “employing the inputs” used for marketing functions); *Certain Active Matrix Panels*, ID at 178 (Nov. 15, 2024) (holding that off-the-shelf purchases are not qualifying investments without evidence connecting those purchases to an employment of labor or capital). Mr. Jarosz will testify that other categories set out in Tab 10 of his report have not been shown to be capital, and excluding just those would reduce Ms. Rowe’s claimed labor and capital investments by up to [REDACTED]%. JDX-0005C. Ms. Rowe did exclude some expenses in her [REDACTED] alternative summation, but without addressing how the remaining expenses relate to employment of labor and capital. *See* JDX-0002C at Exs. 1.2, 10.1.

Further, Mr. Jarosz will testify that industry benchmarks are evidence that Ms. Rowe overcounted labor investments. *See* JDX-0005C at Tabs 8-9 (Rowe suggesting annual investments between [REDACTED] for Samsung relating to the DI Products whereas Oura’s analogous investments (rounded) were [REDACTED]).

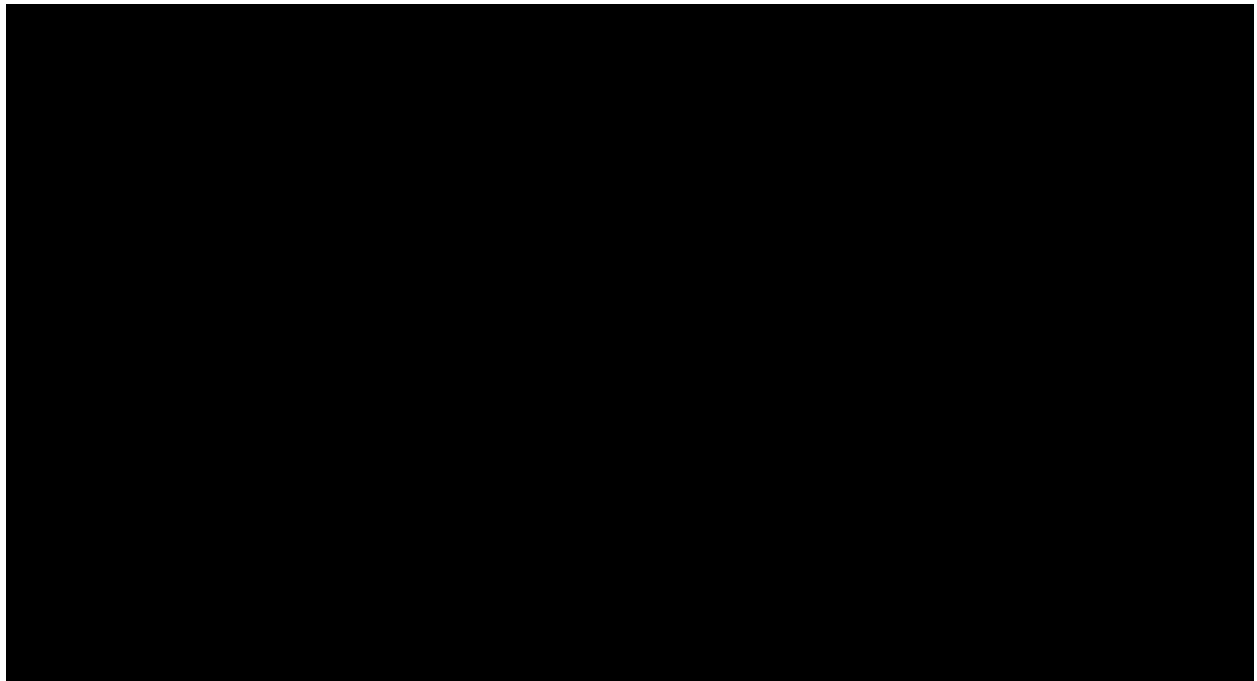
E. Lack of Significance

Samsung failed to show that its claimed investments (which improperly overcount investments, as set out above) are significant relative to relevant metrics. *Lelo*, 786 F.3d at 883; *see also Certain Printing and Imaging Devices & Components Thereof*, Inv. No. 337-TA-690, Comm’n Op. at 26 (Feb. 17, 2011) (investments must be “assessed with respect to the nature of the activities and how they are ‘significant’ to the articles protected by the intellectual property right”). Mr. Jarosz will testify that Samsung’s reliance on cherry-picked data pulled from one U.S.-based subsidiary precludes a “review of all relevant considerations” and the “context dependent” analysis the Federal Circuit requires. *Wuhan Healthgen Biotechnology Corp. v. Int’l Trade Comm’n*, 127 F.4th 1334, 1339 (Fed. Cir. 2025). Samsung does not “articulate why [any] proffered comparison was useful in determining quantitative significance of [its] investments *as to the products at issue....*” *Certain Power Converter Modules and Computing Systems Containing the Same*, Inv. No. 337-TA-1370, Comm’n Op. at 94 (Feb. 26, 2025) (EDIS ID 844518); *Certain Environmentally-Protected LCD Displays*, ID at 289 n.77 (complainant has burden to show that comparisons are “relevant...for determining significance”).

Further, Mr. Jarosz will testify that the significance ratios Samsung generated for the more probative metrics are generally below the [REDACTED] benchmark the Commission consistently finds is a minimum for significance. *See Lelo*, 786 F.3d at 882, 885; *Certain Electronic Candle Products and Components Thereof*, Inv. No. 337-TA-1195, Comm’n Op. at 18 (September 13, 2021) (EDIS ID 751478); *Certain Foodservice Equipment and Components Thereof*, Inv. No. 337-TA-1166,

ID at 90 (June 4, 2021) (EDIS ID 745962). An investigation recently refused to grant summary determination (which was unopposed) where Samsung’s “raw numbers” were high but were not significant because they were “less than six percent of Samsung’s total global investments in labor and capital associated with the Samsung DI Products” and “[l]ess than six percent is so low.” *Certain Elec. Devices, Including Mobile Phones, Tablets, Laptops, Components Thereof, and Prods. Containing the Same*, Inv. No. 337-TA-1376, Order No. 32 at 24 (July 12, 2024).

Mr. Jarosz will also testify that Ms. Rowe’s own calculations show [REDACTED] investments, particularly in view of the growing smart ring, smartwatch, and smart phone industries. *See* RX-0290.28; RX-0291.25; RX-0292.23; JX-0016C (Martinez) at 29:15-22; JDX-0053C (Rowe) at 95:18-22, 101:8-16, 106:2-10; JDX-0002C at Exs. 1.0-1.1, 9.0-9.1. [REDACTED] in investments show lack of significance in the domestic industry. *Lelo*, 786 F.3d at 883. For example, Mr. Jarosz will testify that the evidence shows [REDACTED], including as represented in the following figure he created (independent of SEA’s [REDACTED], addressed further below):



1. Samsung's Proposed Comparisons

Ms. Rowe proposes several comparisons to show significance, but the evidence will show that they instead show *insignificance*.

a) Investment-to-Revenue

Ms. Rowe compares the claimed investments to SEA's U.S. mobile product revenues. JDX-0002C.96. No ratio is above █% (except the DI Ring, which was improperly allocated), and Mr. Jarosz will testify that Ms. Rowe's own ratios show insignificance. *Id.* at .96 and Ex. 9.0; *see also id.* Ex. 9.1 (showing ratios of █ for plant & equipment investments). Ms. Rowe presents no theory for how her comparisons show significance.

Mr. Jarosz will also testify that this comparison metric is not probative of significance here for the reasons explained by Commissioner Schmidlein in *Certain Replacement Automotive Lamps*, Comm'n Op. at 44-46: when claimed investments are estimated through sales-based allocations, dividing the investments by the DI Products' revenues (for the significance ratio) cancels out the patent-specific information, rendering it useless for evaluating the significance of investments in a given patent-protected product. This is why Ms. Rowe's proposed investment-to-revenue ratios are the same (█%) for the DI Phones and DI Watches and *would* be identical for the DI Ring without the error identified above. Mr. Jarosz will testify that Samsung did not mitigate this issue by using worldwide rather than U.S. revenue.

Samsung never grapples with how █ its claimed investments (especially for the DI Watches and Ring) are compared to Samsung's overall revenues. RX-0390.50 (2025 revenue for SEC was █); JX-0016C (Martinez) at 104:8-15 (admitting that █). Mr. Martinez testified that Samsung's business is █ relatively insignificant. JX-0016C (Martinez) at 172:6-16. Samsung also does not compare its

claimed investments to SEA's total revenues (rather than the cherry-picked mobile revenues), which is more appropriate given the cross-fertilization of SEA's expenses across different product groups (as discussed above regarding allocation). *See* JDX-0005C at Tab 13.

b) Investment-to-COGS

Ms. Rowe opined that comparing U.S. COGS to the claimed investments is a relevant measure of "the context of foreign costs," which Mr. Jarosz will testify is not. JDX-0002C.102. She compares the labor and capital (comprehensive) with SEA's U.S. COGS, showing ratios of [REDACTED] (DI Phone), [REDACTED] (DI Watch), and [REDACTED] (DI Ring). JDX-0002C.103 and Ex. 9.0. Mr. Jarosz will testify that the low percentages of the DI Phone and DI Watch investments show they are not significant, and the DI Ring is affected by Ms. Rowe's accounting error (discussed above). She also compares the DI Product plant and equipment investments to SEA's U.S. COGS costs. JDX-0002C.97. The ratios are less than [REDACTED] showing no significance. *Id.* at 97 and Ex. 9.1.

Mr. Jarosz will testify that comparing the claimed investments to U.S. COGS (as Ms. Rowe does) has a structural deficiency because the claimed investments (as allocated) were generated through revenue data, meaning that any variation in this ratio is driven by differences in the markup above COGs, such as extraneous factors including pricing strategy, demand conditions, or competitive market dynamics. He will further testify that Ms. Rowe's use of a U.S. COGS benchmark overstates the relative significance of U.S. investment, whereas a more appropriate comparison would be using worldwide COGS considered alongside measures of worldwide activity (discussed more below).

He will explain that the COGS comparison used does not capture foreign manufacturing costs or other costs related to the DI Products, such as R&D, design, engineering, and testing, because the COGS reflects only the [REDACTED] to SEA for goods imported into and sold within the U.S. JX-0016C (Martinez) at 80:3-25; JX-0022C (D. Kim) at 100:15-101:3. Samsung did not

provide evidence of what non-U.S. expenses are accounted for in COGS, and the COGS benchmark is subject to Samsung's (unknown) internal sales-based allocation. *See* JX-0016C (Martinez) at 80:3-81:5, 52:4-12 ([REDACTED]), 83:7-11. Samsung also did not produce (or acknowledge) an agreement with SEC concerning SEA's purchase of the DI Products, which could explain how the COGS number is generated or modified. JDX-0053C (Rowe) at 61:15-63:7 (admitting [REDACTED]). To the extent Samsung says that Oura has not shown any evidence to the contrary, that is neither factually true nor legally proper—it is Samsung's burden to prove its case, not Oura's.

Although Ms. Rowe compares her COGS ratio to Oura's efforts in a separate investigation, Mr. Jarosz will testify that Oura showed that nearly all of its activities were related to the DI Products, and Oura's significance metrics did consider worldwide (not just U.S.) COGS.

c) Investment-to-OPEX

Ms. Rowe compares the DI Product plant and equipment investments to SEA's operating costs for the DI Products. JDX-0002C.97. The ratios are [REDACTED] (DI Phone), [REDACTED] (DI Watch), and [REDACTED] (DI Ring), showing they are not significant. *Id.* at 97 and Ex. 9.1. She also compares the labor and capital (comprehensive), with ratios of [REDACTED] (DI Phone), [REDACTED] (DI Watch), and [REDACTED] (DI Ring), which are substantially different than her other measures of significance, indicating that these are not probative. *Id.* at 98 and Ex. 9.0. Mr. Jarosz will testify these do not show significance.

The U.S. OPEX numbers Ms. Rowe relies on are not detailed reports of the operating expenses for each product; rather, they (like COGS) are another result of black-box sales-based allocations performed internally by Samsung that it failed to describe. JDX-0002C.96 n. 566; JX-0016C (Martinez) at 82:4-23, 83:7-11; JX-0022C (D. Kim) at 65:23-67:18 (testifying that [REDACTED]); JDX-0053C (Rowe) at 229:9-18, 241:12-245:15 (unable to explain how [REDACTED])

For example, Samsung failed to explain whether its internal allocation of OPEX was based on unit sales or dollar sales, and if dollar sales are used, what measure (e.g., gross sales or net revenue) served as the allocation base. Nor does the record establish whether different SEA cost centers used different allocation-factor bases depending on the scope of the work performed (e.g., revenues for mobile-only cost centers or those inclusive of other product groups).

It is further not clear which SEA investments (although they are only SEA's costs) are counted within operating expenses, so the comparison is not useful. JX-0022C (D. Kim) at 57:23-62:21 (no knowledge of [REDACTED], [REDACTED], 101:7-101:20. Accordingly, as with the Revenue and COGS ratios, they are not reliable probes of investments that are also generated through sales-based allocations. And the OPEX column is for limited SEA operating expenses only, excluding [REDACTED], [REDACTED] that are related to the COGs. *See* JX-0016C (Martinez) at 51:20-52:12, 81:22-24.

As a result, as Mr. Jarosz will testify, the comparison is not useful and cannot be fully evaluated or rebutted. Further, this comparison merely shows DI Product investments compared with SEA's (estimated) operating expenses for those products—and comparing claimed investments in the DI Products to operating expenses for the DI Product expenses does not show the necessary context for SEA's investments. At the very most, this comparison shows the relative significance of U.S. commercialization costs that qualify as plant and equipment or labor and capital under Section 337 to the U.S. commercialization costs that do not. So it cannot probe whether the investment is significant to the DI Product, because the category of costs (which could be as trivial as putting a manual in a box) that make up SEA's qualifying investments are compared

with the same category of costs (e.g., the administrative overhead for the printers and the box shipments). JX-0016C (Martinez) at 148:5-149:8 (“[REDACTED] [REDACTED]”). Ms. Rowe does not provide the context to show why this comparison shows that the claimed investments are significant under Section 337.

d) Investment-to-All Mobile Investments

Ms. Rowe compares the DI Product plant and equipment investments to SEA’s mobile plant and equipment investments. JDX-0002C.99. The ratios are [REDACTED] and [REDACTED] (DI Phone) and a combined just [REDACTED] and [REDACTED] (DI Watch and Ring, which is improperly aggregated and below a reasonable significance threshold) for 2022 and 2024, respectively. *Id.* at 99 and Ex. 9.1. She also compares the DI Product labor and capital (comprehensive) investments to SEA’s mobile (comprehensive) investments. JDX-0002C.100. The ratios are [REDACTED] and [REDACTED] (DI Phone) and a combined [REDACTED] and [REDACTED] (DI Watch and Ring, which is improperly aggregated and below a reasonable significance threshold) for 2022 and 2024, respectively. *Id.* at 100 and Ex. 9.0.

As discussed for the investment-to-revenue ratio, which is not probative here because it relies on factors that are baked into the sales-based allocation, this particular comparison is not especially meaningful because it is comparing estimated investments with the amount excluded from that estimate, all of which is just a function of which products sold well. But Ms. Rowe’s comparison does show that the DI Products’ revenue allocation factors (JDX-0002C at Ex. 5.0), which is their revenue compared to the [REDACTED], are larger than their investment allocation factors, which is their investment compared to non-DI mobile product investments, indicating that the DI Products have a lower investment-to-revenue ratio than other products [REDACTED].

e) U.S.-to-Worldwide Revenues

Ms. Rowe compares SEA revenues from the DI Products with SEA’s overall U.S. mobile product revenues. JDX-0002C.92. But the DI Watch is only [REDACTED] with the DI Ring at only [REDACTED].

Id. She also compares SEA’s U.S. revenues from the DI Products with Samsung’s global net revenues from those products. JDX-0002C.105. Revenue comparison does not show significance. *Certain Soft Projectile Launching Devices, Components Thereof, Ammunition, and Products Containing Same*, Inv. No. 337-TA-1325, Comm’n Op. at 31-32 (Dec. 18, 2025) (EDIS ID 867953) (“*Certain Soft Projectile*”). Mr. Jarosz will also testify that the revenue metrics for the DI Watch and DI Ring are de minimis, and that the marked declines in the DI Products’ revenue signals decreasing significance. Ms. Rowe’s global revenue comparison at most shows that the U.S. is one of Samsung’s largest markets for the DI Products, but it does not show the significance of U.S. investments, and further suggests that other markets (including Korea) may be larger and have larger localization and commercialization investments than SEA. JX-0016C (Martinez) at 38:2-10 ([REDACTED]).

f) Headcount

Ms. Rowe applied her allocation factor to the employee count within the [REDACTED] groups, generating a total of [REDACTED] employees allegedly doing work on the DI Products by November 2025: [REDACTED] for the DI Phone, [REDACTED] for the DI Watch, and [REDACTED] for the DI Ring (although these total [REDACTED]). JDX-0002C.101-02. In sum, her (overinclusive) allocation attributes [REDACTED] doing work on the DI Ring and just [REDACTED] doing work for the DI Watch in 2025—among the [REDACTED] SEA employees that Ms. Rowe opined did work on mobile products. JDX-0002C (Rowe) ¶224 n.589. Using Ms. Rowe’s math (*id.*), this would be [REDACTED] for the DI Watch and [REDACTED] for the DI Ring. This is not significant in any context. And [REDACTED] employees ([REDACTED]) is not a significant number when compared to SEA’s workforce, even if limited to the [REDACTED] employees that are allegedly working on the DI Phones.

Further, Mr. Martinez testified that the spreadsheet Ms. Rowe relied on does not provide an accurate estimate of employees. JX-0016C (Martinez) at 110:5-111:10. Ms. Rowe also does

not explain how much of their work is related to the DI Products because she assigned them purely based on [REDACTED] and a blanket sales-based allocation, and there is no evidence of time-based or good-faith allocations. JDX-0002C at Ex. 9.3; *see also* JX-0016C (Martinez) at 33:2-4 ([REDACTED]), 26:17-25 (employees within groups may “[REDACTED]”). Further, Mr. Jarosz will testify that Ms. Rowe’s own presentation of allocable headcount shows a [REDACTED] in investments over the nearly four years accounted for, which is evidence of *insignificance*:

	Dec. 2022	Dec. 2023	Dec. 2024	Nov. 2025
DI Phone	[REDACTED]			
DI Watch				
DI Ring				
Total				

2. Comparisons Samsung Failed to Present

Although the Commission has not found any one comparison dispositive, a complainant must show sufficient context to prove a significant EDI. *Wuhan Healthgen Biotechnology Corp. v. Int’l Trade Comm’n*, 127 F.4th 1334, 1339 (Fed. Cir. 2025). Samsung failed to use comparisons showing this fuller context, and in many cases failed to provide evidence necessary to do them.

a) Comparisons to Investments in Other Products

Samsung did not compare its claimed investments with those in non-DI Products. There is no evidence of SEA’s (or Samsung’s) investments relating to products outside the mobile services group, such as home appliances, computers, and televisions. For example, Samsung touted its more than [REDACTED] U.S. facilities, but admits that just [REDACTED] are related—in any way at all—to the DI Products (without describing how). RX-0959C.8. This comparison would show (as Ms. Rowe’s own sales-based allocation suggests) that at least SEA’s investments in the DI Watches and DI Rings are a

“minor component” of SEA’s business. *Certain Environmentally-Protected LCD Displays*, ID at 289 (Apr. 16, 2024) (EDIS ID 821125). Samsung also did not explain how the DI Ring and DI Watch investments could be considered significant in light of its DI Phone investments.

b) Comparisons to Foreign Metrics

The Commission recognized that a company—even if its whole enterprise is solely directed toward a DI product—“may not have a domestic industry if the bulk of its investment are overseas.” *Certain Wearable Elec. Devices with ECG Functionality & Components Thereof*, Inv. No. 337-TA-1266, Comm’n Op. at 24 (Jan. 20, 2023) (EDIS ID 788332). There is no dispute that the bulk of Samsung’s work related to the DI Products is done overseas.

Mr. Jarosz will testify that worldwide metrics are more appropriate than U.S.-only metrics. Samsung is a global company, the DI Products are sold globally, and SEA is just one of many local commercialization arms, suggesting that worldwide comparisons would be helpful to understand the significance of SEA’s investments in the DI Products. RX-0390.58-65 (reciting wholly owned subsidiaries all over the world); *see also* JX-0022C (D. Kim) at 91:9-92:4, 94:18-95:2 (testifying that [REDACTED] [REDACTED]); CX-1469C (“[REDACTED] [REDACTED]”).

Mr. Jarosz will testify that the following worldwide benchmarks show that SEA’s investments in the DI Products are not significant (JDX-0005C at Tabs 17-25, showing full calculations):

- Claimed investments compared with worldwide revenues from sales of the DI Products:

PUBLIC VERSION

	Plant & Equipment	Labor & Capital	Labor & Capital (comprehensive) (per patent group rather than per product)
DI Phone			
DI Watch			
DI Ring			

- Claimed investments compared with worldwide COGS for the DI Products:

	Plant & Equipment	Labor & Capital	Labor & Capital (comprehensive) (per patent group rather than per product)
DI Phone			
DI Watch			
DI Ring			

- Claimed investments compared with worldwide OPEX allocated to the DI Products:

	Plant & Equipment	Labor & Capital	Labor & Capital (comprehensive) (per patent group rather than per product)
DI Phone			
DI Watch			
DI Ring			

The Commission has recognized that comparing domestic investment to worldwide revenues is probative, including because it indicates the domestic value-add and any associated profit margin. *See Certain Carburetors & Prods. Containing Such Carburetors*, Inv. No. 337-TA-1123, Comm’n Op. at 18 (October 28, 2019) (EDIS ID 692517); *Certain Electronic Candle Products*, Comm’n Op. at 18 n. 22 (Kearns, C.).

Samsung also admits that some SEA activities impact the DI Products outside of the U.S. *See* JX-0016C (Martinez) at 37:12-14 ([REDACTED]), 54:12-25 ([REDACTED]), 144:7-21 ([REDACTED]). Moreover, [REDACTED]. JX-0016C (Martinez) at 43:2-44:5, 44:15-46:7 ([REDACTED]). Accordingly, comparing SEA’s investments to just U.S. revenue overestimates the comparative investment and does not reflect significance.

c) Comparisons to Investment Counterparts

Samsung has also failed to show significance for its subsection (A) and (B) investments by comparing them with foreign investments in those subsections. *See Certain Crafting Machines & Components Thereof*, Inv. No. 337-TA-1426, Comm’n Op. at 15 (July 16, 2026) (EDIS ID 888252) (finding significance where domestic investments were “significantly greater than its corresponding foreign investments in the same categories of activities”); *Certain Video Capable Elec. Devices, Including Computers, Streaming Devices, Televisions, Cameras & Components and Modules Thereof*, Inv. No. 337-TA-1379, ID at 97 (Jan. 29, 2025) (EDIS ID 842988) (comparing Samsung’s global R&D investments with its domestic R&D investments). There is no evidence of investments made by Samsung’s commercialization counterparts to SEA or counterparts to SEA’s [REDACTED]. *See, e.g.*, JX-0016C (Martinez) at 37:1-21 (admitting [REDACTED]).

[REDACTED], 41:12-42:23
(identifying some counterparts to SEA).

Further, there is no evidence in the record that allows the Commission to compare the claimed investments with other plant & equipment or labor & capital efforts that contribute to the DI Products (including design, development, manufacture, etc.). Samsung admits that everything related to the DI Products, from design through manufacture, is done outside the U.S. *See* JX-0016C (Martinez) at 50:14-51:19 ([REDACTED]
[REDACTED]); JDX-0002C (Rowe) ¶¶205-06 ([REDACTED]
[REDACTED]). There is no evidence in the record of expenses incurred by Samsung's foreign cost centers (or even what those cost centers are). But to assess the "full picture" of Samsung's expenses, without seeing just a subset of cherry-picked expenses related to U.S. commercialization (which is what Samsung claims here), there must be evidence of Samsung's foreign investments, including evidence of "foreign manufacturing expenses." *Certain Crafting Machines*, Comm'n Op. at 10 n.4 (Commissioner Kearns) (warning against "a situation in which the only DI Product labor in the United States is a small amount of sales and marketing").

Samsung's comparisons to *U.S.* COGS and OPEX are not sufficient to do this at least for the reasons stated above. In addition, Mr. Jarosz will testify that simply considering U.S. COGS and OPEX is insufficient because they exclude foreign manufacturing costs allocated to DI Products sold outside the U.S. and non-manufacturing investments in the DI Products sold in the U.S., including research, design, development, engineering, project management, and any other cost that supports the DI Products that are not the U.S. commercialization costs exclusively handled by SEA. Mr. Jarosz will testify that failing to consider the non-U.S. contributions to the

DI Products will introduce significant distortions, especially when they are compared to COGS that are shrunk to apply to only U.S. sales.

Ms. Rowe also did not compare the claimed U.S. headcount to non-U.S. employees who play a role in the DI Products, making it impossible to compare and show significance for the headcount of U.S. employees. As another example, Samsung did not identify (much less provide lease information or square footage) non-U.S. properties used for activities or services related to the DI Products, making it impossible to compare and show significance for the facility expenditures and footprint in the U.S.

d) Comparisons to Marketplace Participants

Although Ms. Rowe opined on the U.S. smart watch, smart ring, and smart phone markets, she did not present any meaningful comparison with Samsung's claimed investments and the investments others in the marketplace have made. But Mr. Jarosz will also testify that Samsung's claimed investments are not significant compared to marketplace participant Oura. For example, he will testify that Oura's investments far outstrip Samsung's investments in its smart ring, including [REDACTED] U.S. facility investment in 2025 as compared with Samsung's [REDACTED] plant and equipment investment in the same year. RX-0413C. Mr. Jarosz will further testify that Samsung's U.S. labor and capital investment-to-worldwide-revenue ratios are dwarfed by Oura's analogous investment-to-revenue ratios, showing that Samsungs investments are comparatively insignificant.

3. Qualitative Significance

SEA is just the [REDACTED] for SEC. CX-1590C (Sheppard) at 13:17-22. Mr. Martinez testified that [REDACTED]
[REDACTED]. See JX-0016C (Martinez) at 50:14-51:2-19 ([REDACTED]) y

[REDACTED]; 46:8-47:13 ([REDACTED]
[REDACTED]); JDX-0002C (Rowe) ¶205 (Mr. Martinez
confirmed that “[REDACTED]”); RX-0519C ([REDACTED]
[REDACTED]
[REDACTED]); RX-0384C (Sheppard) at 120:5-121:2 ([REDACTED]
[REDACTED]); RX-0385C (Sheppard) at 73:3-75:7 ([REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED]; JX-0021C (J. Lee) at 40:15-41:11 (“[REDACTED]
[REDACTED]). Samsung’s other U.S. subsidiary, SRA, often works on Samsung products,
but Samsung does not claim that it made any domestic investments in the DI Products. JX-0016C
(Martinez) at 47:14-48:13.

Put differently, Samsung admits that the investments it claims for EDI are *only* those relating to the U.S. commercialization of products that have been researched, designed, developed, and manufactured outside of the U.S. and by non-SEA entities. And those efforts are very limited; for example, SEA [REDACTED] before their U.S. launch, demonstrating that [REDACTED]. CX-1590C at 41:13-42:4. The evidence shows that SEA makes few contributions to the DI Products (although it does [REDACTED]
[REDACTED]). *Digital Video-Capable Device*, ID at 164 (“[T]he true question is how important the United States is to the DI Product, not how important the DI Product is to the complainant.”).

Mr. Martinez further testified that [REDACTED]
[REDACTED]. JX-0016C
(Martinez) at 49:20-51:1. For example, Mr. Martinez admitted that [REDACTED]

[REDACTED]. JX-0016C (Martinez) at 147:4-12; CX-1590C (Sheppard) at 93:19-95:12 (t [REDACTED]
[REDACTED]
[REDACTED]). And much of the work done for the remaining portion of products, those imported into the [REDACTED] facility, is often [REDACTED]
[REDACTED]. JX-0016C (Martinez) at 148:11-149:8, 153:3-154:22. As another example, the [REDACTED] facility mainly houses employees who work [REDACTED]
[REDACTED]
[REDACTED]. CX-1590C (Sheppard) at 97:8-98:5. Further many of the facilities are primarily just for [REDACTED]
[REDACTED]. CX-1590C (Sheppard) at 108:6-109:3; JX-0016C (Martinez) at 50:8-13 (explaining that [REDACTED]
[REDACTED]).

Further, Samsung's investments in the DI Products do not contribute to the marketplace, especially given that Samsung's interest is merely in keeping its own Galaxy ecosystem attractive to current mobile users, and not in advancing the technology or market forward generally. Martinez 1468 Dep. at 237:13-19 ("[REDACTED]
[REDACTED]"). Ms. Rowe opines that Samsung's efforts are important to the wearable market but does not tie these to the DI Products or show how Samsung's U.S. efforts support the industry. Ms. Rowe further makes comparisons to Oura that are not analogous, least of which because the core of Oura's business is the ring, and Oura is the market leader and originator of the smart ring market. *Infra* Section X.C.

Mr. Jarosz will further testify that Samsung’s claimed investments are not significant in view of the size of the relevant markets. For example, he will explain that the U.S. smartphone market is very large, which drives up the sales-based allocation factor Samsung and Ms. Rowe use, even though those large U.S. sales are attributable to non-U.S. work (including product design and development that happens in Korea). But when viewed in the broader context of the DI Products and their global chain, the real contribution by SEA is relatively minor.

Samsung claims that the domestic investments are necessary to bring the DI Products to the U.S., but Mr. Jarosz will testify that this “but-for” cause of the DI Products being available is not sufficient to prove significance. *See Certain Mobile Devices with Multifunction Emulators*, Inv. No. 337-TA-1170, ID at 182 (Mar. 16, 2021) (EDIS ID 738549) (rejecting complainant’s assumption that commercialization, without more explanation of importance, is a domestic industry). Moreover, the evidence shows that these efforts could be handled by SEC employees, who have [REDACTED]. *See JX-0016C* (Martinez) at 54:7-22 (admitting [REDACTED]); Martinez 1468 Dep. at 178:8-179:21 (testifying [REDACTED]). This is not a case where Samsung contends that the DI Products require SEA’s investments to exist; by contrast, Samsung readily admits that [REDACTED]. *Contrast with Certain Video Capable Devices*, ID at 91 (“Without [the] investments, there would be no [DI] products.”). Moreover, the significance of “necessary” claimed investments must still be compared with all other activities that are necessary for the product, such as their design, development, manufacture, and shipment. *Certain Soft Projectile*, Comm’n Op. at 42 (stating that “while design may be important for sales of a product, so are most or all other activities involved in bringing the product to market, including manufacturing.”). For

example, Mr. Jarosz will explain that SEC's R&D expenses were approximately [REDACTED] of revenue in 2025. RX-0390.50, .100.

Mr. Jarosz will also testify that a substantial portion of SEA's U.S. activities purportedly related to the DI Products would likely exist in the same form and scale even without the DI Products, which is supported by the existence of the same [REDACTED] predating many of the DI Products. *See* RX-0381C; RX-0382C.

F. Samsung's Opinions and Records Are Not Reliable

Finally, the evidence Samsung provided for assessing significance are not the most useful or reliable for doing so. For example, Samsung claims rental investments, but does not identify how the facilities support the DI Products or why the investments seem disproportionate to the activities there—including [REDACTED], and [REDACTED]. JX-0016C (Martinez) at 172:11-24 (unable to provide context for [REDACTED] facility and admitting no knowledge of how the work in [REDACTED] relates to the DI Products, if at all); 173:22-174:13 (unable to provide context about [REDACTED] and admitting on behalf of Samsung no knowledge about whether the work in the Irving facility has to do with DI Products). Further, the financial record Samsung collected for production in this case and identify as the sole source of its claimed investments under Section 337 are not reliable for that purpose and are not accompanied by evidence that could be used to explain or support those records. *See* JX-0022C (D. Kim) at 95:8-98:12 (refusing to admit that entries in RX-0391, including [REDACTED] [REDACTED] were duplicates); JDX-0053C (Rowe) at 228:15-229:8 (admitting Samsung [REDACTED] without explanation).

Ms. Rowe's opinion heavily relies on information she said she learned from SEA employee Mr. Martinez during an hour-long phone call. JDX-0053C (Rowe) at 24:1-4, 29:20-22 (could not

recall Mr. Martinez lacking knowledge on anything asked during the call), 194:14-195:7 (discussed more than one hundred expense groupings for categories outside of Mr. Martinez's normal Product Management responsibilities, in addition to reviewing "all" of Exhibits 4.0 and 5.0 with him); 34:8-11 (testifying their phone call covered the "number of costs identified line by line"). Mr. Martinez was deposed on these issues for several hours and could not provide related information to Oura's counsel. For example, Ms. Rowe testified that Mr. Martinez told her that [REDACTED]. JDX-0053C (Rowe) at 119:2-120:18. But Mr. Martinez had testified that activities within these cost centers [REDACTED] [REDACTED]. JX-0016C (Martinez) at 162:8-164:17. He further did not identify [REDACTED] as a relevant business unit (whereas he did identify [REDACTED] for mobile products) during his deposition and identified "[REDACTED] as a separate business unit from [REDACTED] [REDACTED] *Id.* at 97:5-25.

G. Conclusion

In conclusion, Samsung has failed to carry its burden to show that its claimed investments are a reliable accounting of investments in the DI Products and that they are significant relative to probative benchmarks.

X. REMEDY AND BONDING

No remedy is warranted because Samsung has failed to prove a violation of Section 337. However, the evidence will show and Mr. Jarosz will testify that any imposed remedy should have carveouts, and no bond (or a bond of 0%) is warranted during the PRP.

A. Remedy Exemptions

As Oura's witnesses will testify, a transition period and warranty exemptions are appropriate to protect the U.S. industry. Mr. Jarosz will testify that a twelve-month transition period is warranted because Oura has concrete plans to [REDACTED]

[REDACTED]

[REDACTED] See, e.g., JX-0029C (Vincent) at 22:7-20; 62:22-63:13; RX-0417C, RX-0419C; RX-0452C; RX-0263C.6-7, .9, .29; RX-0557C; JX-0030C (Schleich) at 62:21-63:1. A transition period should be entered to allow Oura to shift production

[REDACTED]

[REDACTED].

There should be a warranty carveout for the Oura rings to allow for replacements promised under warranty obligations. *Certain Personal Data and Mobile Commc'ns Devices and Related Software*, Inv. No. 337-TA-710, Comm'n Op. at 72-73 (Dec. 29, 2011) (EDIS ID 467457) (providing exemption to allow for replacements permitted under warranty obligations). As Oura's witnesses and Mr. Jarosz will testify, Oura provides at least twelve months, and often longer (based on the retailer and a grace period), of a guarantee to replace damaged or malfunctioning rings with either refurbished or newly imported rings. JX-0028C (Murrin) at 39:24-40:6 ([REDACTED]), 40:14-25 ([REDACTED]), 41:23-42:7 ([REDACTED]), 75:9-76:22 ([REDACTED]), 46:19-47:1 ([REDACTED]); RX-0456C. As Mr. Jarosz and Oura's witnesses may testify, [REDACTED]. JX-0028C (Murrin) at 95:4-19; RX-0457C; RX-0458C; RX-0459C.5; RX-0460C; RX-0461C. At the time of product purchase, some consumers pay for an additional Extended Protection Plan that extends the replacement option to two or three years and expands conditions for replacement. RX-0389; RX-0388; RX-0455C.10.

Mr. Jarosz will testify that consumers lose the benefits they expected through the standard and extended warranties if they cannot receive replacement rings due to an exclusion order. *See* JX-0028C (Murrin) at 91:25-93:2, 79:14-22 (agreeing that w [REDACTED]). Mr. Jarosz will also testify that users who lose their health data because they could not quickly replace their Oura ring will be significantly harmed, and a refund—which is not Oura’s policy—would not compensate users who have lost the utility of their collected data and are not able to continue collecting that data with a substitute ring or must incur switching costs associated with starting over with another brand. *See, e.g.,* JX-0028C (Murrin) at 91:25-93:2 ([REDACTED]). Mr. Jarosz will further testify that Oura’s large partner network, number of users, algorithmic maturity and performance, and subscription model—including a prepaid annual subscription—also show the need for a warranty carveout. RX-0260C; RX-0261C; RX-0463C; RX-0464C.

B. Certification Provision

The Commission should not depart from its “standard practice” of issuing a certification provision. *Certain Composite Aerogel Insulation Materials & Methods for Mfg. the Same*, Inv. No. 337-TA-1003, Comm’n Op. at 62-63, 2018 WL 8648369, *40 (EDIS Doc. ID 637154) (Feb. 22, 2018). Mr. Jarosz will testify that the technology and economic realities of this case especially warrant a certification provision. Finally, a certification provision will ensure that no remedy should be applied against any rings that are contracted for use or purchase by the U.S. government, including Oura’s forthcoming [REDACTED] *Certain Personal Data*, Comm’n Op. at 75 (“Military and other U.S. government sales, however, are exempted from exclusion orders by statute. 19 U.S.C. § 1337(l); 28 U.S.C. § 1498(a).”); JX-0030C (Schleich) at 62:13-63:6, 64:11-22.

C. Bonding

Samsung failed to meet its burden to prove that it has experienced harm from Oura's accused products warranting a bond, and it has not proposed—much less shown—any bond amount. *See Certain Rubber Antidegradants, Components Thereof, and Prods. Containing Same*, Inv. No. 337-TA-533, Comm'n Op., at 39-40 (July 21, 2006) (EDIS ID 259120) (“[T]he complainant has the burden of supporting any proposition it advances, including the amount of the bond.”); *Certain Magnetic Tape Cartridges and Components Thereof*, Inv. No. 337-TA-1058, Comm'n Op. at 73 (Apr. 9, 2019) (EDIS ID 672595). Accordingly, no bond should be entered.

Samsung's sole assertion of harm from Oura's products is that Oura's Gen 3 and 4 rings compete with its DI Ring (which, as set out above, does not have a domestic industry and consequently is not eligible for protection by a bond). Mr. Jarosz will testify that although Samsung's DI Ring harms Oura by taking away sales, which Oura showed in the 1468 Investigation, Samsung has not shown evidence of the inverse; *i.e.*, that Samsung will be harmed enough by Oura's Gen 3 or Gen 4 ring to warrant a bond. *Certain Light-Based Physiological Measurement Devices and Components Thereof*, Inv. No. 337-TA-1276, Comm'n Op. at 122-23 (Nov. 14, 2023) (EDIS ID 808521) (no bond where complainant made only “vague assertions as to the ‘competitive status of the parties’”). As discussed in connection with EDI, any lost sales of the DI Ring to the Oura Gen 4 (which Samsung has not proven) would anyway be de minimis to Samsung given its overall U.S. revenue. Mr. Jarosz will also testify that Samsung is not the market leader and so sales that may have gone to Oura's Gen 4 may go to many of the alternatives on the market. RX-0291.138. Further, he will testify that the DI Ring is not generally a substitute for the Oura Gen 4 because the DI Ring can be used only with Android phones, which only about one-third of U.S. consumers have, and particularly designed for use by consumers with a Samsung phone. RX-0290.29, .112. Indeed, Mr. Martinez testified that one of the primary reasons behind

launching the DI Ring was to [REDACTED].
Martinez 1468 Dep. at 206:9-19 (testifying that [REDACTED]
[REDACTED]
[REDACTED]); 237:13-19 ([REDACTED]
[REDACTED]
[REDACTED]).

Moreover, Samsung addresses only the Gen 3 (no longer sold) and Gen 4 ring without contending that Oura's Gen 5 and forthcoming [REDACTED] ring (with expected commercial release [REDACTED]) will compete. Samsung has thus failed to prove that these products, which Oura's witnesses will testify will replace Gen 4, will cause harm.

Mr. Jarosz will also testify that a 0% rate is warranted even if the Commission determines that a bond is appropriate because Samsung has not shown evidence supporting a reasonable bond rate. Mr. Jarosz will testify that the traditional price differential test used to identify a bond amount that will compensate for price-based competitive advantage is not reliable here, and also that it cannot be used for setting a bond above 0% because by any reasonable calculation the price of the Oura ring is higher than the DI Ring, especially when taking into account the full cost of Oura's ownership that includes subscription pricing (which Samsung does not offer), and given the higher pricing of the Gen 5 ring (had Samsung properly identified it) and Samsung's discounts reduce the average selling price of the DI Ring to around \$[REDACTED]. JDX-0002C (Rowe) at Ex. 7.4; *see also Certain Laparoscopic Surgical Staplers, Reload Cartridges, & Components Thereof*, Inv. No. 337-TA-1167, Comm'n Op. at 81 (Dec. 20, 2021) (EDIS ID 758899) (setting the bond at 0% where the accused products were more expensive than the DI products); *Certain Cameras, Camera Sys., and Accessories Used Therewith*, Inv. No. 337-TA-1400, ID at 279 (July 10, 2025) (EDIS 857630)

(“In such circumstances, even when there is direct competition, the Commission typically sets the bond at zero percent.”). Mr. Jarosz will testify that Samsung’s licenses are too broad and otherwise unrelated to support a bond, and Samsung’s corporate witness admitted [REDACTED] [REDACTED]. JX-0018C (M. Kim) at 61:10-67:10. Mr. Jarosz will further testify that a 100% bond would be excessive and unjustified, particularly given Samsung’s failure to assert or support any reasonable bond. *Certain L-Tryptophan, L-Tryptophan Products, and Their Methods of Production*, Inv. No. 337-TA-1005, Comm’n Op. at 52-53 (Jan. 11, 2018) (finding the ALJ “correctly recommended a zero percent bond” where complainants “fail[ed] to satisfy their burden to support a 100% bond or to properly explain why a reasonable royalty or price differential would be impractical”); *Certain Elec. Devices, Including Streaming Players, Televisions, Set Top Boxes, Remote Controllers, & Components Thereof*, Inv. No. 337-TA-1200, Comm’n Op. at 42-43 (Dec. 3, 2021) (EDIS ID 757848) (imposing zero percent bond where complainant “has not carried its burden of establishing the need for a bond or the amount of any bond”). Samsung has provided no rebuttal facts or opinions. JX-0018C (M. Kim) at 67:3-10 (corporate witness unable to propose a bond).

XI. CONCLUSION

For the reasons stated above and as will be established at the evidentiary hearing, there should be no finding of violation of Section 337 against the Oura Respondents.

CERTIFICATE OF COMPLIANCE

I hereby certify that the foregoing brief complies with the word-count limit under Order No. 11 because it contains 59,948 words as calculated by Microsoft Word's word-count feature. This total excludes the captions, table of contents, table of authorities, table of abbreviations, table of asserted claims, words in figures, the word count certificate, the signature block, and the certificate of service pursuant to Ground Rule 1.7.

Dated: August 10, 2026

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*Counsel for Respondents Ouraring Inc. and
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**CONTAINS CONFIDENTIAL BUSINESS INFORMATION SUBJECT TO
PROTECTIVE ORDER**

Dated: August 10, 2026

Respectfully submitted,

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I hereby certify that a copy of the foregoing **RESPONDENTS' PRE-HEARING BRIEF** was served upon the following as indicated below on August 10, 2026:

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